



Engineering Standards and Procedures Manual

October 1, 2015

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Town of Huntersville Engineering Standards and Procedures Manual

The Town of Huntersville's Engineering Standards and Procedures Manual (ESAPM) is provided as a resource that will assist in ensuring compliance with all Town requirements related to proposed land development activities inside the Town limits and within its Extraterritorial Jurisdiction (ETJ).

It is the Town's goal that the ESAPM present clear and concise technical requirements, policies, and procedures while providing the guidance and details necessary for an effective and efficient process.

The ESAPM is intended as a supplement to the Town Zoning Ordinance, Subdivision Ordinance, and Water Quality Design Manual. County, State, and Federal agencies may also have additional requirements not provided for or referenced within this manual. This manual does not relieve the design professional of the responsibility to correctly incorporate the provided information. It is the Design Engineer's responsibility to provide technical adequacy of the design using engineering judgment, experience, and sufficient knowledge in providing all related design elements.

The Town of Huntersville's Director of Engineering & Public Works shall be responsible for incorporating revisions as deemed appropriate based on a continual review of the ESAPM. Copies of the manual will be made available for a fee through the Huntersville Engineering & Public Works Department. The ESAPM is also available for online viewing on the Town of Huntersville website.

(<http://www.huntersville.org/Departments/EngineeringPublicWorks/EngineeringPlanReview/tabid/201/Default.aspx>)

Where discrepancies exist between this manual and any adopted Town Ordinance, the Ordinance shall govern. The latest revision of the "NCDOT Standard Specifications for Roads and Structures" and the "NCDOT Design Manual" shall apply to all roadway and storm drainage construction unless otherwise specified herein this manual.

This manual was created to capture most, but not all, scenarios related to development within the Town of Huntersville. Town of Huntersville Engineering & Public Works reserves the right to enforce standards not included within this manual, which uphold the Town's initiative to maintain a safe environment for its citizens.

I. Administrative Procedures

A. Introduction

Processes and procedures for various plan review and development standards are discussed in this section. Each section provides information on the process, standard, or the plan review agency to contact regarding that process.

B. Electronic Plan Management

The Town along with Mecklenburg County utilizes an electronic plan management (EPM) process. The EPM system is a web-based work flow tool that allows architects, engineers, and designers the ability to oversee the plan review and permitting process. This EPM process is required for all new or proposed modifications to existing plans. Additional plan submittal processes/requirements can be found on the Mecklenburg County Land Development webpage at <http://charmeck.org/mecklenburg/county/WaterandLandResources/LandDevelopment/Pages/default.aspx>.

Along with the submission of plans through the EPM process, an application for plan review is required. For plan review applications, contact the Town of Huntersville Planning Department at 704-875-7000 or at <http://www.huntersville.org/Departments/Planning.aspx>.

C. Engineering Plan Review Checklist

The engineering plan review checklist is a detailed list of the items to be reviewed by the Town Engineering and Public Works Department or designee. The preliminary plan must include, at a minimum, the information described in the Town's Subdivision Ordinance and/or other applicable ordinances. A copy of the engineering plan review checklist is included in the Appendix.

Note: The Town Planning Department maintains their own plan review checklists. Additionally, the duration of the plan review varies by review agency. Plan calculations and details are reviewed at this stage.

D. Fees

Fees for plan review are set and collected by the Mecklenburg County Land Use and Environmental Service Agency (LUESA) and/or Town of Huntersville Planning Department. Fees vary by the type and size of development and are updated on an as needed basis. Plan review fees can be found at the below web addresses:

<http://charmeck.org/mecklenburg/county/WaterandLandResources/LandDevelopment/Pages/Land%20Development%20Fees.aspx>

<http://huntersville.org/Departments/Planning/PermitsProcess.aspx>

E. Driveway Permits

Town Driveway Permit

A Town of Huntersville Driveway Permit is required for all new or proposed modifications to connections to Town streets except an individual single family residence. A copy of the Town Driveway Permit Application is in the Appendix. The Town fee for a driveway permit is \$200. If a property owner is proposing to do work within Town maintained right-of-way, a Town Encroachment Permit may be required. Contact the Town Engineering and Public Works Department at 704-766-2220 to confirm if a permit is needed.

Note: Two signed original copies of the driveway permit application along with two sets of plans are required for submission to the Town Engineering and Public Works Department. A separate encroachment permit is not needed if a driveway permit has been obtained.

NCDOT Driveway Permit

When accesses and/or driveways to North Carolina Department of Transportation (NCDOT) maintained facilities are proposed or are proposed to be modified, contact the NCDOT Division 10 District 2 office at (980) 523-0000. Forms are available on the web at <http://www.ncdot.gov/>. The Town will review the NCDOT driveway permit applications for accesses proposed within the Town of Huntersville.

F. Encroachment Permits

The Town of Huntersville requires that an encroachment permit be obtained when construction activity, including installation of temporary or permanent structures, is proposed under, on, or over property in which the Town has property rights. Property rights include but are not limited to street rights of way, utility easements, or other owned property. An Encroachment Permit is required regardless of any other approvals (excluding a driveway permit), such as building permits or Planning Department entitlements.

Encroachment Permit applications are processed through the Town of Huntersville Engineering and Public Works Department. A copy of the Town of Huntersville Encroachment Agreement is included in the Appendix.

G. PE Certification Process for Subdivisions and Streets

The Town of Huntersville requires that all streets proposed to be taken over by the Town for maintenance be reviewed, inspected, and certified by a licensed professional engineer registered in the state of North Carolina for adequate construction. A preconstruction meeting with the Town Engineer and/or Town Construction Engineer is required. Review of street construction by the certifying Engineer is required throughout the construction process. PE Certification is required for all developments in which the first submittal of the Town sketch plan or construction plans (if no sketch plan was submitted) occurred after the adoption of this manual on November 16th, 2009. A copy of the PE Certification requirements is included in the Appendix.

All sketch plans and construction plans submitted to the Town for subdivision approval must have the following statement on the cover sheet of the plan set:

The Town of Huntersville requires that all streets proposed to be taken over by the Town for maintenance be reviewed, inspected, and certified by a licensed professional engineer registered in the state of North Carolina for adequate construction. Review of street construction by the certifying Engineer is required throughout the construction process. Refer to the Town of Huntersville Engineering Standards and Procedures Manual for additional information including the required certification form.

H. Bonding

The following list contains information regarding the bonding process including minimum amounts, duration, and security type.

1. Release of the final subdivision plat will not occur until the improvements required for the area of the final plat are constructed and a final inspection has been performed and found to be in conformance with the plans approved by the Town, or a security has been posted with the Land Development Bond Coordinator of the applicable department and all required documents are received in their entirety.
2. Securities shall be posted for a minimum of six months with a two year maximum. The security shall be posted and remain in force until the construction is complete and found to be in conformance with the plans approved by the Town. The security will be reevaluated when an extension to the security is being considered.
3. Upon receipt of a notice from the bond holder, a final inspection will be made by the Town Engineer to check completeness of the project.
4. One type of security may be replaced by another type of security in certain situations. The amount of the replacement security will be based on the Town's Engineer Estimate of the work remaining. If the estimate of work results in a lower amount, the replacement security will be treated as a reduction. Certain situations will require an increase in a security and in such cases the replacement security shall be required to equal the higher amount.

5. If Mecklenburg County and the Town have different bonding or security requirements, the higher minimum amount shall apply.
6. A one-time reduction in security will be allowed if requested in writing by the principal party of the security. Additional reductions may be approved at the discretion of the Town Engineer. However, the security shall never be less than 15 percent of the total bond or \$20,000 for the Town unless approved by the Town Engineer.
7. Securities in the form of a Letter of Credit must be drawn on a full service bank in Mecklenburg County.
8. The applicant will be required to post a maintenance bond per Town requirements before the security for completion of subdivision is released.

I. Final Inspection

A final inspection of all streets to be turned over to the Town for Maintenance must be inspected by the Town or Town designated inspector. Contact the Town Engineer or Town Construction Engineer for scheduling of final inspections.

When a phase/map of a subdivision reaches 75% occupancy, the phase/map will be considered eligible for acceptance by the Town. The Town may consider acceptance at an occupancy as low as 60%; however, in the case of early acceptance (<75%), the developer will be required to post a 2 year maintenance bond. The procedures for requesting a final inspection are as follows:

1. Submit an executed "Request for Final Inspection Form", along with a "PE Certification for Subdivisions and Streets" form. (refer to Appendix).
2. A representative from the Town and/or Mecklenburg County will proceed with the Final Inspection.
3. Necessary repairs will be marked in the field, and indicated on a punchlist, which shall be valid for a period of thirty days.
4. When the necessary repairs have been completed, the Town should be contacted to verify the repairs have been completed.
5. When all conditions have been met, the developer may then proceed with requesting the Town to accept the streets for maintenance.

J. Street Maintenance Acceptance

The Town of Huntersville may consider the acceptance of privately owned streets upon the written request of the owning entity. Streets will only be accepted in their entirety or by block and street construction must be completed at time of petition.

To initiate the acceptance procedure, the following information shall be submitted to the Town Manager's Office:

1. A letter to the Town Manager requesting that the Town consider accepting subject streets for public maintenance; and
2. One copy of the recorded map(s) of the subject street(s); and
3. One completed "Application for Street Maintenance Acceptance" form(s) of the subject Street(s) or, in the case of existing streets petitioned by residents, a "Street Maintenance Acceptance Petition" signed by eighty (80) percent of the street's residents;
4. For new streets petitioned by subdivider, written verification from Town Engineering Department that the street(s) have been constructed according to the required standard and completed.

Once the required information and form have been submitted, an inspection of the subject street(s) shall be conducted by the Town. The Town shall notify the owning entity of all construction

deficiencies identified by the Town. Upon correction of the deficiencies, the Board of Commissioners shall then consider the acceptance of the subject street(s).

Petitions for acceptance will be presented to the Board of Commissioners on a quarterly basis. The Board of Commissioners of the Town of Huntersville shall not adopt any resolution accepting a new street unless:

- The Board receives a report from the Town Manager that all conditions of street acceptance are met.
- The Board of Commissioners determines that such street corresponds in its location and aligns with a street shown on preliminary subdivision plat formally approved by the Planning Board of the Town of Huntersville or that said street was established as a public street prior to the adoption of this policy and therefore not subject to this policy.

The street acceptance policy includes streets, curbs, gutters, sidewalks, and all items located within the right-of-way. A copy of the Street Acceptance Policy and application form are found in the Appendix.

K. Street Lights

Request Procedure

For neighborhood street lights, individual street lights, or to have the Town pay for street light bills, neighborhoods must complete and submit a request form provided by the Town of Huntersville. The Town will record and log each request at the time it is submitted to the Town Manager's office. The lighting must be located within a public street right-of-way. Streetlights are installed and billing paid by the Town on a chronological basis, as funding is available.

After a field survey has been completed, a financial spreadsheet estimating the cost of the project shall be completed by the Town Manager, or designated official. After the request is received and reviewed by Town staff, the contact person on the request form will be notified of any action regarding their request.

For neighborhood lighting requests, the neighborhood will be provided with a street lighting design for signature and approval. At least 75 percent of the residents on the streets are required to approve the plan before authorization will be given to the Utility Company to install streetlights. It is recommended that every resident review the street lighting design whether or not that signature is required. Only one signature per household is required.

When a completed petition is submitted to the Town Manager's office, streetlights will be installed, as funding is available. A copy of the Street Lighting Request form and policy is located in the Appendix.

II. Design Criteria

A. Introduction

The following sections present minimum design criteria for the design of public streets, storm drainage, street lighting, street and roadway signage for traffic regulation and street identification, and landscaping.

B. Street Design

For use in designing Residential and Retail/Mixed-Use Public Streets

Posted Speed Limit	25	30	35	40	45
Stopping Sight Distance* (feet)	155	225	285	350	415
Intersection Sight Distance - Left-Turn Movement From Stop*and** (feet)	280	365	425	485	545
Intersection Sight Distance - Right-Turn From Stop*and** (feet)	240	315	370	420	475
Minimum Horizontal Radius (Normal Crown) (feet)	200	430	675	980	1470
Minimum K value for Crest Vertical Curves	11	24	37	56	81
Minimum K value for Sag Vertical Curves	25	43	58	75	94
Maximum Longitudinal Grade	10 percent				
Maximum Longitudinal Grade within 125 feet of intersection (measured from intersecting street nearest edge of pavement of travel way)	5 percent				
Intersection Angle Range	75 to 105 degrees				

* Values will need to be adjusted for grades of more than +/- 3 percent

** Values to be adjusted for streets with more than two total lanes; measurements to be taken 14.5' from travel lane

For developments located in TOD, TND, and TC zoning jurisdictions, lower posted speed limits may be allowable.

Design standards that apply for the ETJ are taken from the latest edition of the NCDOT design manual *Subdivision Roads*. Any revisions to *Subdivision Roads* will supersede the design standards given in the Town of Huntersville Engineering Standards and Procedures Manual for ETJ streets. However, under no circumstances shall an NCDOT/ETJ standard be less restrictive than what is required by the Town of Huntersville.

Provisions of adequate stopping sight distance may require use of larger K values than the minimums listed above. The Town of Huntersville reserves the right to prescribe more stringent sight distance standards and/or means to achieve adequate sight distance than those listed above. Recordation of sight distance easements may be required on plats prior to approval.

The minimum distance between two horizontal curves is 50 feet. Longer distances may be needed based on the specifics of the roadway design.

Minimum curb and right-of-way radius measured from face of curb (when intersecting streets have different classification, use the more restrictive):

- Residential Local Street – 20 feet
- Residential Local Street to Residential Alley – 10 feet
- Residential Collector – 25 feet
- Retail/Mixed-Use Local – 25 feet
- Retail/Mixed-Use Collector – 25 feet
- Industrial Local and Collector – 35 feet

For minimum intersection separation, refer to block length minimums in the Zoning or Subdivision Ordinance. When distances in those ordinances are not specified, use the following criteria:

- Along local streets – 125 feet
- Along collector streets – 350 feet
- Along thoroughfares – to be determined by Town and/or NCDOT on a case-by-case basis

Intersection offsets/separation from a thoroughfare, at signalized intersections, or at intersections that may become signalized in the future may need to be greater than these minimums and will be determined by the Town and/or NCDOT on a case-by-case basis.

Design criteria for arterial streets shall be established jointly by the Town Engineer and the NCDOT on a case-by-case basis using the latest edition of the American Association of State Highway and Transportation Officials (AASHTO) A Policy on Geometric Design of Highway and Streets and/or NCDOT Roadway Design Manual.

Intersection corner easements – A minimum 35 foot x 35 foot triangular maintenance easement (measured along right-of-way lines) shall be provided at each intersection corner where any street type intersects a collector or thoroughfare. A minimum 15 foot x 15 foot triangular maintenance easement (measured along right-of-way lines) shall be provided at each intersection corner where two local streets intersect. An additional 10 foot x 70 foot triangular maintenance easement shall be provided at intersections connecting to NCDOT maintained roadways (measured along right-of-way lines). Driveways (no formal right-of-way) to serve a single project may be required to provide triangular maintenance easements as determined on a case by case basis. Other triangular maintenance easements or sight distance requirements may be required by the NCDOT or the Town at all intersections.

Refer to the NCDOT Subdivision Roads Minimum Construction Manual for development criteria for sites located within the Town's Extraterritorial Jurisdiction (ETJ) within these areas governed by the Town of Huntersville Engineering Standards and Procedures Manual and the NCDOT Subdivision Roads Minimum Construction Standards Manual. The more restrictive standard shall apply.

Sidewalks and Driveways

1. Planting strip adjacent to sidewalk shall be graded to one quarter inch per foot (min.) up to one and one quarter inch per foot (max.), except where excessive natural grades make this requirement impractical. In such cases, the Town Engineer may authorize a suitable grade.
2. Sidewalk widths shall be a minimum of five feet unless otherwise specified.

3. Accessible ramps are required where sidewalks intersect curbing at any street intersection and curbed driveway connections.

Roundabouts

Refer to the Manual on Uniform Traffic Control Devices (MUTCD) for roundabout signage and pavement markings.

C. Storm Drainage

1. In addition to this manual, all storm drainage design shall conform to the standards and specifications as provided in the Charlotte-Mecklenburg Storm Water Design Manual, and NCDOT Standards Specifications for Roads and Structures. If conflicts occur, the more restrictive standard shall govern.
2. Reinforced concrete pipe may be used in all storm drain applications. High Density Polyethylene Pipe (HDPE) may be substituted for pipe diameters of 48 inches or less as approved by the Town Engineer. Culverts 60 inches in diameter or greater may be Corrugated Aluminized Metal Pipe (CAMP) or aluminum with a minimum 14 gauge metal subject to approval of the Town Engineer.
3. The minimum cover for all pipes is two feet measured from the final surface. Special applications for less than two feet of cover will be reviewed and approved by the Town Engineer individually. The maximum cover for storm drainage pipes shall at a minimum comply with the requirements of the NCDOT Roadway Design Manual, Part I, Section 5, and "Drainage Design". Storm pipe design that exceeds these criteria may be approved at the discretion of the Town Engineer.
4. All storm drain structures over three feet six inches in height must have steps in accordance with standard details set forth in this manual.
5. All graded creek banks and slopes shall be at a maximum of two feet horizontal to one foot vertical (2:1) and not to exceed ten feet without terracing or the slopes shall be designed by a Professional Geotechnical Engineer and approved by the Town Engineer on a case by case basis.
6. Adequate storm drainage shall be provided throughout the development by means of storm drainage pipes or properly graded channels. All pipes shall be of adequate size and capacity, as approved by the Town Engineer, to carry all storm water in its drainage area.
7. In accordance with the Town Zoning Ordinance, the Town Engineer or duly authorized designee shall review the drainage plan for compliance with the standards contained in the current edition of the Town of Huntersville Engineering Standards and Procedures Manual and the Charlotte-Mecklenburg Storm Water Design Manual and all other relevant and appropriate standards established by the Town Engineering Department.
8. Sub-surface drainage shall be provided where the ground water level is likely to be near the surface. In capillary soils, the water level should be four to six feet below the surface to prevent the rise of moisture into the subgrade. Subdrains shall be used to lower ground water in low areas in the street.

9. All Storm Drainage Easements must extend down stream of flared end sections to an appropriate property line or buffer. Overlapping of storm drainage easements shall be approved by the Town Engineer on a case by case basis.
10. Storm Drainage Easements shall be provided for all storm drainage pipes and shown on site plans, construction plans and plats with widths specified in detail 314.1. The following note shall be placed on all grading plans and plats; "The purpose of the storm drainage easement (SDE) is to provide storm water conveyance. Buildings are not permitted in the easement area. Any other objects which impede storm water flow or system maintenance are also prohibited."
11. In areas where the Floodway Regulations are applicable, the Future Conditions Flood Fringe Line, FEMA Flood Fringe Line, Community Encroachment Line, and FEMA Encroachment Line shall be shown on the preliminary plan and the final plat. An application for a Floodlands Development Permit shall be submitted to Mecklenburg County in accordance with the requirements set forth in the Town/County Floodway Regulations.

D. Utilities

Avoid placement of sewer manholes in gutter pans, the crown of the road, wheel paths, wheelchair ramps, and over stormwater lines.

Avoid placement of water lines under roadway pavement.

Water valves shall not be placed in curbing.

E. Street Lighting

1. Light Spacing

Street lights shall be spaced a minimum of 250 feet to 300 feet on average, with measurements on one side of the road taken independently from the other, to be eligible for transfer to the Town street lighting account. There is no guarantee that the Town will accept the lights – refer to street lighting ordinance.

2. Major Pedestrian Areas within Public Rights-of-Way

If an engineering evaluation indicates major pedestrian activities exist within public right-of-way, the Engineering and Public Works Department may determine that special lighting should be specified. A lighting plan and cost estimate for the special lighting would be developed, and if funds are available to implement the plan, then the Engineering and Public Works Department would coordinate the authorization and installation of the special lighting plan.

F. Signage

All regulatory, warning, and guide roadway signage shall be consistent with the Manual on Uniform Traffic Control Devices (MUTCD), the North Carolina Supplement to the MUTCD or as specified in this manual. All street name markers are also to be designed in accordance with 700 series standard drawings. All street name markers shall be nine inch tall extruded aluminum blades and utilize high intensity white prismatic reflective sheeting.

Certain street name markers within the Town limits shall include the Town logo. The Town logo shall be included to all street name markers that meet the following criteria:

- 1) Town Center zoning district
- 2) Intersections of Town streets with major or minor thoroughfares. Major and minor thoroughfares are defined on the Charlotte Regional Transportation Planning Organization (formerly the Mecklenburg-Union Metropolitan Planning Organization) Thoroughfare plan. (A PDF of the thoroughfare plan map can be found on the web at <http://www.crtpo.org/>.)

G. Landscaping

Refer to Article 7 Part A of the Town of Huntersville Zoning Ordinance regarding landscaping requirements.

While landscaping can be installed at street intersections, it shall not block the sight distance of vehicles at the intersection. Sight distance for an intersection shall be calculated in accordance with Section II. B. of this manual. Trees should not be planted within 40 feet of an intersection radius return measured along the street along the main or side street of intersections or commercial development driveways.

Trees shall not be planted in permanent drainage easements or within 10 feet of a masonry drainage structure. (This does not apply to Stormwater BMP's.)

H. Cluster Box Units (CBU's)

Mail cluster box units shall be placed outside of the line of sight (determined by intersection sight distance measurements), sight distance triangles and intersection corner easements. They shall not be placed between the subdivision entrance and its first street intersection. It is best to avoid placing CBU's on the main entrance road to a subdivision, however, special cases may apply.

When locating CBU's near on-street parking, do not place units directly adjacent to the on-street parking. CBU's shall be behind the sidewalk in such cases.

When placing CBU's within the green zone, units shall be oriented perpendicular to the street.

Access easements shall be required for all CBU's located outside of the right-of-way and/or common open space.

The ultimate goal in determining locations for mail cluster box units is to avoid placing the CBU in any way which encourages driving on the wrong side of the street and/or hinders handicap accessibility. The above standards are included to supplement the requirements of the United States Postal Service and shall be followed in addition to USPS standards.

III. Specifications and Special Provisions

A. General Notes

The following specifications and special provisions are intended to be used in conjunction with Town of Huntersville Standard Drawings, NCDOT Roadway Standard Drawings, and NCDOT Standard Specifications for Roads and Structures for all development within the Town of Huntersville and the Town of Huntersville ETJ unless otherwise directed by the Town Engineer.

1. Unless otherwise specified in this manual, **all work and materials shall conform to the latest edition of the North Carolina Department of Transportation Standard Specifications for Roads and Structures.**
2. All backfill material shall be non-plastic in nature, free from roots, vegetative matter, waste, construction material or other objectionable material. Said material shall be capable of being compacted by mechanical means and the material shall have no tendency to flow or behave in a plastic manner under the tamping blows or proof rolling.
3. Materials deemed by the inspector as unsuitable for backfill purposes shall be removed and replaced with select backfill material.
4. Compaction requirements shall be attained by the use of mechanical compaction methods. Each six inch layer of backfill shall be placed loose and thoroughly compacted into place.
5. ALL concrete used in the public right-of-way for streets, curb and gutter, sidewalks and drainage structures, etc. shall have a minimum compressive strength of 3600 PSI at 28 days. This requirement shall be provided regardless of any lesser compressive strength specified in the North Carolina Department of Transportation Standard Specifications for Roads and Structures. The contractor shall prepare concrete test cylinders in accordance with Section 1000 of the North Carolina Department of Transportation Standard Specifications for Roads and Structures at the direction of the project inspector. All equipment and cylinder molds shall be furnished by the contractor. It shall be the responsibility of the contractor to protect the cylinders until such time as they are transported for testing. Testing for projects shall be performed by an independent testing lab, at no cost to the Town. The contractor shall provide equipment and perform tests on concrete for a maximum slump and air content as defined in Section 1000 of the North Carolina Department of Transportation Standard Specifications for Roads and Structures. These tests shall be performed at a frequency established by the inspector. Materials failing to meet specifications shall be removed by the contractor.
6. Concrete or asphalt shall not be placed until the air temperature measured at the location of the paving operation is at 35 degrees Fahrenheit and rising by 10:00 a.m. Concrete or paving operations should be suspended when the air temperature is 40 degrees Fahrenheit and descending. The contractor shall protect freshly placed concrete or asphalt in accordance with Sections 420 (Concrete Structures), 600 (Asphalt Bases And Pavements), and 700 (Concrete Pavements And Shoulders) of the North Carolina Department of Transportation Standard Specifications for Roads and Structures when the air temperature is at or below 35 degrees Fahrenheit and the concrete has not obtained an age of 72 hours.
7. Plant all street trees in the middle of the planting strip unless otherwise noted on the standard detail.

Grading

1. Proposed street rights-of-way shall be graded to their full width for ditch type streets and a minimum of eight feet behind the curb for curb and gutter sections.
2. Fill embankments shall be constructed in accordance with section 235 of the North Carolina Department of Transportation Standard Specifications for Roads and Structures and placed in successive lifts not to exceed more than six inches in depth for the full width of the cross-section, including the width of the slope area. No stumps, trees, brush, rubbish or other unsuitable materials or substances shall be placed in the right-of-way. Each successive six inch layer shall be thoroughly compacted by the sheepsfoot tamping roller, 10-ton power roller, pneumatic-tired roller, or other methods approved by the Town Engineer. Embankments over and around all pipe culverts shall be of select material, placed and thoroughly tamped and compacted as directed by the Town Engineer or his representative.

Roadway Base

1. All roadways shall be constructed with a base course as detailed on the applicable Town of Huntersville Standard Detail Drawing.
2. The material for the aggregate base course (ABC) shall be in conformance with Section 520 – Aggregate Base Course of the North Carolina Department of Transportation Standard Specifications for Roads and Structures.
3. An asphalt concrete base course, as detailed on the Standard Detail Drawing may be substituted in lieu of an aggregate base course and shall be in accordance with all applicable articles of the North Carolina Department of Transportation Standard Specifications for Roads and Structures.
4. Asphalt concrete base course (ACBC) shall be used for widening strips less than five feet in width.

Roadway Intermediate and Surface Course

1. Plant mixed asphalt shall conform in all respects to Section 610 of the NCDOT Standard Specifications for Roads and Structures.
2. The final one and one half inch lift of asphalt surface course for residential subdivision streets shall be withheld until a minimum of 75 percent of the development is occupied (occupied means a certificate of occupancy has been issued) (All documentation to be provided by the developer and approved by the Town Engineer or designee). All known base failures shall be repaired prior to application of the final one and one half inch lift of asphalt surface course.
3. The Town Engineering and Public Works Department shall be given at least a 48 hour notification to inspect the first lift of surface course deficiencies. Prior to application of the final layer of asphalt, all deficiency repairs are to be monitored and accepted by the Town Engineer or designee.

4. The Town Engineering and Public Works Department shall be notified prior to using recycled plant mixes.
5. Failure to meet any of the requirements of this manual may result in the delay or prevention of street acceptance by the Town of Huntersville or NCDOT.

Sidewalks and Driveways

1. Sidewalks shall be constructed with concrete having a minimum compressive strength of not less than 3600 P.S.I. concrete. The sidewalk shall be at least six inches thick where sidewalk crosses a driveway and at least four inches thick in all other locations. The subgrade shall be compacted to 95 percent of the maximum density obtainable with the Standard Proctor Test. The surface of the sidewalk shall be steel trowel and light broom finished and cured with an acceptable curing compound. Tooled joints shall be provided at intervals of not less than five feet and expansion joints at intervals of not more than 45 feet. The sidewalk shall have a lateral or cross slope of one-quarter inch per foot.
2. Planting strip adjacent to sidewalk shall be graded to ¼ inch per foot (min.) up to 1 ¼ inch per foot maximum, except where excessive natural grades make this requirement impractical. In such cases, the Town Engineer may authorize a suitable grade.
3. Sidewalk widths shall be a minimum of five feet unless otherwise specified.
4. Approval of sidewalk construction plans must be obtained as part of the plan review process. A recorded permanent public sidewalk easement is required for all sidewalk located outside public right-of-way; the width of the easement shall be specified by the Town. The sidewalk easement must be recorded with the Mecklenburg County Register of Deeds prior to issuance of a certificate of occupancy for the corresponding building(s).
5. Accessible ramps are required where sidewalks intersect curbing at all street intersections and curbed driveway connections.

B. 100 Series Drawings – Miscellaneous Concrete Infrastructure

Drawings in this series include details for curb and gutter, sidewalks, driveways, accessible ramps, culvert crossings, and street tapers. The following list provides information in addition to that included in the standard drawings in this series.

1. All curb and gutter shall be backfilled with soil approved by the Inspector within 48 hours after construction to prevent erosion.
2. All concrete shall be cured with 100 percent Resin Base, white pigmented curing compound which meets ASTM Specifications C-309, Type 1, applied at a uniform rate at one gallon to 400 square feet within 24 hours of placement of the concrete.
3. Straight forms shall not be used for forming curb and gutter in curves.
4. All excess concrete on the front edge (lip) of gutter shall be removed when curb and gutter is poured with a machine.

C. 200 Series Drawings – Street Sections

Drawings in this series include details for street typical sections including pavement design, cul-de-sacs, parallel parking space location/layout, alleys, and hammerheads.

1. All asphalt cuts shall be made with a saw when preparing street surfaces for patching or widening strips.
2. All subgrade shall be compacted to 100% of the maximum density obtainable with the Standard Proctor Test to a depth of twelve (12) inches, and a density of 95% Standard Proctor for depths greater than twelve (12) inches. All tests shall be performed by developer at no cost to the Town.
3. Paper joints shall be used to seal the ends of an asphalt pour so that future extensions can be made without causing rough joints.
4. When placing asphalt against existing surfaces, a straight edge shall be used to prevent “humping” at that location.
5. Stone shall be primed if paving is not complete within seven days following stone base approval.
6. Surfaces shall be tacked when asphalt is being placed over existing asphalt streets or adjoining concrete, storm drain and sanitary sewer structures.
7. Sweeping of the stone base and/or application of a tack coat may be required near intersections. These requirements will be established by the Town/NCDOT Inspector based on field conditions.
8. A canvas cover or other suitable cover shall be required for transporting plant mix asphalt during cool weather when the following conditions are present:
 - a. Air temperature is below 60 degrees Fahrenheit.
 - b. Length of haul from plant to job is greater than five (5) miles.
 - c. Other occasions at the Inspector’s discretion when a combination of factors indicates that material should be covered in order to assure proper placement temperature.
9. Roadside ditches shall conform to NCDOT standards unless otherwise specified by Town along Town maintained roads.

D. 300 Series Drawings – Storm Drainage

Drawings in this series include NCDOT standards approved for use, catch basins, wingwalls, riprap aprons, flared end section pipe, riprap plunge pools, trench drains, paved ditches, subdrains, overlapping of easements, minimum drainage easements, and grading at drop inlets. The following list provides information in addition to that included in the standard drawings in this series.

1. All concrete shall be at least 3600 PSI. Prior approval from the Town Engineer shall be obtained in order to use pre-cast storm drainage structures in any street right-of-way.
2. Concrete pipe used within the street right-of-way shall be a minimum of Class III Reinforced Concrete Pipe, with a minimum diameter of fifteen inches (eighteen inches minimum on cross drain culverts). Installation of Class IV or higher concrete pipe shall be identified on the As-Built Plan and the Town Inspector shall be given documentation and notification of this information prior to construction.
3. Concrete mortar joints shall be used for joining all concrete pipes. The pipe shall be clean and moist when mortar is applied. The lower portions of the bell or groove shall be filled with mortar sufficient to bring the inner surface flush and even when the next joint is fitted into place. The remainder of the joint shall then be filled with mortar and a bead or ring of mortar formed around the outside of the joint. The application of mortar may be delayed until fill is completed when the pipe is larger than thirty inches.
4. Performed joint sealer, which conforms to AASHTO specification M-198 for Type B flexible plastic gaskets, may be used in lieu of the mortar joining method.
5. Under no circumstances shall water be permitted to rise in un-backfilled trenches after the pipe has been placed.

High Density Polyethylene Pipe (HDPE)

1. All trenches in the street right-of-way shall be backfilled with suitable material immediately after the pipe is laid. The fill around all pipes shall be placed in layers not to exceed six inches and each layer shall be compacted thoroughly.
2. Any installation within the maintenance limits of the Town is subject to the approval of the Town Engineer/Public Works Director.
3. The product used shall be corrugated exterior/smooth interior pipe (Type S), conforming to the requirements of AASHTO Specification M294 (latest edition) for Corrugated Polyethylene Pipe.
4. Bell and spigot joints shall be required on all pipes inside the right-of-way. Bells shall cover at least two full corrugations on each section of pipe. The bell and spigot joint shall have an "O" ring rubber gasket meeting ASTM F477 with the gasket factory installed, placed on the spigot end of the pipe. Pipe joints shall meet all requirements of AASHTO M294.
5. All HDPE pipe installed must be inspected and approved by the Town's Inspector prior to any backfill being placed. The Town Engineer or his designee must be present during the backfilling operation.

6. Backfill material used to install HDPE pipe within the street right-of-way shall be Select Material, Class II-IV, as defined by Section 1016-3 of the North Carolina Department of Transportation Standard Specifications for Roads and Structures. Upon submittal of written certification of material suitability by a licensed geotechnical engineer, NCDOT Class I Select Material may be used. All backfill material shall be approved by the Town inspector prior to placement of the material within the Town street right-of-way.
7. The minimum length of HDPE pipe permitted for use shall be four feet. HDPE flared end sections are not allowed.
8. All HDPE pipe installed shall be third party certified and shall bear the Plastic Pipe Institute's (PPI) certificate sticker.

Installation of Reinforced Concrete and Corrugated Metal Pipe

1. All backfill shall be non-plastic in nature, free from roots, vegetative matter, waste, construction material or other objectionable material. Said material shall be capable of being compacted by mechanical means and shall have no tendency to flow or behave in a plastic manner under the tamping blows or proof rolling.
2. Materials deemed by the Engineer as unsuitable for backfill purposes shall be removed and replaced with select backfill material.
3. Backfilling of trenches shall be accomplished immediately after the pipe is laid. The fill around the pipe shall be placed in layers not to exceed eight inches, each layer shall be thoroughly compacted to 95 percent of the maximum density obtainable with the Standard Proctor Test (a density of 100 percent Standard Proctor is required for the top eight inches).
4. Compaction requirements shall be attained by the use of mechanical compaction methods. Each layer of backfill shall be placed loose and thoroughly compacted in place.

E. 400 Series Drawings – Stormwater BMP

Drawings in this series include bioretention, flow splitters, wetponds, wetlands, grass swales, grass channels, infiltration ditches, observation wells, buffer strips, sand filters, and level spreaders.

F. 500 Series Drawings – Erosion Control

Drawings in this series include sediment traps, skimmers, pipe slope drains, silt ditches/fences, inlet protection, check dams, construction entrances, filter berm basins, dewatering, stream crossings, slope stability, seeding schedules, construction within creek banks, baffles, embankments, and brick storm structures.

1. The contractor shall do that which is necessary to control erosion and to prevent sedimentation damage to all adjacent properties and streams in accordance with the appropriate Mecklenburg County Erosion and Sedimentation Control Ordinance.

G. 600 Series Drawings – Trees

Drawings in this series include tree plantings/protection, tree pits, irrigation, valve boxes shrub plantings, medians, root crown depths, planting notes, bridging tree roots, asphalt curb placement at existing trees, and rock chimneys.

H. 700 Series Drawings – Miscellaneous

Drawings in this series include concrete control monuments, handrails, street name signs, end of road devices and markers, parking standards, accessible parking signage, roundabout signage, emergency vehicle median crossovers, bicycle racks and bicycle lockers.

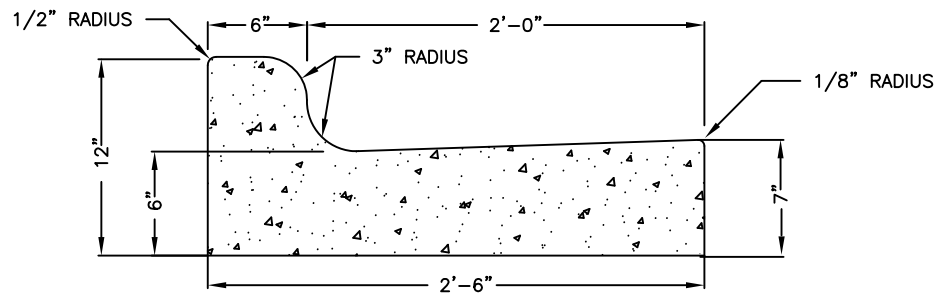
I. Traffic Control

The contractor shall maintain two-way traffic at all times when working within existing streets. The contractor shall place and maintain signs, danger lights, and barricades and furnish watchmen or flagmen to direct traffic in accordance with the latest edition Work Area Traffic Control Handbook (WATCH), Work in the right-of-way of State System Streets may require additional traffic control provisions.

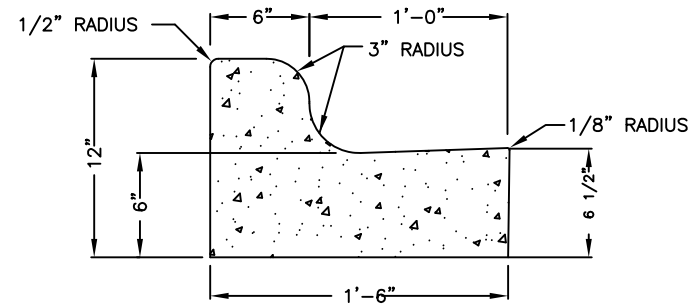
Refer to the Work Area Traffic Control Handbook (WATCH) for traffic control needs for work within the road right-of-way.

References

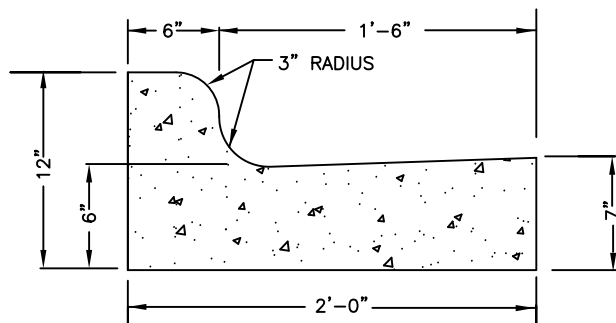
1. North Carolina Department of Transportation, most recent edition, Standard Specifications for Roads and Structures.
2. North Carolina Department of Transportation, most recent edition, Roadway Standards Drawings.
3. City of Charlotte Department of Transportation, most recent edition, Work Area Traffic Control Handbook (WATCH).
4. City of Charlotte Storm Water Services-Mecklenburg County Storm Water Services most recent edition, Charlotte-Mecklenburg Storm Water Design Manual.
5. American Association of State Highway and Transportation Officials most recent edition, A Policy on Geometric Design of Highways and Streets.
6. North Carolina Department of Transportation, Roadway Design Manual, latest edition.
7. North Carolina Department of Environment and Natural Resources most recent edition, Erosion and Sediment Control Planning and Design Manual.
8. Charlotte-Mecklenburg BMP Design Manual, latest edition.
9. Mecklenburg County Storm Water Services, most recent edition, Administrative Manual for Implementation of the Post-Construction Storm Water Ordinance.
10. Mecklenburg County Board of County Commissioners, most recent edition, Mecklenburg County Soil and Sedimentation Control Ordinance.
11. Manual of Uniform Traffic Control Devices for Streets and Highways, Federal Highway Administration, latest edition.



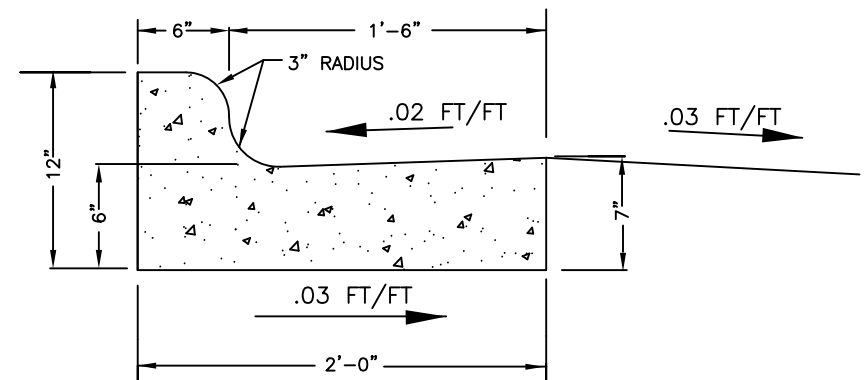
STANDARD 2'-6" CURB AND GUTTER



1'-6" STANDARD CURB AND GUTTER



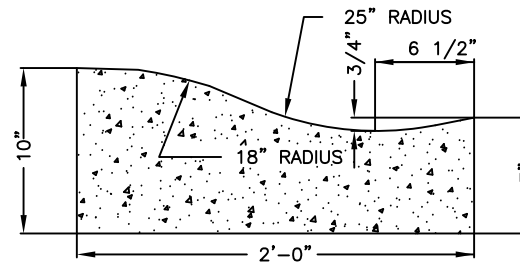
2'-0" STANDARD CURB & GUTTER



SLOPE FOR VARIABLE
SUPERELEVATION RATES

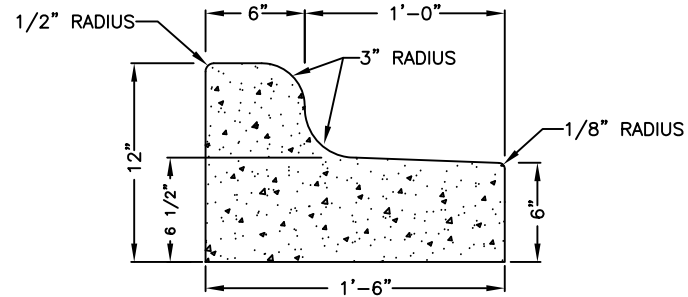
NOT TO SCALE

2'-0" VALLEY GUTTER



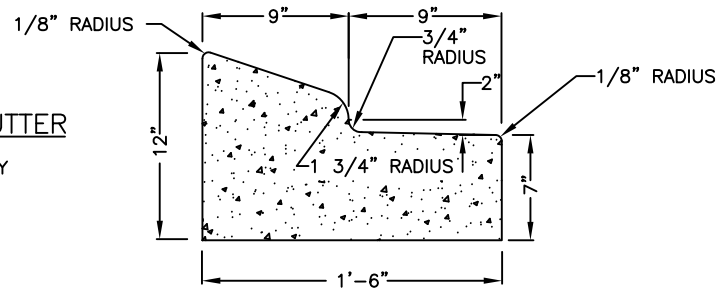
1'-6" MEDIAN CURB AND GUTTER

TO BE USED IN MEDIANS WHEN LANES ARE SLOPED FROM ISLAND OR AS SPECIFIED BY THE ENGINEER.

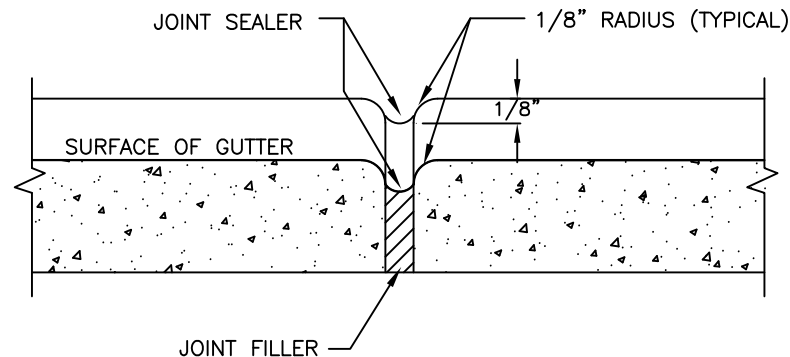


1'-6" MOUNTABLE CURB AND GUTTER

TO BE USED IN MEDIANS ONLY: WHEN SPECIFIED BY THE ENGINEER.



NOT TO SCALE



TRANSVERSE EXPANSION JOINT

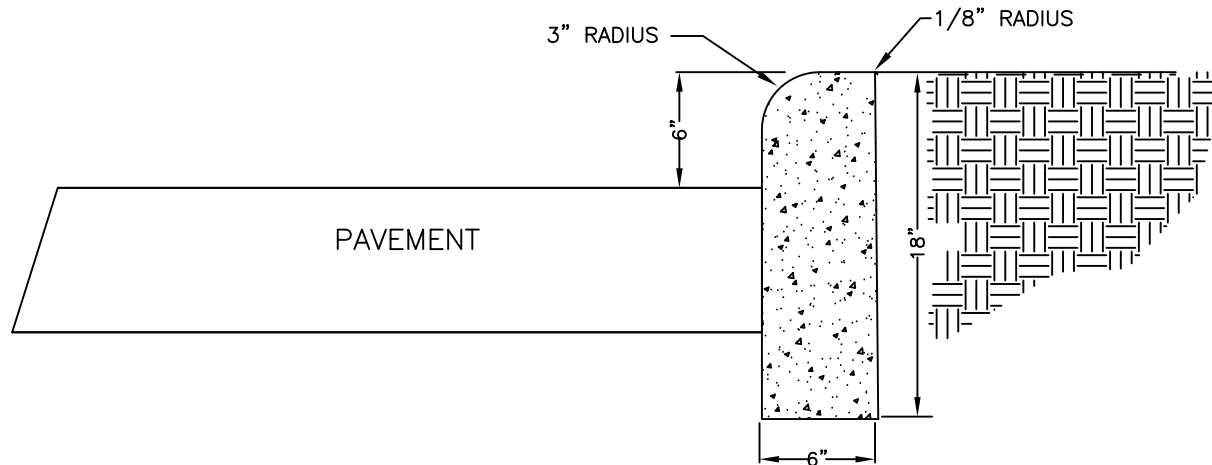
NOTES:

1. CONTRACTION JOINTS SHALL BE SPACED AT 10-FOOT INTERVALS. FOR VALLEY GUTTER, A 10-FOOT SPACING MAY BE USED WHEN A MACHINE IS USED. JOINT SPACING MAY BE ALTERED BY THE TOWN ENGINEER TO PREVENT UNCONTROLLED CRACKING.
2. CONTRACTION JOINTS MAY BE INSTALLED BY THE USE OF TEMPLATES OR FORMED BY OTHER APPROVED METHODS. WHERE SUCH JOINTS ARE NOT FORMED BY TEMPLATES, A MINIMUM DEPTH OF 1 1/2" SHALL BE OBTAINED.
3. ALL EXPANSION JOINTS SHALL BE SPACED AT 90-FOOT INTERVALS, AND ADJACENT TO ALL RIGID OBJECTS. JOINTS SHALL MATCH LOCATIONS WITH JOINTS IN ABUTTING SIDEWALK.
4. CONCRETE COMPRESSIVE STRENGTH SHALL BE 3600 P.S.I. IN 28 DAYS.
5. CURB SHALL BE DEPRESSED AT INTERSECTIONS TO PROVIDE FOR FUTURE ACCESSIBLE RAMPS.
6. TOP 6" OF SUBGRADE BENEATH THE CURB AND GUTTER SHALL BE COMPACTED TO 100% STANDARD PROCTOR DENSITY.

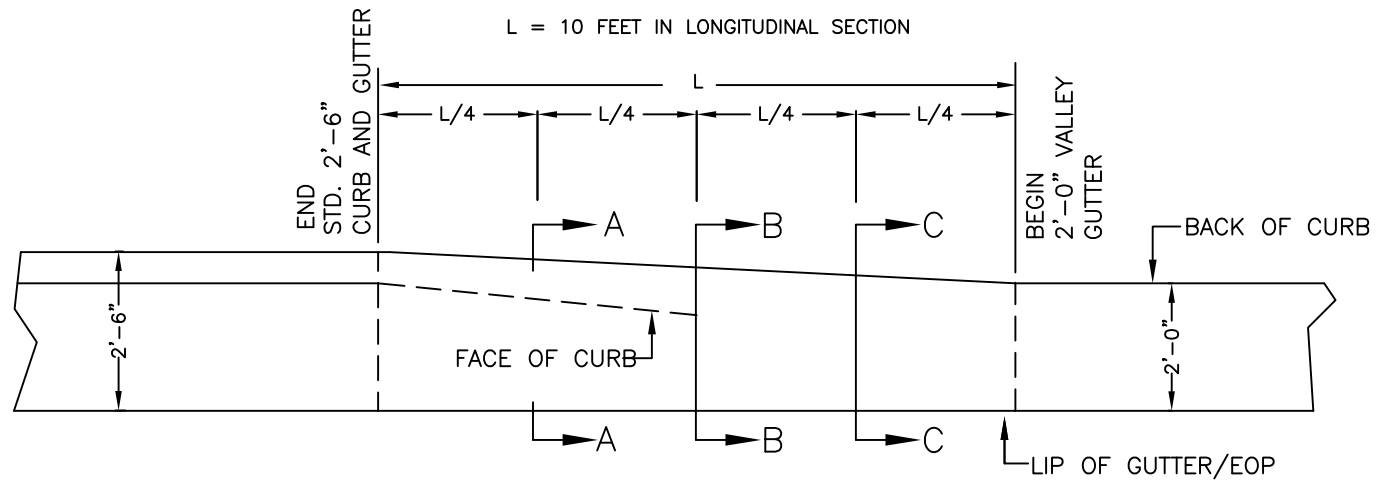
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NOTES:

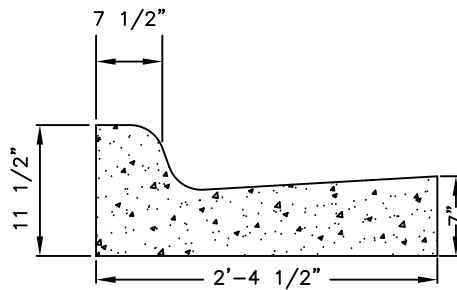
1. CONTRACTION JOINTS SHALL BE SPACED AT 10-FOOT INTERVALS. JOINT SPACING MAY BE ALTERED BY THE ENGINEER TO PREVENT UNCONTROLLED CRACKING.
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3. ALL EXPANSION JOINTS SHALL BE SPACED AT 90-FOOT INTERVALS, AND ADJACENT TO ALL RIGID OBJECTS. JOINTS SHALL MATCH LOCATIONS WITH JOINTS IN ABUTTING SIDEWALK.
4. CONCRETE COMPRESSIVE STRENGTH SHALL BE 3600 P.S.I. IN 28 DAYS.
5. CURB SHALL BE DEPRESSED AT INTERSECTIONS TO PROVIDE FOR FUTURE ACCESSIBLE RAMPS.
6. TOP 6" OF SUBGRADE BENEATH THE CURB SHALL BE COMPACTED TO 100% STANDARD PROCTOR DENSITY.
7. DETAIL MAY BE USED FOR PRIVATE DRIVES, PARKING LOTS, AND INTERIOR CIRCULATION DRIVE.



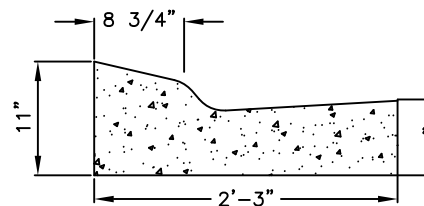
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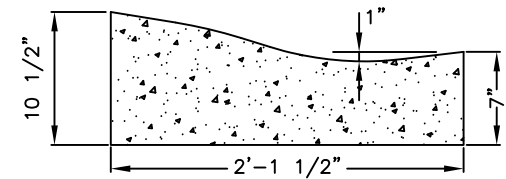
PLAN VIEW



SECTION A-A



SECTION B-B

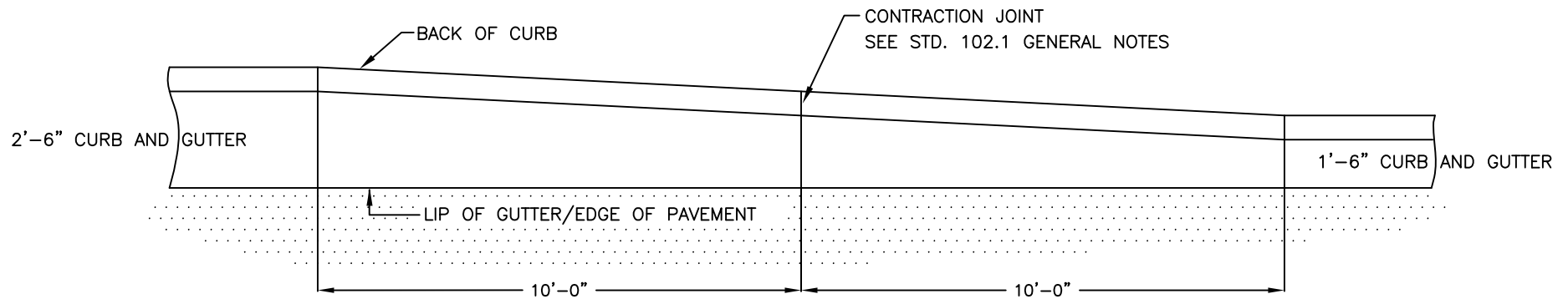


SECTION C-C

NOTES:

1. TRANSITION IS NOT TO BE LOCATED WITHIN THE CURB RADIUS.

NOT TO SCALE

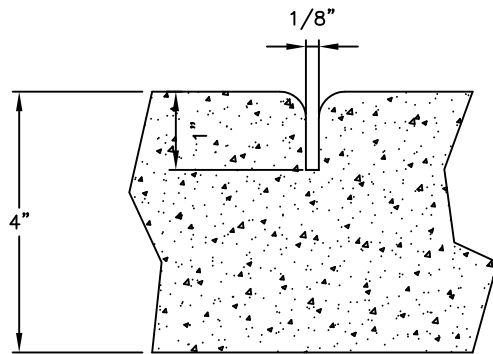


PLAN VIEW

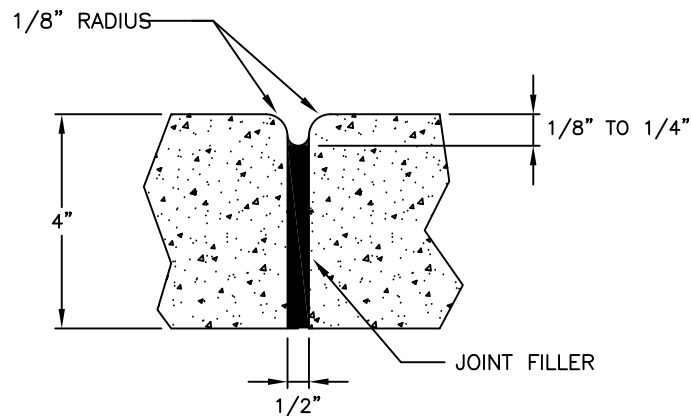
NOTES:

1. TRANSITION TO BE ALONG BACK OF CURB.

NOT TO SCALE



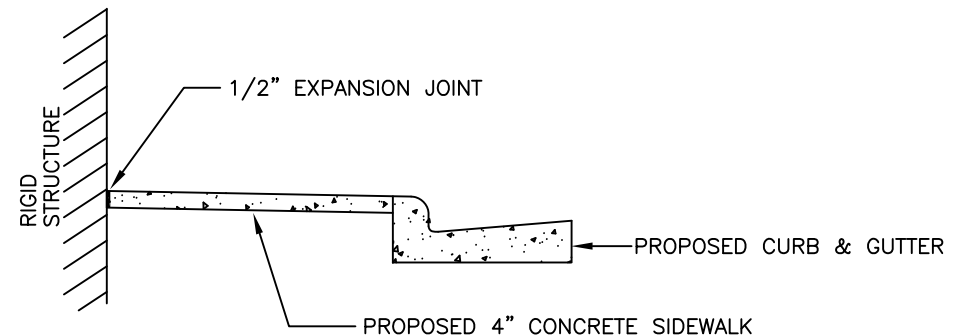
GROOVE JOINT IN SIDEWALK



TRANSVERSE EXPANSION
JOINT IN SIDEWALK

GENERAL NOTES:

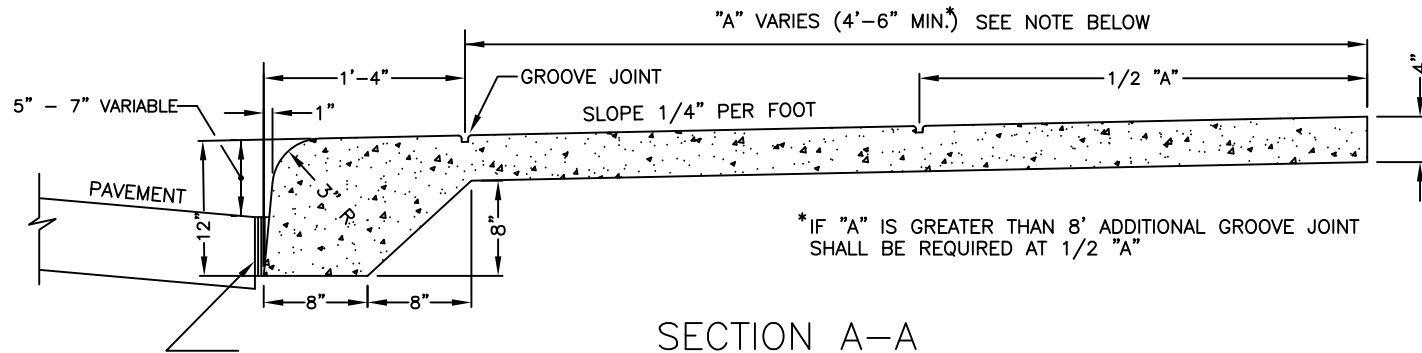
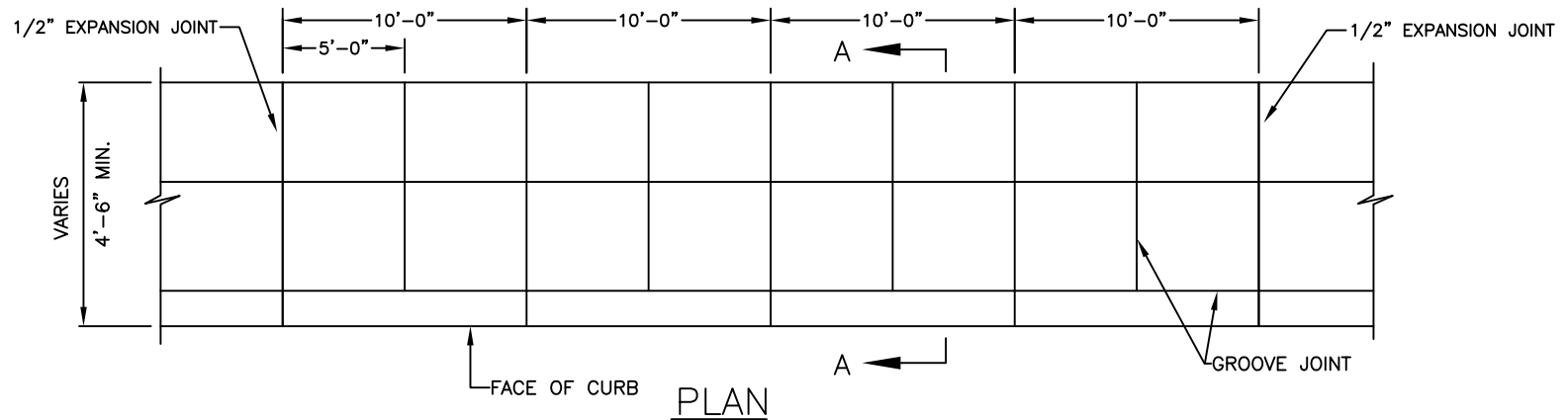
1. A GROOVE JOINT 1" DEEP WITH 1/8" RADII SHALL BE REQUIRED IN THE CONCRETE SIDEWALK AT 5' INTERVALS. ONE 1/2" EXPANSION JOINT WILL BE REQUIRED AT 45' INTERVALS NOT TO EXCEED 50' AND MATCHING EXPANSION/CONSTRUCTION JOINT IN ADJACENT CURB. A SEALED 1/2" EXPANSION JOINT WILL BE REQUIRED WHERE THE SIDEWALK JOINS ANY RIGID STRUCTURE.
2. SIDEWALK AT DRIVEWAY ENTRANCES TO BE 6" THICK.
3. WIDTH OF SIDEWALK ON TOWN STREETS SHALL BE A MINIMUM OF 5'.
4. WIDTH OF SIDEWALKS ON NON-TOWN STREETS SHALL BE A MINIMUM OF 5'.
5. SIDEWALK TO BE POURED TO END OF RADIUS AT INTERSECTING STREETS.
6. CONCRETE COMPRESSIVE STRENGTH SHALL BE 3600 PSI. IN 28 DAYS.
7. ZONING CONDITIONS MAY REQUIRE ADDITIONAL WIDTH SIDEWALKS WHICH SHALL SUPERSEDE THESE STANDARD DIMENSIONS SHOWN.



DETAILS SHOWING EXPANSION JOINTS

FOR CONCRETE SIDEWALK

NOT TO SCALE



TWO 1/2" THICK PIECES BITUMINOUS FIBER REQUIRED
IF SUBBASE IS CONCRETE. MUST BE SEALED WITH
APPROVED JOINT SEALER.

GENERAL NOTES:

1. A GROOVE JOINT 1" DEEP WITH 1/3" RADII SHALL BE REQUIRED IN THE CONCRETE SIDEWALK AT 5' INTERVALS. ONE 1/2" EXPANSION JOINT WILL BE REQUIRED AT 40' INTERVALS. A 1/2" EXPANSION JOINT WILL BE REQUIRED WHERE THE SIDEWALK JOINS ANY RIGID STRUCTURE.
2. ALL CONCRETE TO BE 3600 P.S.I. COMPRESSIVE STRENGTH.
3. SEE STANDARD 106.1 FOR DETAIL OF EXPANSION JOINT AND GROOVE JOINT.
4. SEE STANDARD 117.1 FOR DETAIL OF DRIVEWAY.
5. MONOLITHIC CURB AND SIDEWALK TO BE CONSTRUCTED ONLY WHEN REPLACING GRANITE CURB OR AT LOCATIONS APPROVED BY THE APPROPRIATE TOWN ENGINEER.

NOT TO SCALE

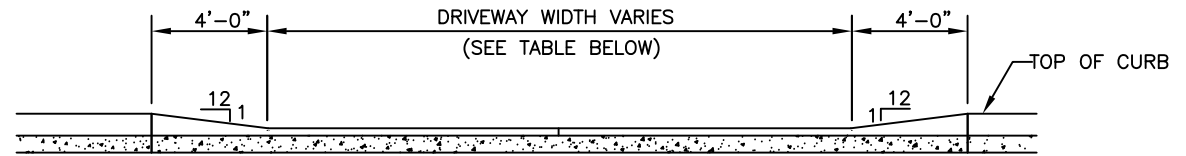
NOTE:

1. 1/2" EXPANSION JOINTS REQUIRE INSTALLATION OF ONE 1/2" THICK PIECE OF BITUMINOUS FIBER THROUGH THE ENTIRE SLAB.
2. TO LIMIT STORM WATER FLOW DOWN DRIVEWAYS, USE STANDARD 110.1 FOR DRIVEWAYS NEAR LOW POINTS.
3. ALL DRIVEWAYS MUST MEET THE CURRENT TOWN DRIVEWAY REGULATIONS AND NCDOT REQUIREMENTS FOR SPACING, SIGHT DISTANCE AND OFFSETS FROM PROPERTY LINES AND INTERSECTIONS.
4. "A" BREAKOVER SHALL BE 8% OR LESS (A = ALGEBRAIC DIFFERENCE).
5. PRIOR APPROVAL IS REQUIRED ON GRADES EXCEEDING WHAT ARE SHOWN.

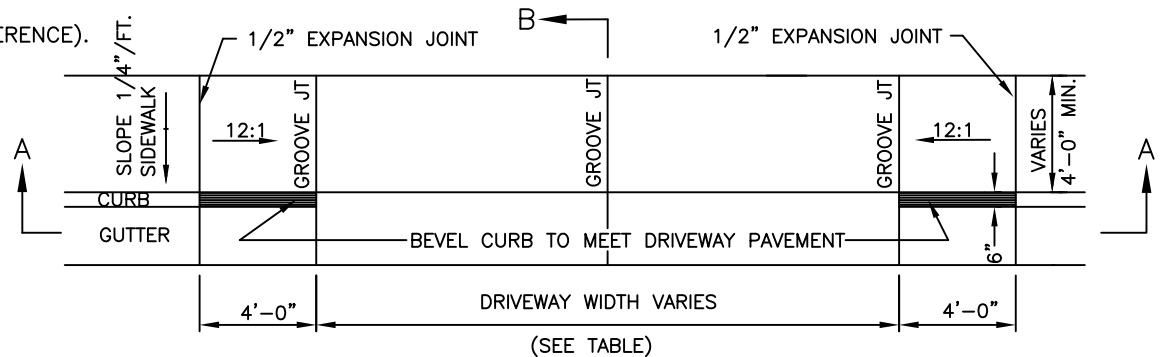
GENERAL NOTES:

ALL CONCRETE TO BE 3600 P.S.I. COMPRESSIVE STRENGTH.

ALL CURB, CURB AND GUTTER AND SIDEWALKS ARE TO BE REMOVED TO THE NEAREST JOINT BEYOND NEW CONSTRUCTION OR CUT WITH A SAW AND REMOVED. SAW CUT OR JOINT TO BE PERPENDICULAR TO EDGE OF EXISTING PAVEMENT.
SEE STD. NO 102.1 FOR DETAIL OF EXPANSION JOINT AND GROOVE JOINT.

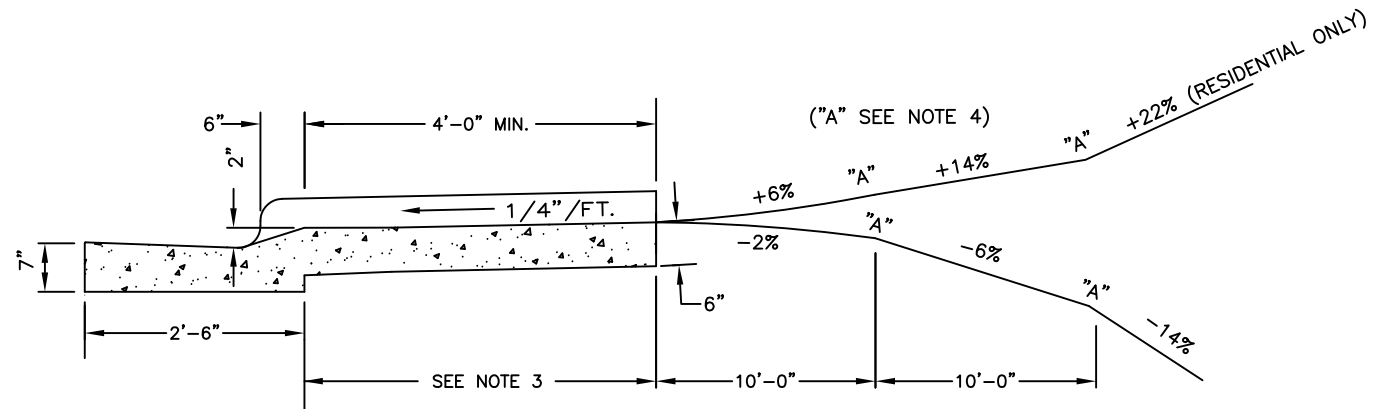


SECTION A - A



PLAN

DRIVEWAY CLASSIFICATION		
TYPE DRIVEWAY	MINIMUM	MAXIMUM
TYPE I—RESIDENTIAL	15'	30'
ONE—WAY TYPE II COMMERCIAL	20'	30'
TWO—WAY TYPE II COMMERCIAL	26'	50'



SECTION B — B

NOT TO SCALE

NOTE:

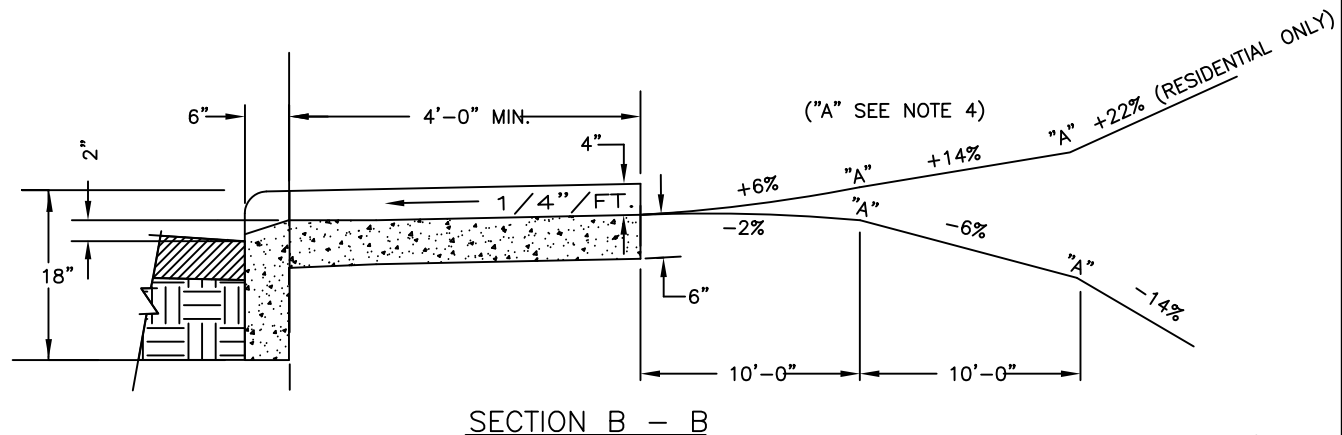
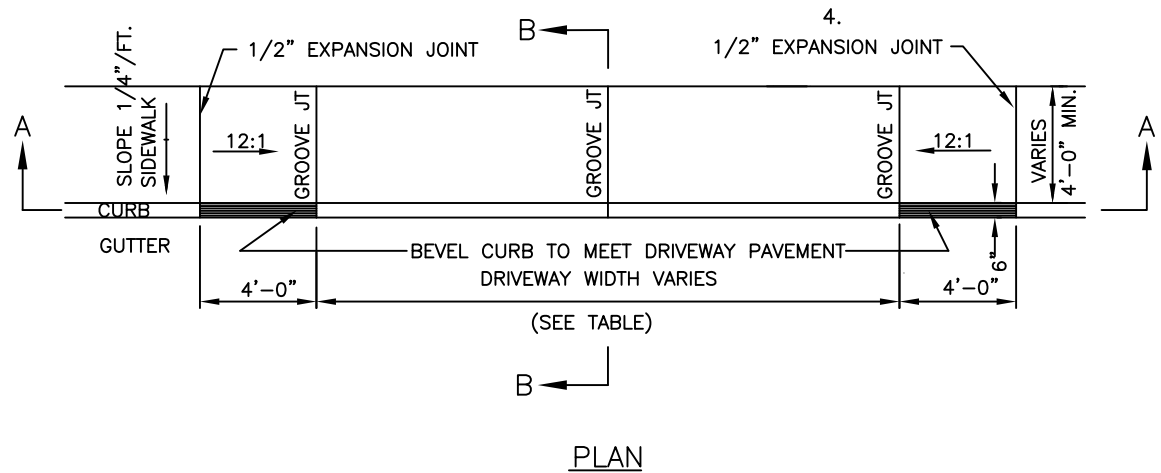
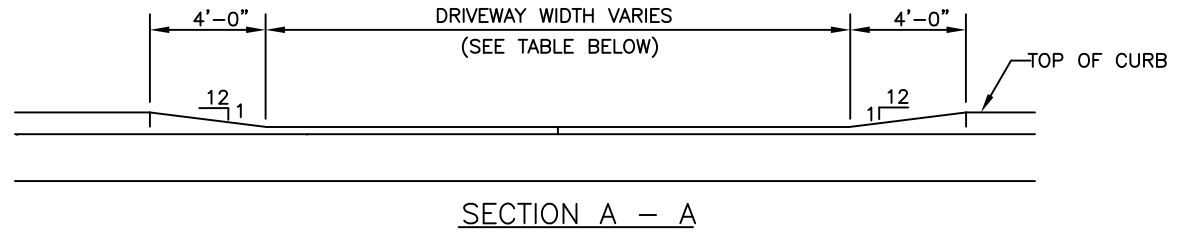
- 1/2" EXPANSION JOINTS REQUIRE INSTALLATION OF ONE 1/2" THICK PIECE OF BITUMINOUS FIBER MATERIAL THROUGH THE ENTIRE SLAB.
- TO LIMIT STORM WATER FLOW DOWN DRIVEWAYS, USE STANDARD 110.1 FOR DRIVEWAY LOWPOINT.
- ALL DRIVEWAYS MUST MEET THE CURRENT TOWN DRIVEWAY REGULATIONS AND NCDOT REQUIREMENTS FOR SPACING, SIGHT DISTANCE AND OFFSETS FROM PROPERTY LINES AND INTERSECTIONS.
- "A" BREAKOVER SHALL BE 8% OR LESS (A = ALGEBRAIC DIFFERENCE).
- PRIOR APPROVAL IS REQUIRED ON GRADES EXCEEDING WHAT ARE SHOWN.

GENERAL NOTES:

ALL CONCRETE TO BE 3600 P.S.I. COMPRESSIVE STRENGTH.

ALL CURB OR CURB AND GUTTER AND SIDEWALKS ARE TO BE REMOVED TO THE NEAREST JOINT BEYOND NEW CONSTRUCTION OR CUT WITH A SAW AND REMOVED. SAW CUT OR JOINT TO BE PERPENDICULAR TO EDGE OF EXISTING PAVEMENT. SEE STD. NO 102.1 FOR DETAIL OF EXPANSION JOINT AND GROOVE JOINT.

DRIVEWAY CLASSIFICATION		
TYPE DRIVEWAY	MINIMUM	MAXIMUM
TYPE I-RESIDENTIAL	15'	30'
ONE-WAY TYPE II COMMERCIAL	20'	30'
TWO-WAY TYPE II COMMERCIAL	26'	50'



NOT TO SCALE

SECTION A - A

SECTION A - A

PLAN

SECTION B - B

PLAN

SECTION B - B

NOTES

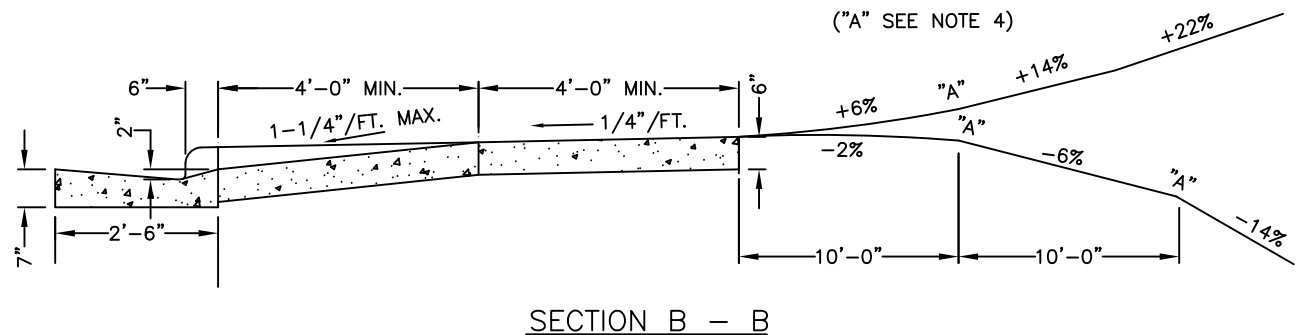
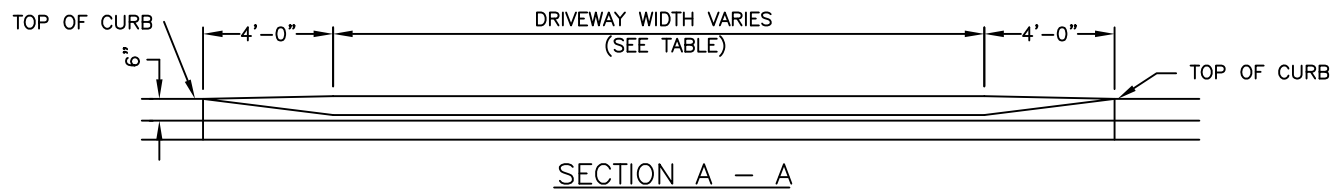
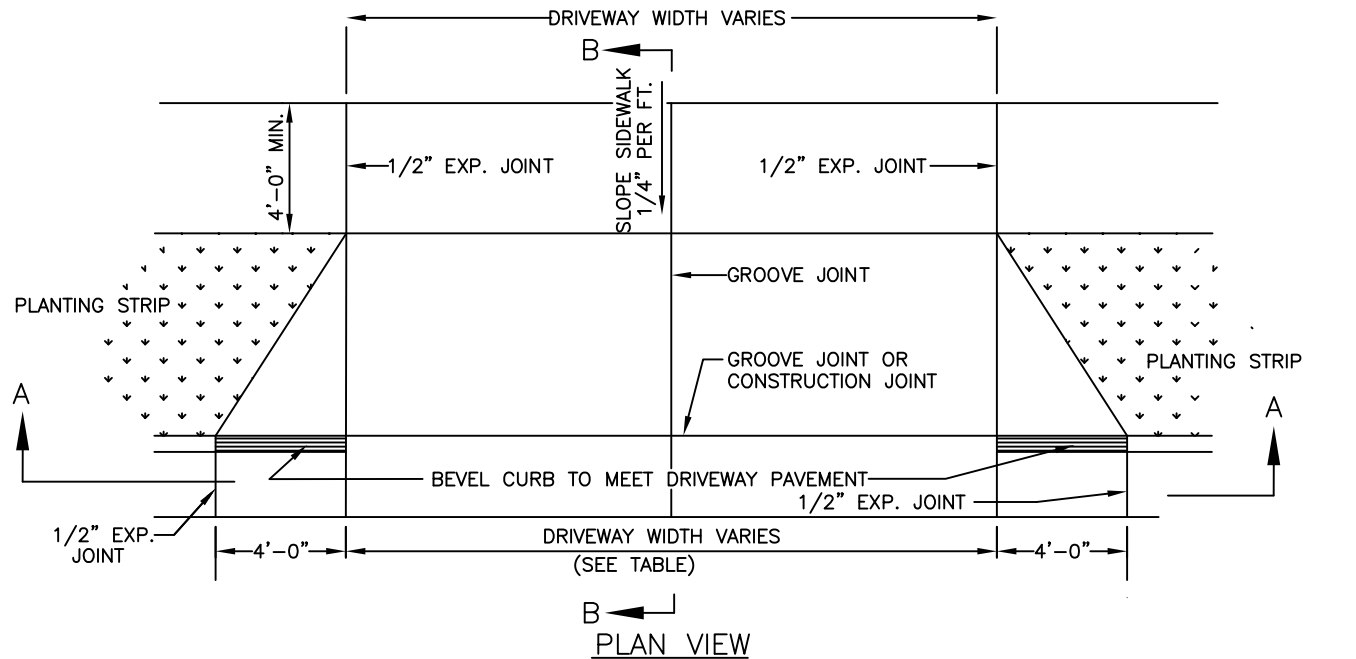
1. USED AT LOW POINTS IN ROADWAYS WITH 2'-6" CURB AND GUTTER OR 6" X 18" CURB AS DIRECTED BY ENGINEER.
2. SEE STANDARDS 108.1 & 109.1 FOR ADDITIONAL DETAILS.
3. ALL DRIVEWAYS MUST MEET THE CURRENT TOWN DRIVEWAY REGULATIONS AND NCDOT REQUIREMENTS FOR SPACING, SIGHT DISTANCE AND OFFSETS FROM PROPERTY LINES AND INTERSECTIONS.

NOT TO SCALE

NOTES:

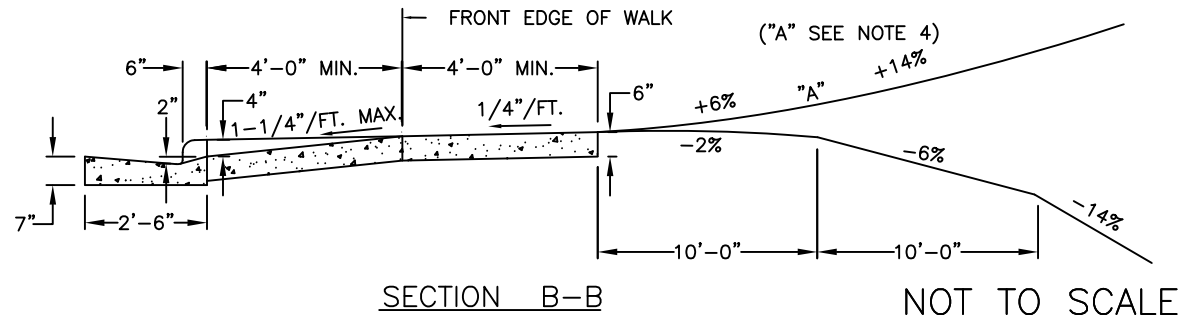
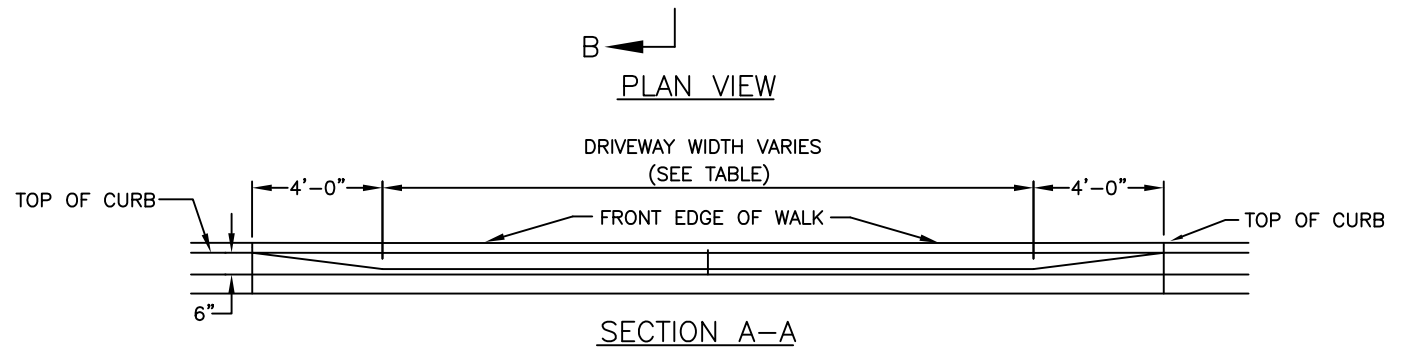
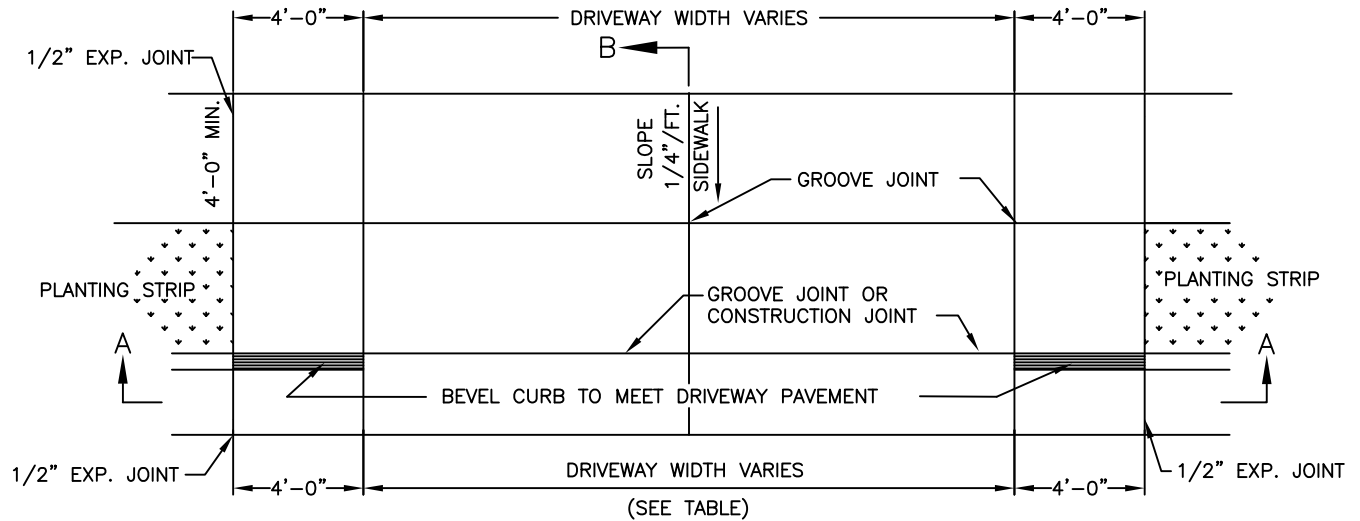
1. ALL CONCRETE TO BE 3600 P.S.I.
2. ALL CURB OR CURB AND GUTTER AND SIDEWALK ARE TO BE REMOVED TO THE NEAREST JOINT BEYOND NEW CONSTRUCTION OR CUT WITH A SAW AND REMOVED. SAW CUT OR JOINT TO BE PERPENDICULAR TO EDGE OF EXISTING PAVEMENT. SEE STD. NO. 102.1 FOR JOINT DETAIL.
3. ALL DRIVEWAYS MUST MEET THE CURRENT TOWN DRIVEWAY REGULATIONS AND NCDOT REQUIREMENTS FOR SPACING, SIGHT DISTANCE AND OFFSETS FROM PROPERTY LINES AND INTERSECTIONS.
4. "A" BREAKOVER SHALL BE 8% OR LESS (A = ALGEBRAIC DIFFERENCE).
5. PRIOR APPROVAL IS REQUIRED ON GRADES EXCEEDING WHAT ARE SHOWN.

DRIVEWAY CLASSIFICATION		
TYPE DRIVEWAY	MINIMUM	MAXIMUM
TYPE I - RESIDENTIAL	15'	30'



NOTES:

1. ALL CONCRETE TO BE 3600 P.S.I.
2. ALL CURB OR CURB AND GUTTER AND SIDEWALK ARE TO BE REMOVED TO THE NEAREST JOINT BEYOND NEW CONSTRUCTION OR CUT WITH A SAW AND REMOVED. SAW CUT OR JOINT TO BE PERPENDICULAR TO EDGE OF EXISTING PAVEMENT. SEE STD. NO. 102.1 FOR JOINT DETAIL.
3. ALL DRIVEWAYS MUST MEET THE CURRENT TOWN DRIVEWAY REGULATIONS AND NCDOT REQUIREMENTS FOR SPACING, SIGHT DISTANCE AND OFFSETS FROM PROPERTY LINES AND INTERSECTIONS.
4. "A" BREAKOVER SHALL BE 8% OR LESS (A = ALGEBRAIC DIFFERENCE).
5. PRIOR APPROVAL IS REQUIRED ON GRADES EXCEEDING WHAT ARE SHOWN.



NOT TO SCALE

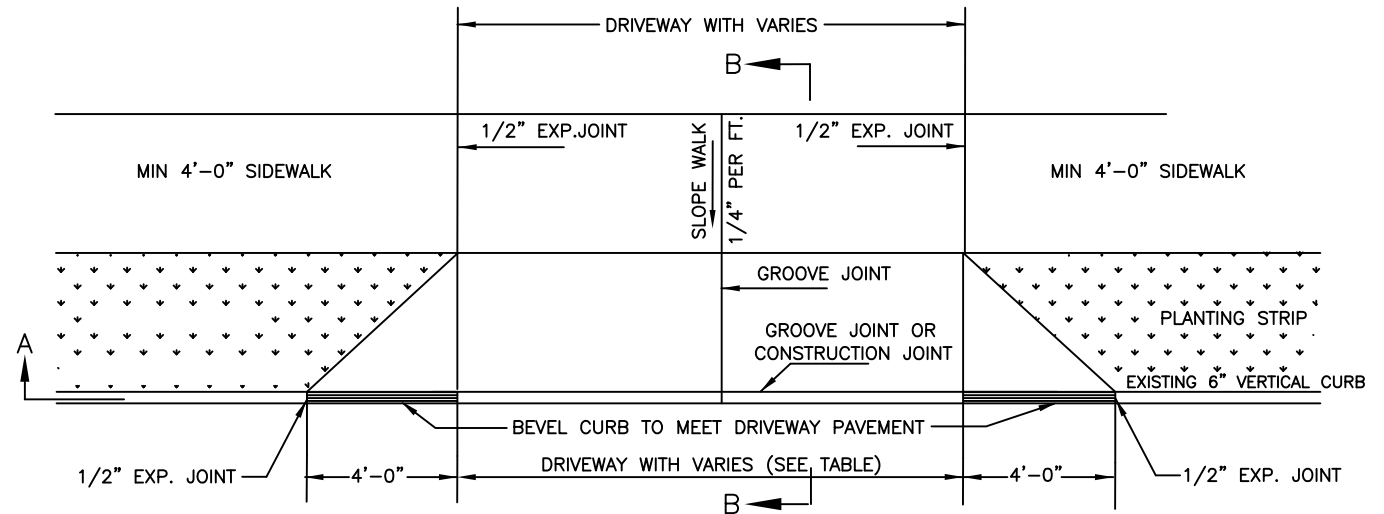
DRIVEWAYS CLASSIFICATION		
TYPE DRIVEWAYS	MINIMUM	MAXIMUM
ONE-WAY TYPE II - COMMERCIAL	20'	30'
TWO-WAY TYPE II - COMMERCIAL	26'	50'*

* NEED MORE THAN ONE CONTRACTION JOINT IN CENTER.

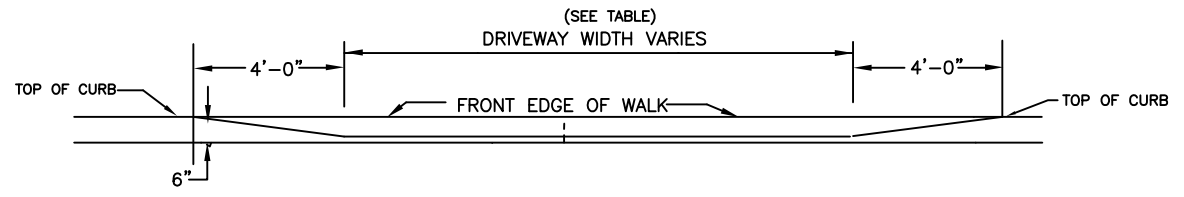
NOTES:

1. ALL CONCRETE TO BE 3600 P.S.I.
2. ALL CURB OR CURB AND GUTTER AND SIDEWALK ARE TO BE REMOVED TO THE NEAREST JOINT BEYOND NEW CONSTRUCTION OR CUT WITH A SAW AND REMOVED. SAW CUT OR JOINT TO BE PERPENDICULAR TO EDGE OF EXISTING PAVEMENT. SEE STD. NO. 102.1 FOR JOINT DETAIL.
3. ALL DRIVEWAYS MUST MEET THE CURRENT TOWN DRIVEWAY REGULATIONS AND NCDOT REQUIREMENTS FOR SPACING, SIGHT DISTANCE AND OFFSETS FROM PROPERTY LINES AND INTERSECTIONS.
4. "A" BREAKOVER SHALL BE 8% OR LESS.
5. PRIOR APPROVAL IS REQUIRED ON GRADES EXCEEDING WHAT ARE SHOWN.

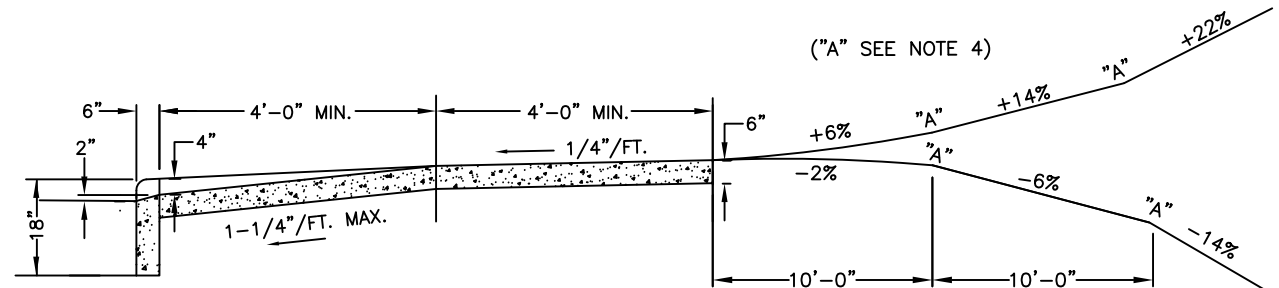
DRIVEWAYS CLASSIFICATION		
TYPE DRIVEWAYS	MINIMUM	MAXIMUM
ONE-WAY TYPE I -	15'	30'



PLAN VIEW



SECTION A-A (ALONG FLOW LINE)

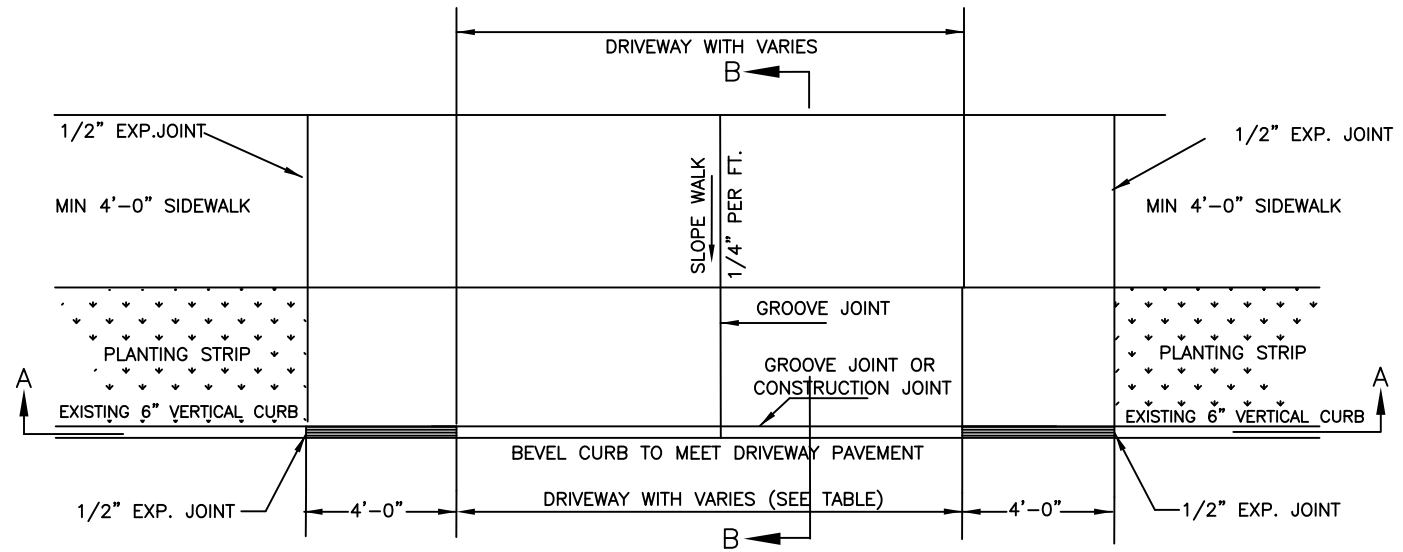


SECTION B-B

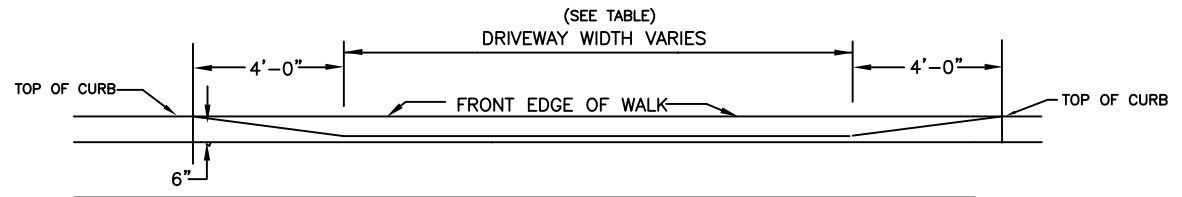
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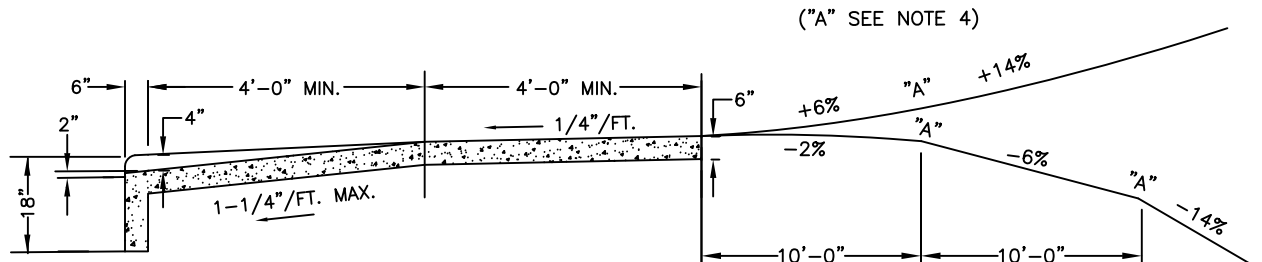
1. ALL CONCRETE TO BE 3600 P.S.I.
2. ALL CURB OR CURB AND GUTTER AND SIDEWALK ARE TO BE REMOVED TO THE NEAREST JOINT BEYOND NEW CONSTRUCTION OR CUT WITH A SAW AND REMOVED. SAW CUT OR JOINT TO BE PERPENDICULAR TO EDGE OF EXISTING PAVEMENT. SEE STD. NO. 102.1 FOR JOINT DETAIL.
3. ALL DRIVEWAYS MUST MEET THE CURRENT TOWN DRIVEWAY REGULATIONS AND NCDOT REQUIREMENTS FOR SPACING, SIGHT DISTANCE AND OFFSETS FROM PROPERTY LINES AND INTERSECTIONS.
4. "A" BREAKOVER SHALL BE 8% OR LESS (A = ALGEBRAIC DIFFERENCE).
5. PRIOR APPROVAL IS REQUIRED ON GRADES EXCEEDING WHAT ARE SHOWN.



PLAN VIEW



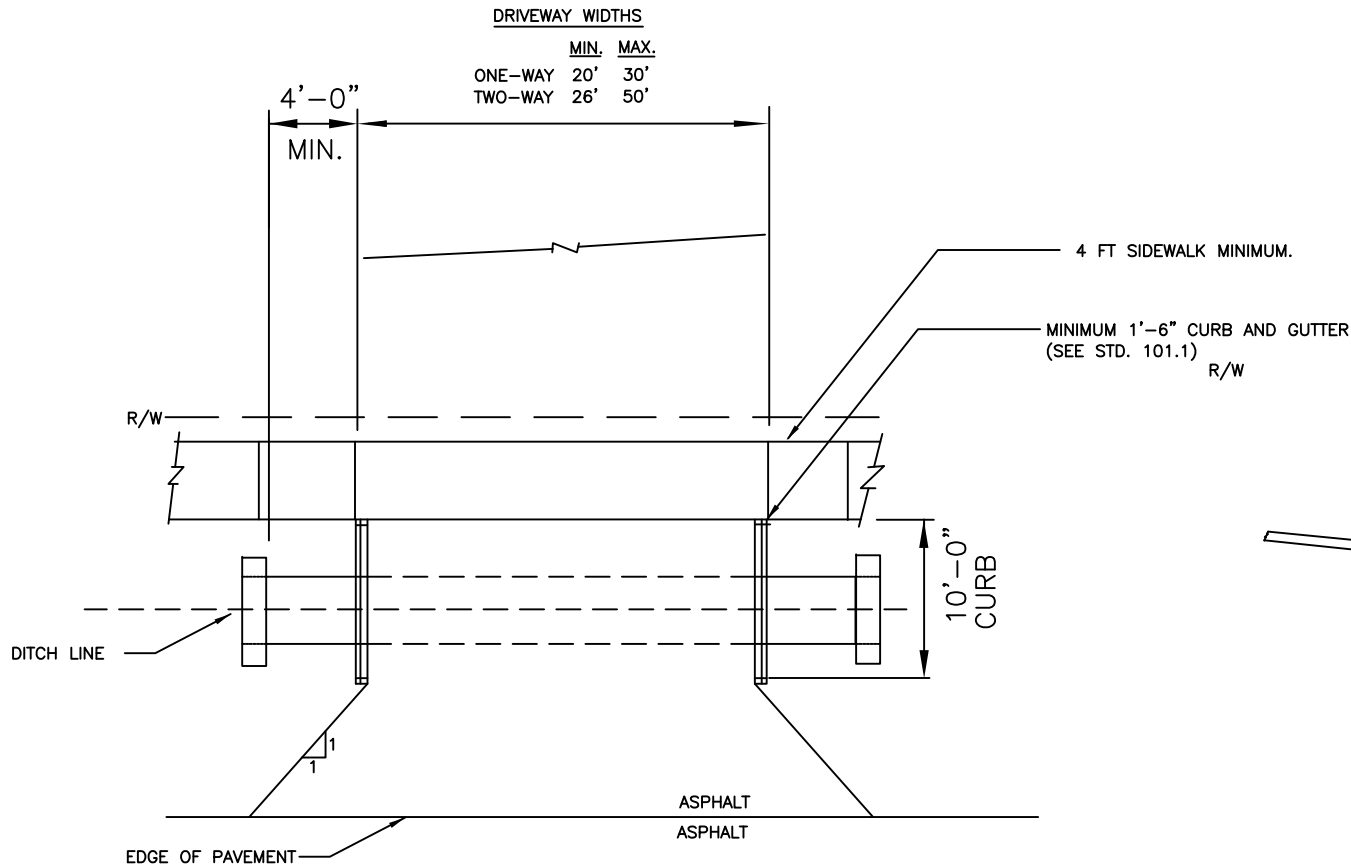
SECTION A-A (ALONG FLOW LINE)



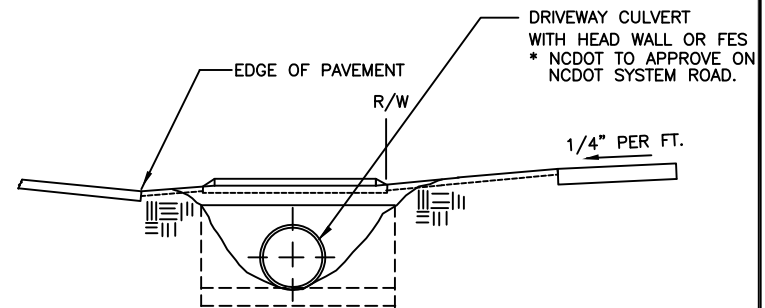
SECTION B-B

NOT TO SCALE

DRIVEWAYS CLASSIFICATION		
TYPE DRIVEWAYS	MINIMUM	MAXIMUM
ONE-WAY TYPE II-COMMERCIAL	20'	30'
TWO-WAY TYPE II-COMMERCIAL	26'	50'*



PLAN



SECTION

NOTE:

1. TO BE USED ON ROADS WITHOUT CURB AND GUTTER AND WHERE CURB AND GUTTER IS NOT BEING INSTALLED (MUST MEET BOTH CRITERIA).
2. ALL CONCRETE TO BE 3600 P.S.I. COMPRESSIVE STRENGTH.
3. USE OF THIS STANDARD FOR RESIDENTIAL DRIVEWAY CONSTRUCTION AT THE DISCRETION OF THE TOWN ENGINEER ONLY.

NOT TO SCALE

GENERAL NOTES:

ALL CONCRETE TO BE 3600 P.S.I. COMPRESSIVE STRENGTH.

A 1/2" EXPANSION JOINT WILL BE REQUIRED WHERE WHERE THE SIDEWALK JOINS ANY RIGID STRUCTURE. SEE STANDARD 106.1.

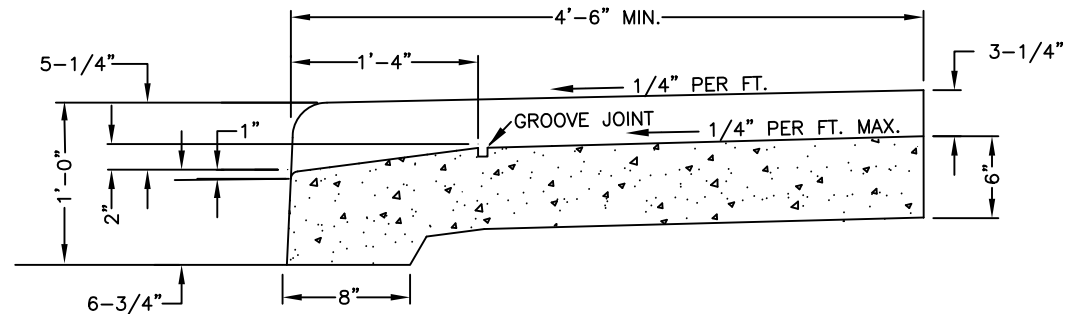
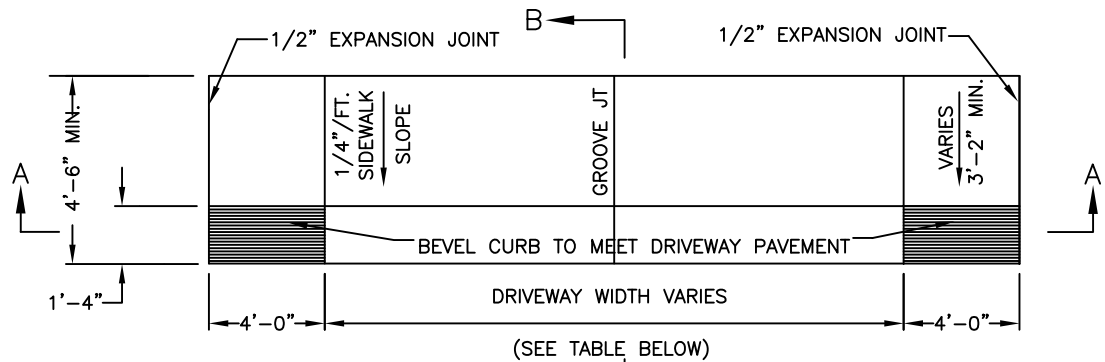
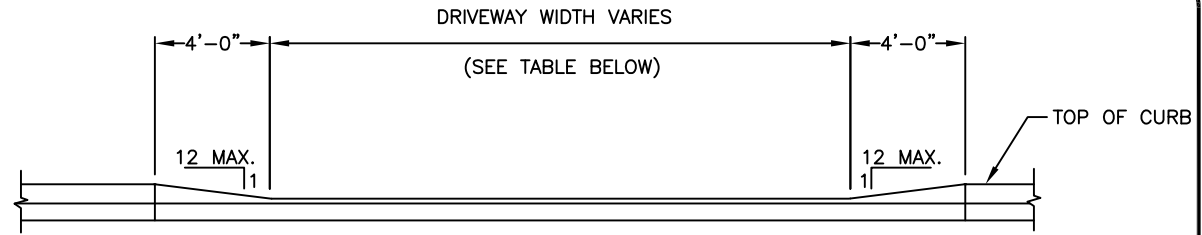
THIS DETAIL TO BE USED ONLY IN CONJUNCTION WITH MONOLITHIC SIDEWALK AS ON STANDARD NO. 107.1

NOTES:

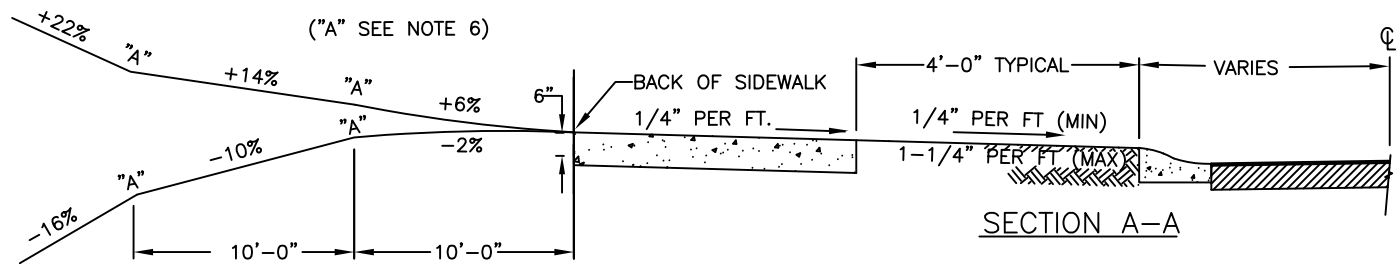
1. ALL DRIVEWAYS MUST MEET THE CURRENT TOWN DRIVEWAY REGULATIONS AND NCDOT REQUIREMENTS FOR SPACING, SIGHT DISTANCES, AND OFFSETS FROM PROPERTY LINES AND INTERSECTIONS.

DRIVEWAY CLASSIFICATIONS

TYPE DRIVEWAY	MINIMUM	MAXIMUM
TYPE I- RESIDENTIAL	15'	30'
ONE-WAY - TYPE II COMMERCIAL	20'	30'
TWO-WAY - TYPE III COMMERCIAL	26'	50'

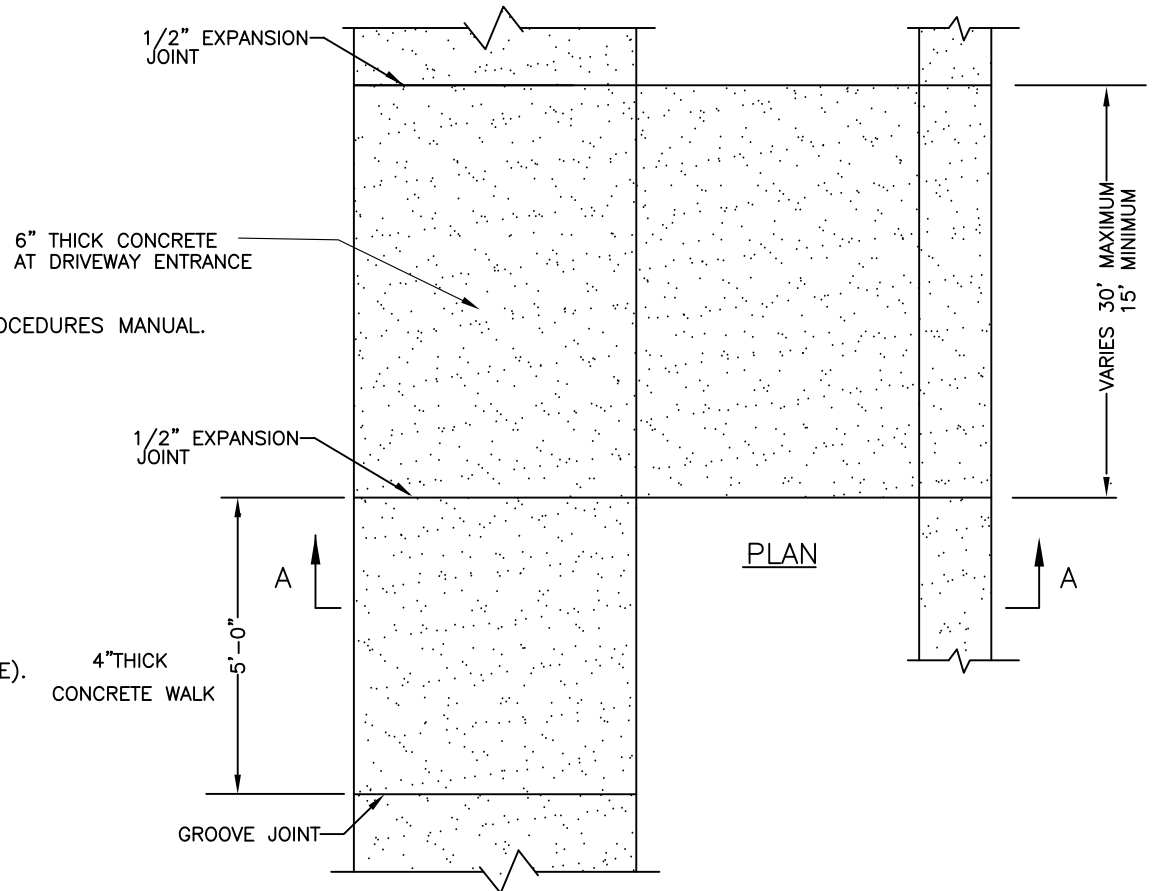


NOT TO SCALE



NOTES:

1. THE ELEVATION OF THE SIDEWALK SHALL BE NOT LESS THAN SIX INCHES OR MORE THAN EIGHTEEN INCHES ABOVE THE ROADWAY CROWN. THIS ELEVATION DIFFERENTIAL SHALL BE CONSISTENT WITHIN EACH BLOCK.
2. ALL CONCRETE TO BE 3600 PSI STRENGTH.
3. ALL CONSTRUCTION PRACTICES, INCLUDING COMPACTION, CURING, FINISHING, ETC. SHALL BE IN ACCORDANCE WITH THE TOWN OF HUNTERSVILLE ENGINEERING STANDARDS AND PROCEDURES MANUAL.
4. PLANTING STRIP SHALL BE GRADED WITH A CROSS SLOPE BETWEEN 1/2 IN. PER FOOT AND 1 1/4 IN. PER FOOT EXCEPT WHERE EXCESSIVE NATURAL GRADES MAKE THIS REQUIREMENT IMPRACTICAL. IN SUCH CASES, THE TOWN ENGINEER MAY AUTHORIZE A SUITABLE GRADE.
5. ALL DRIVEWAYS MUST MEET THE CURRENT TOWN DRIVEWAY REGULATIONS AND NCDOT REQUIREMENTS, INCLUDING BUT NOT LIMITED TO SPACING, SIGHT DISTANCE, AND OFFSETS FROM PROPERTY LINES AND INTERSECTIONS.
6. "A" BREAKOVER SHALL BE 8% OR LESS (A = ALGEBRAIC DIFFERENCE).
7. PRIOR APPROVAL IS REQUIRED ON GRADES EXCEEDING WHAT ARE SHOWN.

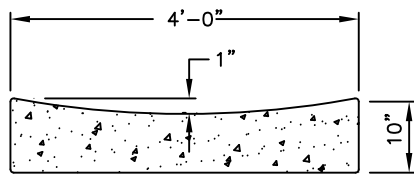


NOT TO SCALE

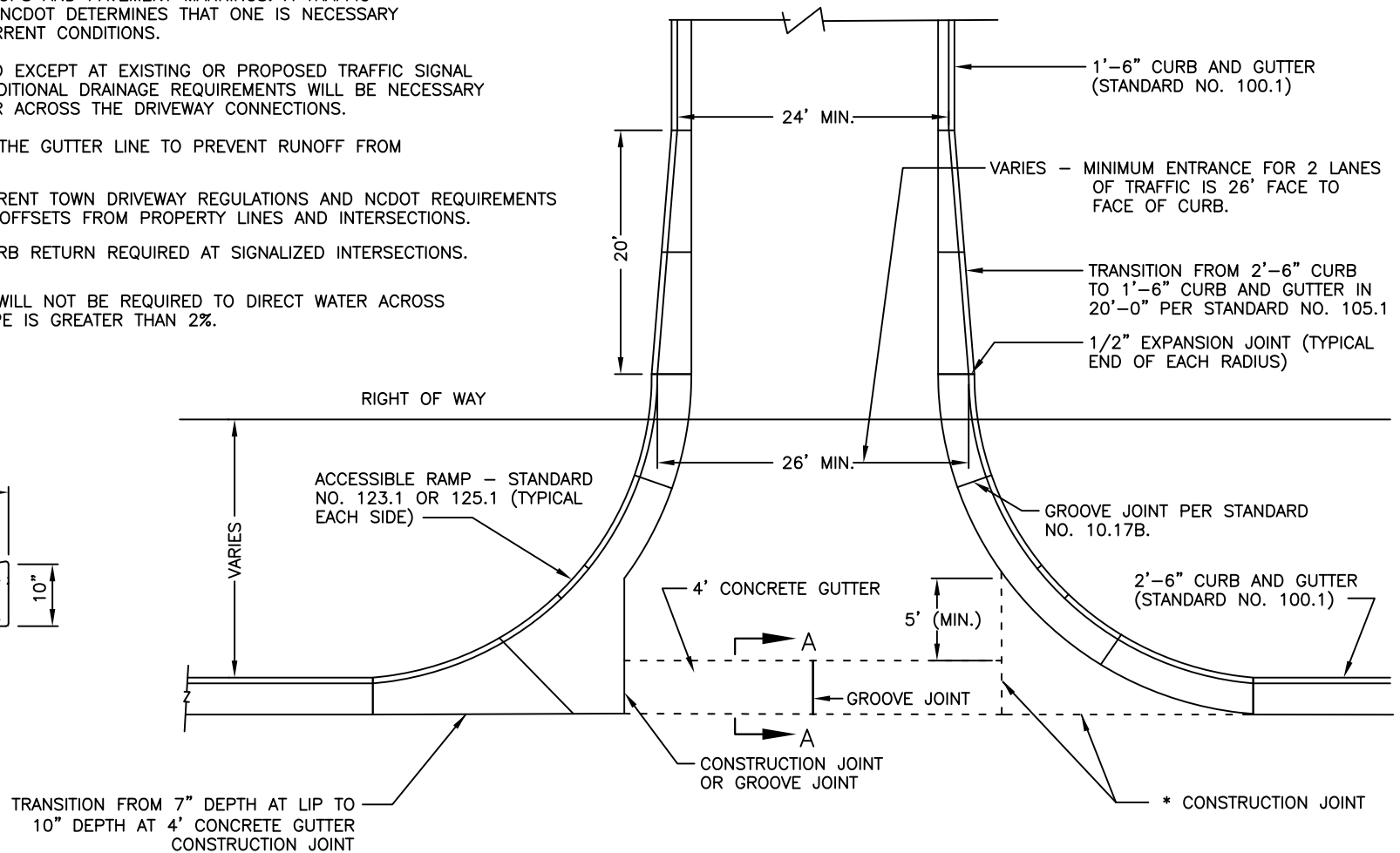
NOTES:

1. WHERE A TYPE III DRIVEWAY IS APPROVED BY THE NORTH CAROLINA DEPARTMENT OF TRANSPORTATION (NCDOT) THAT CONNECTS TO AN EXISTING SIGNALIZED INTERSECTION, OR AT A LOCATION WHERE A TRAFFIC SIGNAL INSTALLATION IS PROPOSED BY NCDOT BASED ON A TRAFFIC IMPACT/SIGNAL WARRANT STUDY, A FULL DEPTH ASPHALT PAVEMENT (2-1/2" S-9.5 B/C AND 6" B-25.0 B/C) IS REQUIRED. THIS PAVEMENT DESIGN IS REQUIRED IN THE DRIVEWAY EASEMENT (100-FOOT MINIMUM) TO MAINTAIN DETECTOR LOOPS AND PAVEMENT MARKINGS. A TRAFFIC SIGNAL WILL BE INSTALLED ONLY IF NCDOT DETERMINES THAT ONE IS NECESSARY BASED ON A TRAFFIC STUDY OF CURRENT CONDITIONS.
2. A CONCRETE GUTTER IS TO BE USED EXCEPT AT EXISTING OR PROPOSED TRAFFIC SIGNAL LOCATIONS. AT THESE LOCATIONS ADDITIONAL DRAINAGE REQUIREMENTS WILL BE NECESSARY TO ELIMINATE THE NEED FOR GUTTER ACROSS THE DRIVEWAY CONNECTIONS.
3. THE DRIVEWAY MUST RISE 6" FROM THE GUTTER LINE TO PREVENT RUNOFF FROM ENTERING DRIVEWAY.
4. ALL DRIVEWAYS MUST MEET THE CURRENT TOWN DRIVEWAY REGULATIONS AND NCDOT REQUIREMENTS FOR SPACING, SIGHT DISTANCE, AND OFFSETS FROM PROPERTY LINES AND INTERSECTIONS.
5. TWO (2) ACCESSIBLE RAMP PER CURB RETURN REQUIRED AT SIGNALIZED INTERSECTIONS.

* FOUR (4) FOOT GUTTER AND WINGS WILL NOT BE REQUIRED TO DIRECT WATER ACROSS DRIVE IF THE DRIVEWAY GUTTER SLOPE IS GREATER THAN 2%.

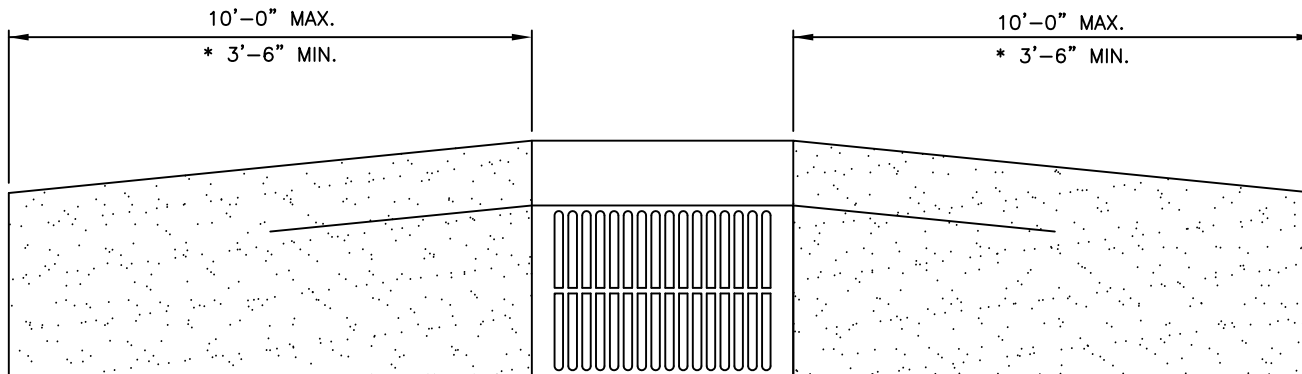


SECTION A-A



PLAN

NOT TO SCALE

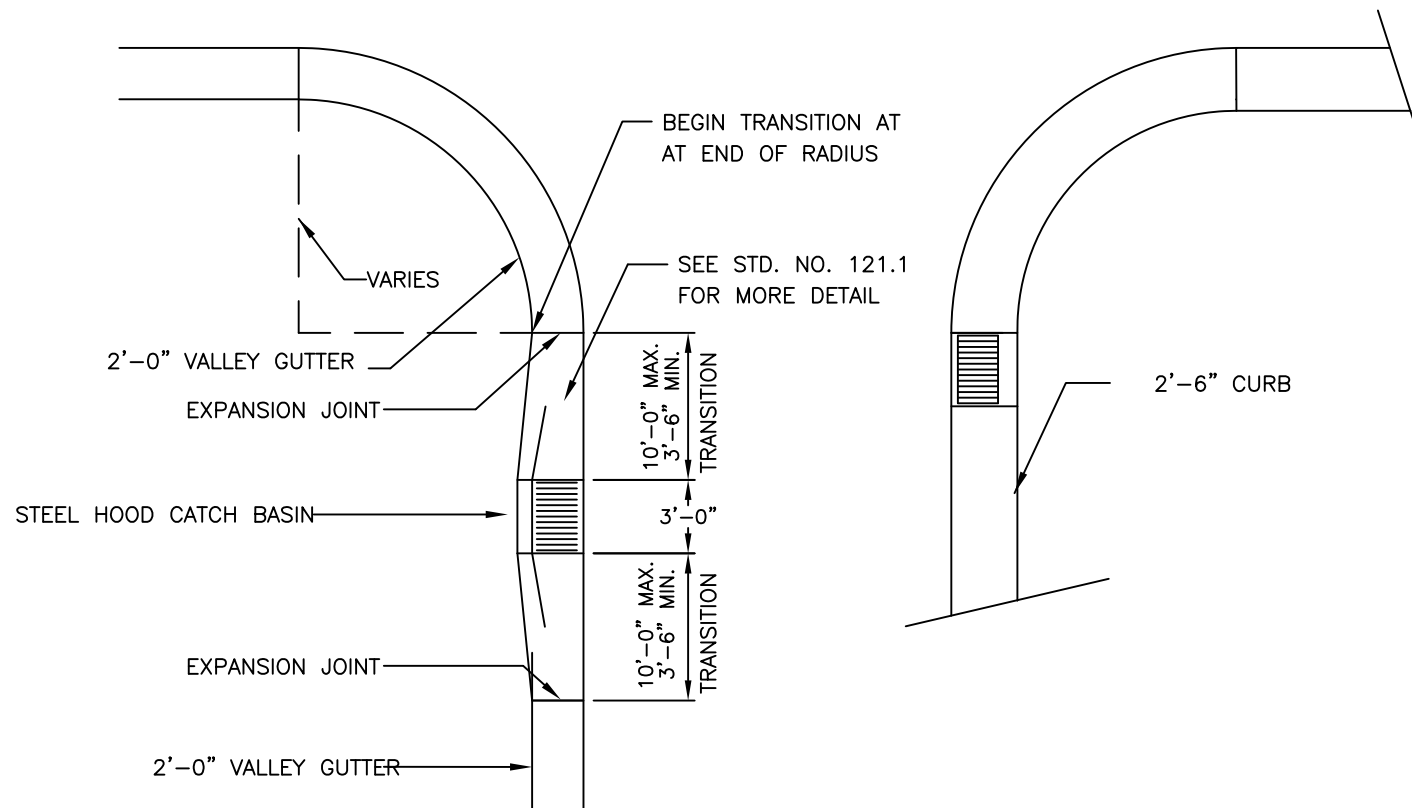


PLAN

NOTE:

- * TRANSITION FROM 2'-6" STANDARD CURB TO VALLEY CURB
AT A DRAINAGE INLET ONLY.
- SEE STANDARD 104.1 FOR CROSS SECTION GEOMETRY.

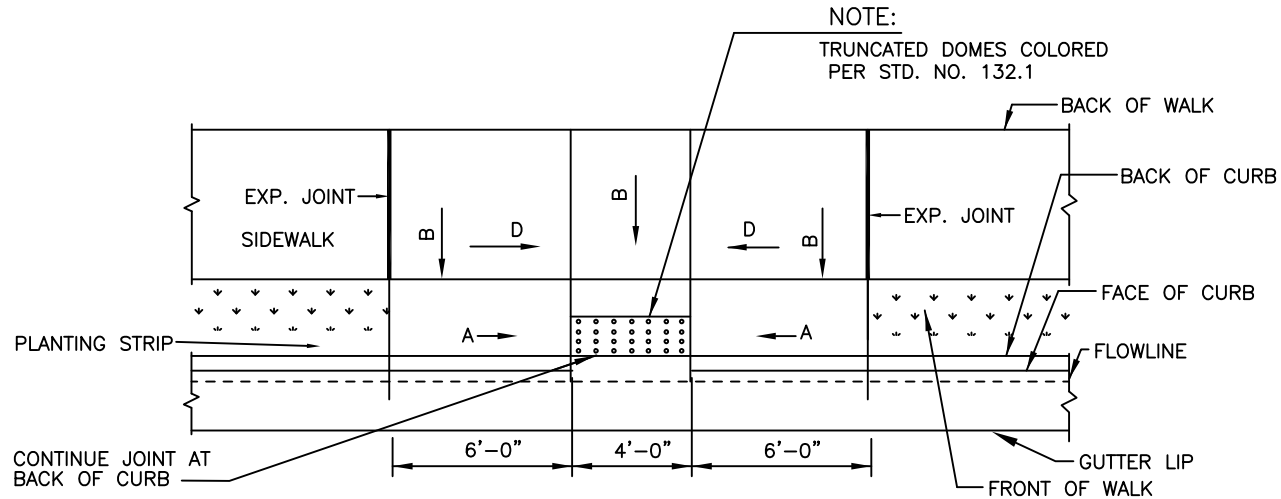
NOT TO SCALE



NOTE:

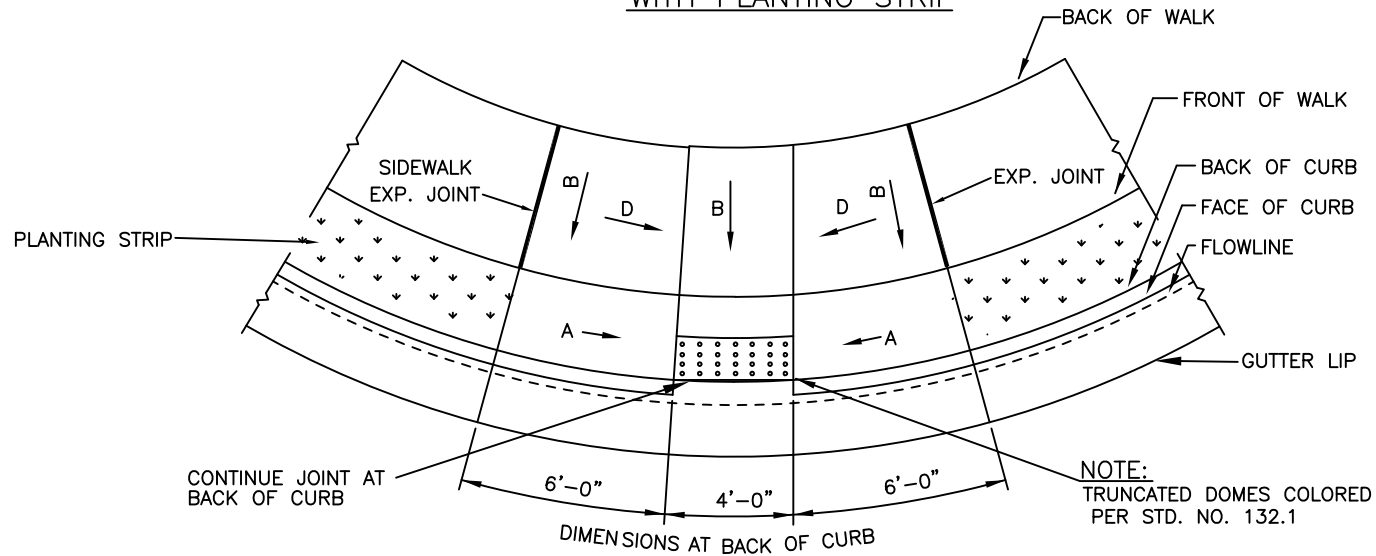
1. WHERE 2'-6" CURB AND GUTTER IS USED, CATCH BASINS MAY BE LOCATED AT END OF RADIUS.
2. RADIUS AT INTERSECTION MAY VARY.

NOT TO SCALE



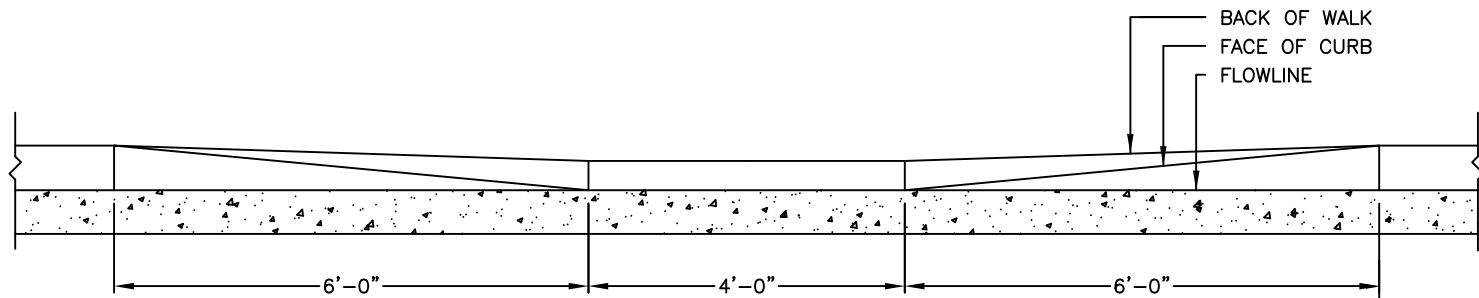
PLAN VIEW-PARALLEL RAMP
WITH PLANTING STRIP

SLOPE "A"	12:1
SLOPE "B"	1/4"/FT
SLOPE "D"	3/8"/FT

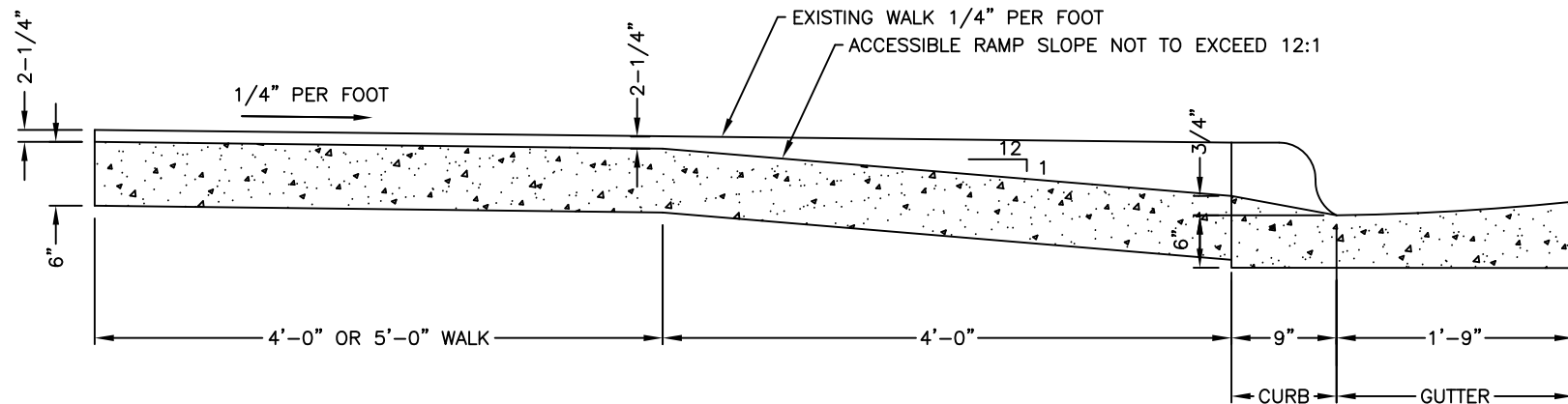


PLAN VIEW-DIAGONAL RAMP WITH PLANTING STRIP

NOT TO SCALE

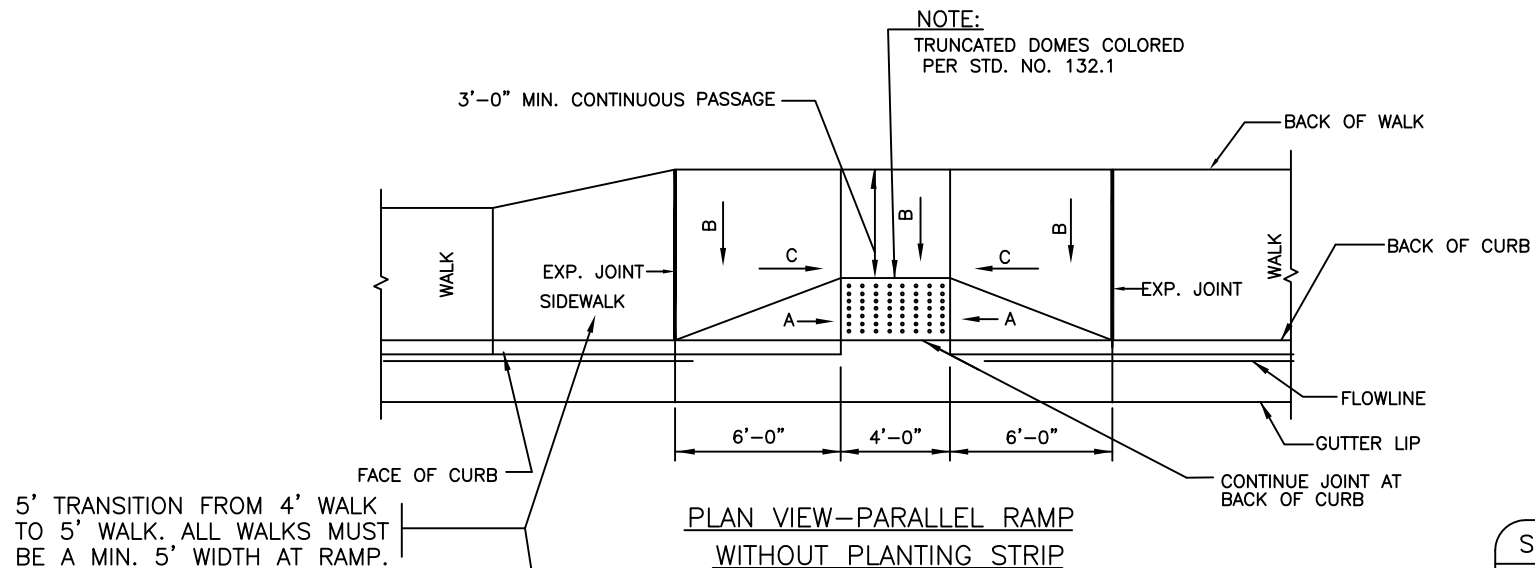


ELEVATION

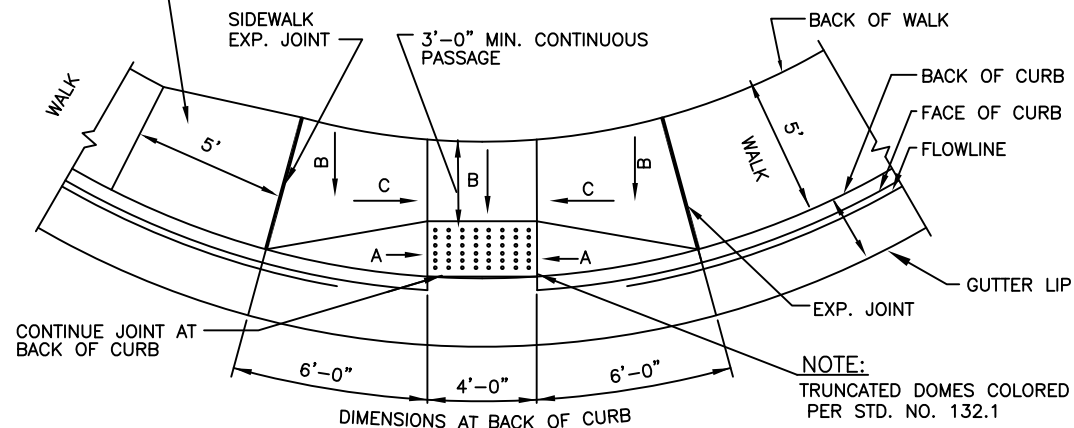


TYPICAL RAMP SECTION

NOT TO SCALE

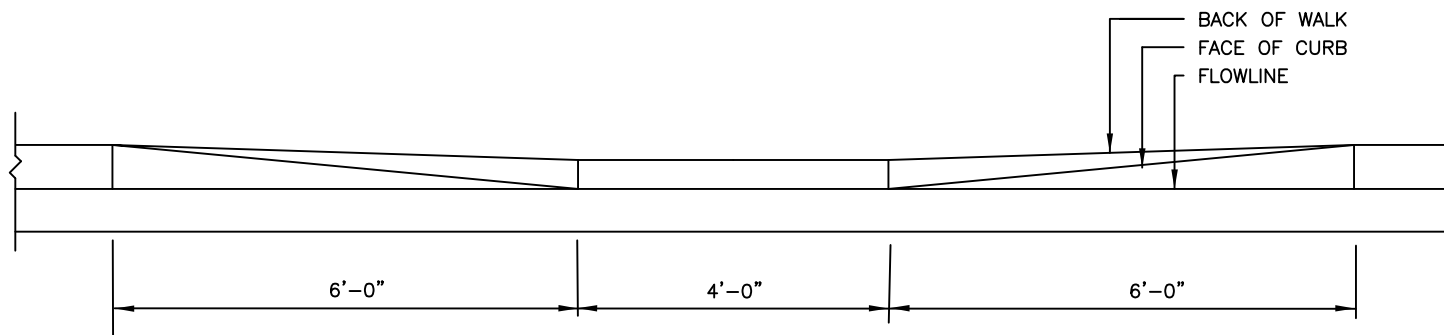


SLOPE "A"	12:1
SLOPE "B"	1/4"/FT
SLOPE "C"	5/8"/FT

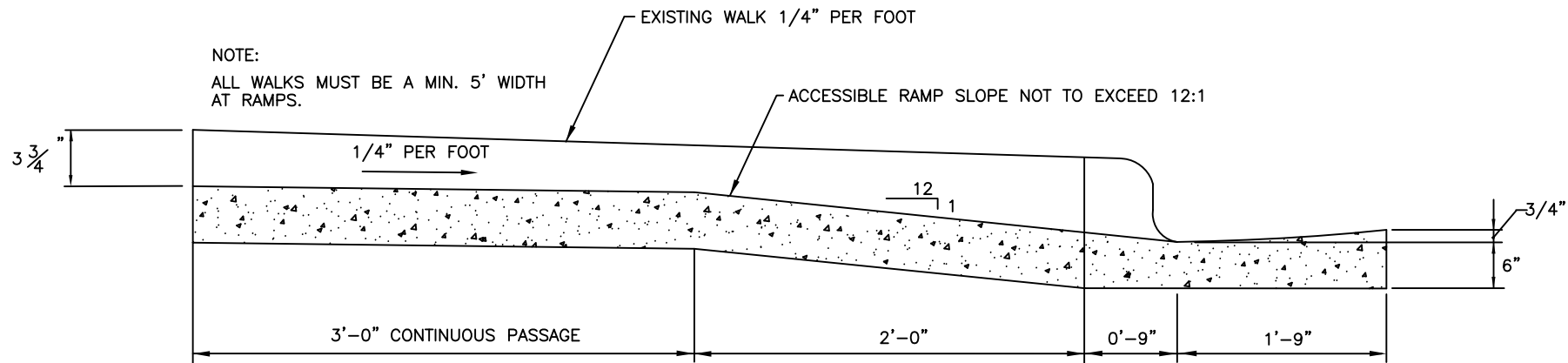


PLAN VIEW-DIAGONAL RAMP WITHOUT PLANTING STRIP

NOT TO SCALE

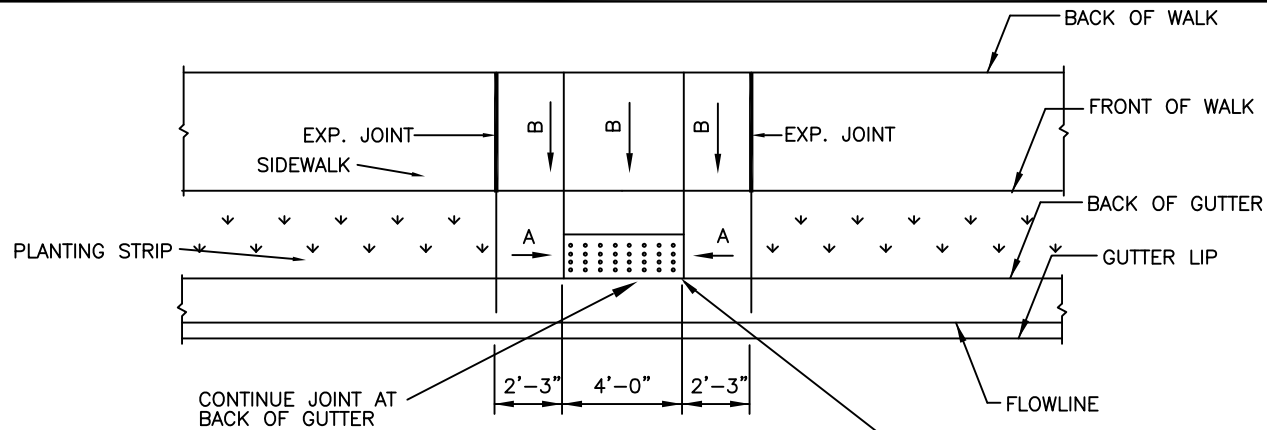


SECTION THROUGH FLOWLINE



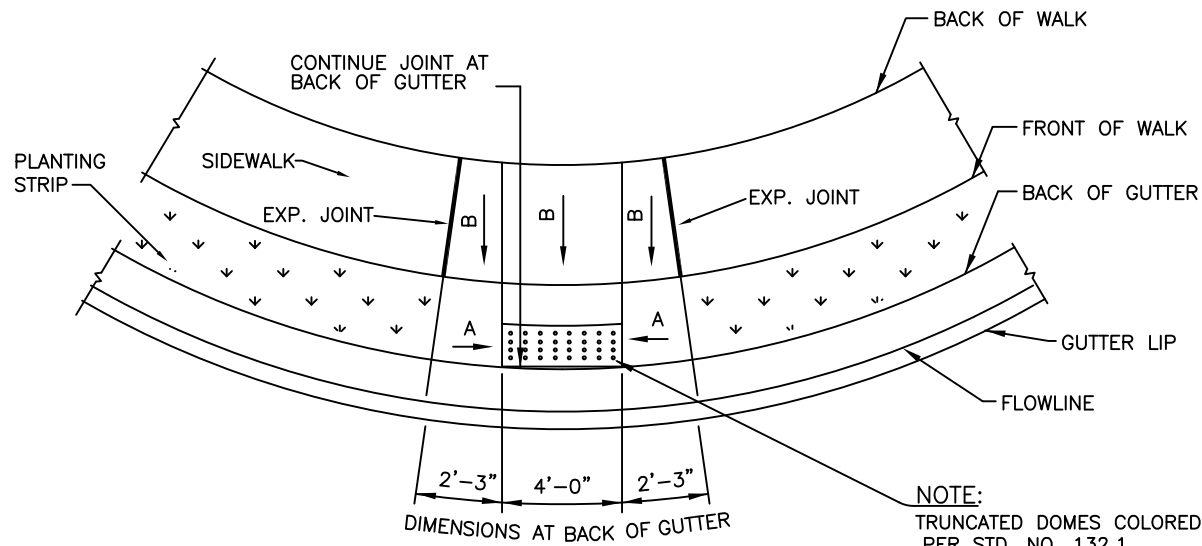
TYPICAL RAMP L SECTION

NOT TO SCALE



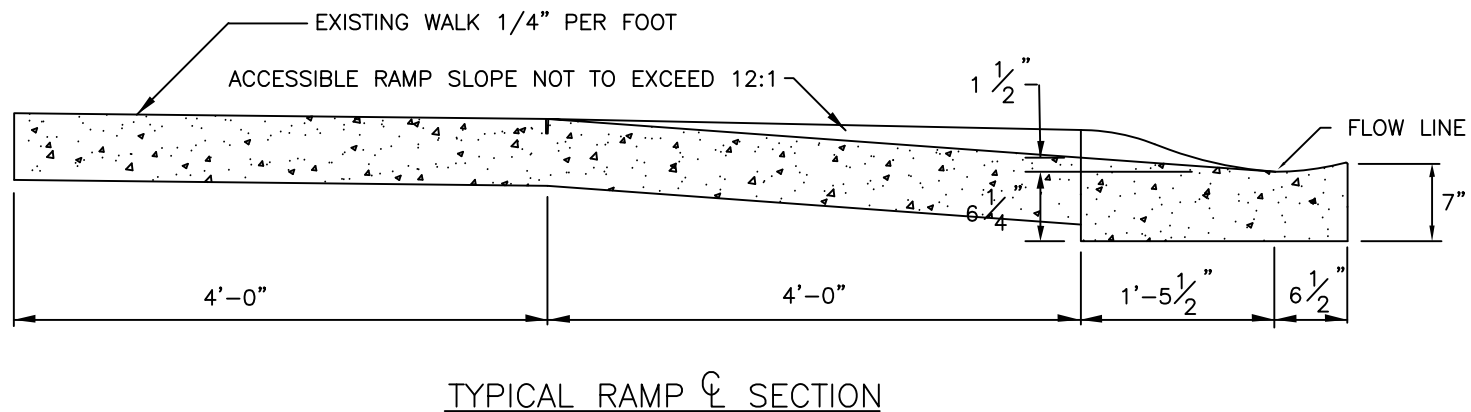
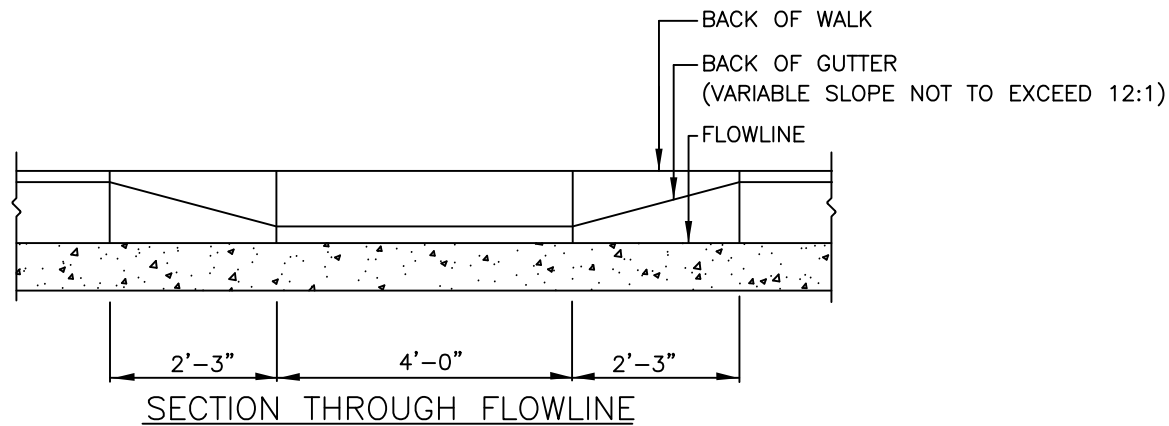
SLOPE "A"	12:1
SLOPE "B"	1/4"/FT

PLAN VIEW-PARALLEL
RAMP WITH PLANTING STRIP

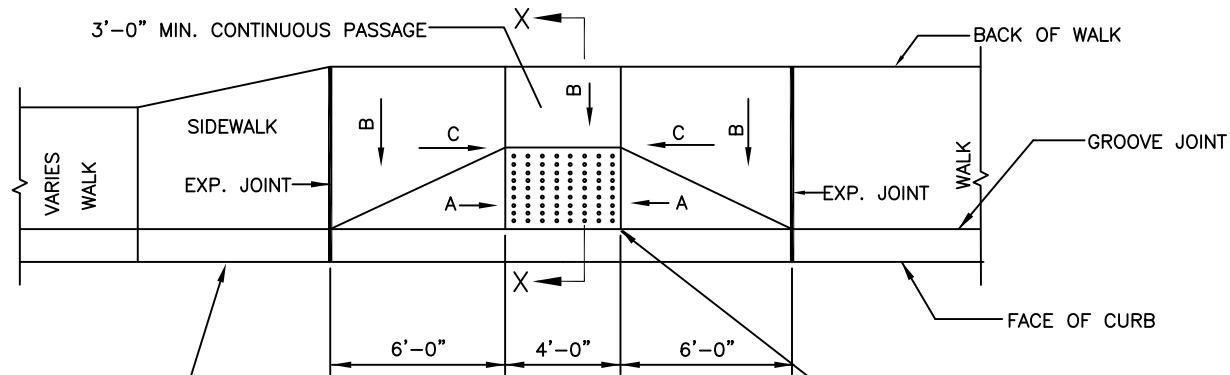


PLAN VIEW-DIAGONAL RAMP
WITH PLANTING STRIP

NOT TO SCALE



NOT TO SCALE

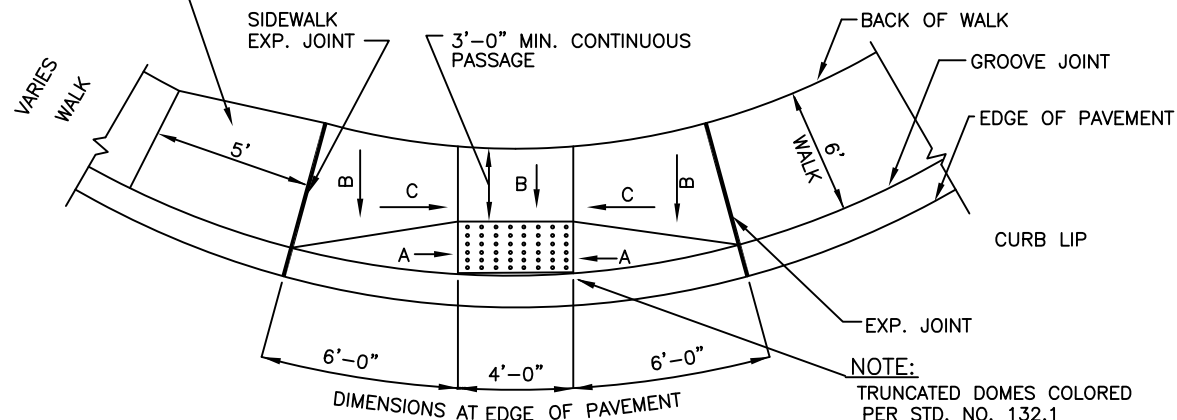


NOTE:
TRUNCATED DOMES COLORED
PER STD. NO. 132.1

PLAN VIEW-PARALLEL RAMP

5' TRANSITION FROM 4' WALK
TO 6' WALK. ALL WALKS MUST
BE A MIN. 6' WIDTH AT RAMP.

SLOPE "A"	12:1
SLOPE "B"	1/4"/FT
SLOPE "C"	1/2"/FT

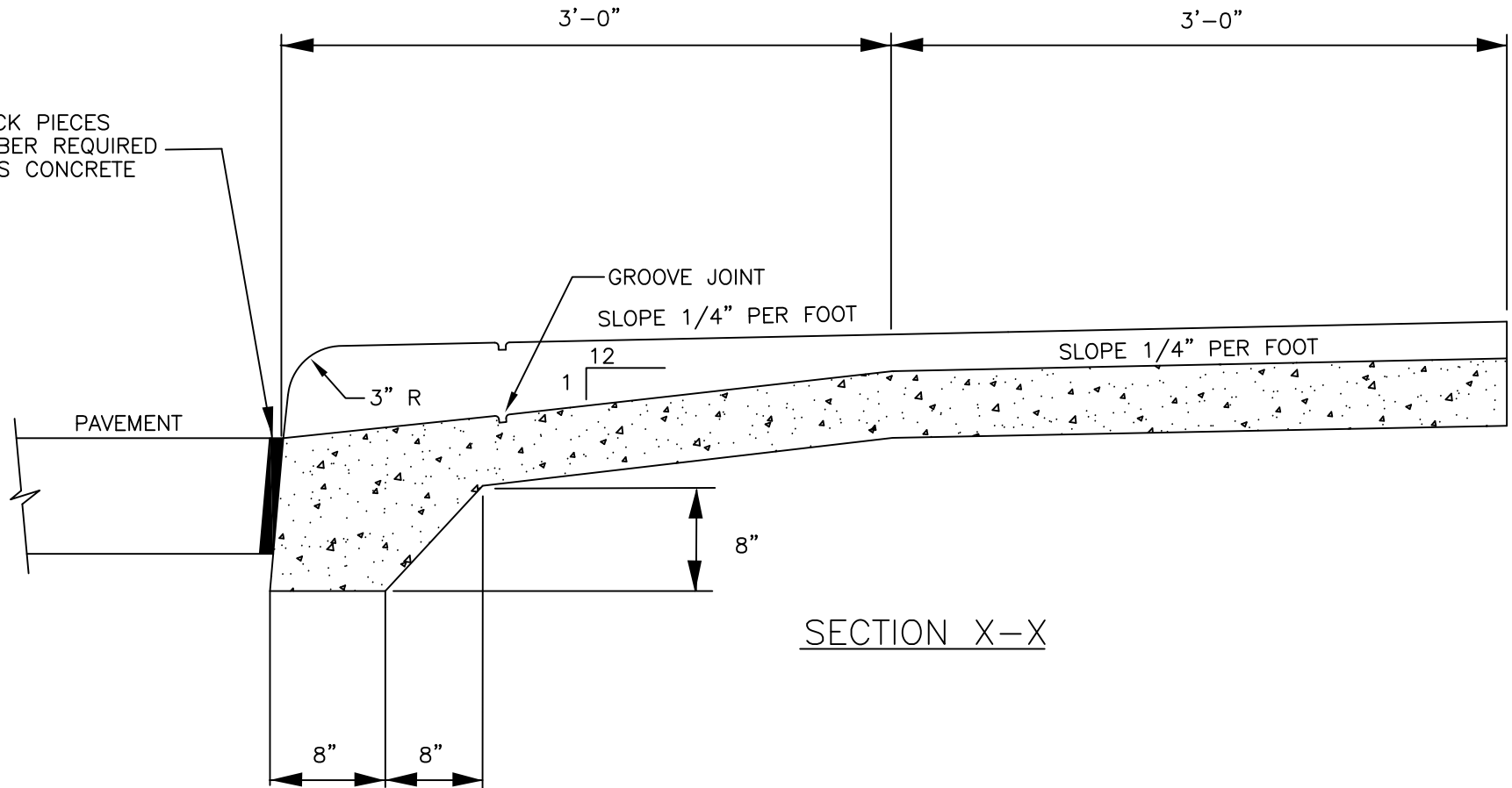


NOTE:
TRUNCATED DOMES COLORED
PER STD. NO. 132.1

PLAN VIEW-DIAGONAL RAMP

NOT TO SCALE

TWO 1/2" THICK PIECES
BITUMINOUS FIBER REQUIRED
IF PAVEMENT IS CONCRETE

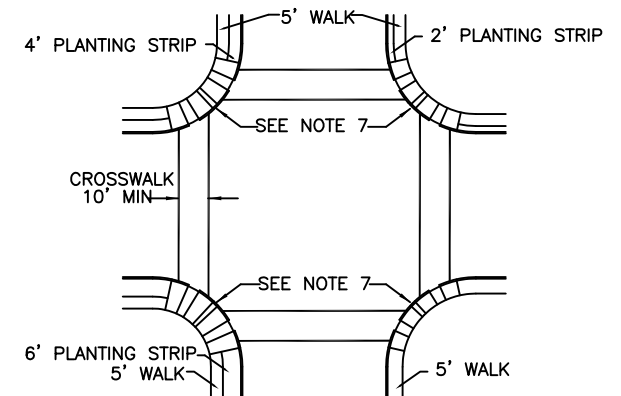
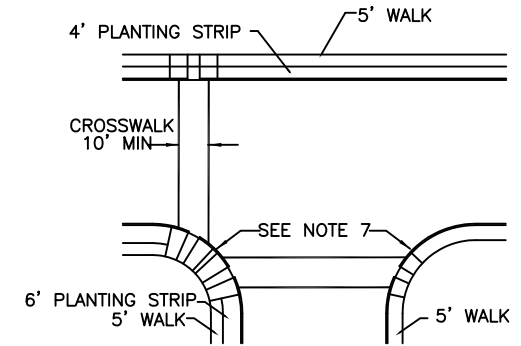
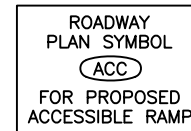
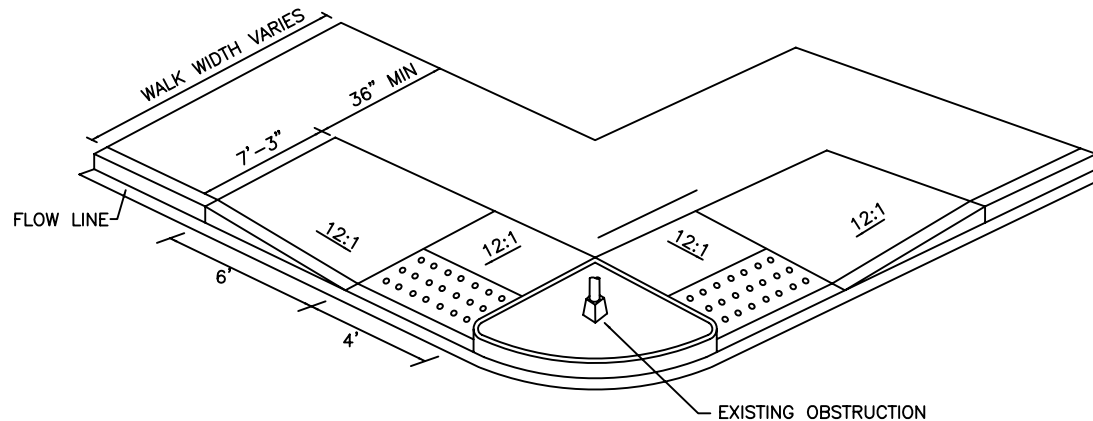


SECTION X-X

NOT TO SCALE

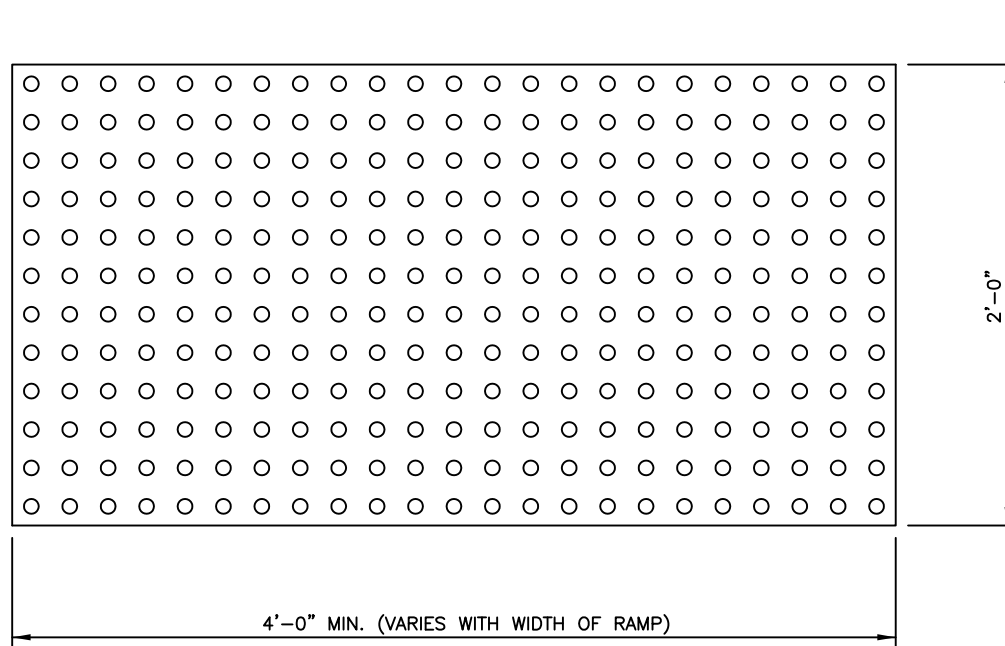
NOTES:

1. RAMP AND WING SLOPES SHALL NOT BE STEEPER THAN 12:1.
2. GUTTER FLOW LINE AND PLAN PROFILE SHALL BE MAINTAINED THROUGH THE RAMP AREA.
3. THE SURFACE OF THE RAMP SHALL BE FLUSH WITH THE FLOWLINE OF THE CURB AND GUTTER.
4. THE RAMP OPENING (AT THE FULLY DEPRESSED CURB) SHALL BE LOCATED WITHIN THE PARALLEL BOUNDARIES OF THE CROSSWALK MARKINGS. THE RAMP CENTERLINE SHALL BE LOCATED AT THE CORNER RADIUS CENTERLINE UNLESS OTHERWISE DIRECTED BY THE ENGINEER. DIAGONAL CURB RAMPS SHALL HAVE A SEGMENT OF STRAIGHT CURB AT LEAST 24 INCHES LONG LOCATED ON EACH SIDE OF THE WING SLOPE AND WITHIN THE CROSSWALK MARKINGS.
5. THE WING AND RAMP SURFACES SHALL BE 3600 PSI CONCRETE WITH A SIDEWALK FINISH IN ACCORDANCE WITH CURRENT EDITION NCDOT STANDARD SPECIFICATIONS FOR ROADS AND STRUCTURES.
6. DRAINAGE STRUCTURES, MAST ARMS, LIGHT POLES AND OTHER OBSTRUCTIONS SHALL NOT BE PLACED IN LINE WITH RAMPS. LOCATION OF THE RAMP SHALL TAKE PRECEDENCE OVER LOCATION OF OBSTRUCTIONS EXCEPT WHERE EXISTING OBSTRUCTIONS ARE BEING UTILIZED IN THE NEW CONSTRUCTION.
7. AT ALL LOCATIONS, NOT LESS THAN 2 FEET OF FULL HEIGHT CURB SHALL BE PLACED BETWEEN THE RAMPS.
8. SEE STANDARD DRAWING 132.1 FOR DETECTABLE WARNING INSTALLATION.



TYPICAL LOCATION OF ACCESSIBLE RAMPS AND PEDESTRIAN CROSSWALKS ON

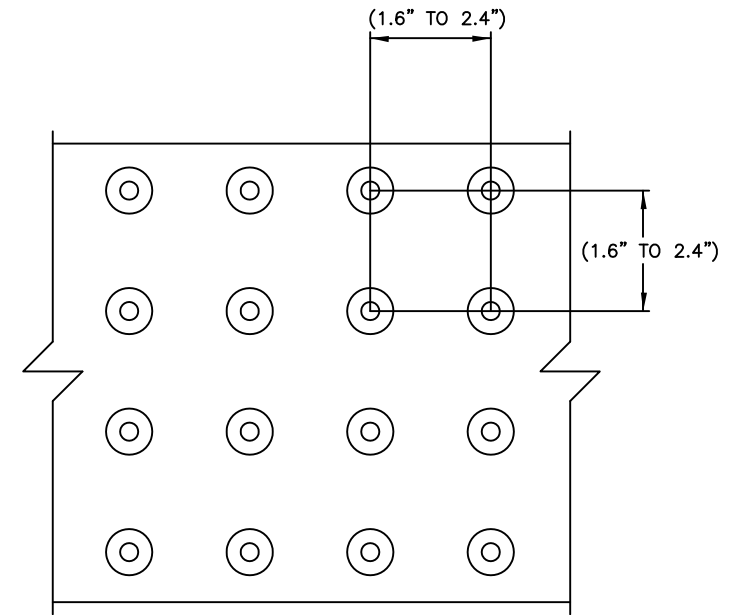
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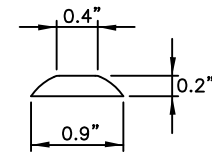
TRUNCATED DOME PLAN VIEW

NOTES:

1. ALL DETECTABLE WARNING DEVICES USED IN NEW CONSTRUCTION SHALL BE OF A RIGID PRECAST OR EMBEDDED PRODUCT APPROVED BY THE TOWN ENGINEER. RETRO FIT MATS WILL ONLY BE ALLOWED ON EXISTING RAMPS WITH PRIOR APPROVAL OF THE TOWN ENGINEER FOR MATERIAL TYPE AND INSTALLATION (IE. RESURFACING).
2. WIDTH OF DETECTABLE WARNING AREA SHALL BE A MINIMUM OF 4 FEET AND VARY WITH WIDTH OF RAMP.
3. LENGTH OF DETECTABLE WARNING AREA SHALL BE 2 FEET REGARDLESS OF SECTION WIDTH.
4. DETECTABLE WARNING AREA CAN BE SQUARE WHERE USED IN A CURB RADIUS.
5. DETECTABLE WARNING DOMES SHALL BE ALIGNED ON A SQUARE GRID IN THE PREDOMINANT DIRECTION OF OF TRAVEL TO PERMIT WHEELS TO ROLL BETWEEN DOMES.
6. DETECTABLE WARNING AREA SHALL BE COLORED BLACK IN ALL LOCATIONS.
7. IF PAVERS ARE TO BE USED, PAVERS SHALL BE 6" THICK AND CAST FROM 5000 psi CONCRETE.
8. MATS ARE TO BE RIGID WITH TURN DOWN EDGES EMBEDDED IN CONCRETE TO ELIMINATE TRIP HAZARD.



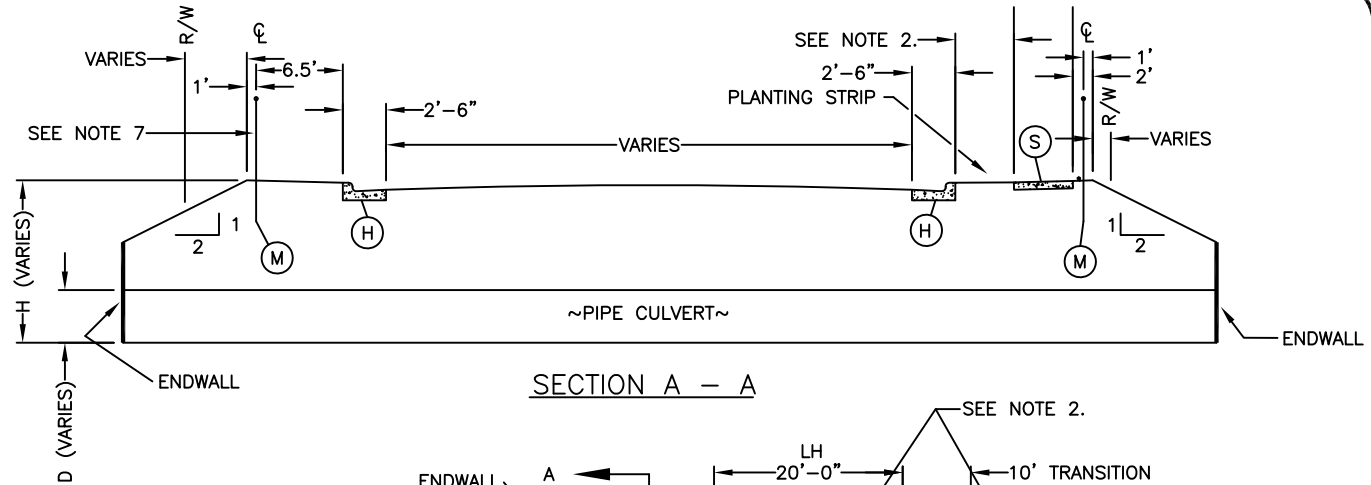
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TRUNCATED DOME SECTION

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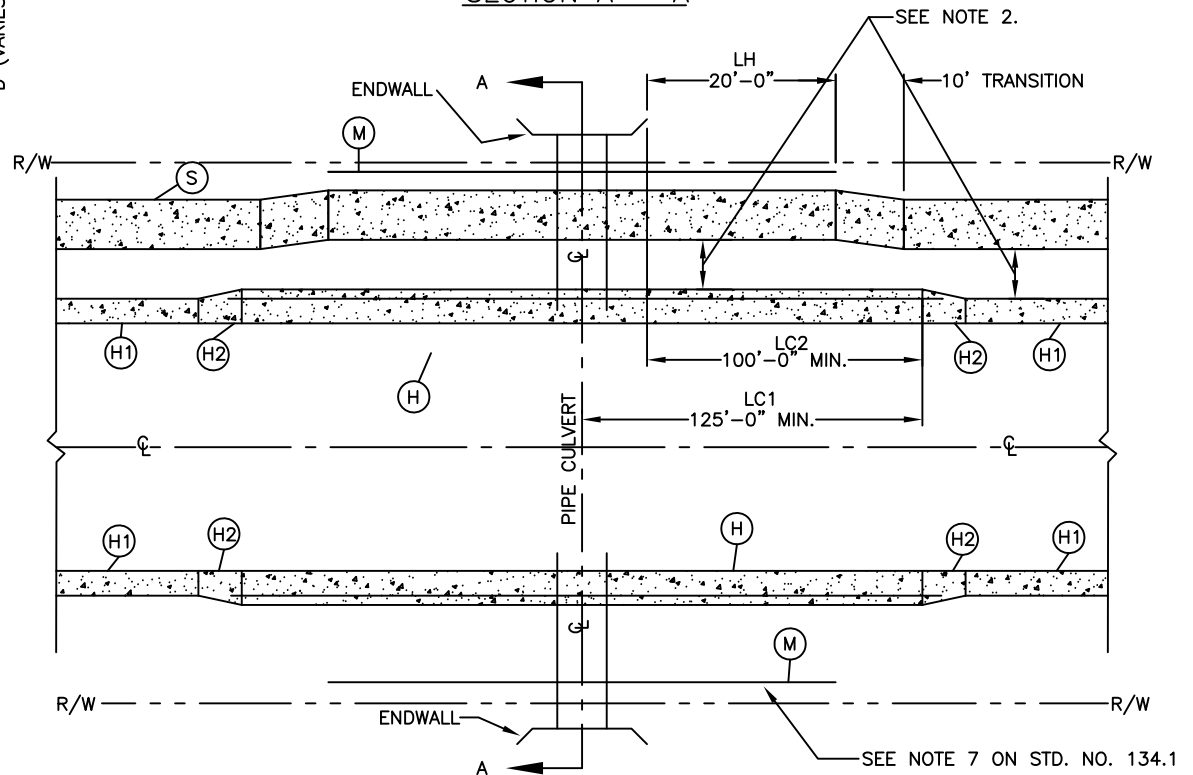
- (H) 2'-6" CURB AND GUTTER, STD. 100.1
- (M) HANDRAIL, STD. 700.1 & 701.1
- (S) 5'-0" SIDEWALK
- (H1) 2'-0" VALLEY GUTTER, STD. 101.1
- (H2) CURB TRANSITION 2'-6" CURB AND GUTTER TO 2'-0" VALLEY GUTTER, STD. 104.1



- LH = DISTANCE FROM END OF WINGWALL TO END OF HANDRAIL.
- LC1 = DISTANCE FROM ϕ OF CULVERT TO END OF 2'-6" CURB AND GUTTER.
- LC2 = DISTANCE FROM END OF WINGWALL TO END OF 2'-6" CURB AND GUTTER.

NOTES:

1. SEE STD. NO. 134.1 FOR GENERAL NOTES AND CLEAR ZONE DISTANCES
2. PLANTING STRIP WIDTH TO BE IN ACCORDANCE WITH CROSS SECTION PER TOWN REQUIREMENTS



NOT TO SCALE

GENERAL NOTES:

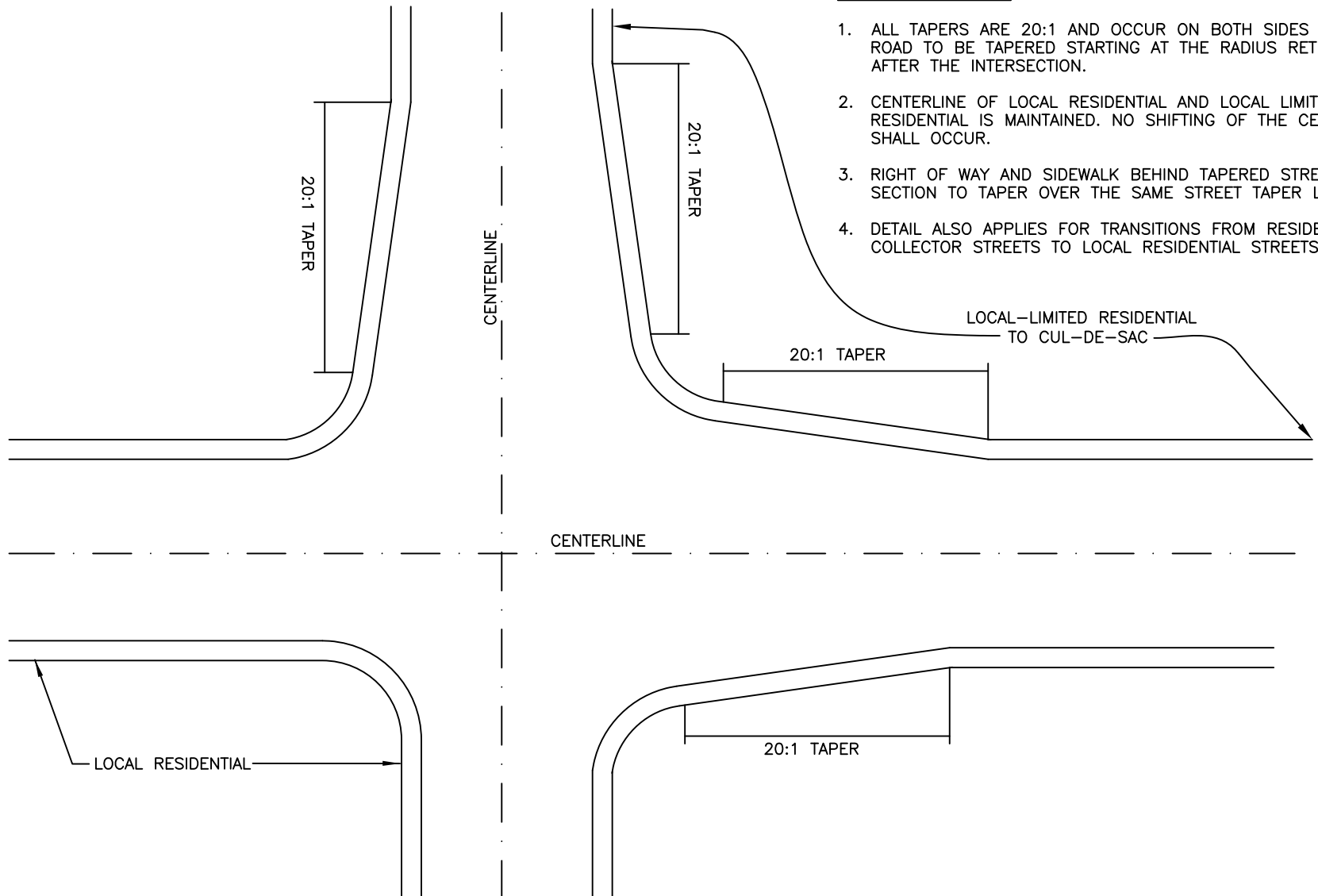
1. UNLESS OTHERWISE DETERMINED BY THE TOWN ENGINEER, THE MEASURES ILLUSTRATED SHALL BE USED WHEN CULVERT DIAMETER, D, IS GREATER THAN OR EQUAL TO 24 INCHES AND WHEN THE DIFFERENCE IN ELEVATION BETWEEN THE CULVERT INVERT AND THE TOP OF SLOPE, H, IS GREATER THAN OR EQUAL TO 5 FEET.
2. INSTALLATION OF 2'-6" CURB AND GUTTER MAY NOT BE REQUIRED WHEN AN ADEQUATE CLEAR ZONE IS PROVIDED FOR VEHICLES WITH A MAXIMUM OF 6:1 SLOPE (SEE TABLE 1).
3. INSTALLATION OF HANDRAIL MAY NOT BE REQUIRED WHEN A 10-FOOT PEDESTRIAN CLEAR ZONE IS PROVIDED BEHIND THE SIDEWALK WITH A MAXIMUM OF 6:1 SLOPE. WHERE NO SIDEWALK IS REQUIRED, INSTALLATION OF HANDRAIL MAY NOT BE REQUIRED WHEN A 15-FOOT PEDESTRIAN CLEAR ZONE IS PROVIDED BEHIND THE CURB WITH A MAXIMUM OF 6:1 SLOPE.
4. FOR CULVERT CROSSINGS WITHOUT ENDWALLS, LH AND LC2 SHALL BE MEASURED FROM THE OUTSIDE OF THE NEAREST WALL OF THE CULVERT BARREL.
5. FOR MULTIPLE BARREL CULVERT CROSSINGS, LC1 SHALL BE MEASURED FROM THE CENTERLINES OF THE OUTBOARD CULVERT BARRELS.
6. WHEN NECESSARY, AS DETERMINED BY THE TOWN ENGINEER, ADDITIONAL MEASURES MAY BE REQUIRED.
7. INSTALLATION OF HANDRAIL IS REQUIRED ON BOTH SIDES OF STREET IF SIDEWALK IS REQUIRED ON BOTH SIDES.
8. INSTALLATION OF HANDRAIL IS REQUIRED ON BOTH SIDES OF STREET IF NO SIDEWALK IS REQUIRED EXCEPT WHEN A 15-FOOT PEDESTRIAN CLEAR ZONE IS PROVIDED BEHIND THE CURB WITH A MAXIMUM OF 6:1 SLOPE.
9. INSTALLATION OF HANDRAIL IS REQUIRED ON THE SIDEWALK SIDE OF STREET IF SIDEWALK IS ONLY REQUIRED ON ONE SIDE OF STREET. PROVIDE HANDRAIL OR 15 FOOT CLEAR ZONE ON SIDE WITHOUT SIDEWALK.
10. DESIGN ADT IS CALCULATED ASSUMING A TRIP GENERATION OF 10 DAILY TRIPS PER SINGLE FAMILY DWELLING UNIT.

TABLE 1.
CLEAR ZONE DISTANCES
LOCAL, COLLECTOR, AND COMMERCIAL STREETS

DESIGN ADT	CLEAR ZONE FROM EDGE OF PAVEMENT	
	TANGENT SECTION	CURVE (WITHIN 125' OF CULVERT)
UNDER 750	10'	15'
750 - 1500	12'	18'
1501 - 6000	14'	21'
OVER 6000	16'	24'

SEE STD. NO. 133.1 FOR PLAN AND CROSS SECTIONAL SCHEMATICS.

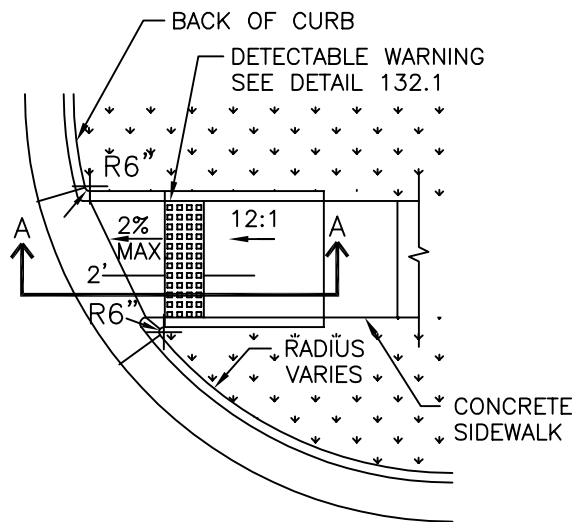
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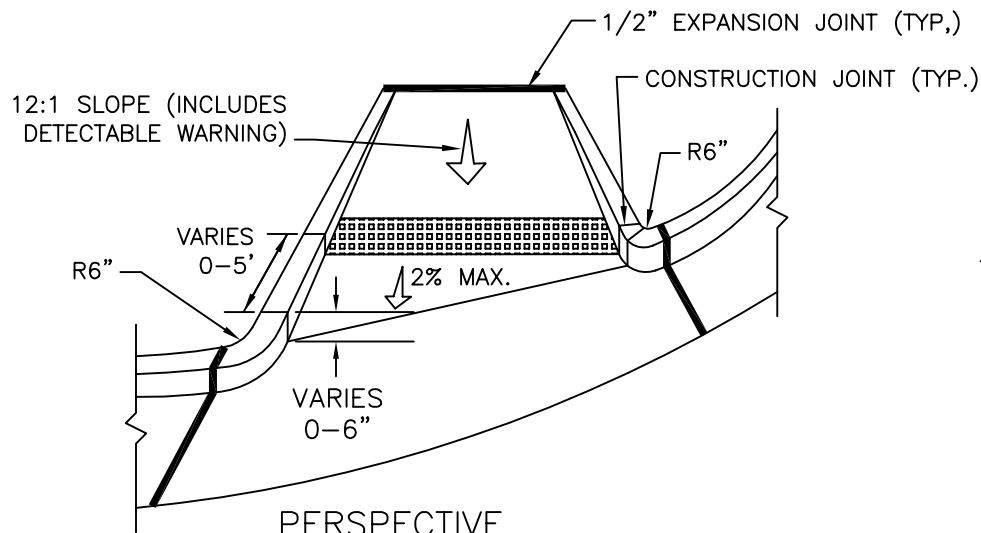
GENERAL NOTES:

1. ALL TAPERS ARE 20:1 AND OCCUR ON BOTH SIDES OF THE ROAD TO BE TAPERED STARTING AT THE RADIUS RETURN AFTER THE INTERSECTION.
2. CENTERLINE OF LOCAL RESIDENTIAL AND LOCAL LIMITED RESIDENTIAL IS MAINTAINED. NO SHIFTING OF THE CENTERLINE SHALL OCCUR.
3. RIGHT OF WAY AND SIDEWALK BEHIND TAPERED STREET SECTION TO TAPER OVER THE SAME STREET TAPER LENGTH.
4. DETAIL ALSO APPLIES FOR TRANSITIONS FROM RESIDENTIAL COLLECTOR STREETS TO LOCAL RESIDENTIAL STREETS.

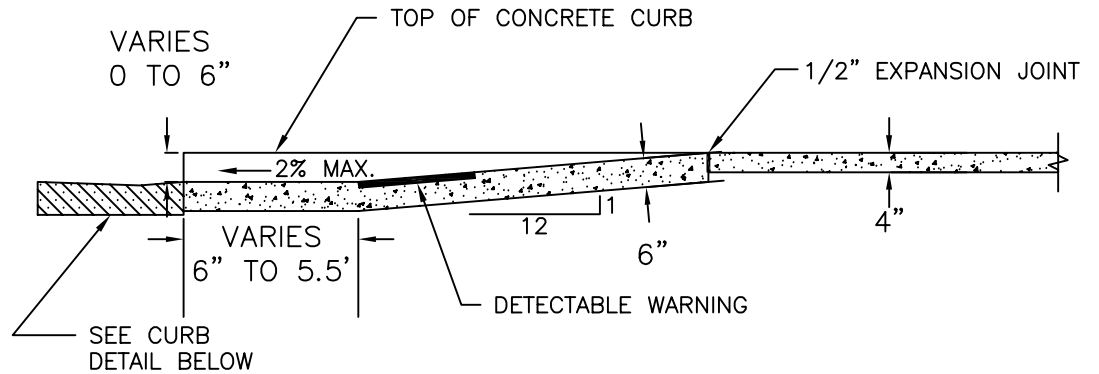
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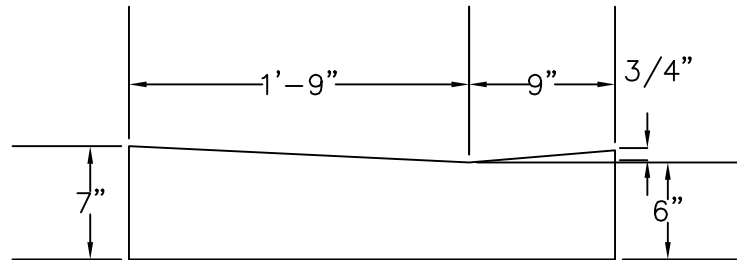
PLAN



PERSPECTIVE



SECTION A-A

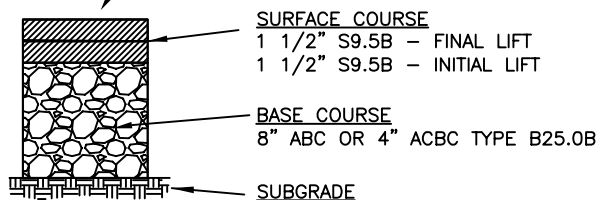
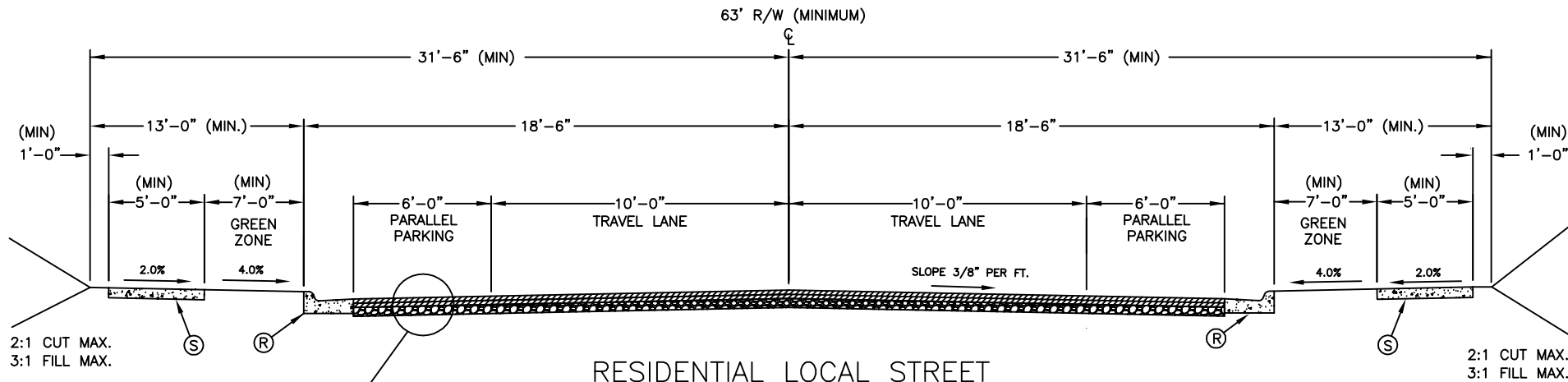


CURB DETAIL

NOTES:

1. DIRECTIONAL RAMPS MAY BE USED WHEN A PLANTING STRIP IS PROVIDED. DO NOT USE THIS DETAIL IF THERE IS HARDSCAPE INSTEAD OF A PLANTING STRIP.
2. ALL CONCRETE SHALL BE AT LEAST 3600 PSI.

NOT TO SCALE



TYPICAL PAVEMENT SECTION

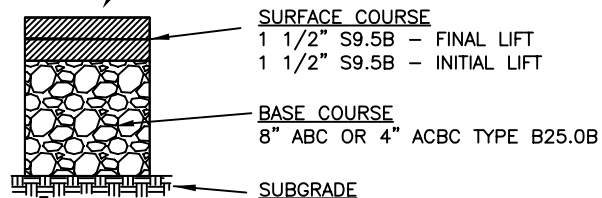
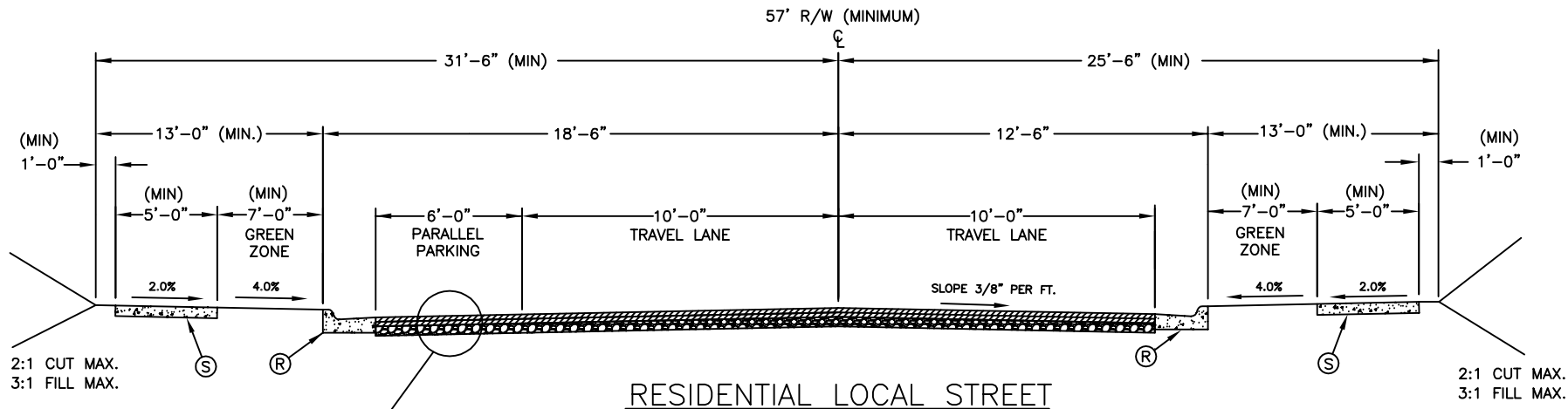
NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. REFER TO STANDARD DRAWING 285.1 FOR PARALLEL PARKING LAYOUT.

KEY

- (R) 2'-6" STD. CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK

NOT TO SCALE



TYPICAL PAVEMENT SECTION

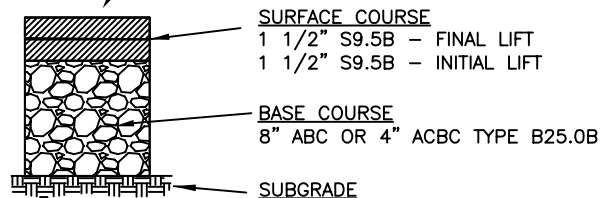
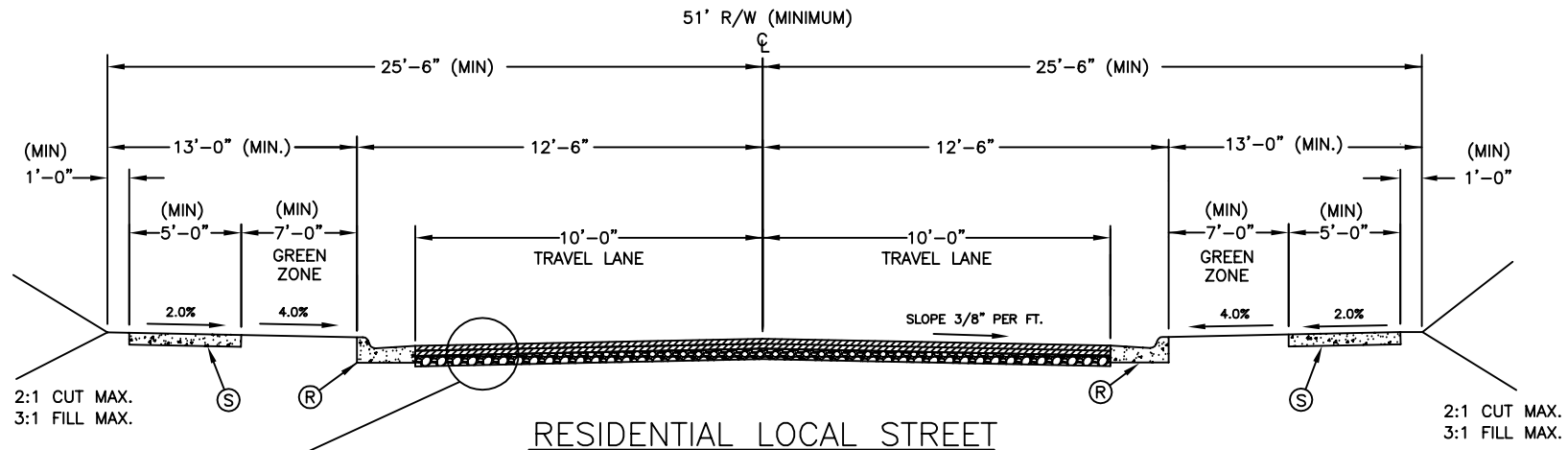
KEY

- (R) 2'-6" STD. CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK

NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. 2'-0" VALLEY GUTTER MAY BE SUBSTITUTED FOR 2'-6" CURB AND GUTTER ON THE SIDE OF THE STREET WITHOUT PARALLEL PARKING. THIS REDUCES THE MINIMUM RIGHT-OF-WAY BY SIX INCHES. 2'-0" VALLEY GUTTER MAY NOT BE SUBSTITUTED FOR 2'-6" CURB AND GUTTER ON THE SIDE OF THE STREET WITH PARALLEL PARKING.
4. REFER TO STANDARD DRAWING 285.1 FOR PARALLEL PARKING LAYOUT.

NOT TO SCALE



TYPICAL PAVEMENT SECTION

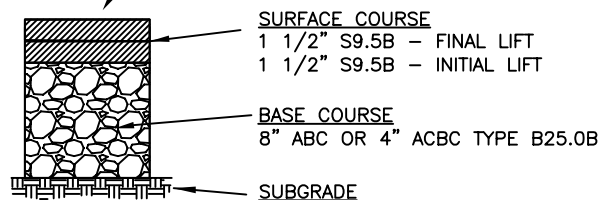
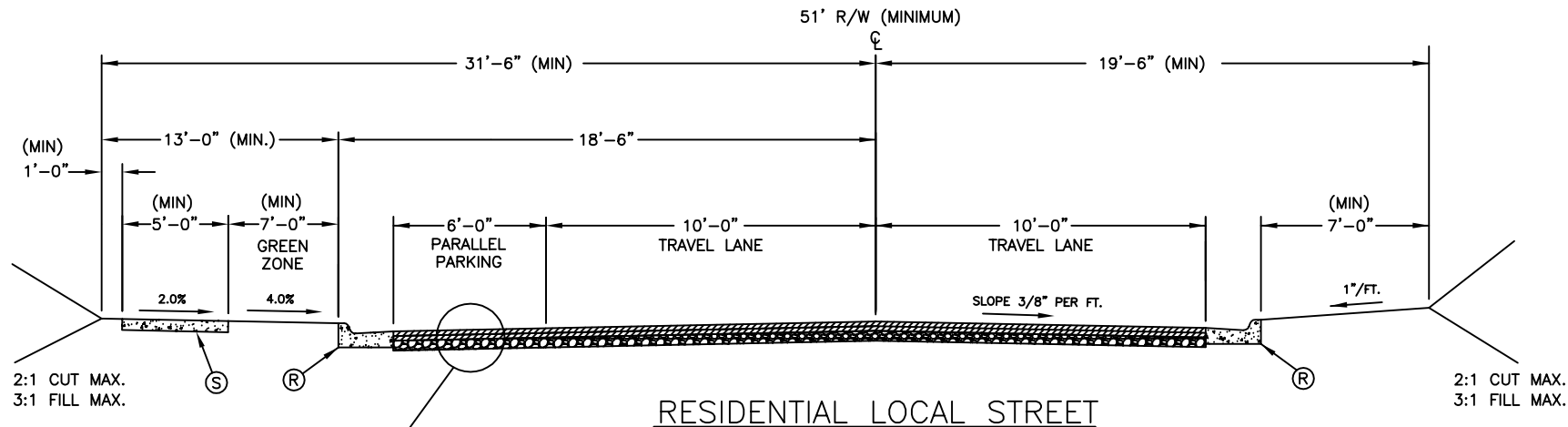
NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. 2'-0" VALLEY GUTTER MAY BE SUBSTITUTED FOR 2'-6" CURB AND GUTTER. THIS REDUCES THE MINIMUM RIGHT-OF-WAY BY ONE FOOT.

KEY

- (R) 2'-6" STD. CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK

NOT TO SCALE



TYPICAL PAVEMENT SECTION

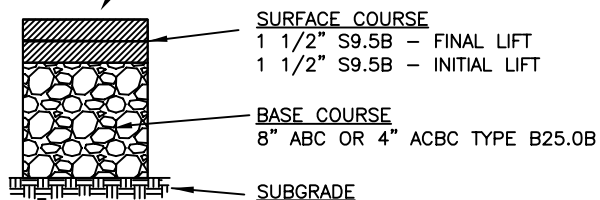
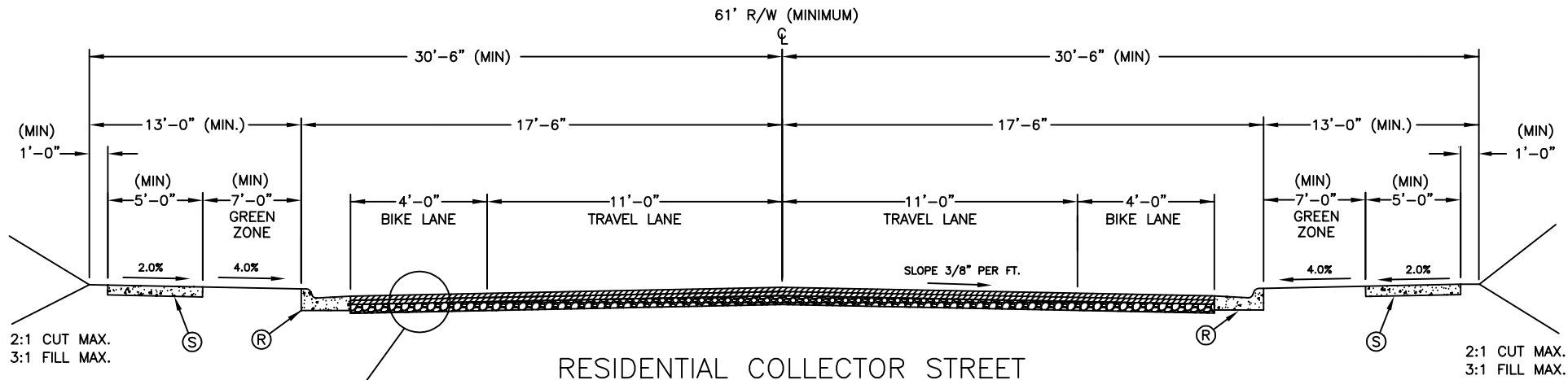
KEY

- (R) 2'-6" STD. CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK

NOTES:

1. SIDEWALK SHALL BE PROVIDED ON THE BUILDING SIDE OF STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. 2'-0" VALLEY GUTTER MAY BE SUBSTITUTED FOR 2'-6" CURB AND GUTTER ON THE SIDE OF THE STREET WITHOUT PARALLEL PARKING. THIS REDUCES THE MINIMUM RIGHT-OF-WAY BY SIX INCHES. 2'-0" VALLEY GUTTER MAY NOT BE SUBSTITUTED FOR 2'-6" CURB AND GUTTER ON THE SIDE OF THE STREET WITH PARALLEL PARKING.
4. REFER TO STANDARD DRAWING 285.1 FOR PARALLEL PARKING LAYOUT.

NOT TO SCALE



TYPICAL PAVEMENT SECTION

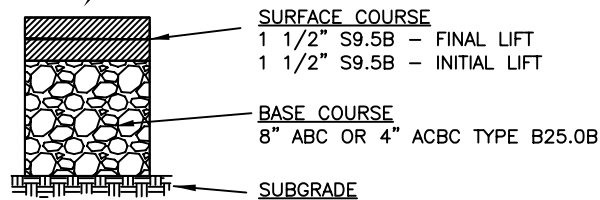
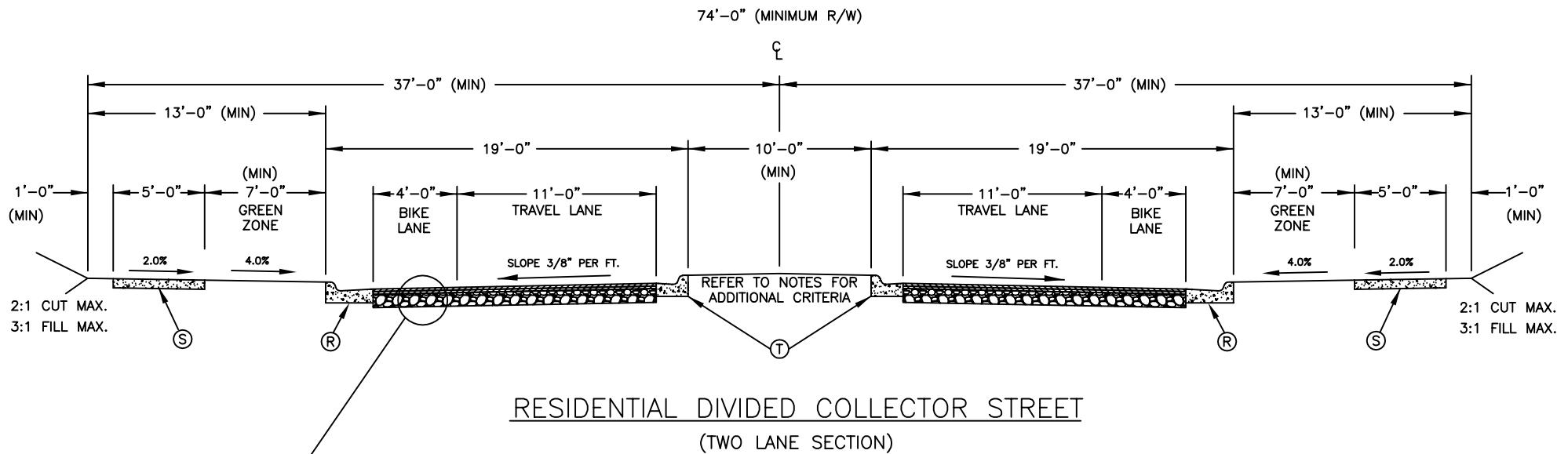
KEY

- (R) 2'-6" STD. CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK

NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. BIKE LANE TO BE STRIPED.

NOT TO SCALE



TYPICAL PAVEMENT SECTION

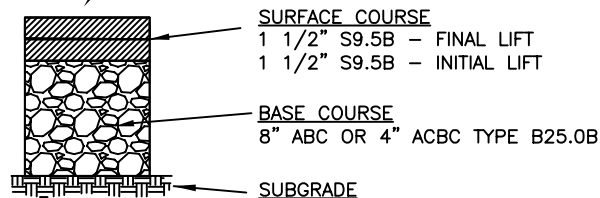
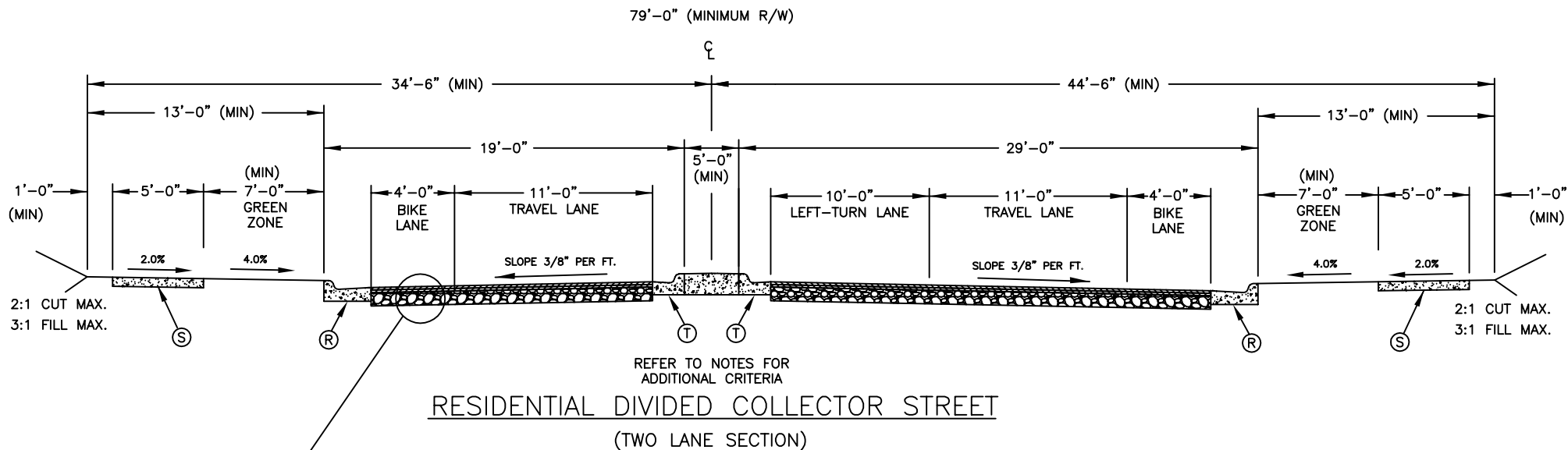
KEY

- (R) 2'-6" STD. CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK
- (T) 1'-6" MEDIAN CURB AND GUTTER

NOTES:

1. CURB RETURN RADIUS DIMENSIONS AT INTERSECTIONS MAY VARY DEPENDING ON MEDIAN WIDTH AND WILL BE APPROVED ON A CASE BY CASE BASIS.
2. SUBDRAINS ARE REQUIRED ON ALL MEDIANS. (TO BE TIED INTO STORM DRAINAGE SYSTEM.) REFER TO SUBDRAIN STANDARD DETAIL 312.1.
3. MEDIAN PLANTINGS SHALL NOT OBSTRUCT INTERSECTION SIGHT DISTANCE REQUIREMENTS.
4. A TEN (10) FOOT WIDE MEDIAN IS REQUIRED FOR SMALL MATURING TREES. A TWENTY (20) FOOT WIDE MEDIAN IS REQUIRED FOR LARGE MATURING TREES.
5. BIKE LANE TO BE STRIPED.
6. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.

NOT TO SCALE



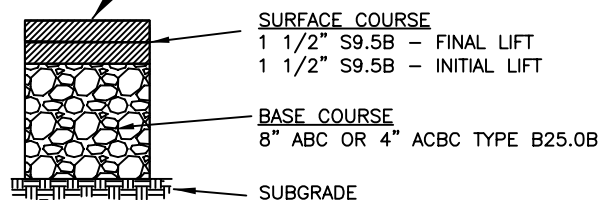
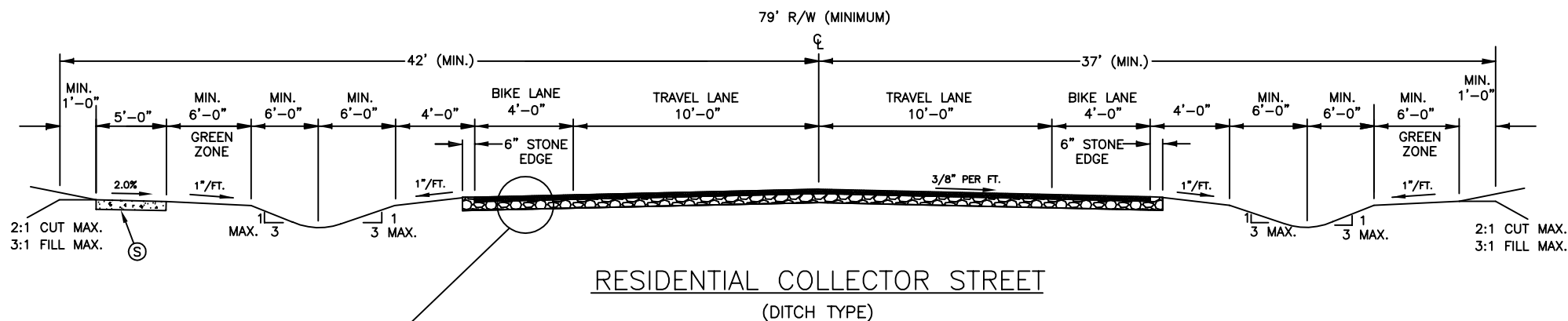
KEY

- (R) 2'-6" STANDARD CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK
- (T) 1'-6" MEDIAN CURB AND GUTTER

NOTES:

1. CURB RETURN RADIUS DIMENSIONS AT INTERSECTIONS MAY VARY DEPENDING ON MEDIAN WIDTH AND WILL BE APPROVED ON A CASE BY CASE BASIS.
2. SUBDRAINS ARE REQUIRED ON ALL MEDIANS. (TO BE TIED INTO STORM DRAINAGE SYSTEM.) REFER TO SUBDRAIN STANDARD DETAIL 312.1.
3. MEDIAN PLANTINGS SHALL NOT OBSTRUCT INTERSECTION SIGHT DISTANCE REQUIREMENTS.
4. TEN (10) FOOT WIDE MEDIANS CAN ACCOMMODATE SMALL MATURING TREES. TWENTY (20) FOOT WIDE MEDIAN IS REQUIRED FOR LARGE MATURING TREES.
5. MONOLITHIC CONCRETE MEDIANS WITH BEVELED EDGES AND A MINIMUM WIDTH OF 6 FEET CAN BE USED IN LIEU OF LANDSCAPED MEDIANS.
6. BIKE LANE TO BE STRIPED.
7. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.

NOT TO SCALE



TYPICAL PAVEMENT SECTION

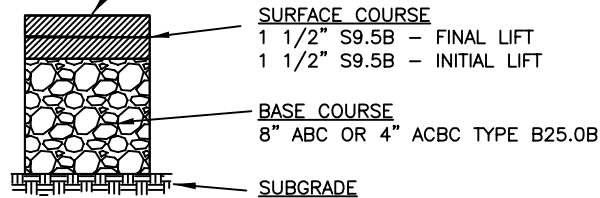
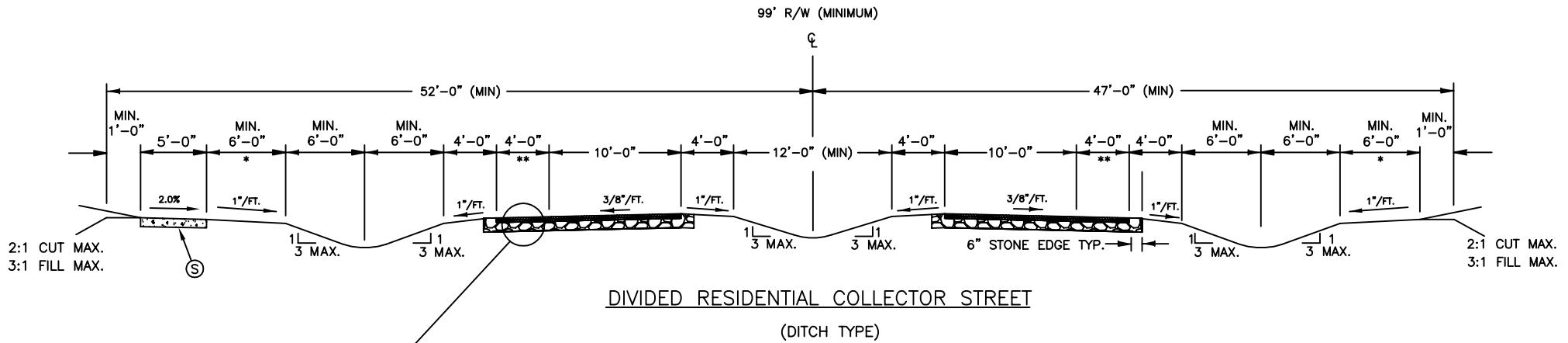
KEY

(S) 4" CONCRETE SIDEWALK

NOTES:

1. SIDEWALK LOCATED OUTSIDE OF STREET RIGHT-OF-WAY SHALL BE LOCATED IN A PERMANENT SIDEWALK EASEMENT EXTENDING 1 FOOT BEYOND BACK OF SIDEWALK.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. TREES TO BE PLACED IN GREEN ZONE 3.5 FEET FROM FACE OF SIDEWALK.
4. BIKE LANE TO BE STRIPED.

NOT TO SCALE



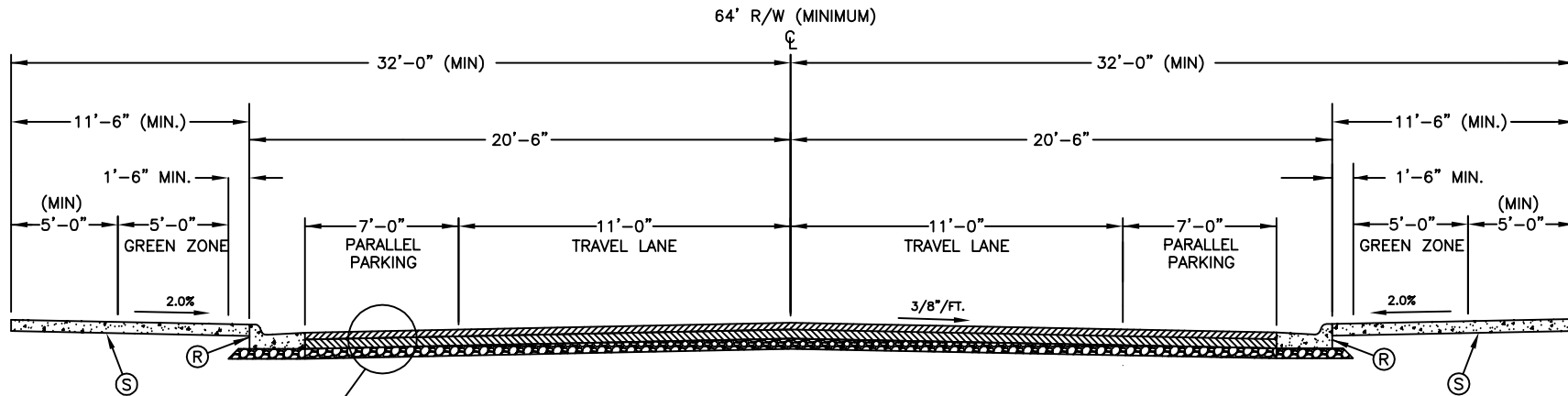
NOTES:

1. SIDEWALK LOCATED OUTSIDE OF STREET RIGHT-OF-WAY SHALL BE LOCATED IN A PERMANENT SIDEWALK EASEMENT EXTENDING 1 FOOT BEYOND BACK OF SIDEWALK.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. TREES TO BE PLACED IN GREEN ZONE 3.5 FEET FROM FACE OF SIDEWALK.
4. BIKE LANE TO BE STRIPED.

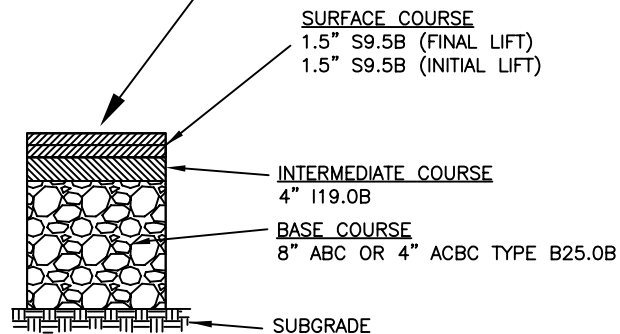
KEY

- Ⓢ 4" CONCRETE SIDEWALK
- * GREEN ZONE
- ** BIKE LANE

NOT TO SCALE



RETAIL/MIXED USE LOCAL STREET



TYPICAL PAVEMENT SECTION

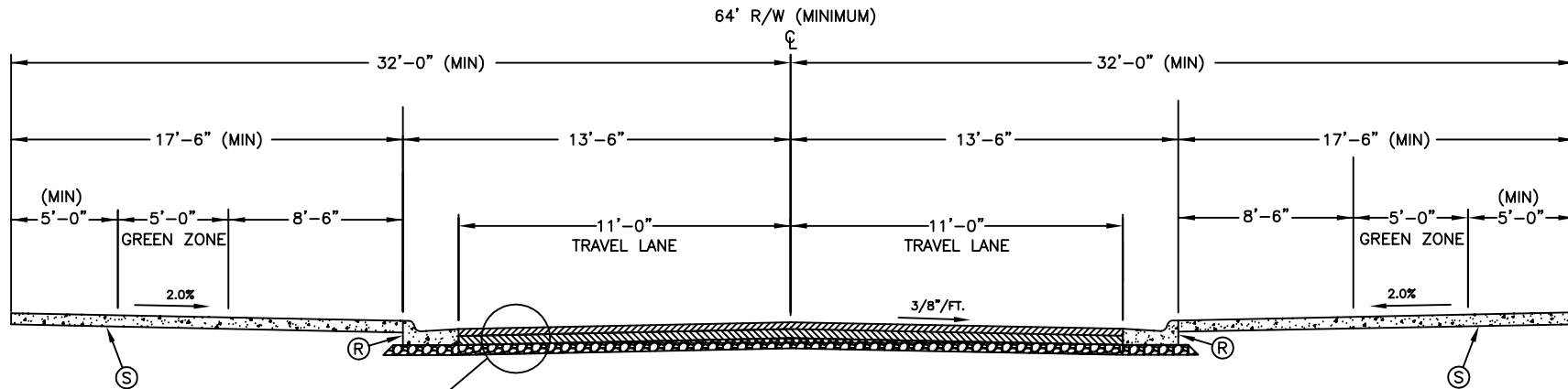
KEY

- (R) 2'-6" STD. CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK

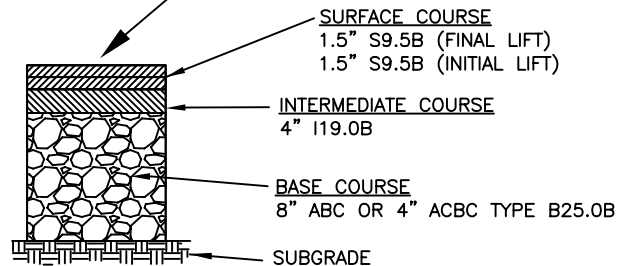
NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. REFER TO STANDARD DRAWINGS 602.1 - 605.1 REGARDING SIDEWALK AROUND TREE GRATES. TREE GRATES SHALL BE PROVIDED WHEN TREES ARE LOCATED IN THE GREEN ZONE.
4. BASE COURSE TO EXTEND SIX INCHES BEYOND BACK OF CURB THEN TAPER OUT AT A FORTY-FIVE DEGREE ANGLE.
5. REFER TO STANDARD DRAWING 285.1 FOR PARALLEL PARKING LAYOUT.

NOT TO SCALE



RETAIL/MIXED USE LOCAL STREET



TYPICAL PAVEMENT SECTION

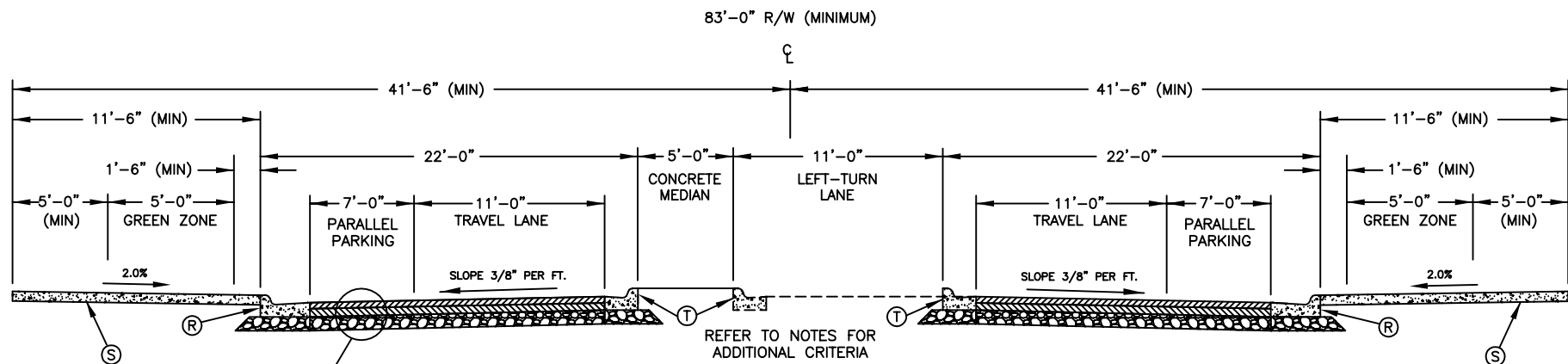
KEY

- (R) 2'-6" STD. CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK

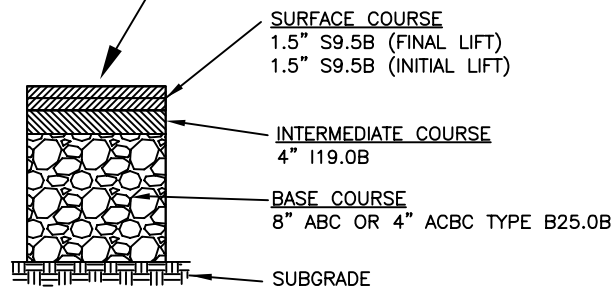
NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. REFER TO STANDARD DRAWINGS 602.1 - 605.1 REGARDING SIDEWALK AROUND TREE GRATES. TREE GRATES SHALL BE PROVIDED WHEN TREES ARE LOCATED IN THE GREEN ZONE.
4. BASE COURSE TO EXTEND SIX INCHES BEYOND BACK OF CURB THEN TAPER OUT AT A FORTY-FIVE DEGREE ANGLE.
5. DRAWING TO BE USED IN CONJUNCTION WITH STANDARD 220.1 AND 285.1.

NOT TO SCALE



RETAIL/MIXED USE LOCAL STREET (TWO LANE SECTION)



TYPICAL PAVEMENT SECTION

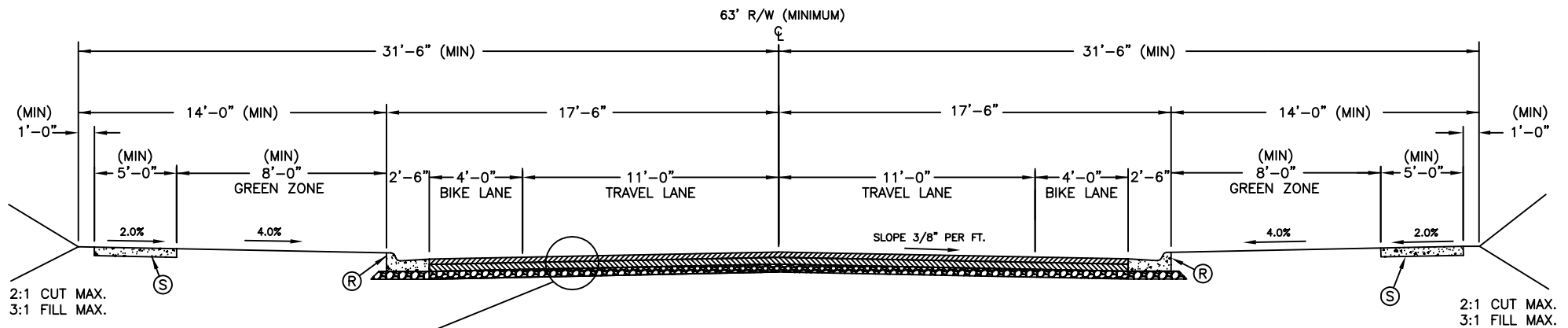
KEY

- (R) 2'-6" STANDARD CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK
- (T) 1'-6" MEDIAN CURB AND GUTTER

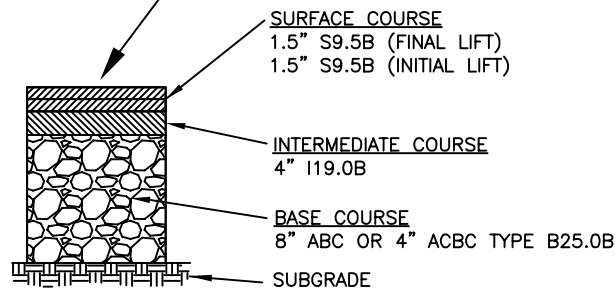
NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. REFER TO STANDARD DRAWINGS 602.1 – 605.1 REGARDING SIDEWALK AROUND TREE GRATES. TREE GRATES SHALL BE PROVIDED WHEN TREES ARE LOCATED IN THE GREEN ZONE.
4. FOR MEDIAN DIVIDED FACILITIES, A MINIMUM SIXTEEN (16) FOOT WIDE MEDIAN WITH ONE FOOT SIX INCH CURB AND GUTTER IS NEEDED. WHERE A LEFT-TURN LANE IS NOT INSTALLED, THE MEDIAN SHALL BE LANDSCAPED.
5. BASE COURSE TO EXTEND SIX INCHES BEYOND BACK OF CURB THEN TAPER OUT AT A FORTY-FIVE DEGREE ANGLE.
6. REFER TO STANDARD DRAWING 285.1 FOR PARALLEL PARKING LAYOUT.

NOT TO SCALE



RETAIL/MIXED USE COLLECTOR STREET



TYPICAL PAVEMENT SECTION

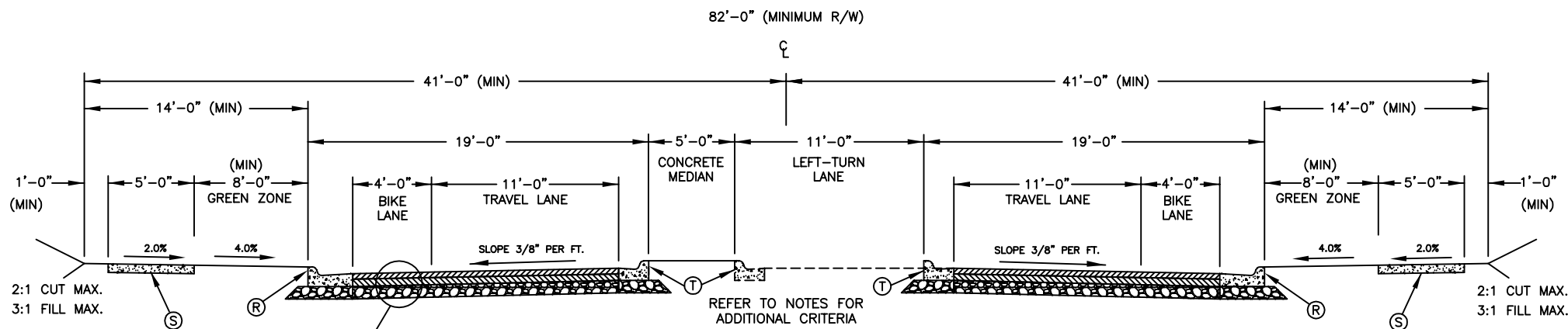
KEY

- (R) 2'-6" STANDARD CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK

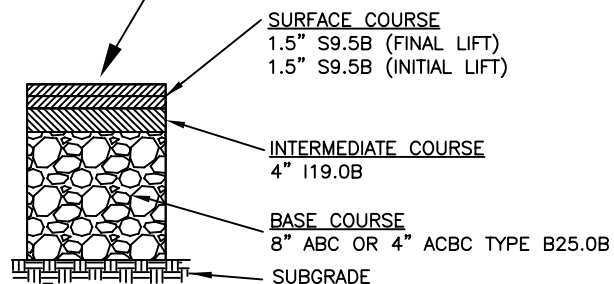
NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. BASE COURSE TO EXTEND SIX INCHES BEYOND BACK OF CURB THEN TAPER OUT AT A FORTY-FIVE DEGREE ANGLE.

NOT TO SCALE



RETAIL/MIXED USE COLLECTOR STREET (TWO LANE SECTION)



TYPICAL PAVEMENT SECTION

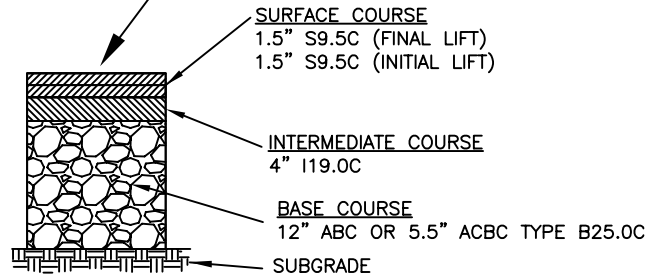
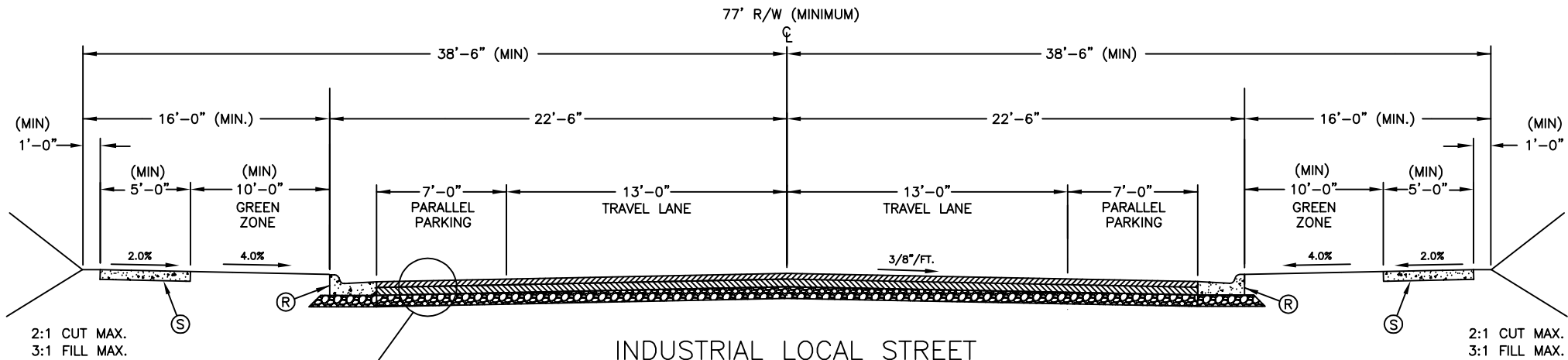
KEY

- (R) 2'-6" STANDARD CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK
- (T) 1'-6" MEDIAN CURB AND GUTTER

NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. FOR MEDIAN DIVIDED FACILITIES, A MINIMUM SIXTEEN (16) FOOT WIDE MEDIAN WITH ONE FOOT SIX INCH CURB AND GUTTER IS NEEDED. IF A LEFT-TURN LANE IS NOT NEEDED, THE MEDIAN SHALL BE LANDSCAPED.
4. BASE COURSE TO EXTEND SIX INCHES BEYOND BACK OF CURB THEN TAPER OUT AT A FORTY-FIVE DEGREE ANGLE.

NOT TO SCALE



TYPICAL PAVEMENT SECTION

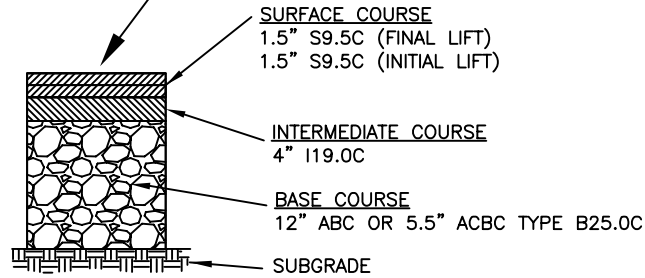
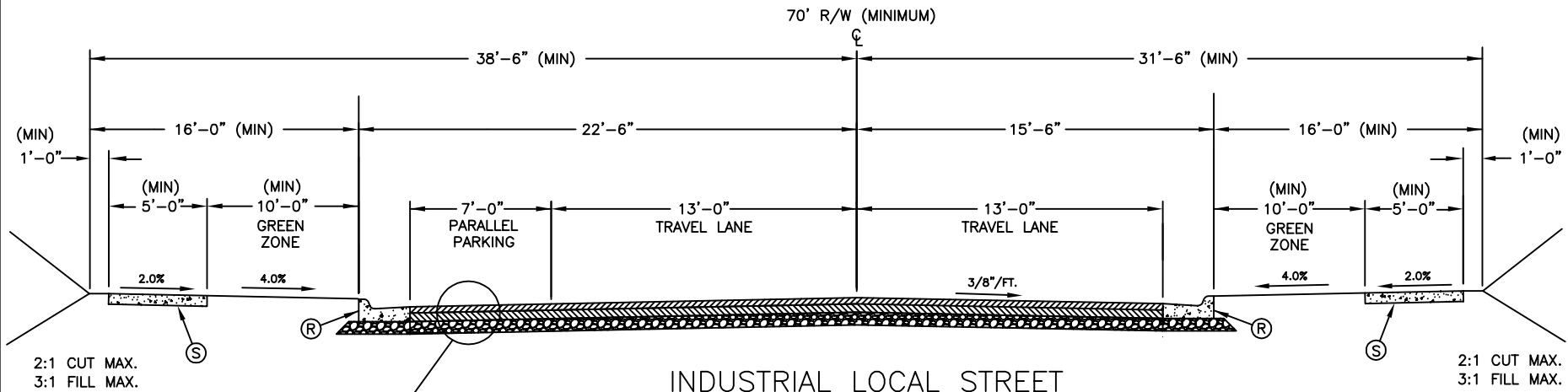
KEY

- (R) 2'-6" STANDARD CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK

NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. BASE COURSE TO EXTEND SIX INCHES BEYOND BACK OF CURB THEN TAPER OUT AT A FORTY-FIVE DEGREE ANGLE.
4. TREE TO BE PLANTED FOUR FEET FROM SIDEWALK.

NOT TO SCALE



TYPICAL PAVEMENT SECTION

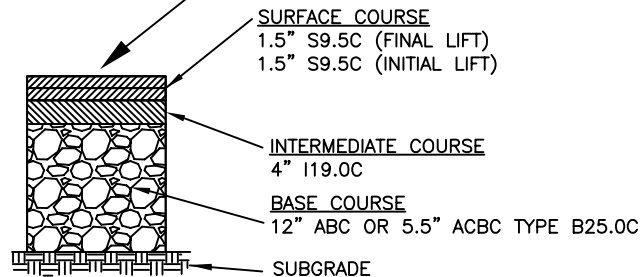
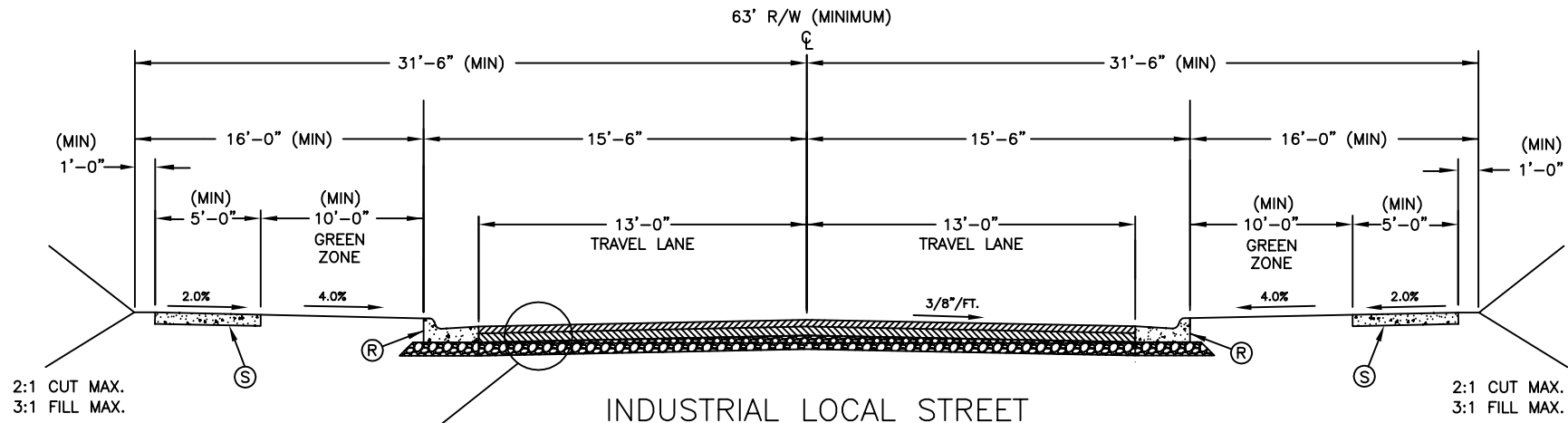
KEY

- (R) 2'-6" STANDARD CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK

NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. BASE COURSE TO EXTEND SIX INCHES BEYOND BACK OF CURB THEN TAPER OUT AT A FORTY-FIVE DEGREE ANGLE.
4. TREE TO BE PLANTED FOUR FEET FROM SIDEWALK.

NOT TO SCALE



TYPICAL PAVEMENT SECTION

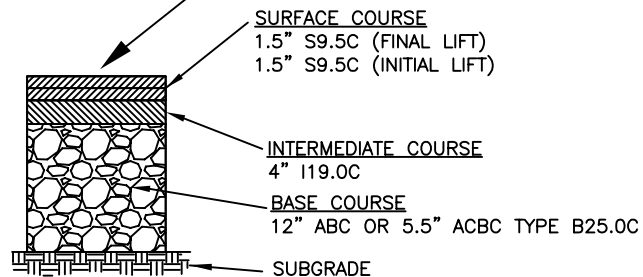
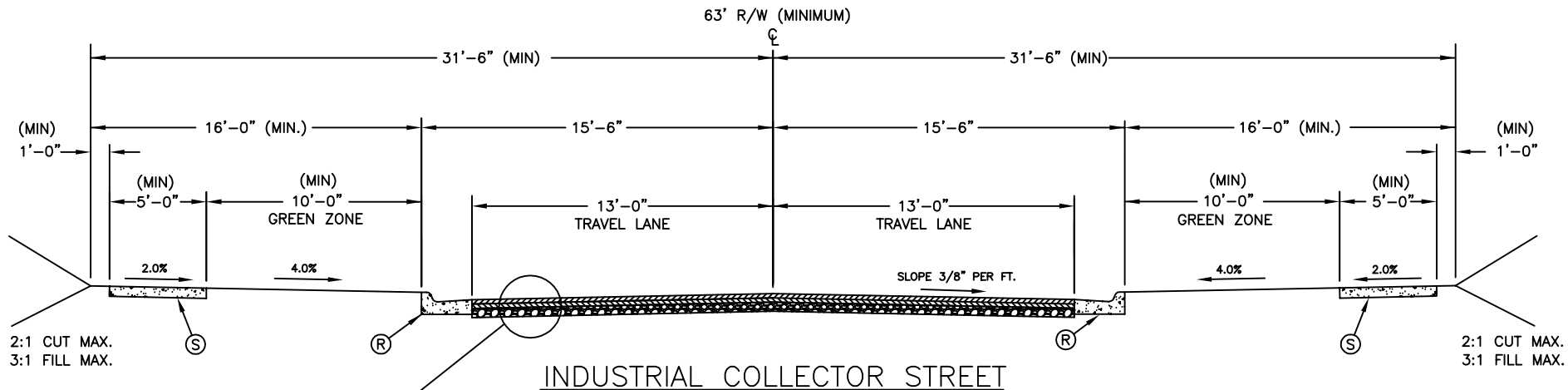
KEY

- (R) 2'-6" STANDARD CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK

NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. BASE COURSE TO EXTEND SIX INCHES BEYOND BACK OF CURB THEN TAPER OUT AT A FORTY-FIVE DEGREE ANGLE.
4. TREE TO BE PLANTED FOUR FEET FROM SIDEWALK.

NOT TO SCALE



TYPICAL PAVEMENT SECTION

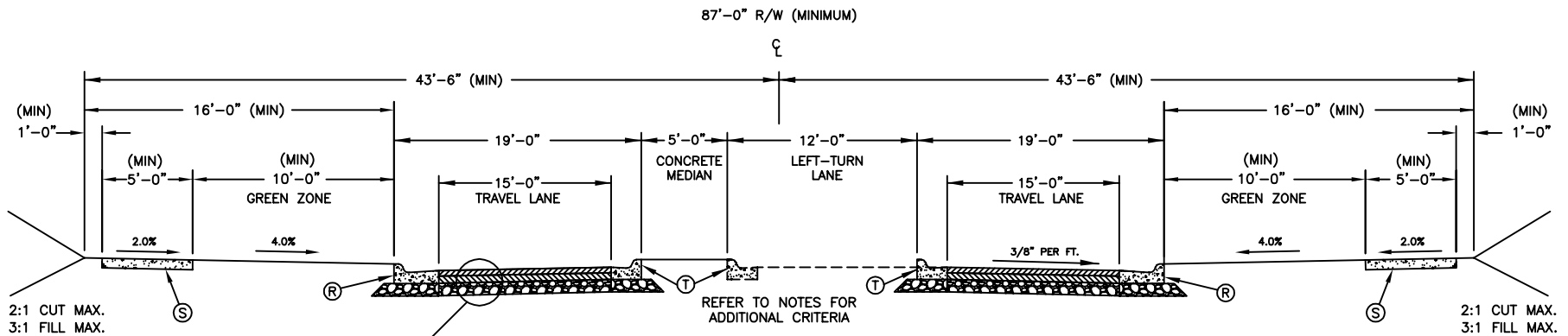
KEY

- (R) 2'-6" STANDARD CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK

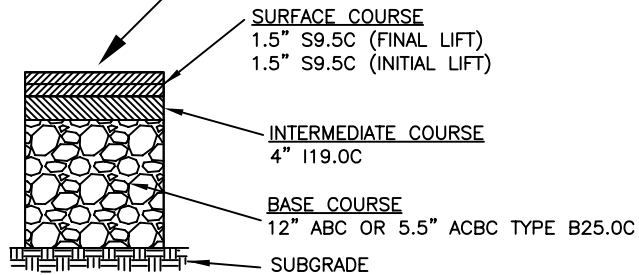
NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. BASE COURSE TO EXTEND SIX INCHES BEYOND BACK OF CURB THEN TAPER OUT AT A FORTY-FIVE DEGREE ANGLE.
4. TREE TO BE PLANTED FOUR FEET FROM SIDEWALK.

NOT TO SCALE



INDUSTRIAL COLLECTOR STREET (TWO LANE SECTION)



TYPICAL PAVEMENT SECTION

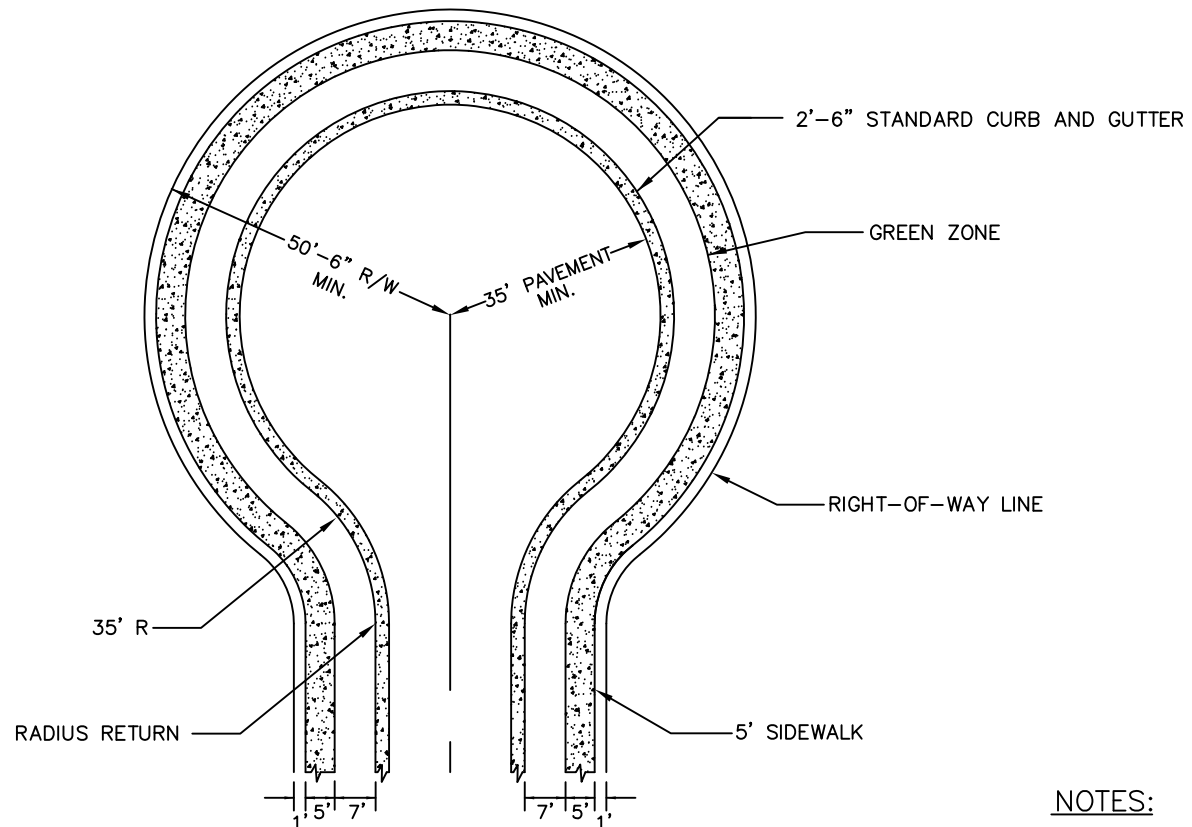
KEY

- (R) 2'-6" STANDARD CURB AND GUTTER
- (S) 4" CONCRETE SIDEWALK
- (T) 1'-6" MEDIAN CURB AND GUTTER

NOTES:

1. SIDEWALK SHALL BE PROVIDED ON BOTH SIDES OF THE STREET.
2. SEE SECTION II. B. FOR ADDITIONAL DESIGN CRITERIA.
3. FOR MEDIAN DIVIDED FACILITIES, A MINIMUM TWENTY (20) FOOT WIDE MEDIAN WITH ONE FOOT SIX INCH CURB AND GUTTER IS NEEDED. IF A LEFT-TURN LANE IS NOT NEEDED, THE MEDIAN SHALL BE LANDSCAPED.
4. BASE COURSE TO EXTEND SIX INCHES BEYOND BACK OF CURB THEN TAPER OUT AT A FORTY-FIVE DEGREE ANGLE.
5. TREE TO BE PLANTED FOUR FEET FROM SIDEWALK.

NOT TO SCALE

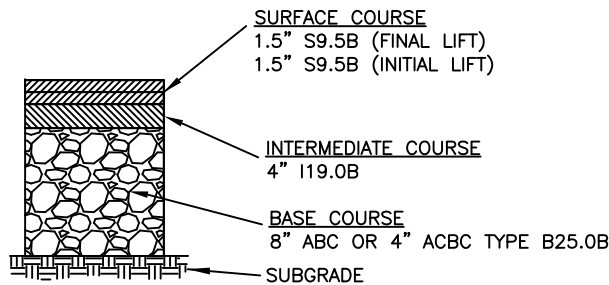


RESIDENTIAL LOCAL STREET

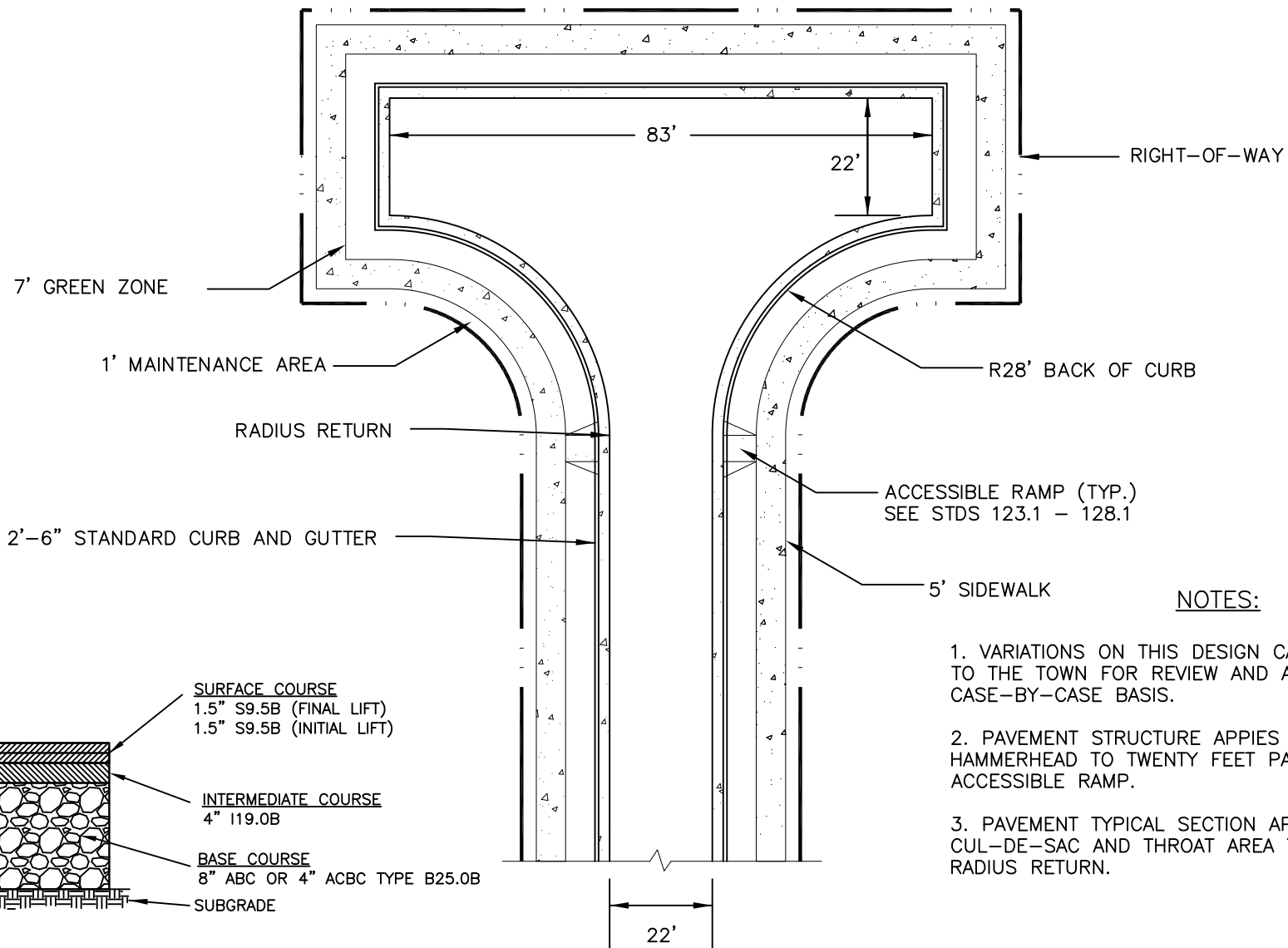
NOTES:

1. VALLEY GUTTER MAY BE USED INSTEAD OF STANDARD CURB AND GUTTER.
2. CENTRAL ISLANDS ARE PERMITTED AS LONG AS A B-40 (DESIGN VEHICLE) STAYS ON THE PAVEMENT WHILE TRAVERSING THE CUL-DE-SAC.
3. THE CROWN FOR THE PAVEMENT SHALL BE 1/4" PER FOOT FROM THE CENTER OF THE CUL-DE-SAC.
4. PAVEMENT TYPICAL SECTION APPLIES TO CUL-DE-SAC AND THROAT AREA TO 25 FEET PAST RADIUS RETURN.

NOT TO SCALE



TYPICAL PAVEMENT SECTION

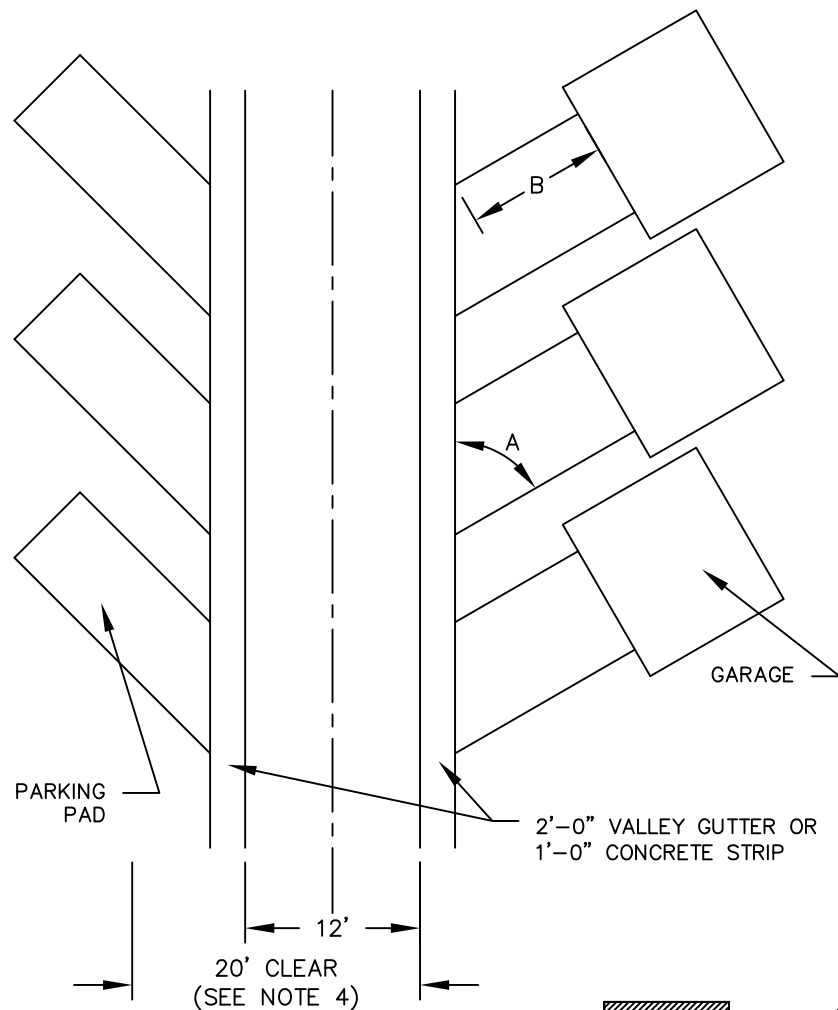


NOTES:

1. VARIATIONS ON THIS DESIGN CAN BE SUBMITTED TO THE TOWN FOR REVIEW AND APPROVAL ON A CASE-BY-CASE BASIS.
2. PAVEMENT STRUCTURE APPLIES WITHIN THE HAMMERHEAD TO TWENTY FEET PAST THE ACCESSIBLE RAMP.
3. PAVEMENT TYPICAL SECTION APPLIES TO CUL-DE-SAC AND THROAT AREA TO 25 FEET PAST RADIUS RETURN.

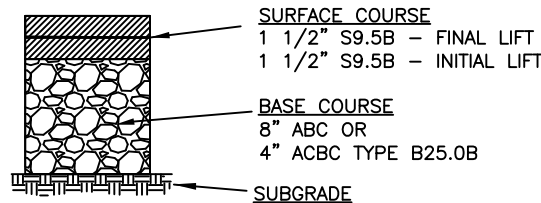
TYPICAL PAVEMENT SECTION

NOT TO SCALE

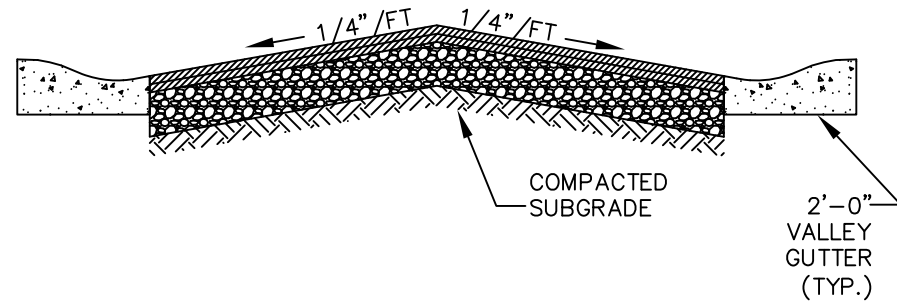


PLAN

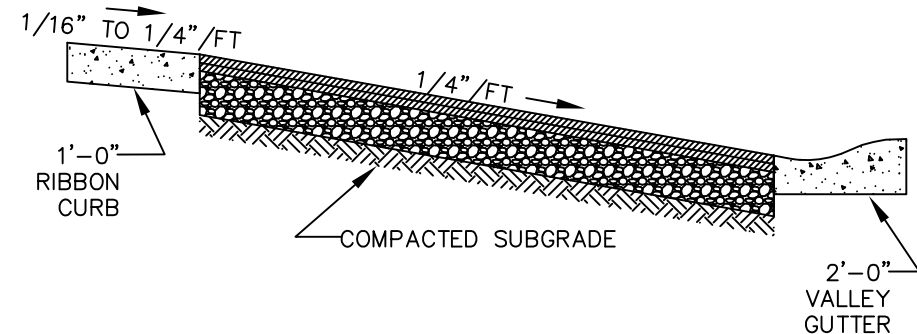
A - LESS THAN 60 DEGREES
B - LESS THAN 3 FEET OR GREATER THAN 20 FEET



TYPICAL PAVEMENT SECTION



ALLEY WITH NORMAL CROWN

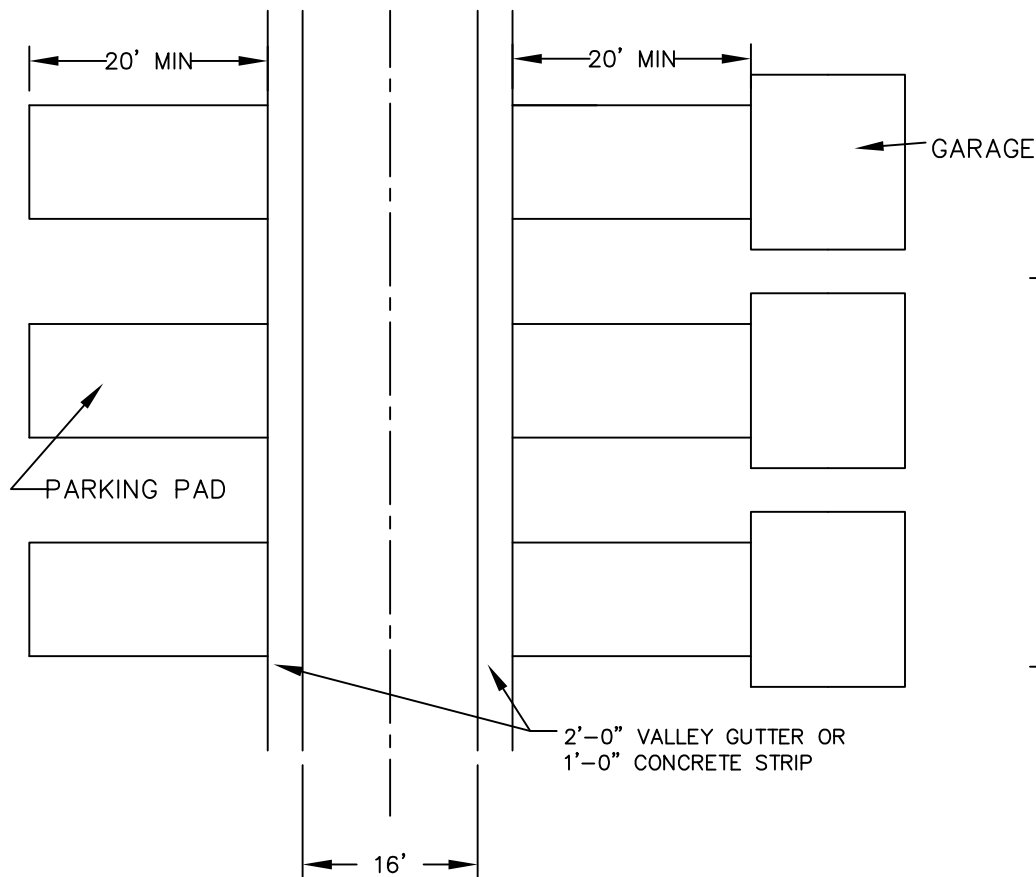


ALLEY WITH NO CROWN

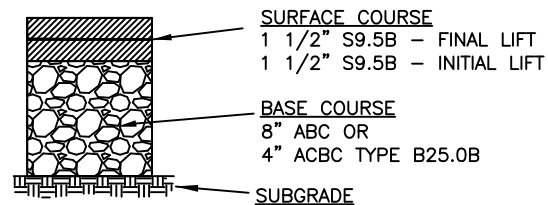
NOTES:

1. ALLEYS SHALL BE CONSIDERED PRIVATE EASEMENTS AND WILL NOT BE ACCEPTED FOR MAINTENANCE.
2. SUBGRADE SHALL BE COMPACTED TO PUBLIC STREET STANDARDS.
3. STORM DRAINAGE (NOT SHOWN) SHALL BE PROVIDED AS NECESSARY.
4. DETAIL APPLIES TO SINGLE- OR DOUBLE-LOADED ALLEYS. FOR SINGLE-LOADED ALLEYS, THERE SHALL BE A 20-FOOT CLEAR ZONE FREE OF CUT SLOPES, OBSTRUCTIONS, HEDGES, ETC. FROM THE LOADED SIDE EDGE OF PAVEMENT.
5. RIBBON CURB TO BE TEN INCHES THICK.

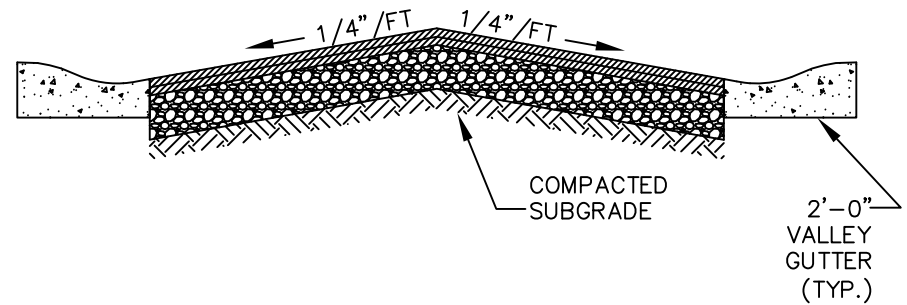
NOT TO SCALE



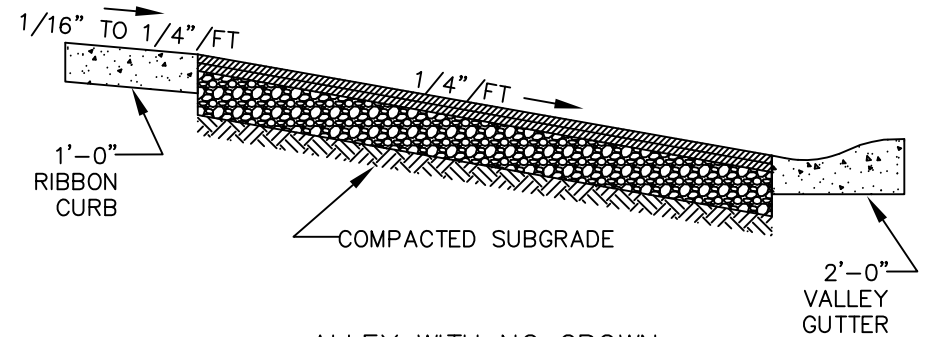
PLAN



TYPICAL PAVEMENT SECTION



ALLEY WITH NORMAL CROWN

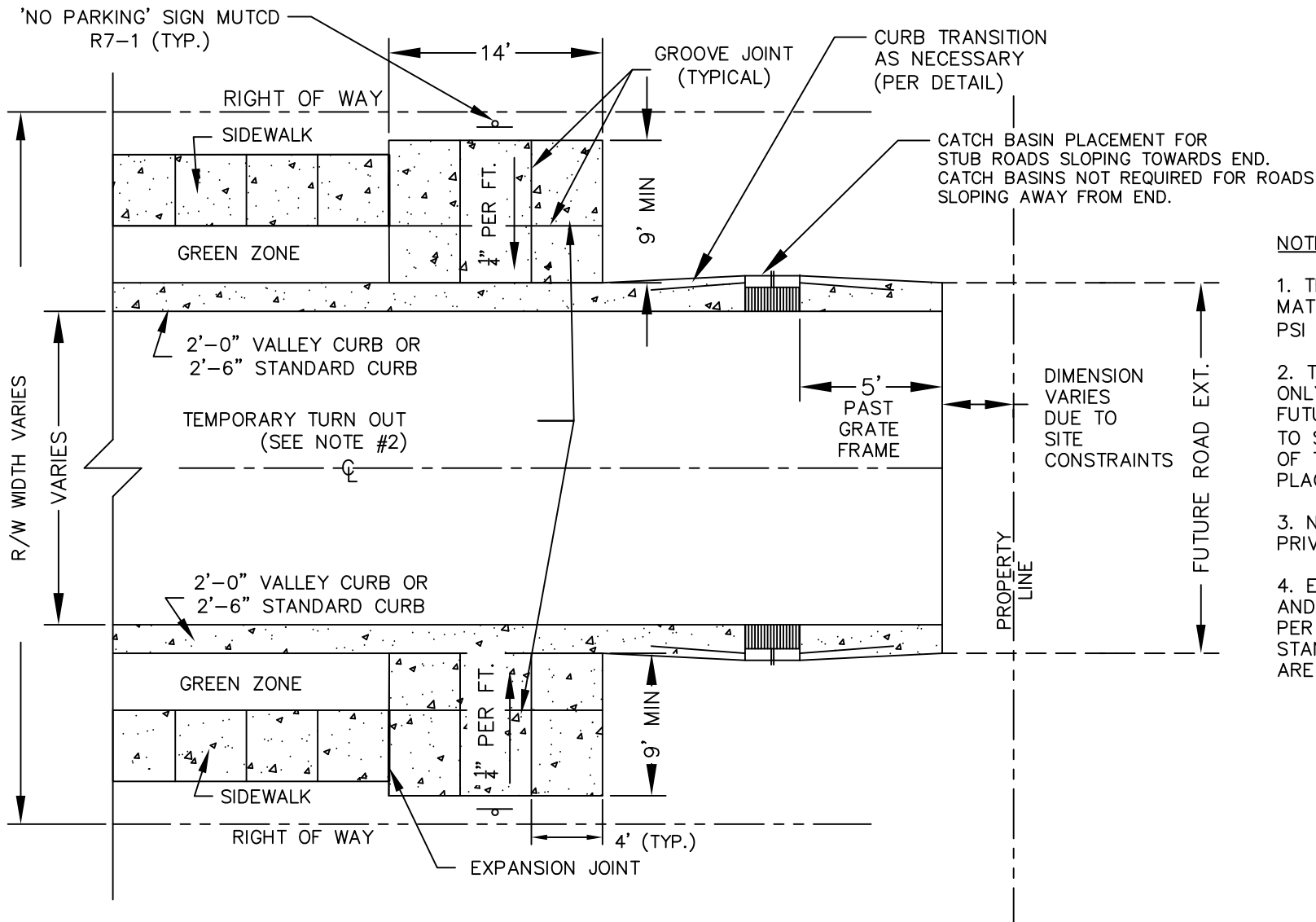


ALLEY WITH NO CROWN

NOTES:

1. ALLEYS SHALL BE CONSIDERED PRIVATE EASEMENTS AND WILL NOT BE ACCEPTED FOR MAINTENANCE.
2. SUBGRADE SHALL BE COMPACTED TO PUBLIC STREET STANDARDS.
3. STORM DRAINAGE (NOT SHOWN) SHALL BE PROVIDED AS NECESSARY.
4. RIBBON CURB TO BE TEN INCHES THICK.

NOT TO SCALE



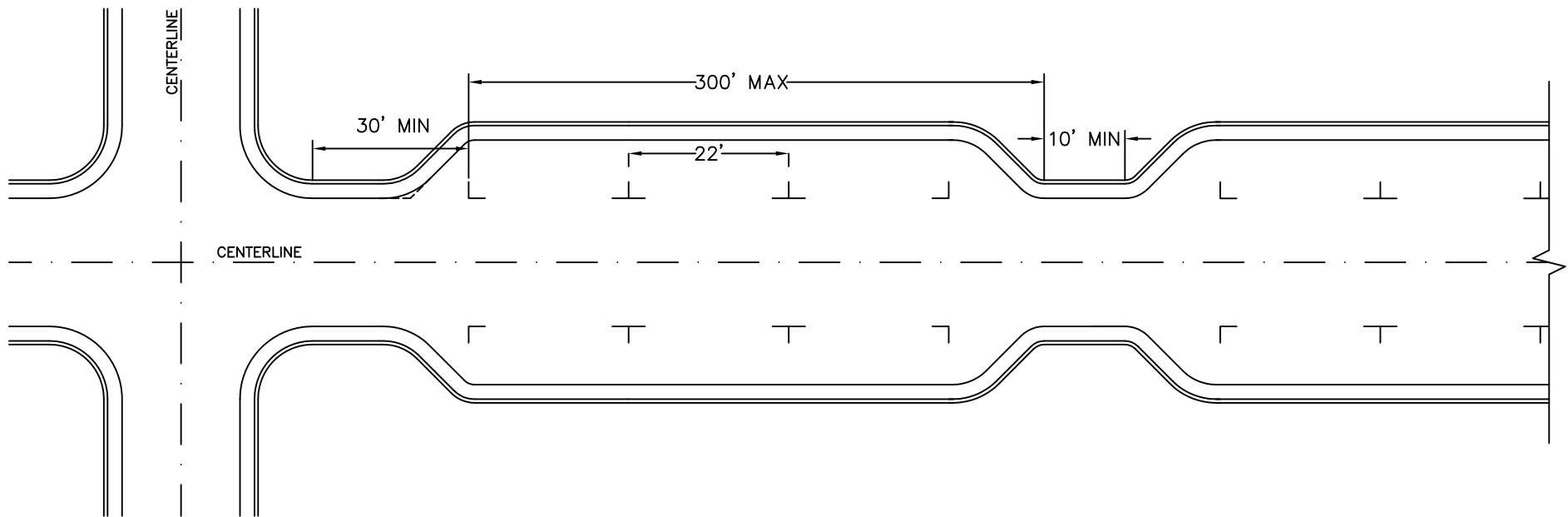
NOTES

1. TEMPORARY TURNAROUND MATERIAL SHALL BE MIN. 3600 PSI CONCRETE, 6" THICK.
2. TEMPORARY INSTALLATION ONLY – TO BE REMOVED WHEN FUTURE DEVELOPMENT CONNECTS TO STREET. "SIDEWALK" PORTION OF TURNAROUND MAY BE LEFT IN PLACE IF NOT DAMAGED.
3. NOT TO BE USED AS A PRIVATE DRIVEWAY.
4. END OF ROADWAY BARRICADE AND END OF ROADWAY MARKER PER TOWN OF HUNTERSVILLE STANDARDS 706.1 THROUGH 709.1 ARE REQUIRED.

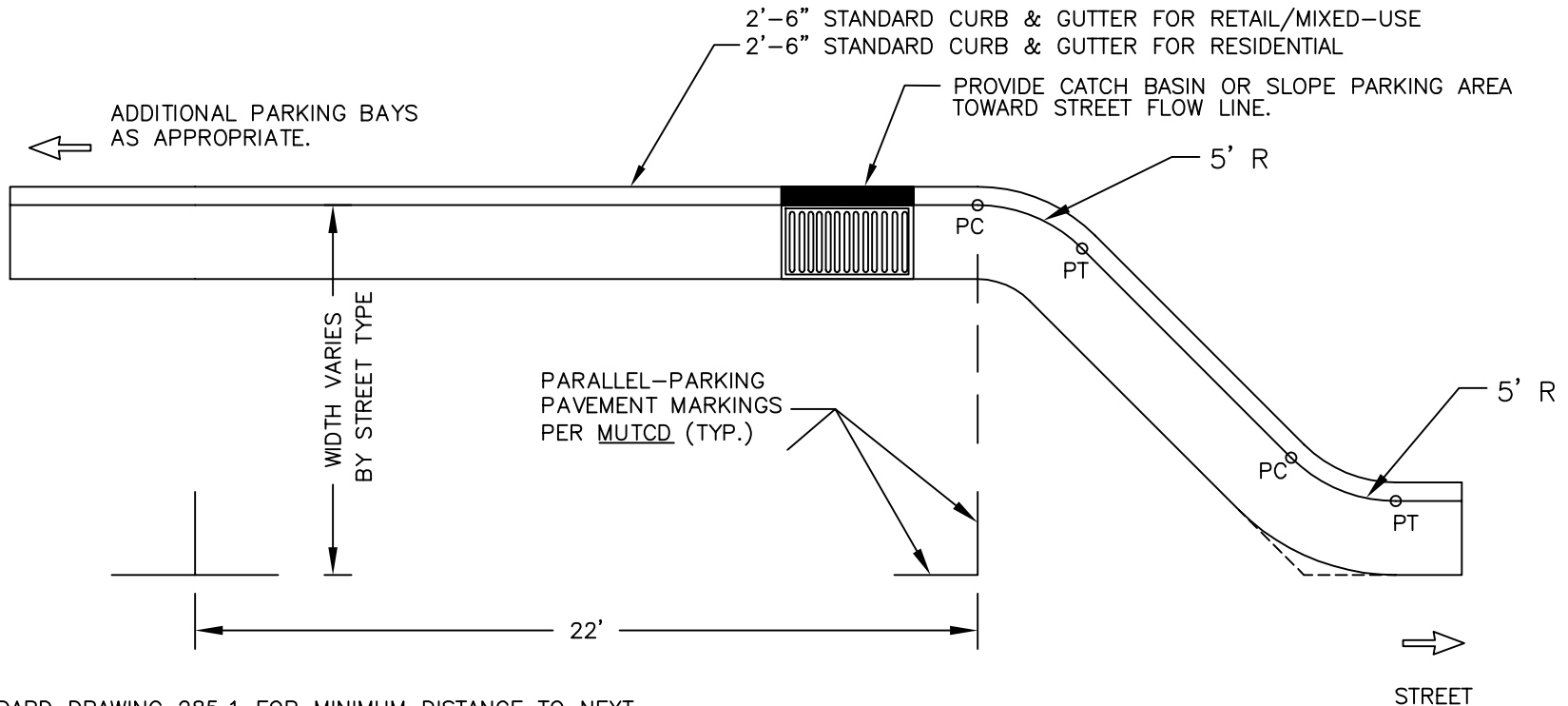
NOT TO SCALE

NOTES:

1. REFER TO STANDARD DRAWINGS 285.2, 285.3, AND 285.4 FOR ADDITIONAL INFORMATION.
2. PARKING STALLS MAY BE ON ONE OR BOTH SIDES OF THE STREET.
3. PAVEMENT MARKINGS TO BE THERMOPLASTIC ON RETAIL/OFFICE/MIXED-USE STREETS.
4. 30' MINIMUM DISTANCE TO FIRST PARKING STALL TO BE MEASURED FROM END OF INTERSECTION RADIUS POINT.



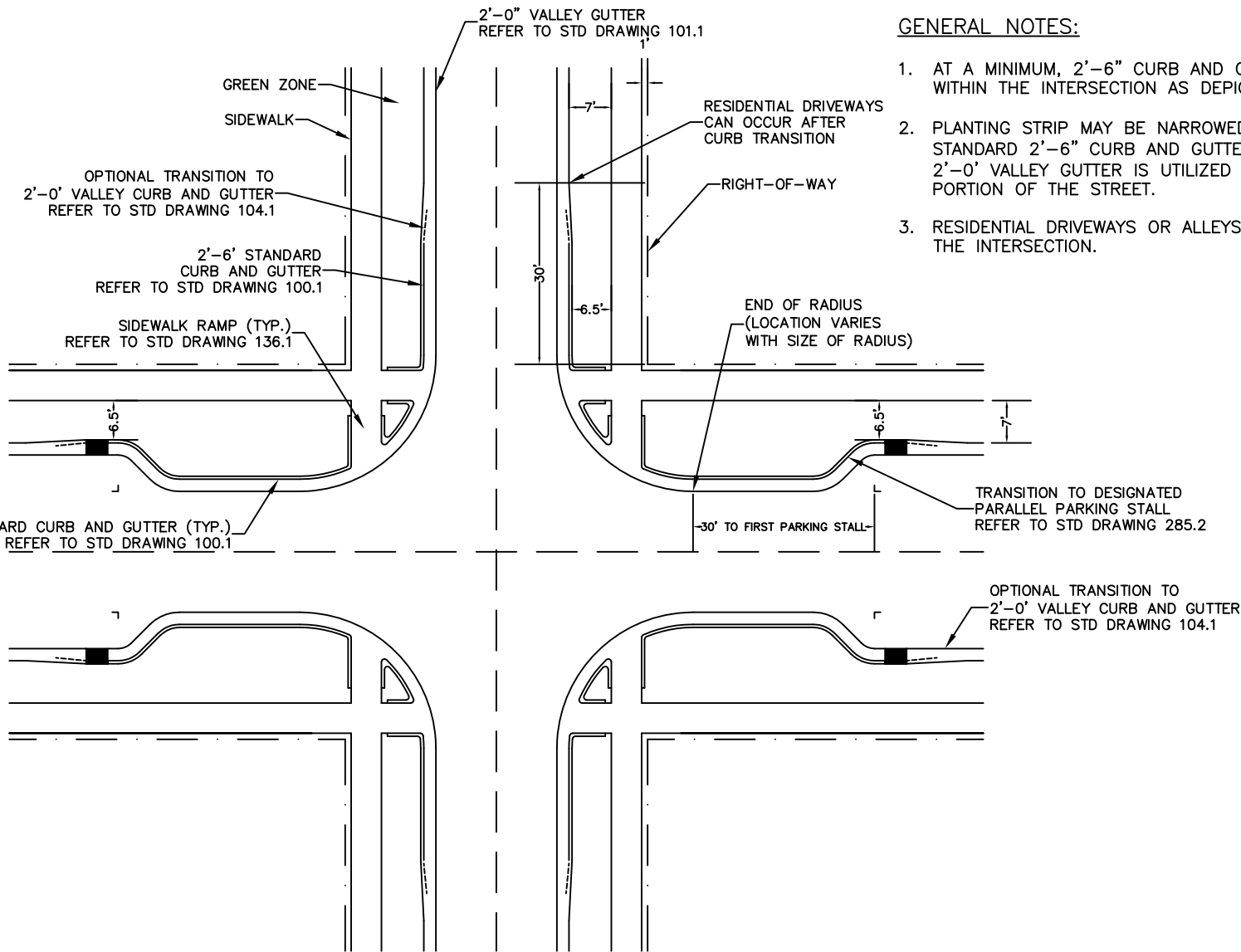
NOT TO SCALE



NOTES:

1. REFER TO STANDARD DRAWING 285.1 FOR MINIMUM DISTANCE TO NEXT INTERVENING STREET.
2. PARALLEL ACCESSIBLE SPACES AND LOADING ZONES TO BE REVIEWED ON A CASE-BY-CASE BASIS.
3. FOR PARKING BAYS THAT ARE 6 FEET IN WIDTH OR GREATER, THE PAVEMENT MARKINGS SHALL BE SET AT ONE (1) FOOT LESS THAT THE STALL WIDTH.
4. GREATER SEPARATION FROM INTERVENING STREETS THAN 30 FEET MAY BE REQUIRED AT TOWN ENGINEER'S DISCRETION.
5. POSITIVE DRAINAGE SHALL BE PROVIDED EITHER BY INSTALLATION OF APPROPRIATE DRAINAGE STRUCTURES OR SLOPE PARKING AREA TO STREET FLOW LINE.

NOT TO SCALE



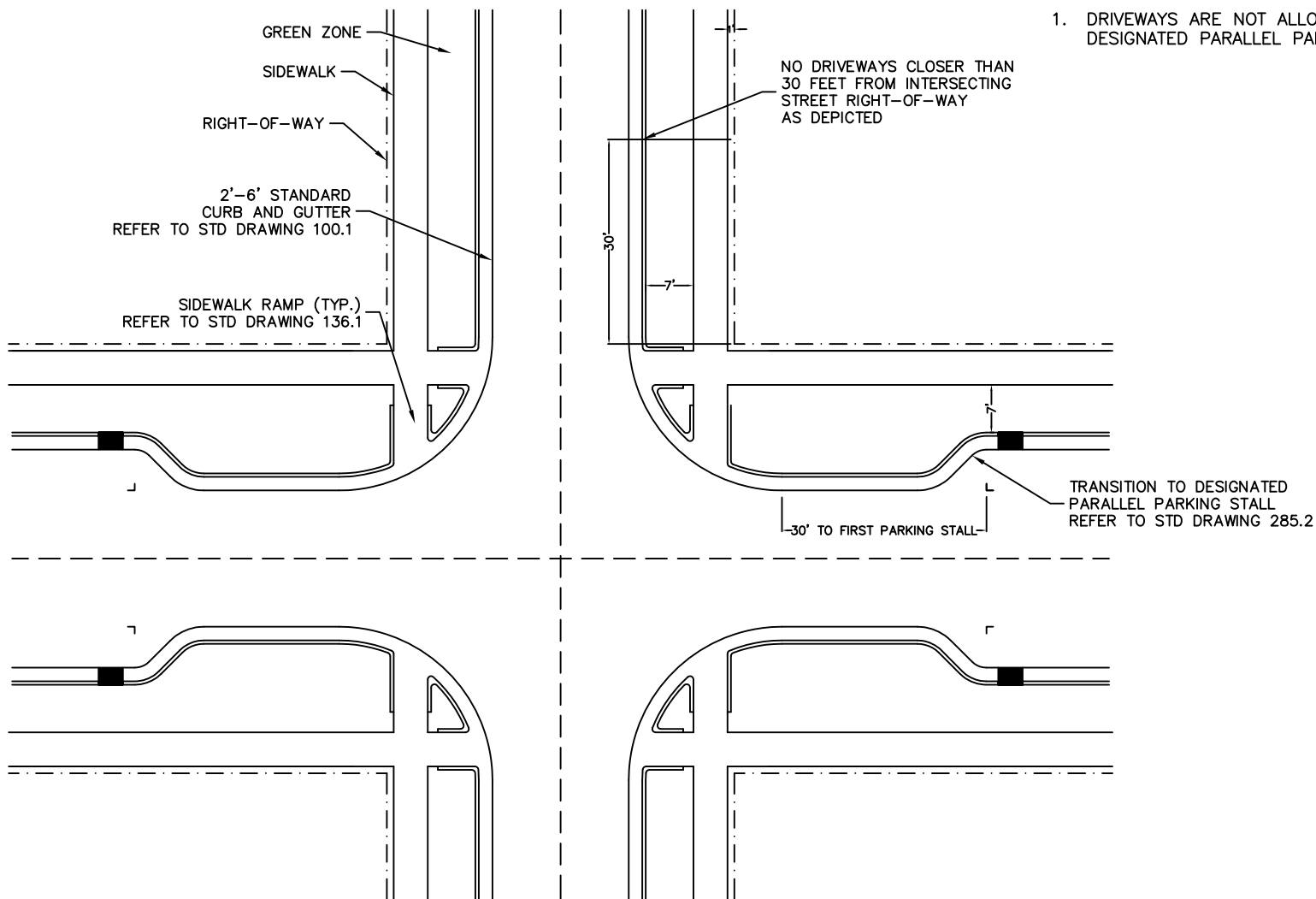
GENERAL NOTES:

1. AT A MINIMUM, 2'-6" CURB AND GUTTER IS REQUIRED WITHIN THE INTERSECTION AS DEPICTED.
2. PLANTING STRIP MAY BE NARROWED TO 6.5' WITHIN THE STANDARD 2'-6" CURB AND GUTTER SECTION AS SHOWN IF 2'-0' VALLEY GUTTER IS UTILIZED FOR THE REMAINING PORTION OF THE STREET.
3. RESIDENTIAL DRIVEWAYS OR ALLEYS ARE NOT ALLOWED WITHIN THE INTERSECTION.

NOT TO SCALE

GENERAL NOTES:

1. DRIVEWAYS ARE NOT ALLOWED WITHIN THE INTERSECTION OR DESIGNATED PARALLEL PARKING AREAS.



NOT TO SCALE

DWG	SHEET TITLE	SPECIAL REQUIREMENTS AND NOTES
300.01	METHOD OF PIPE INSTALLATION – METHOD A	
310.02	PARALLEL PIPE END SECTION–PRECAST CONCRETE FOR 15” TO 24” PIPE	REQUIRED IN RIGHT OF WAY WITHIN THE ETJ
310.03	CROSS PIPE END SECTION–PRECAST CONCRETE FOR 18” TO 30” PIPE	REQUIRED IN RIGHT OF WAY WITHIN THE ETJ
310.10	DRIVEWAY PIPE CONSTRUCTION USING NO SPECIAL END SECTIONS	ONLY AT LOCATIONS APPROVED BY THE TOWN ENGINEER
815.03	PIPE UNDERDRAIN AND BLIND DRAIN	
816.03	GEOCOMPOSITE SHOULDER DRAIN	
838.01	CONCRETE ENDWALL FOR SINGLE AND DOUBLE PIPE CULVERTS 15” THRU 48” PIPE 90° SKEW	NOTE 1
838.02	CONCRETE ENDWALL AND SLUICE GATE 15” THRU 36” PIPE–90° SKEW	NOTE 1
838.04	CONCRETE ENDWALL FOR SINGLE AND DOUBLE PIPE CULVERTS 17”X13” THRU 71”X47” PIPE ARCH 90° SKEW	NOTE 1
838.05	CONCRETE ”L” ENDWALL FOR SINGLE PIPE CULVERTS 15” THRU 48” PIPE	NOTE 1
838.06	CONCRETE ”L” ENDWALL FOR SINGLE PIPE CULVERTS 17”X13” THRU 71”X47” PIPE ARCH	NOTE 1
838.07	CONCRETE ENDWALL FOR SINGLE AND DOUBLE PIPE CULVERTS 40”X31” THRU 66”X51” PIPE ARCH 90° SKEW	NOTE 1
838.08	CONCRETE ”L” ENDWALL FOR SINGLE PIPE CULVERTS 40”X31” THRU 66”X51” PIPE ARCH	NOTE 1
838.10	CONCRETE ENDWALL FOR OUTFALL 4”, 6” OR 8” PIPE	NOTE 1
838.11	BRICK ENDWALL FOR SINGLE AND DOUBLE PIPE CULVERTS 15” THRU 48” 90° SKEW	
838.14	BRICK ENDWALL FOR SINGLE AND DOUBLE PIPE CULVERTS 17”X13” THRU 71”X47” PIPE ARCH 90° SKEW	
838.15	BRICK ”L” ENDWALL FOR SINGLE PIPE CULVERTS 15” THRU 48” PIPE	
838.16	BRICK ”L” ENDWALL FOR SINGLE PIPE CULVERTS 17”X13” THRU 71”X47” PIPE ARCH	
838.17	BRICK ENDWALL FOR SINGLE AND DOUBLE PIPE CULVERTS 40”X31” THRU 66”X51” PIPE ARCH 90° SKEW	
838.18	BRICK ENDWALL FOR SINGLE PIPE CULVERTS 40”X31” THRU 66”X51” PIPE ARCH	
838.20	BRICK ENDWALL FOR OUTFALL 4”, 6” OR 8” PIPE	
838.21	REINFORCED CONCRETE ENDWALL FOR SINGLE 54” PIPE 90° SKEW	NOTE 1 SEE TOWN STANDARD 304.1 & 305.1 FOR SPLASH PAD
838.22	REINFORCED CONCRETE ENDWALL FOR DOUBLE & TRIPLE 54” PIPES 90° SKEW	NOTE 1 SEE TOWN STANDARD 304.1 & 305.1 FOR SPLASH PAD
838.27	REINFORCED CONCRETE ENDWALL FOR SINGLE 60” PIPE 90° SKEW	NOTE 1 SEE TOWN STANDARD 304.1 & 305.1 FOR SPLASH PAD
838.28	REINFORCED CONCRETE ENDWALL FOR DOUBLE & TRIPLE 60” PIPES 90° SKEW	NOTE 1 SEE TOWN STANDARD 304.1 & 305.1 FOR SPLASH PAD
838.33	REINFORCED CONCRETE ENDWALL FOR SINGLE 66” PIPE 90° SKEW	NOTE 1 SEE TOWN STANDARD 304.1 & 305.1 FOR SPLASH PAD
838.34	REINFORCED CONCRETE ENDWALL FOR DOUBLE & TRIPLE 66” PIPES 90° SKEW	NOTE 1 SEE TOWN STANDARD 304.1 & 305.1 FOR SPLASH PAD
838.39	REINFORCED CONCRETE ENDWALL FOR SINGLE 72” PIPE 90° SKEW	NOTE 1 SEE TOWN STANDARD 304.1 & 305.1 FOR SPLASH PAD
838.40	REINFORCED CONCRETE ENDWALL FOR DOUBLE & TRIPLE 72” PIPES 90° SKEW	NOTE 1 SEE TOWN STANDARD 304.1 & 305.1 FOR SPLASH PAD

NOTE 1: FOR ALL STRUCTURES – NCDOT REQUIRES CLASS B CONCRETE (2500PSI). THE TOWN REQUIRES 3600 PSI CONCRETE STRENGTH @ 28 DAYS. 3600 PSI CONCRETE SHALL BE USED IN ALL PROJECTS.

NOT TO SCALE



STANDARD
DRAWING

NCDOT STANDARDS
APPROVED FOR USE

REV. DATE

10/7/09

STD. NO.

300.1

DWG	SHEET TITLE	SPECIAL REQUIREMENTS AND NOTES
838.45	NOTES FOR REINFORCED CONCRETE ENDWALL STANDARD DRAWINGS	NOTE 1 SEE TOWN STANDARDS 304.1 AND 305.1 FOR SPLASH PAD
	838.21 THRU 838.40	
838.51	REINFORCED BRICK ENDWALL FOR SINGLE 54" PIPE 90° SKEW	SEE TOWN STANDARDS 304.1 AND 305.1 FOR SPLASH PAD
838.52	REINFORCED BRICK ENDWALL FOR DOUBLE & TRIPLE 54" PIPES 90° SKEW	SEE TOWN STANDARDS 304.1 AND 305.1 FOR SPLASH PAD
838.57	REINFORCED BRICK ENDWALL FOR SINGLE 60" PIPE 90° SKEW	SEE TOWN STANDARDS 304.1 AND 305.1 FOR SPLASH PAD
838.58	REINFORCED BRICK ENDWALL FOR DOUBLE & TRIPLE 60" PIPES 90° SKEW	SEE TOWN STANDARDS 304.1 AND 305.1 FOR SPLASH PAD
838.63	REINFORCED BRICK ENDWALL FOR SINGLE 66" PIPE 90° SKEW	SEE TOWN STANDARDS 304.1 AND 305.1 FOR SPLASH PAD
838.64	REINFORCED BRICK ENDWALL FOR DOUBLE & TRIPLE 66" PIPES 90° SKEW	SEE TOWN STANDARDS 304.1 AND 305.1 FOR SPLASH PAD
838.69	REINFORCED BRICK ENDWALL FOR SINGLE 72" PIPE 90° SKEW	SEE TOWN STANDARDS 304.1 AND 305.1 FOR SPLASH PAD
838.70	REINFORCED BRICK ENDWALL FOR DOUBLE & TRIPLE 72" PIPES 90° SKEW	SEE TOWN STANDARDS 304.1 AND 305.1 FOR SPLASH PAD
838.75	NOTES FOR REINFORCED BRICK ENDWALL STANDARD DRAWINGS 838.51 THRU 838.70	SEE TOWN STANDARDS 304.1 AND 305.1 FOR SPLASH PAD
838.80	PRECAST CONCRETE ENDWALLS FOR SINGLE 12" THRU 72" PIPE 90° SKEW	
840.00	CONCRETE BASE PAD FOR DRAINAGE STRUCTURES	
840.01	BRICK CATCH BASIN 12" THRU 54" PIPE	
840.02	CONCRETE CATCH BASIN 12" THRU 54" PIPE	
840.03	FRAME, GRATES AND HOOD FOR USE ON STANDARD BASIN 12" THRU 54" PIPE	TYPE F AND G GRATES ARE OPTIONAL WITHIN THE TOWN LIMITS
840.04	CONCRETE OPEN THROAT CATCH BASIN 12" THRU 48" PIPE	NOTE 1 – OPENINGS PERMITTED IN 4 SIDES OUTSIDE OF STREET R/W MANHOLE RING AND COVER REQUIRED IN TOP SLAB SEE STD. 840.54
840.05	BRICK OPEN THROAT CATCH BASIN 12" THRU 48" PIPE	OPENINGS PERMITTED IN 4 SIDES OUTSIDE OF STREET R/W MANHOLE RING AND COVER REQUIRED IN TOP SLAB SEE STD. 840.54
840.14	CONCRETE DROP INLET 12" THRU 30" PIPE	NOTE 1
840.15	BRICK DROP INLET 12" THRU 30" PIPE	
840.16	DROP INLET FRAME AND GRATES FOR USE WITH STANDARD DWGS. 840.14 & 840.15	
840.17	CONCRETE GRATED DROP INLET TYPE "A" 12" THRU 72" PIPE	NOTE 1
840.18	CONCRETE GRATED DROP INLET TYPE "B" 12" THRU 36" PIPE	NOTE 1
840.19	CONCRETE GRATED DROP INLET TYPE "D" 12" THRU 36" PIPE	NOTE 1
840.20	FRAMES AND WIDE SLOT FLAT GRATES	NOT FOR USE IN PEDESTRIAN AREAS
840.22	FRAMES AND WIDE SLOT SAG GRATES	NOT FOR USE IN PEDESTRIAN AREAS
840.24	FRAMES AND NARROW SLOT SAG GRATES	
840.25	ANCHORAGE FOR FRAMES BRICK OR CONCRETE	
840.26	BRICK GRATED DROP INLET TYPE "A" 12" THRU 72" PIPE	
840.27	BRICK GRATED DROP INLET TYPE "B" 12" THRU 36" PIPE	
840.28	BRICK GRATED DROP INLET TYPE "D" 12" THRU 36" PIPE	
840.29	FRAMES AND NARROW SLOT FLAT GRATES	
840.30	DRIVEWAY DROP INLET	

NOTE 1: FOR ALL STRUCTURES – NCDOT REQUIRES CLASS B CONCRETE (2500PSI). THE TOWN REQUIRES 3600 PSI CONCRETE STRENGTH @ 28 DAYS. 3600 PSI CONCRETE SHALL BE USED IN ALL PROJECTS.

NOT TO SCALE



STANDARD
DRAWING

NCDOT STANDARDS
APPROVED FOR USE

REV. DATE

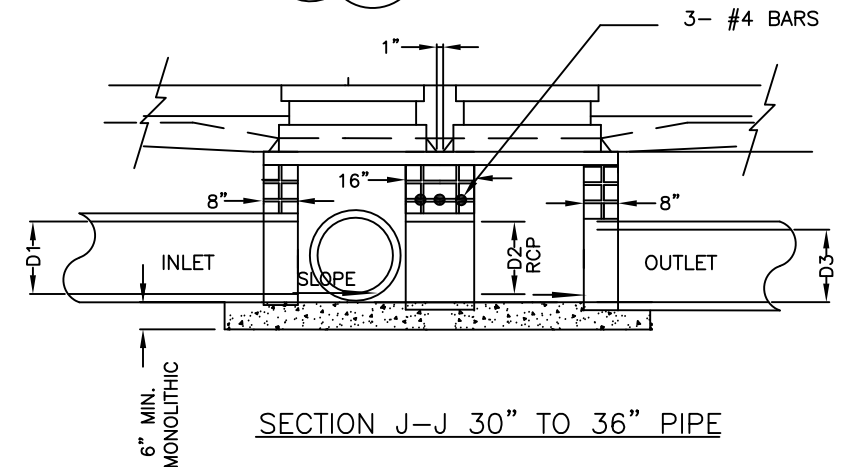
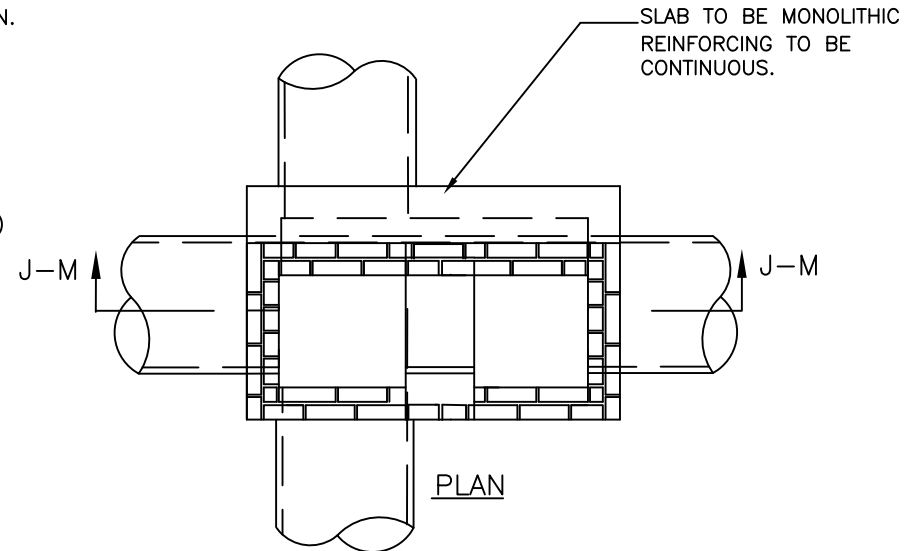
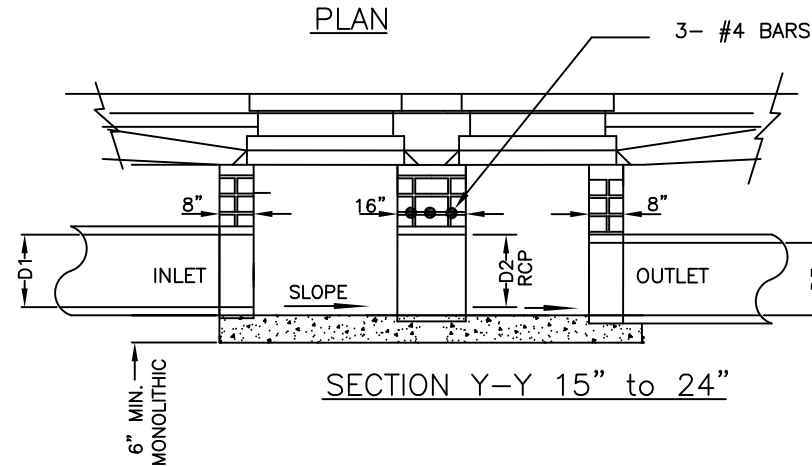
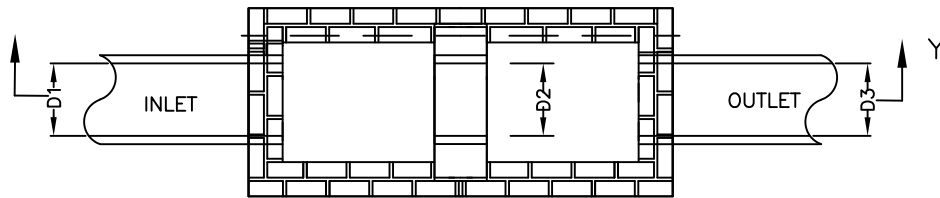
10/7/09

STD. NO.

301.1

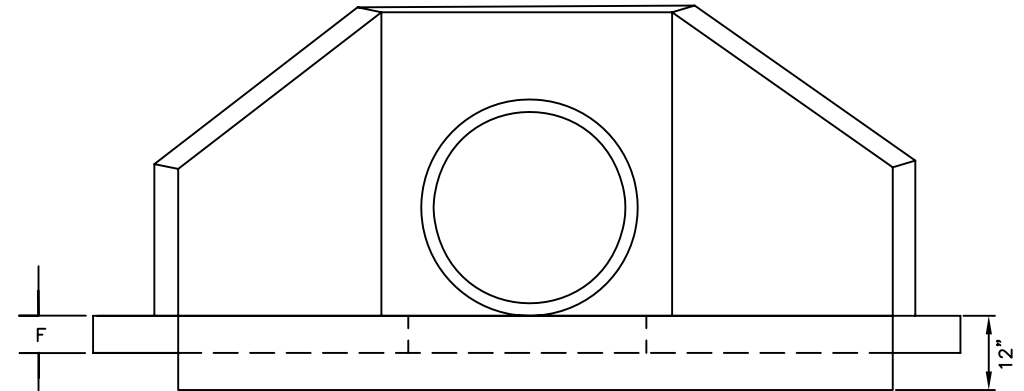
GENERAL NOTES:

1. DOUBLE CATCH BASIN ONLY FOR USE WITHIN TOWN MAINTAINED STREETS. INSTALLATION ON STREETS WITHIN EXISTING/FUTURE NCDOT MAINTAINED RIGHT OF WAY REQUIRES A MINIMUM OF ONE 4 FOOT LONG SECTION OF REINFORCED CONCRETE PIPE BETWEEN CATCH BASINS.
2. SEE NCDOT STANDARD 840.01 FOR DETAILS BASED ON PIPE SIZE PER CROSS-SECTION.
3. CONSTRUCT TWO SINGLE BASINS PER NCDOT STANDARD WITH DOUBLE INTERIOR WALL.
4. ALL CONCRETE TO BE 3600 P.S.I COMPRESSIVE STRENGTH.
5. BASE SLAB SHALL BE MONOLITHIC.
6. SEE STANDARD NUMBERS 120.1 AND 121.1 FOR PLACEMENT OF CATCH BASIN.
7. RCP PIPE SECTION D2 CONNECTING CATCH BASINS SHALL HAVE A MINIMUM DIAMETER SAME AS OF OUTLET PIPE D3.
8. ALL REINFORCING STEEL SHOWN ON NCDOT STANDARDS IS TO BE PROVIDED AS CONTINUOUS MEMBERS. (NO LAPS, USED AS A SINGLE CONTINUOUS BAR IN THE SLAB)



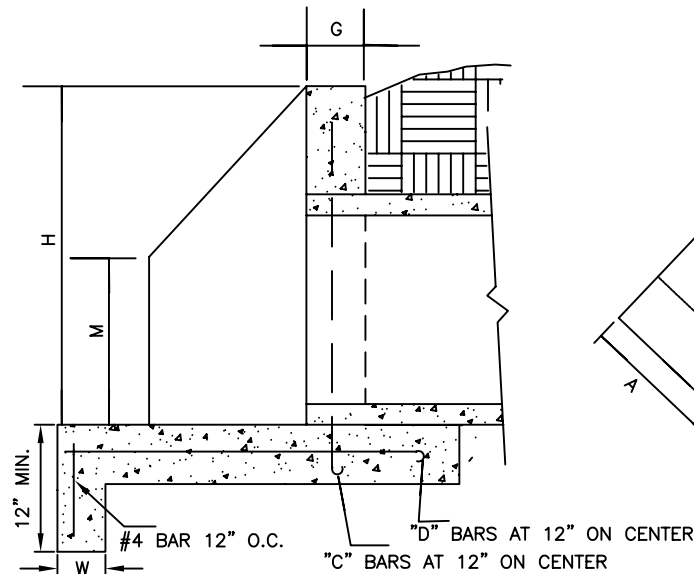
NOT TO SCALE

CONCRETE PIPE			DIMENSIONS										
WALL THK.	OUT DIA.	IN DIA.	MIN. H	A	B	C	E	F	G	W	K	M	
2 1/4"	19 1/2"	15"	27 1/2"	20"	24"	8"	7 1/2"	4"	4"	8"	17"	10"	
2 1/2"	23"	18"	31"	20"	24"	8"	9"	4"	4"	8"	17"	12"	
3"	30"	24"	38"	20"	30"	8"	12"	4"	4"	8"	21"	15"	
3 1/2"	37"	30"	45"	20"	44"	12"	15"	6"	8"	8"	31"	18"	
4"	44"	36"	52"	32"	44"	12"	18"	6"	8"	8"	31"	22"	
4 1/2"	51"	42"	59"	32"	48"	12"	21"	6"	8"	8"	34"	26"	
5"	58"	48"	66"	32"	48"	12"	24"	6"	8"	8"	34"	29"	
5 1/2"	65"	54"	73"	32"	54"	12"	27"	6"	8"	8"	38"	33"	
6"	72"	60"	80"	36"	66"	12"	30"	8"	12"	12"	46"	36"	
6 1/2"	79"	66"	87"	36"	72"	12"	33"	8"	12"	12"	51"	40"	
7"	86"	72"	94"	36"	78"	12"	36"	8"	12"	12"	56"	43"	

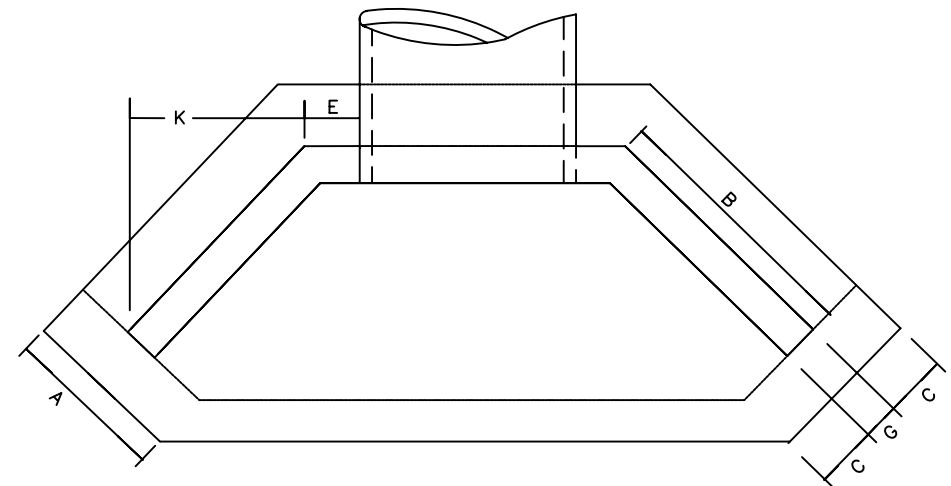


FRONT VIEW

REINFORCING				
DIA.	"C" BAR NO.	"C" BAR LGT.	"D" BAR NO.	"D" BAR LGT.
15"	4	2'-0"	4	1'-11"
18"	4	2'-3"	4	2'-2"
24"	4	2'-9"	4	2'-8"
30"	4	3'-3"	4	3'-2"
36"	4	3'-9"	4	3'-8"
42"	4	4'-3"	4	4'-2"
48"	4	4'-9"	4	4'-8"
54"	4	5'-3"	4	5'-2"
60"	4	5'-9"	4	5'-8"
66"	4	6'-3"	4	6'-2"
72"	4	6'-9"	4	6'-8"



SIDE VIEW



TOP VIEW

NOT TO SCALE

GENERAL NOTES:

1. ALL CORNERS TO BE CHAMFERED 1" IF CONCRETE.
2. THE CONTRACTOR WILL BE REQUIRED TO PLACE 2-#6 BARS "Y" IN THE TOP OF ALL ENDWALL FOR PIPE CULVERTS 42" AND OVER WITH A MINIMUM 3" COVER AND A LENGTH OF 6" LESS THAN ENDWALL.
3. FORMS ARE TO BE USED FOR THE CONSTRUCTION OF THE BOTTOM SLAB.
4. WALL THICKNESS (T) SHOWN IS NOT TO BE INTERPRETED TO MEAN THE THICKNESS ACCEPTABLE, BUT IS USED ONLY IN COMPUTING ENDWALL QUANTITIES.
5. IF CONTRACTOR ELECTS TO USE CONSTRUCTION JOINT AT BOTTOM OF PIPE, AND POURS BASE SEPARATELY, THE TOP OF BASE SHALL BE LEFT ROUGH.
6. ALL CONCRETE TO BE 3600 P.S.I COMPRESSIVE STRENGTH.

NOT TO SCALE

NOTES:

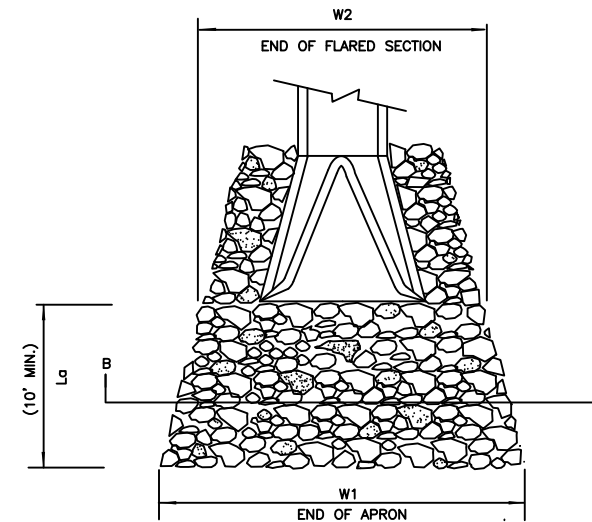
1. CLASS OR MEDIAN SIZE OF RIPRAP AND LENGTH, WIDTH AND DEPTH OF APRON TO BE DESIGNED BY THE ENGINEER.
2. REFER TO THE CHARLOTTE MECKLENBURG STORM WATER DESIGN MANUAL FOR RIPRAP APRON DESIGN STANDARDS.
3. RIPRAP SHOULD EXTEND UP BOTH SIDES OF THE APRON AND AROUND THE END OF THE PIPE OR CULVERT AT THE DISCHARGE OUTLET AT A MAXIMUM SLOPE OF 2:1 AND A HEIGHT NOT LESS THAN TWO THIRDS THE PIPE DIAMETER OR CULVERT HEIGHT.
4. THERE SHALL BE NO OVERFLOW FROM THE END OF THE APRON TO THE SURFACE OF THE RECEIVING CHANNEL. THE AREA TO BE PAVED OR RIPRAPPED SHALL BE UNDERCUT SO THAT THE INVERT OF THE APRON SHALL BE AT THE SAME GRADE (FLUSH) WITH THE SURFACE OF THE RECEIVING CHANNEL. THE APRON SHALL HAVE A CUTOFF OR TOE WALL AT THE DOWNSTREAM END.
5. THE WIDTH OF THE END OF THE APRON SHALL BE EQUAL TO THE BOTTOM WIDTH OF THE RECEIVING CHANNEL. MAXIMUM TAPER TO RECEIVING CHANNEL 5:1.
6. ALL SUBGRADE FOR STRUCTURE TO BE COMPACTED TO 95% OR GREATER.
7. THE PLACING OF FILL, EITHER LOOSE OR COMPACTED IN THE RECEIVING CHANNEL SHALL NOT BE ALLOWED.
8. NO BENDS OR CURVES IN THE HORIZONTAL ALIGNMENT OF THE APRON WILL BE PERMITTED.
9. FILTER FABRIC SHALL BE INSTALLED ON COMPACTED SUBGRADE PRIOR TO PLACEMENT OF RIP RAP.
10. ANY DISTURBED AREA FROM END OF APRON TO RECEIVING CHANNEL MUST BE STABILIZED.

USE USDA NOMOGRAPH FROM NC SEDIMENT AND EROSION CONTROL MANUAL OR CHARLOTTE MECKLENBURG STORM WATER DESIGN MANUAL FOR DESIGN DATA.

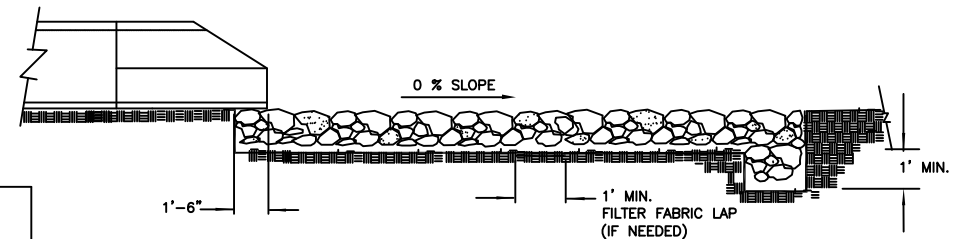
RIPRAP SUMMARY CHART

OUTLET	La	W1	W2	*T	H

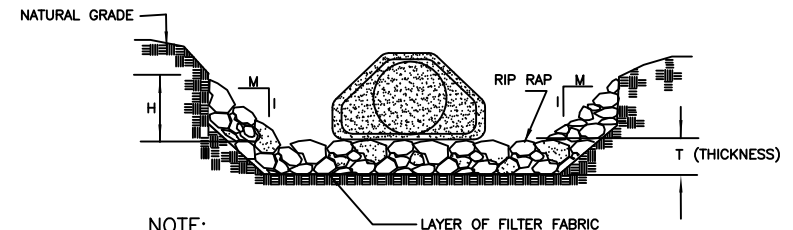
* d50 (see fig 8.06 a&b "NC SEDIMENT AND EROSION CONTROL MANUAL")
dmax = 1.5 x d50
T = 1.5 X dmax.



PLAN



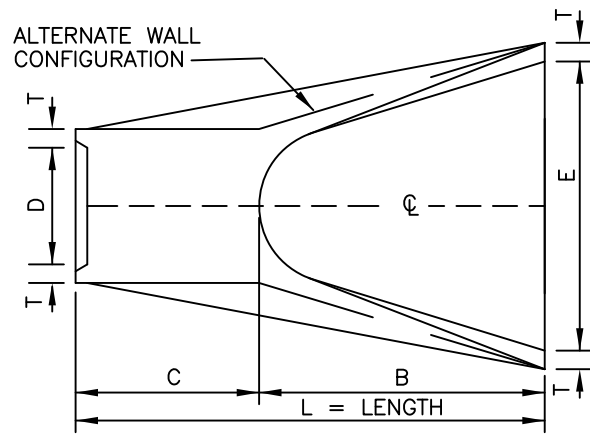
ELEVATION



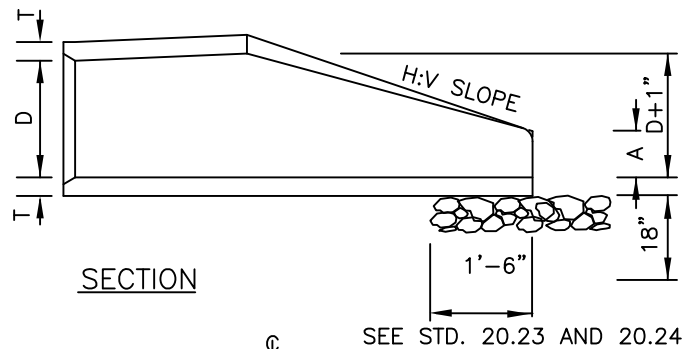
NOTE:
MINIMUM H=2/3 PIPE DIAMETER

SECTION B-B

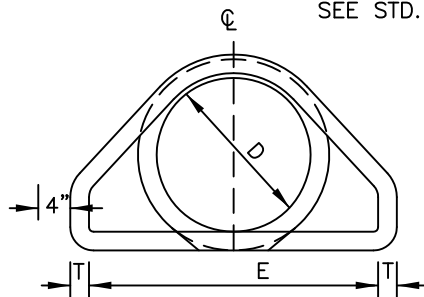
NOT TO SCALE



PLAN



SECTION



END VIEW

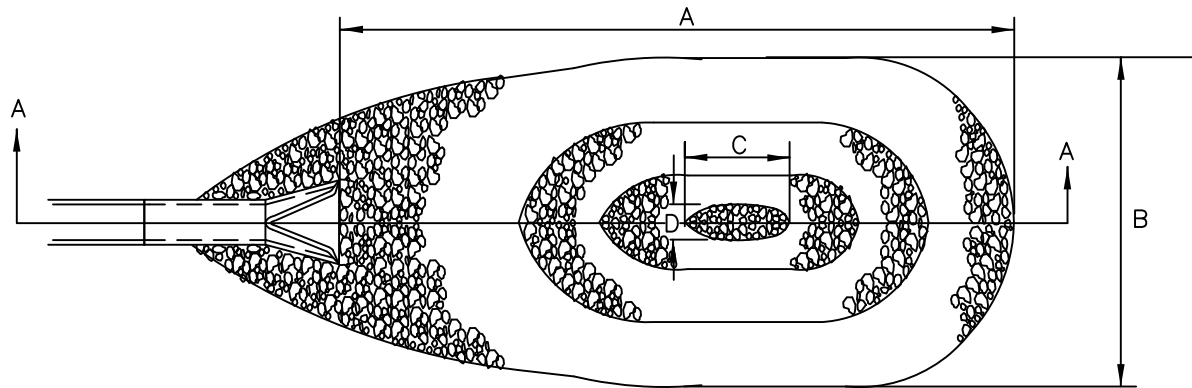
TABLE OF DIMENSIONS

D	T	A	B	C	E	L	H:V	WT.
12"	2-1/4"	4"	2'-0"	4'-1"	2'-0"	6'-1"	3:1	730
15"	2-1/4"	6"	2'-3"	3'-10"	2'-0"	6'-1"	3:1	730
18"	2-1/2"	9"	2'-3"	3'-10"	3'-0"	6'-1"	3:1	1190
24"	3"	10"	3'-8"	2'-6"	4'-0"	6'-2"	3:1	1770
30"	3-1/2"	1'-0"	4'-6"	1'-8"	5'-0"	6'-2"	3:1	2380
36"	4"	1'-3"	5'-3"	2'-11"	6'-0"	8'-2"	3:1	5320
42"	4-1/2"	1'-9"	5'-3"	2'-11"	6'-6"	8'-2"	3:1	5920
48"	5"	2'-0"	6'-0"	2'-2"	7'-0"	8'-2"	3:1	7470
54"	5-1/2"	2'-3"	5'-6"	2'-10"	7'-6"	8'-4"	3:1	8810
60"	6"	2'-6"	5'-0"	3'-3"	8'-0"	8'-3"	3:1	11180
66"	6-1/2"	3'-0"	6'-0"	2'-3"	8'-6"	8'-3"	3:1	12530
72"	7"	3'-0"	6'-6"	1'-9"	9'-0"	8'-3"	3:1	13980

GENERAL NOTES:

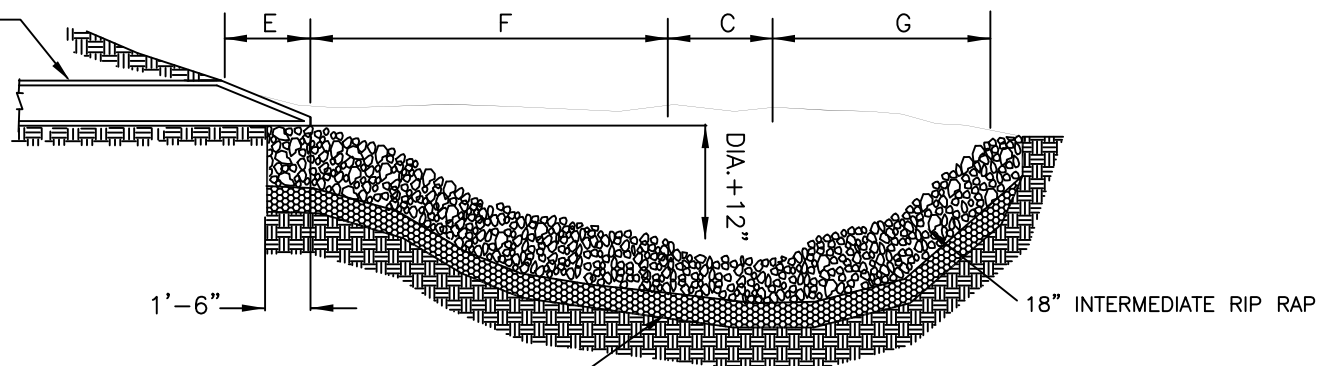
1. SEE FORMER NCDOT STANDARD 310.01 FOR DETAILS.
2. REINFORCEMENT SHALL CONFORM TO THE REQUIREMENTS OF REINFORCED CONCRETE PIPE OF LIKE DIAMETER PER AASHTO M170, TABLE 2, WALL B.
3. ALL CONCRETE TO BE 4000 P.S.I COMPRESSIVE STRENGTH.
4. PROVIDE TONGUE OR SPIGOT JOINT AT INLET END SECTION.
5. PROVIDE GROOVE OR BELL JOINT AT OUTLET END SECTION.
6. THE DIMENSIONS FOR END SECTIONS SHALL SUBSTANTIALLY AGREE WITH THE TABLE. MINOR VARIATIONS WILL BE PERMITTED BASED ON THE MANUFACTURER'S STANDARD FORMS AND TEMPLATES.
7. NOT TO BE USED IN NCDOT MAINTAINED RIGHT OF WAY.

NOT TO SCALE



PLAN

FLARED END SECTION OR END WALL



SECTION A-A

NOTE:

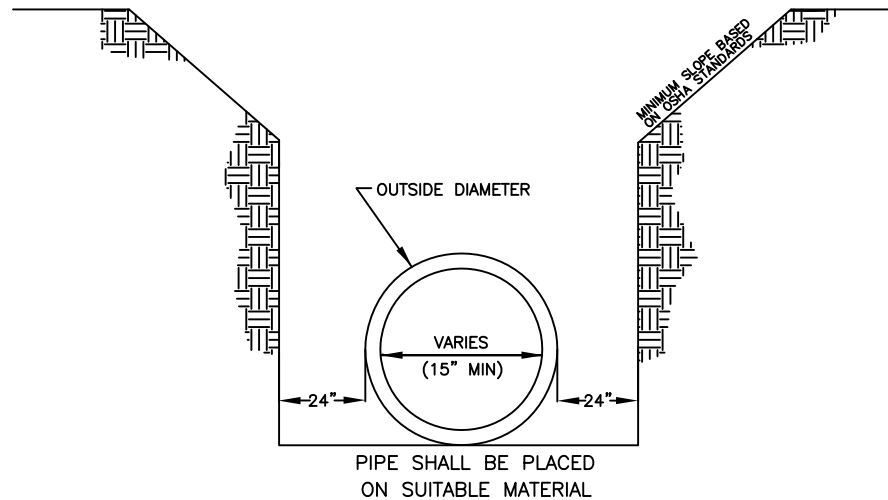
1. THIS DETAIL IS TO ONLY BE USED WHEN OUTFALL HAS A CONTINUOUS FLOW OF WATER AND WITH PRIOR APPROVAL OF THE TOWN ENGINEER.

PIPE SIZE	A	B	C	D	E	F	G	WT. RIP RAP IN TONS
15"	10'	7'	1 1/2'	1'	1'	4 1/2'	3'	6
18"	12'	8'	2'	1'	1'	5'	4'	8
21"	15'	9'	2 1/2'	1 1/2'	1'	7'	4 1/2'	12
24"	17'	10'	2 1/2'	1 1/2'	1'	8'	5 1/2'	15
30"	20'	13'	3'	2'	2'	9'	6'	22
36"	24'	16'	3 1/2'	2'	2'	9 1/2'	7'	33

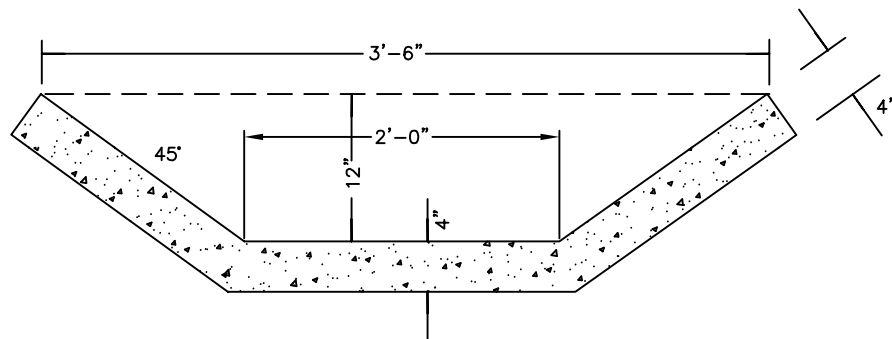
NOT TO SCALE

NOTES:

1. A MINIMUM OF 24" FROM OUTSIDE DIAMETER OF PIPE TO SIDE OF TRENCH MUST BE ALLOWED FOR COMPACTION OF FILL MATERIAL. BACKFILLING OF TRENCHES SHALL BE ACCOMPLISHED IMMEDIATELY AFTER THE PIPE IS LAID. THE FILL AROUND THE PIPE SHALL BE PLACED IN LAYERS NOT TO EXCEED 6". UNDER NO CIRCUMSTANCES SHALL WATER BE PERMITTED TO RISE IN UNBACKFILLED TRENCHES AFTER THE PIPE HAS BEEN PLACED. COMPACTION REQUIREMENTS SHALL BE ATTAINED BY THE USE OF MECHANICAL TAMPS ONLY. EACH AND EVERY LAYER OF BACKFILL SHALL BE PLACED LOOSE AND THOROUGHLY COMPACTED INTO PLACE.
2. ALL BACKFILL MATERIAL SHALL HAVE AN IN PLACE COMPACTED DENSITY OF 95%.
3. STANDARD PROCTOR. THE FINAL 2' BELOW FINISHED GRADE SHALL BE 100%.
4. ALL TRENCHING OPERATIONS SHALL MEET OSHA STANDARDS.
5. BACKFILL MATERIAL BENEATH ROADWAY SHALL BE SELECT BACKFILL MATERIAL.



NOT TO SCALE



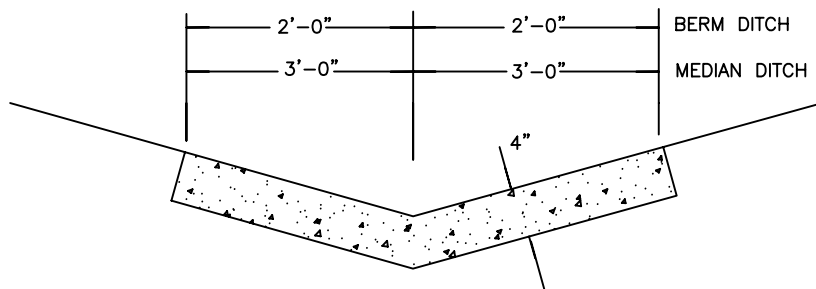
SLOPE DRAIN, BASE DITCH OR BERM DRAINAGE
OUTLET DITCH

GENERAL NOTES:

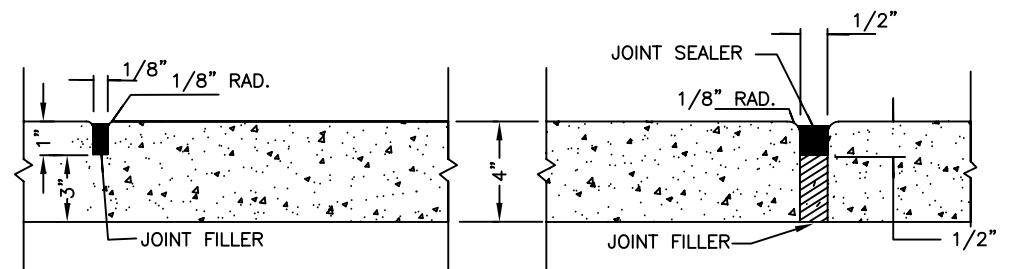
IN THE 4" CONCRETE PAVED DITCHES PLACE 1/2" EXPANSION JOINT AT 30 FT INTERVALS AND AT ALL OTHER POINTS WHERE PROPOSED DITCHES ABUT RIGID OBJECTS. PLACE GROOVED JOINTS 1" DEEP AT 10' INTERVALS BETWEEN EXPANSION JOINTS.

WIDTH AND SHAPE OF PROPOSED 4" CONCRETE PAVED DITCHES SHALL BE AS SHOWN OR AS DIRECTED BY THE ENGINEER.

ALL CONCRETE TO BE 3600 P.S.I. COMPRESSIVE STRENGTH.

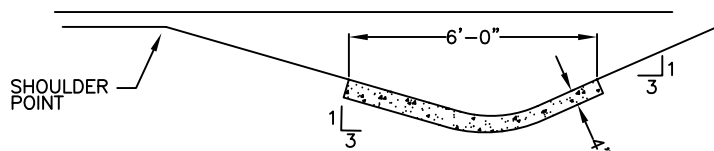


MEDIAN OR BERM DITCH

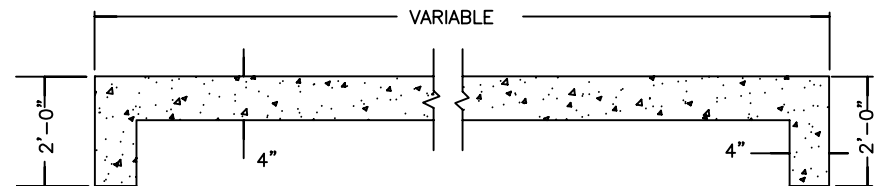


SHOWING GROOVED JOINT

SHOWING EXPANSION JOINT



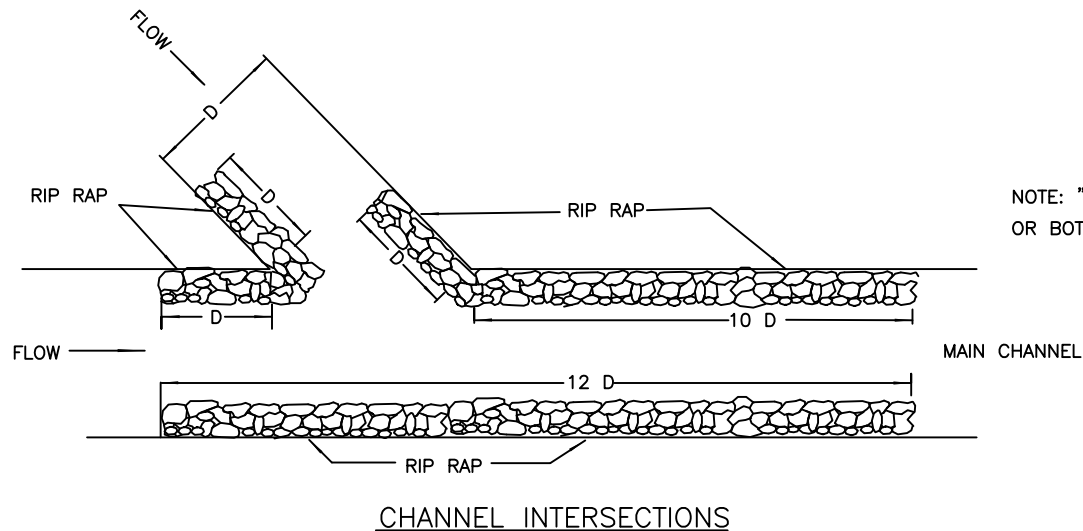
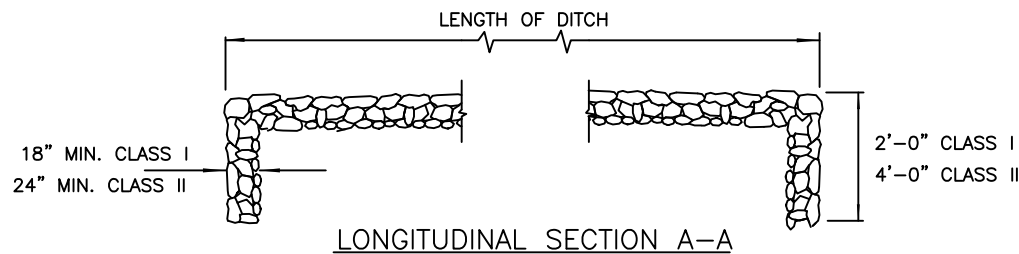
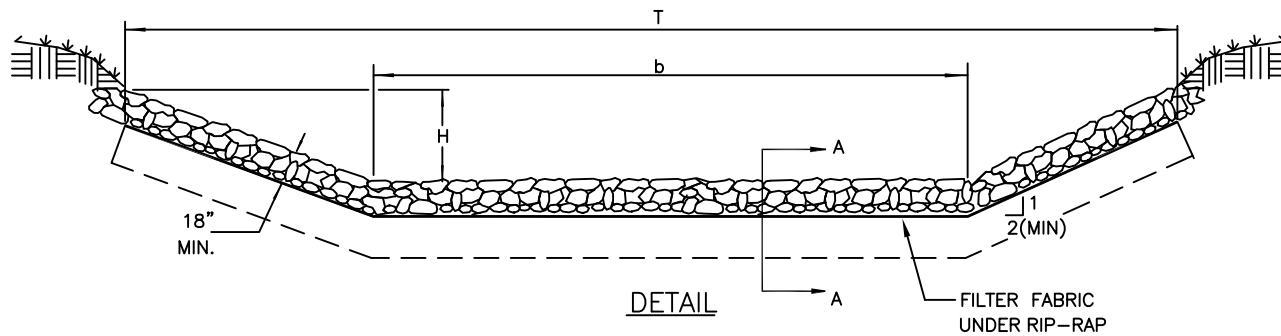
SIDE DITCH



LONGITUDINAL SECTION OF PAVED DITCH

SHOWING 2'-0" CURTAIN WALL REQUIRED AT EACH END

NOT TO SCALE



GENERAL NOTES:

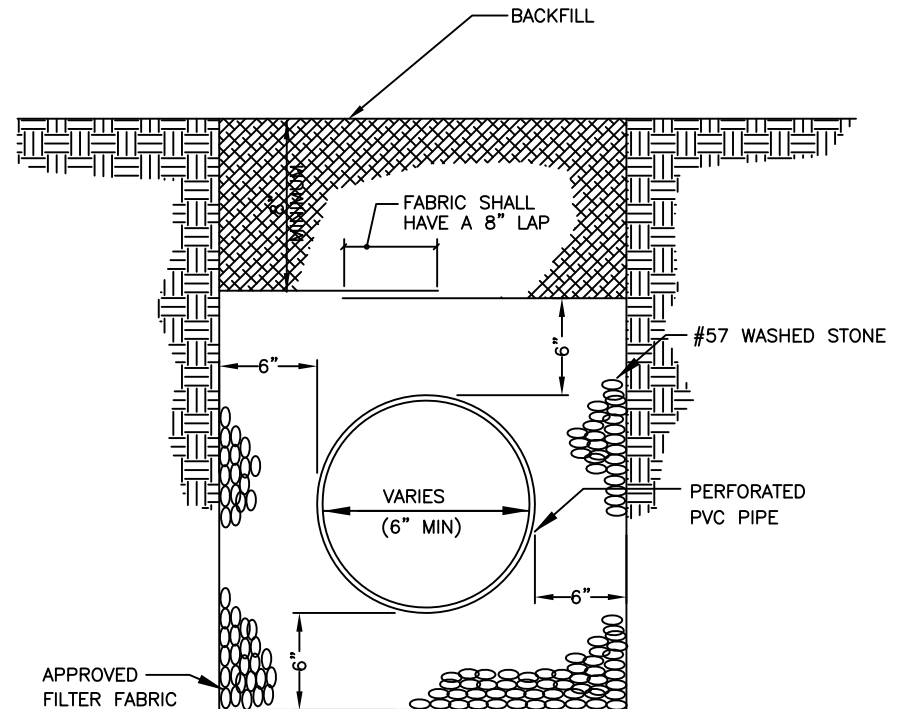
1. IF BEDROCK IS ENCOUNTERED WITHIN THE LIMITS OF THE TOEWALL, BEGIN TOEWALL ON THE BEDROCK OR AS DIRECTED BY THE ENGINEER.
2. WHERE ONLY ONE SIDE REQUIRES RIP RAP CLASS I OR II, LIST STATION AND SIDE OF SAME.
3. CHANNEL AND RIP RAP SIZE TO BE DESIGNED BY THE ENGINEER.
4. DEPENDING ON SOIL CONDITIONS, WASHED STONE AND FILTER FABRIC MAY BE NECESSARY UNDER RIP RAP.
5. CHANNEL DEPTH "H" SHALL INCLUDE A MINIMUM 6" OF FREEBOARD.

NOTE: "D" EQUALS DIAMETER OF PIPE OR BOTTOM WIDTH OF CHANNEL.

NOT TO SCALE

NOTES:

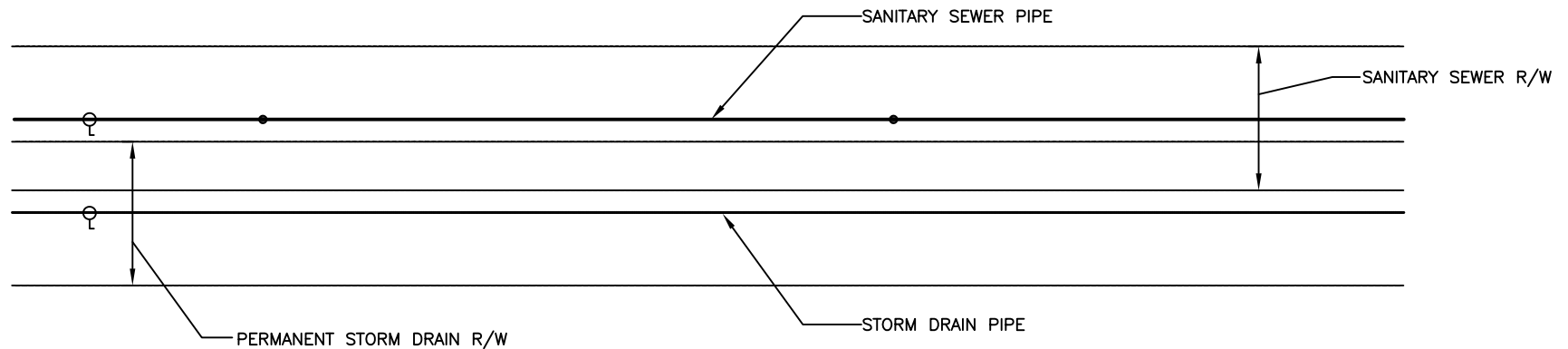
1. A MINIMUM OF 6" FROM OUTSIDE DIAMETER OF PIPE TO SIDE OF TRENCH MUST BE ALLOWED FOR WASHED STONE. THE METHOD OF COMPACTING BACKFILL MATERIAL IS SUBJECT TO APPROVAL BY THE TOWN ENGINEER.
AN APPROVED FILTER FABRIC SHALL BE PLACED AROUND STONE AND OVERLAPPED 8" AT TOP WITHIN STREET RIGHT OF WAY. PIPE SIZE TO BE SHOWN ON PLAN (MINIMUM 6" PIPE). PIPE TO BE SCHEDULE 20 OR 40 PERFORATED PVC.
2. OUTLET PIPE FROM SUBDRAIN SHALL BE NON-PERFORATED UNDER PAVEMENT (INCLUDING SIDEWALKS AND DRIVEWAYS)
3. THE OUTLET PIPES SHALL BE SCHEDULE 80 UNDER ROADWAYS.
4. SEE SITE PLAN FOR SLOPE OF SUBDRAIN AND TIE IN TO STORM DRAINAGE.
5. FILTER FABRIC SHALL BE AN APPROVED, TYPE 2 WATER PERMEABLE, SYNTHETIC FABRIC.



SPECIAL NOTE:

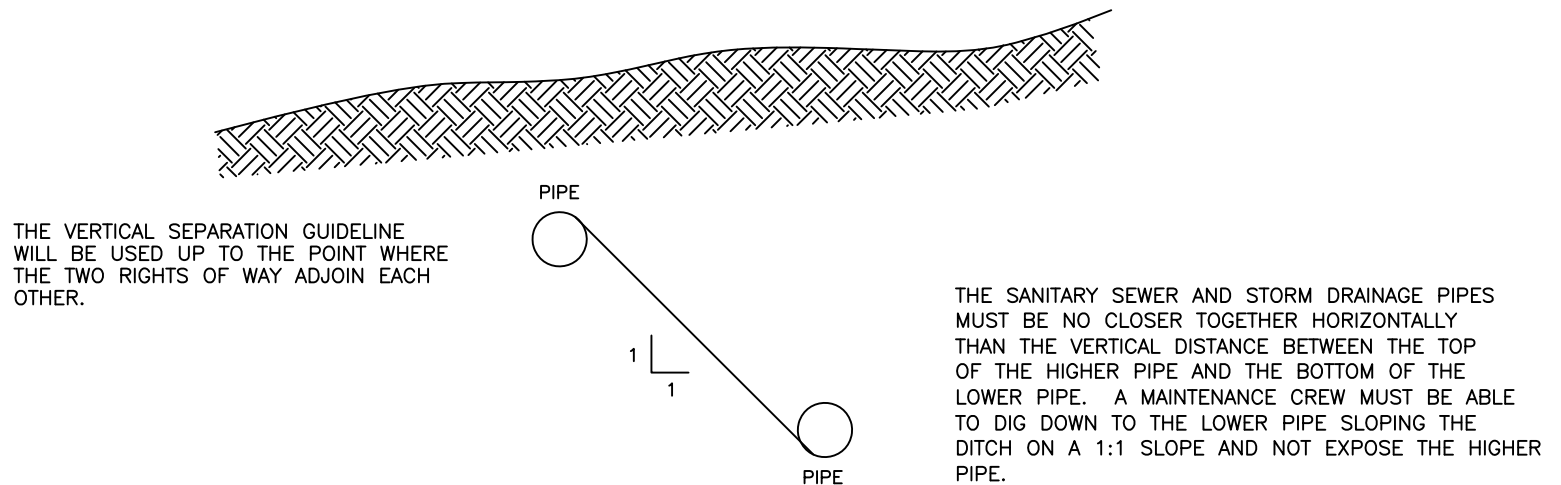
PREFABRICATED DRAINAGE MAY BE USED WITH
APPROVAL OF TOWN ENGINEER.

NOT TO SCALE



THE SANITARY SEWER AND STORM DRAINAGE RIGHTS OF WAY MAY OVERLAP; HOWEVER THE PIPE AND ASSOCIATED STRUCTURES MUST NOT BE IN THE OTHER UTILITY'S RIGHT OF WAY. THE SANITARY SEWER RIGHT OF WAY WIDTHS SHALL BE AS OUTLINED IN C.M.U.D.'S DESIGN MANUAL. THIS DETAIL DOES NOT APPLY TO STORM DRAINAGE UTILIZING OPEN CHANNEL FLOW.

PLAN VIEW



PROFILE VIEW

NOT TO SCALE

GENERAL NOTES:

1. FOR STREAMS CARRYING 500 ACRES OR MORE OF SURFACE RUNOFF, THE EASEMENT REQUIREMENT IS TO BE THE WIDTH OF THE STREAM FROM TOP OF BANK TO TOP OF BANK, PLUS (+) 10' ON EACH SIDE OF STREAM. (40' MINIMUM WIDTH)
2. FOR OPEN CHANNELS THE MINIMUM EASEMENT MUST CONTAIN THE WIDTH OF THE STREAM FROM TOP OF BANK TO TOP BANK.
3. WIDER EASEMENT WIDTHS MAY BE REQUIRED FOR PIPE DEPTHS GREATER THAN TEN FEET.
4. PIPE SYSTEMS AND OPEN CHANNELS ON PRIVATE PROPERTY SHALL BE PLACED IN A STORM DRAINAGE EASEMENT.

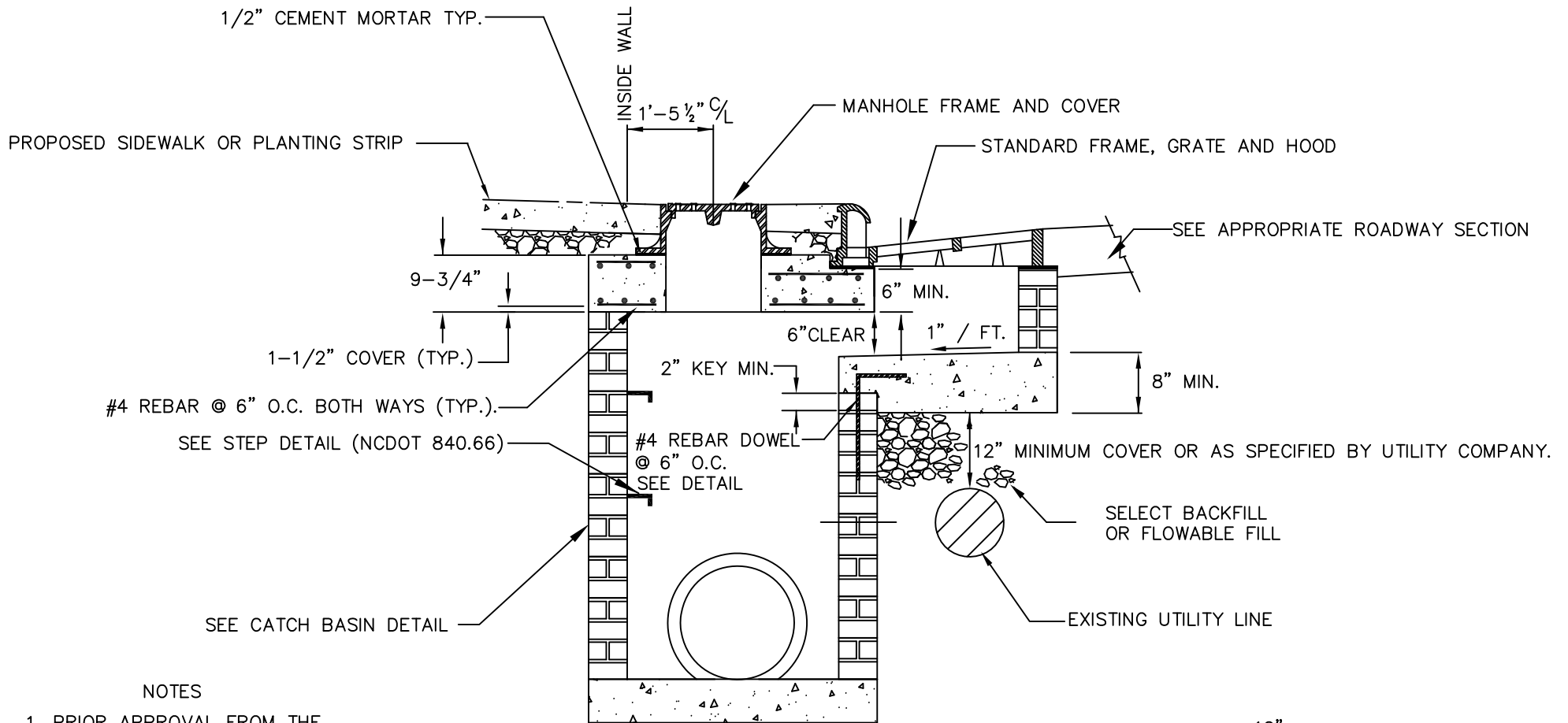
Easement Requirements for Open Storm Drainage Channels

Area in Acreage	Easement Requirement
0-45 ac.	20'
45-120 ac.	30'
120-500 ac.	40'
500 ac.+	see note

Easement Requirements for Storm Drain Pipe

Pipe Size	Easement Requirement
15"	15'
18"	15'
24"	15'
30"	20'
36"	20'
42"	25'
48"	25'
54"+	30'MIN (VARIES)

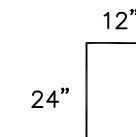
NOT TO SCALE



NOTES

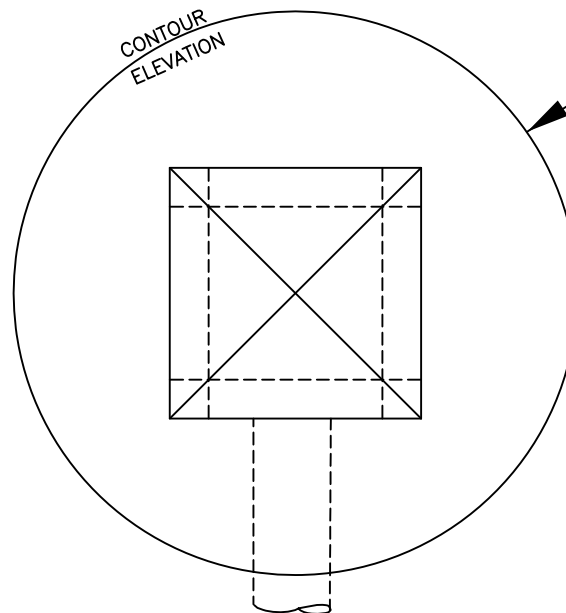
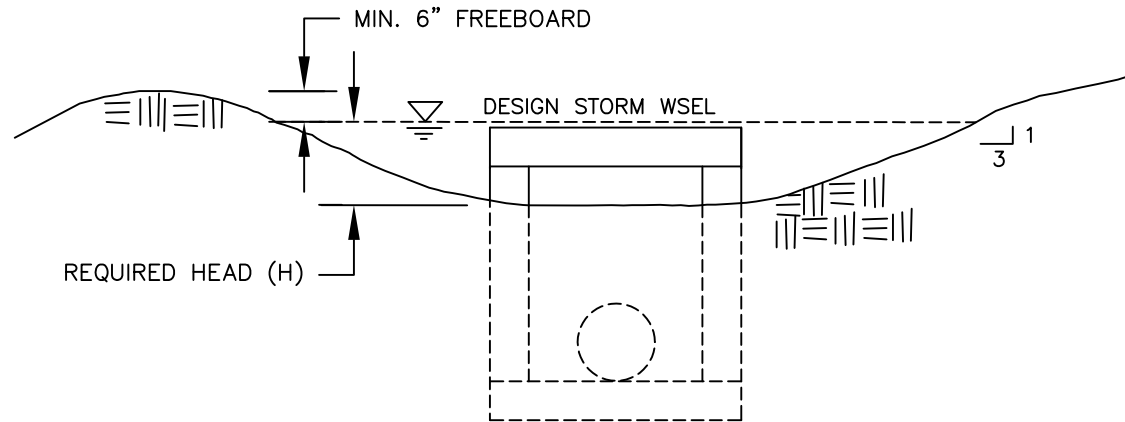
1. PRIOR APPROVAL FROM THE TOWN ENGINEER IS REQUIRED.
2. THIS STRUCTURE IS TO ONLY BE USED ON TOWN MAINTAINED STREETS AND NOT ON NCDOT STREETS WITHOUT THEIR PERMISSION.

OFFSET CATCH BASIN EXISTING
UTILITY CONFLICT



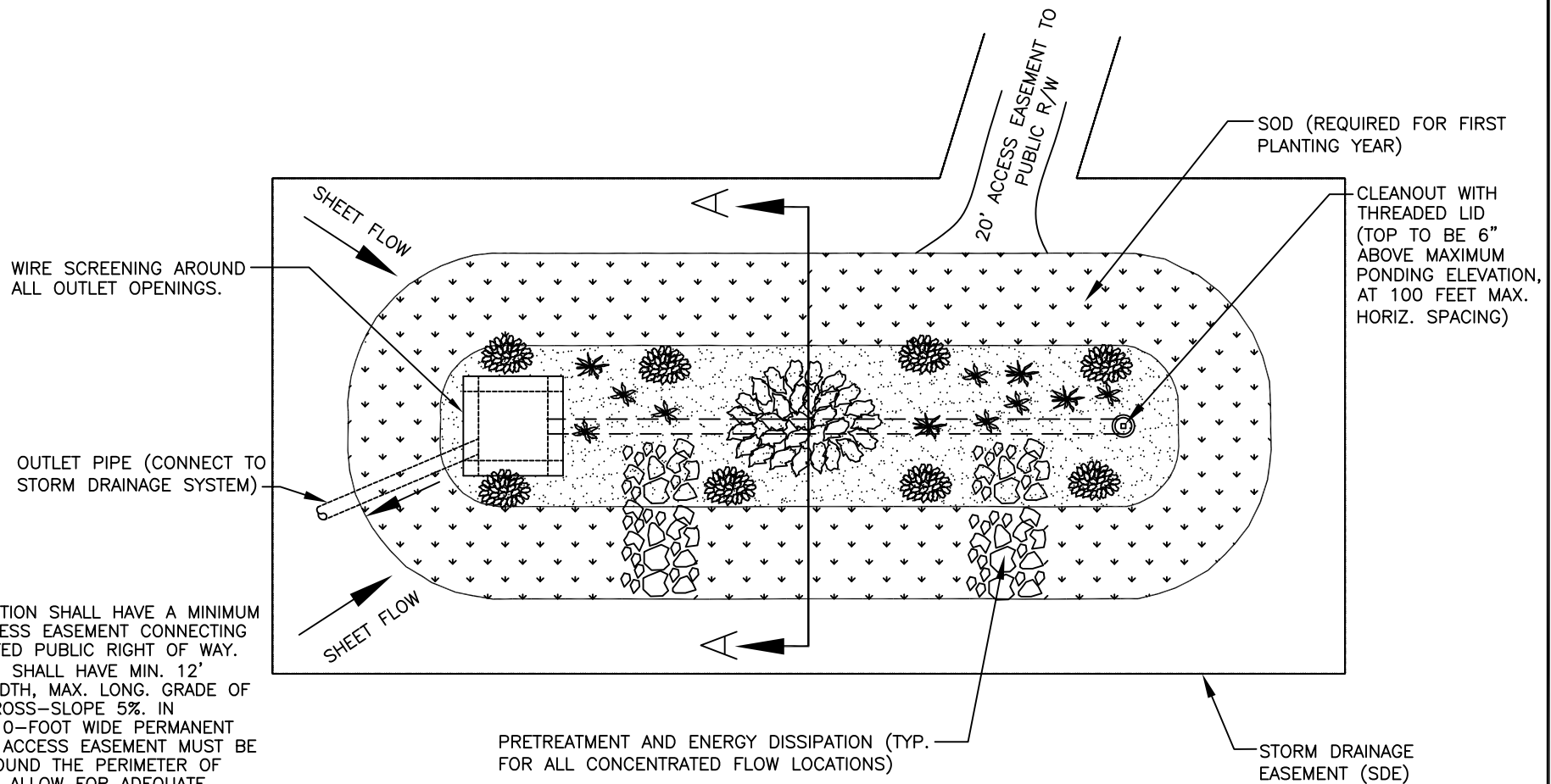
DOWEL DETAIL

NOT TO SCALE



YARD INLET SUMMARY CHART				
YARD INLET	AREA (AC.)	CFS	HEAD H (FT.)	COMMENT

NOT TO SCALE



NOTES:

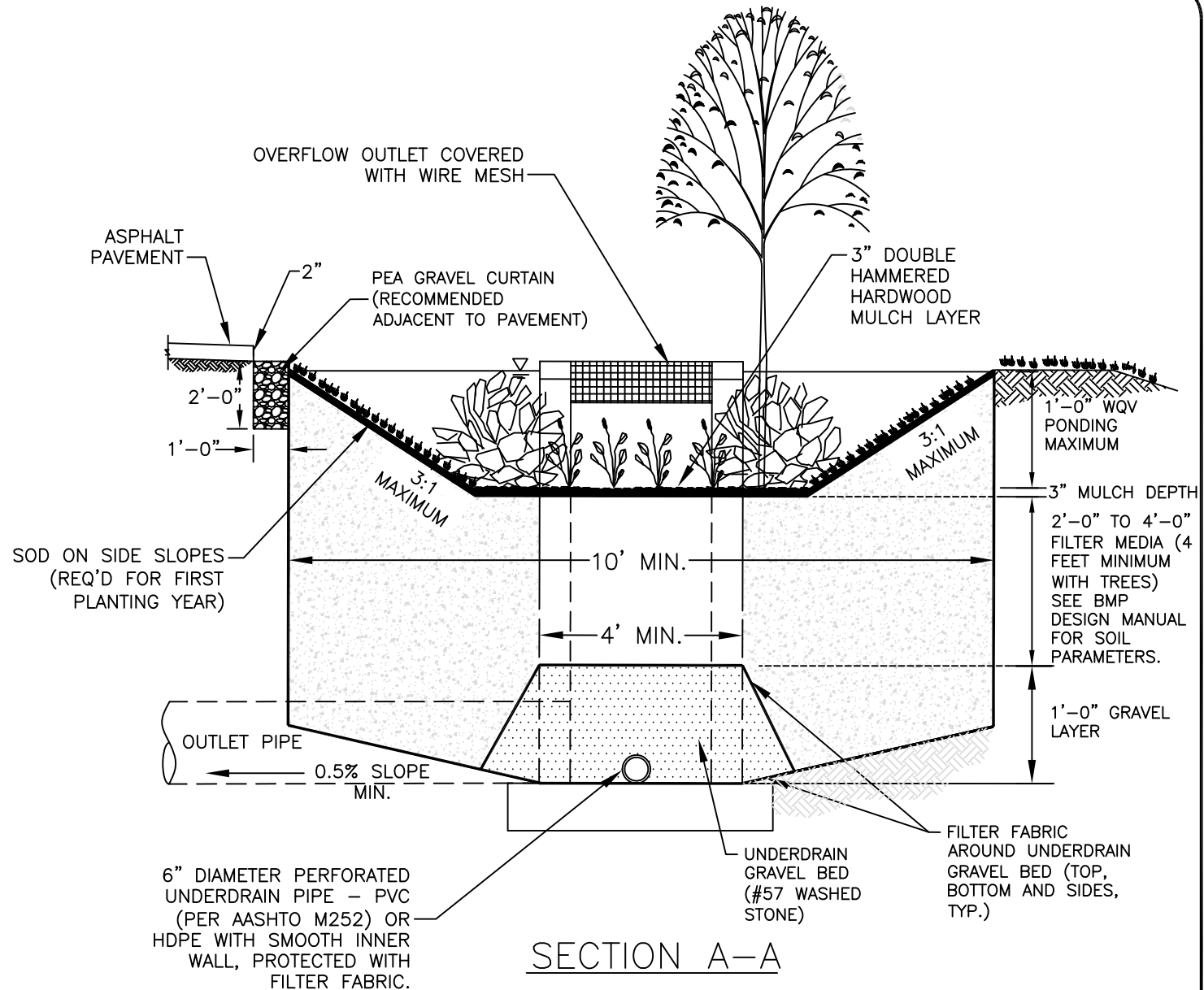
1. ALL BIORETENTION SHALL HAVE A MINIMUM 20 FOOT ACCESS EASEMENT CONNECTING TO A DEDICATED PUBLIC RIGHT OF WAY. ACCESS ROAD SHALL HAVE MIN. 12' STABILIZED WIDTH, MAX. LONG. GRADE OF 15%, MAX. CROSS-SLOPE 5%. IN ADDITION, A 10-FOOT WIDE PERMANENT MAINTENANCE ACCESS EASEMENT MUST BE PROVIDED AROUND THE PERIMETER OF ALL BMPs TO ALLOW FOR ADEQUATE MAINTENANCE AND REPAIR.
2. ALL DRAINAGE AREAS TO A BIORETENTION FACILITY ARE TO BE STABILIZED PRIOR TO INSTALLATION OF AMENDED SOILS, MULCH OR PLANTINGS.
3. AMENDED SOIL WILL ONLY BE PERMITTED WITH A VALID SOIL ANALYSIS REPORT.
4. INSTALL WIRE SCREENING AROUND ALL OUTLET OPENINGS TO PREVENT LOSS OF MULCH.

PLAN

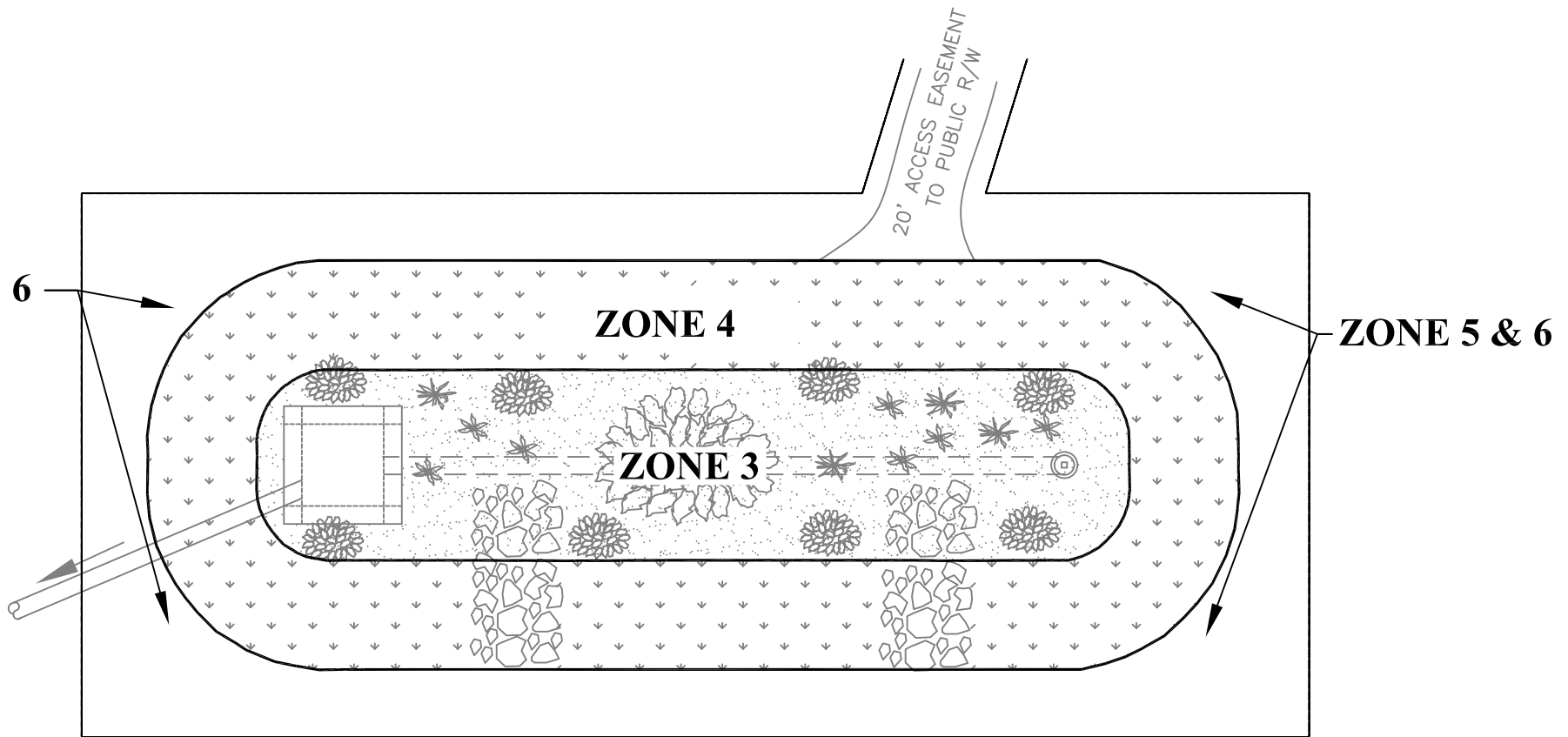
NOT TO SCALE

NOTES:

1. ALL BIORETENTION SHALL HAVE A MINIMUM 20 FOOT ACCESS EASEMENT CONNECTING TO A DEDICATED PUBLIC RIGHT OF WAY. ACCESS ROAD SHALL HAVE MIN. 12' STABILIZED WIDTH, MAX. LONG. GRADE OF 15%, MAX. CROSS-SLOPE 5%. IN ADDITION, A 10-FOOT WIDE PERMANENT MAINTENANCE ACCESS EASEMENT MUST BE PROVIDED AROUND THE PERIMETER OF ALL BMPs TO ALLOW FOR ADEQUATE MAINTENANCE AND REPAIR.
2. ALL DRAINAGE AREAS TO A BIORETENTION FACILITY ARE TO BE STABILIZED PRIOR TO INSTALLATION OF AMENDED SOILS, MULCH OR PLANTINGS.
3. AMENDED SOIL WILL ONLY BE PERMITTED WITH A VALID SOIL ANALYSIS REPORT.
4. INSTALL WIRE SCREENING AROUND ALL OUTLET OPENINGS TO PREVENT LOSS OF MULCH.
5. WRAP PERFORATED UNDERDRAIN WITH FILTER FABRIC PRIOR TO BACKFILLING.
6. UNDERDRAIN PIPE SHOULD HAVE 3/8" PERFORATIONS SPACED AT 6" CENTERS, MIN. 4 HOLES PER ROW. MAX SPACING OF UNDERDRAIN PIPE IS 10 FEET ON CENTER.
7. UNDERDRAIN CLEANOUTS MUST EXTEND A MIN. OF 6 INCHES ABOVE TOP SURFACE OF MULCH LAYER, PREFERABLY TO THE TOP OF DESIGN MAXIMUM PONDING DEPTH IN THE BIORETENTION BASIN.



NOT TO SCALE

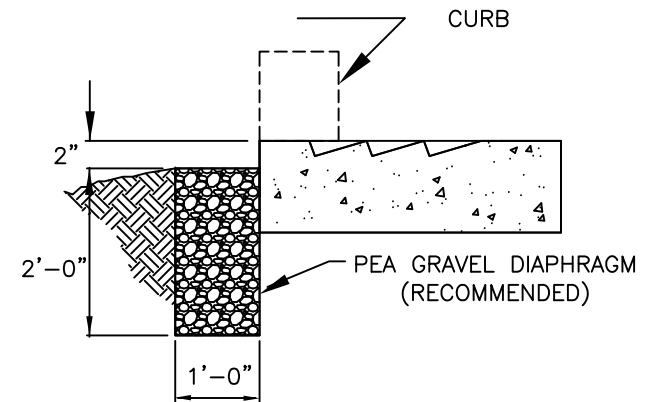
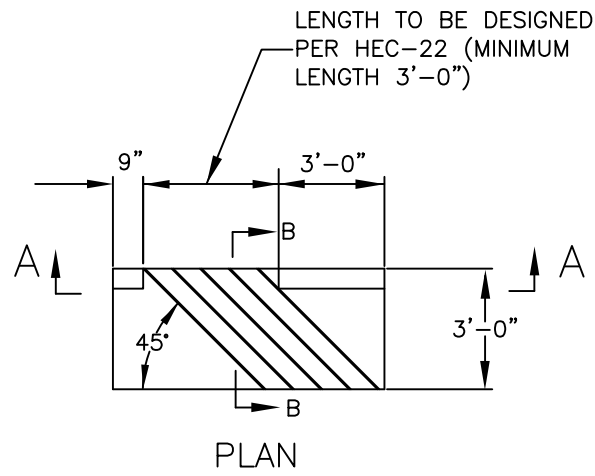
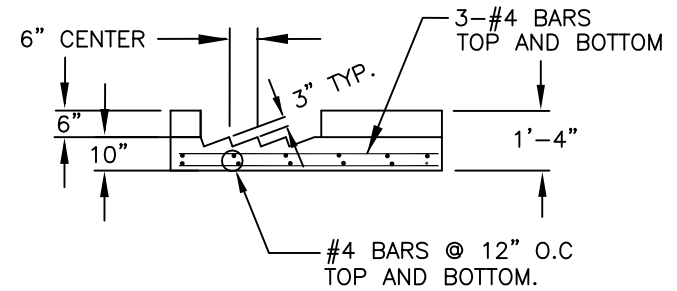
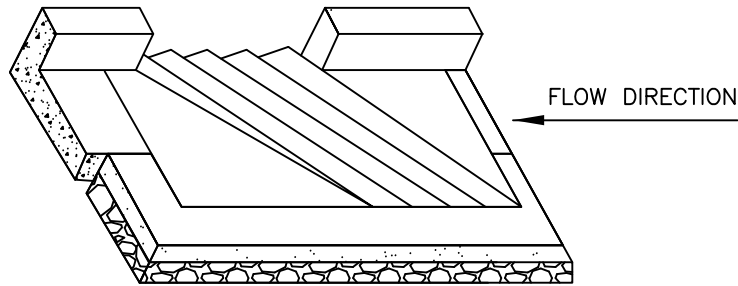


PLAN

NOTES:

1. PLANTING ZONES AND PLANT SELECTION PER THE BMP DESIGN MANUAL, CHAPTER 6 & APPENDICES.
2. ALL PLANTINGS SHALL BE LOCAL NATIVE SPECIES.
3. IRRIGATION MAY BE PROVIDED FOR INITIAL ESTABLISHMENT AND DRY SEASONS.

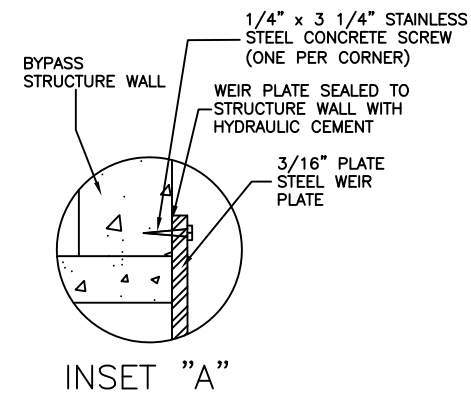
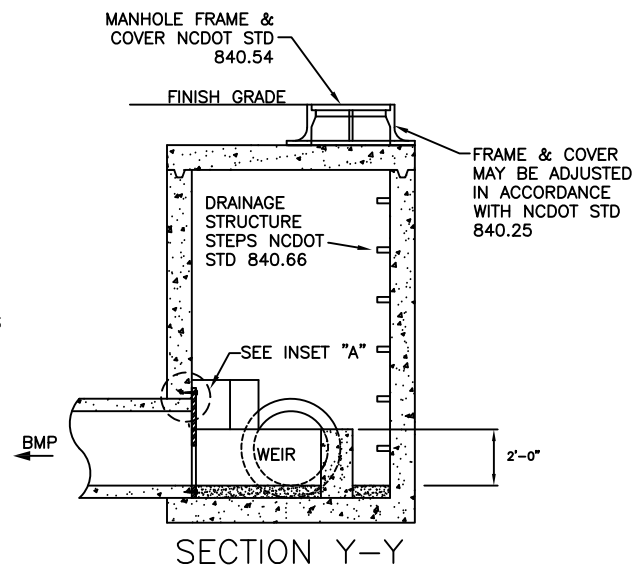
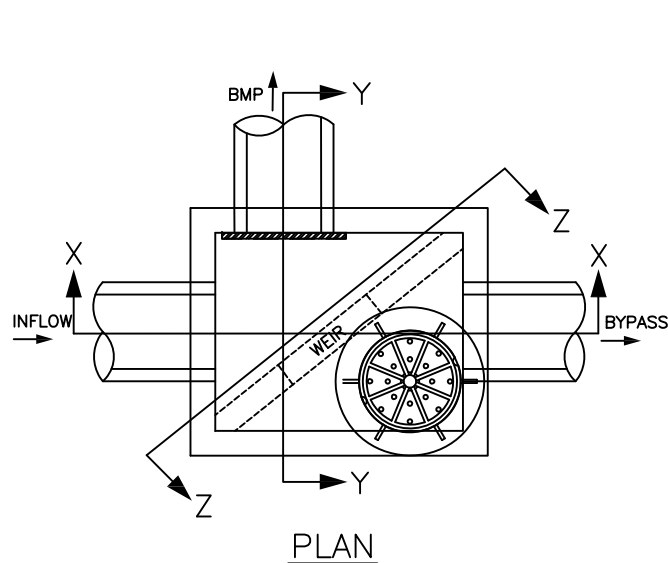
NOT TO SCALE



NOTES

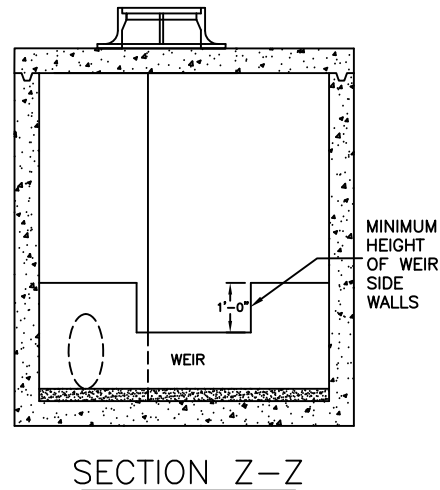
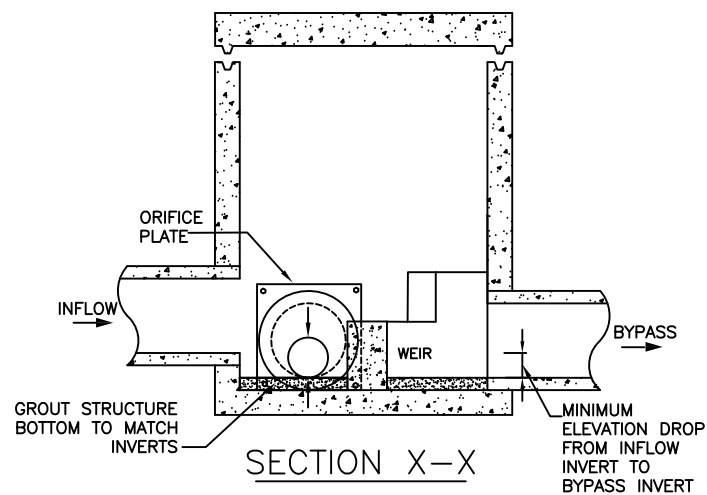
1. ALL CONCRETE 3600 PSI.
2. INLET OPENING IS SIZED USING HEC-22 METHOD.
3. NOT ACCEPTABLE FOR USE IN STREET RIGHT OF WAY WITHOUT PRIOR TOWN/NC DOT APPROVAL.

NOT TO SCALE



NOTES:

1. ALL CONCRETE SHALL BE 3600 PSI.
2. ALL JOINTS ARE TO BE SEALED WATER TIGHT.
3. WEIR IS TO BE POURED-IN-PLACE CONCRETE.
4. REFER TO NCDOT STANDARD DRAWINGS FOR BOX CONSTRUCTION.
5. NOT ACCEPTABLE FOR USE IN STREET RIGHT OF WAY WITHOUT TOWN/NCDOT APPROVAL.

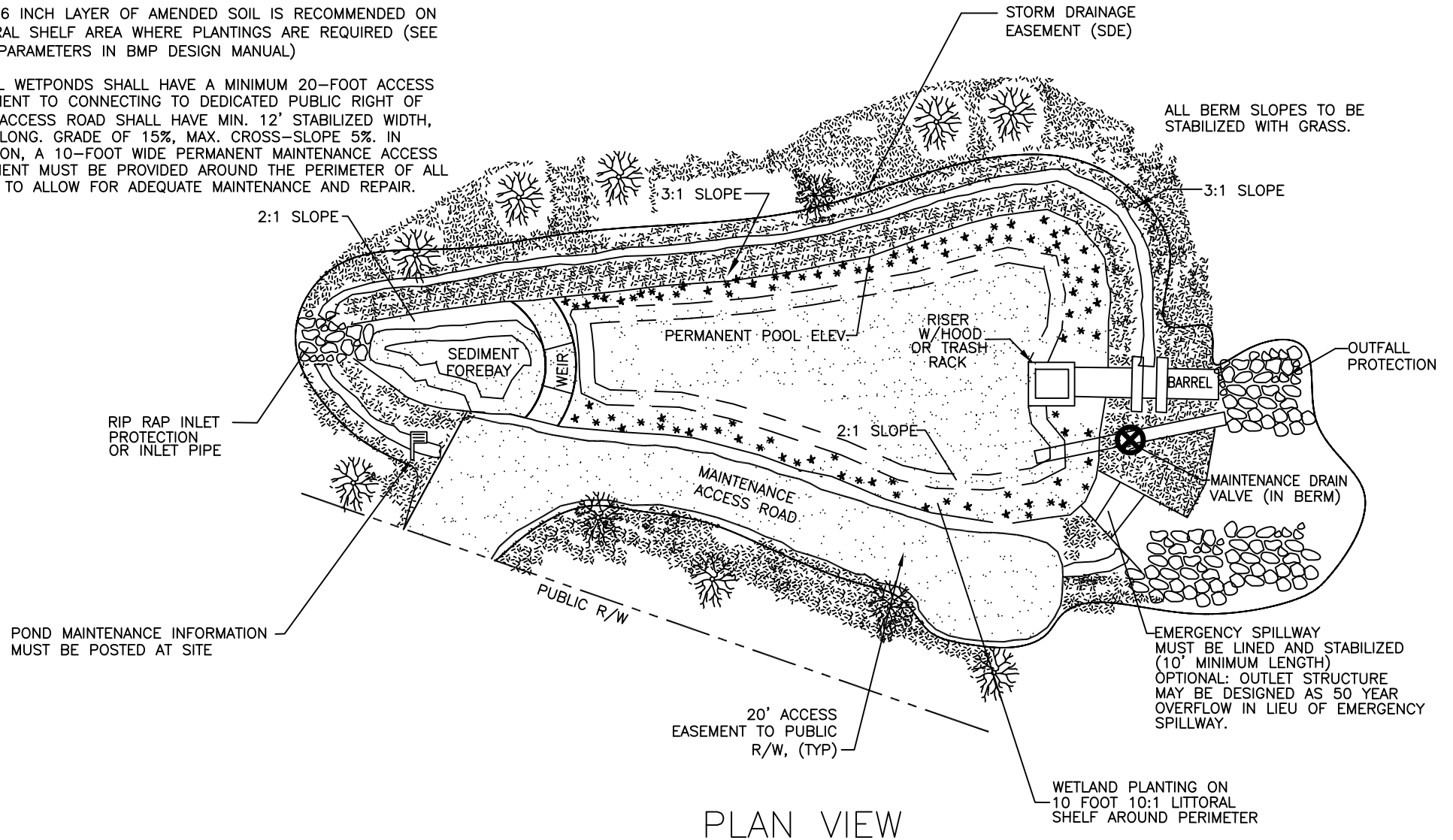


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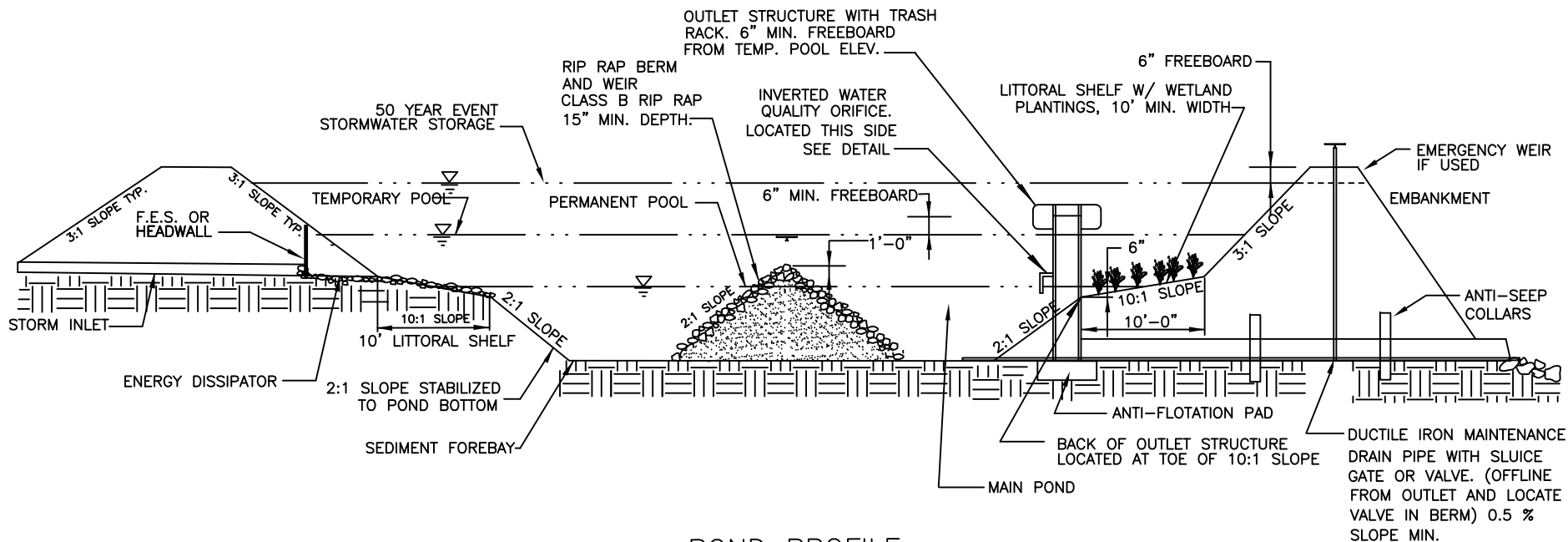
NOTES:

1. 4-6 INCH LAYER OF AMENDED SOIL IS RECOMMENDED ON LITTORAL SHELF AREA WHERE PLANTINGS ARE REQUIRED (SEE SOIL PARAMETERS IN BMP DESIGN MANUAL)

2. ALL WETPONDS SHALL HAVE A MINIMUM 20-FOOT ACCESS EASEMENT TO CONNECTING TO DEDICATED PUBLIC RIGHT OF WAY. ACCESS ROAD SHALL HAVE MIN. 12' STABILIZED WIDTH, MAX. LONG. GRADE OF 15%, MAX. CROSS-SLOPE 5%. IN ADDITION, A 10-FOOT WIDE PERMANENT MAINTENANCE ACCESS EASEMENT MUST BE PROVIDED AROUND THE PERIMETER OF ALL BMPS TO ALLOW FOR ADEQUATE MAINTENANCE AND REPAIR.



NOT TO SCALE



POND PROFILE

NOTES:

1. 4-6 INCH LAYER OF AMENDED SOIL IS RECOMMENDED IN ANY AREA WHERE PLANTINGS ARE REQUIRED (SEE SOIL PARAMETERS IN BMP DESIGN MANUAL).

NOT TO SCALE

20' FOOT MAINTENANCE ACCESS
15% MAXIMUM LONGITUDINAL SLOPE

20' MAINTENANCE ACCESS

12' MIN. STABILIZED
MAINTENANCE ROAD BED
5% MAX. CROSS SLOPE

MAINTENANCE ACCESS ROAD
1' TO 2' ABOVE HIGH WATER ELEVATION

▽ HIGH WATER ELEVATION

3:1 STABILIZED SLOPE

10 FOOT LITTORAL SHELF

▽ PERMANENT POOL ELEVATION

2:1

2:1

SECTION AT MAINTENANCE ROAD ACCESS AND FOREBAY

3:1 STABILIZED SLOPE

20' FOOT MAINTENANCE ACCESS

12' MIN. STABILIZED
MAINTENANCE ROAD BED
5% MAX. CROSS SLOPE

6" MINIMUM FREEBOARD ALONG MAINTENANCE ROAD

▽ HIGH WATER ELEVATION

3:1 STABILIZED SLOPE

STORMWATER OUTLET

▽ PERMANENT POOL ELEVATION

2:1

2:1

10' LITTORAL SHELF

SECTION AT STORMWATER OUTFALL

3:1 STABILIZED SLOPE

20' MAINTENANCE ACCESS

12' MIN. STABILIZED
MAINTENANCE ROAD BED
5% MAX. CROSS SLOPE

MAINTENANCE ACCESS ROAD
1' TO 2' ABOVE HIGH WATER ELEVATION

OUTLET STRUCTURE

▽ HIGH WATER ELEVATION

3:1 STABILIZED SLOPE

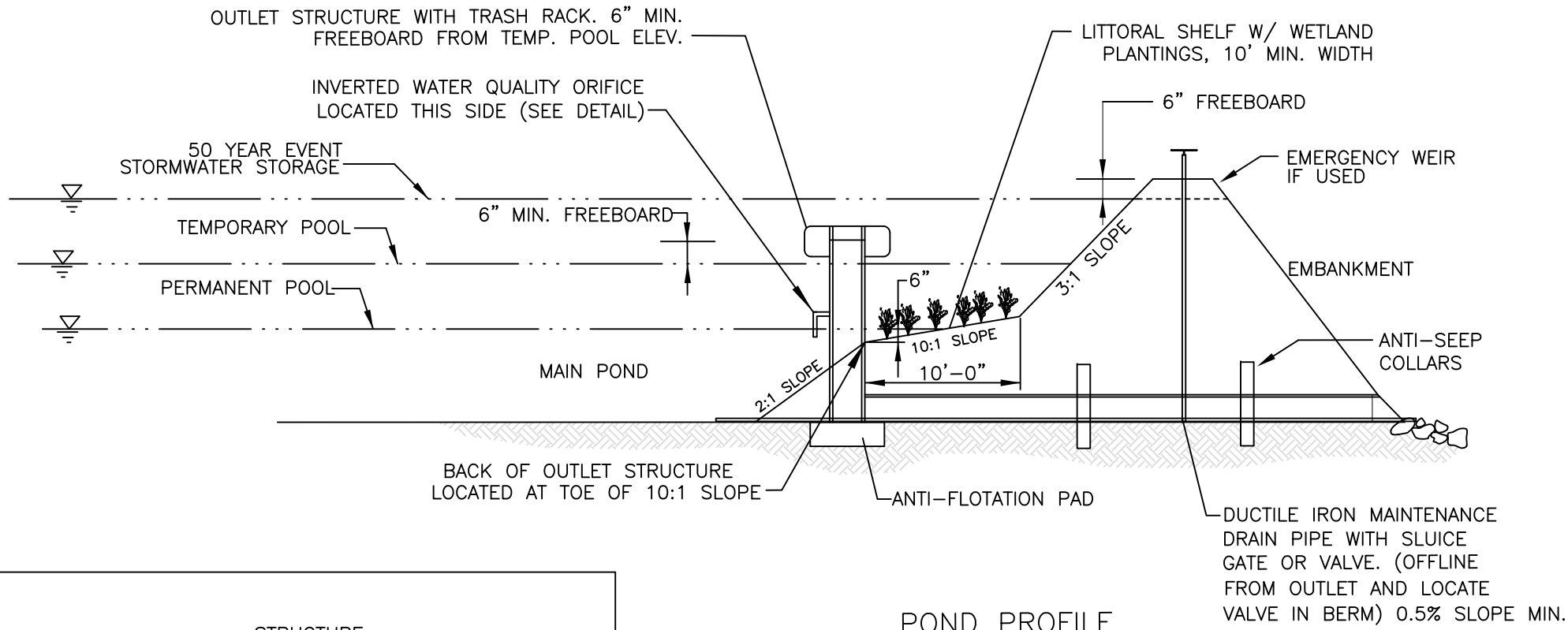
▽ PERMANENT POOL ELEVATION

2:1

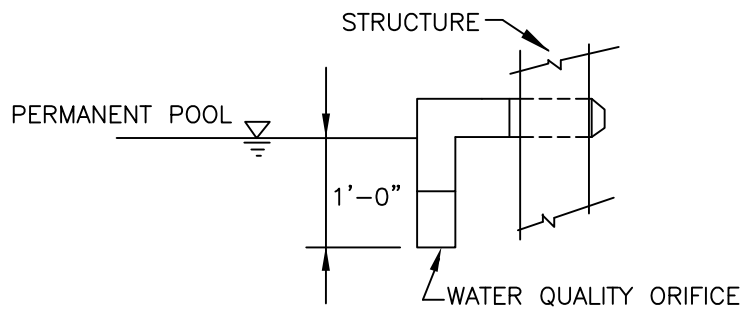
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SECTION AT OUTLET STRUCTURE

NOT TO SCALE



POND PROFILE

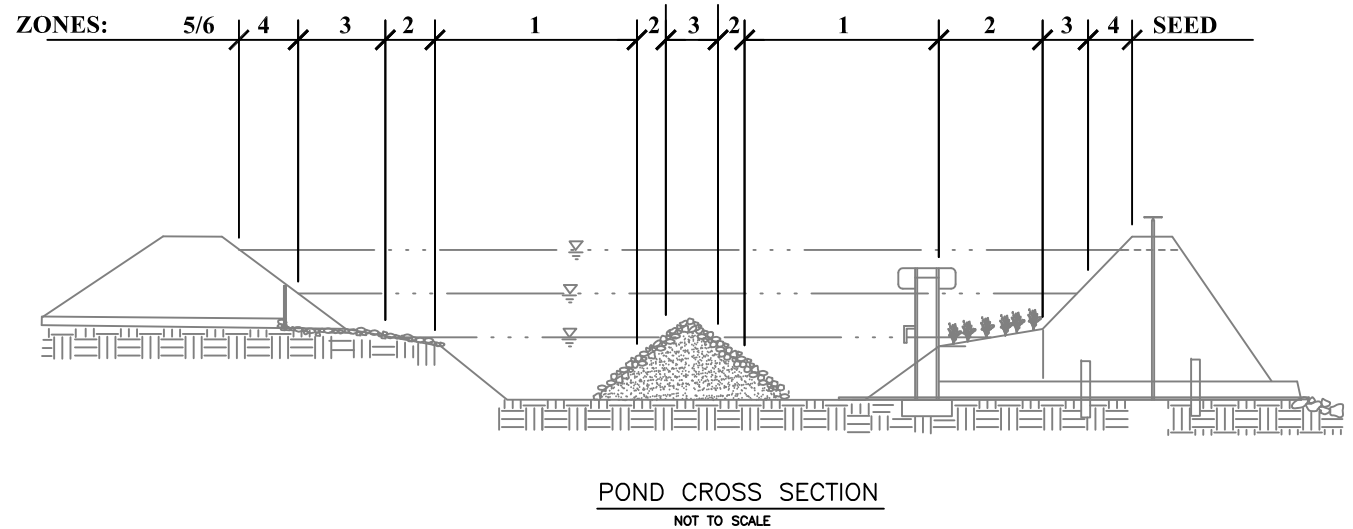
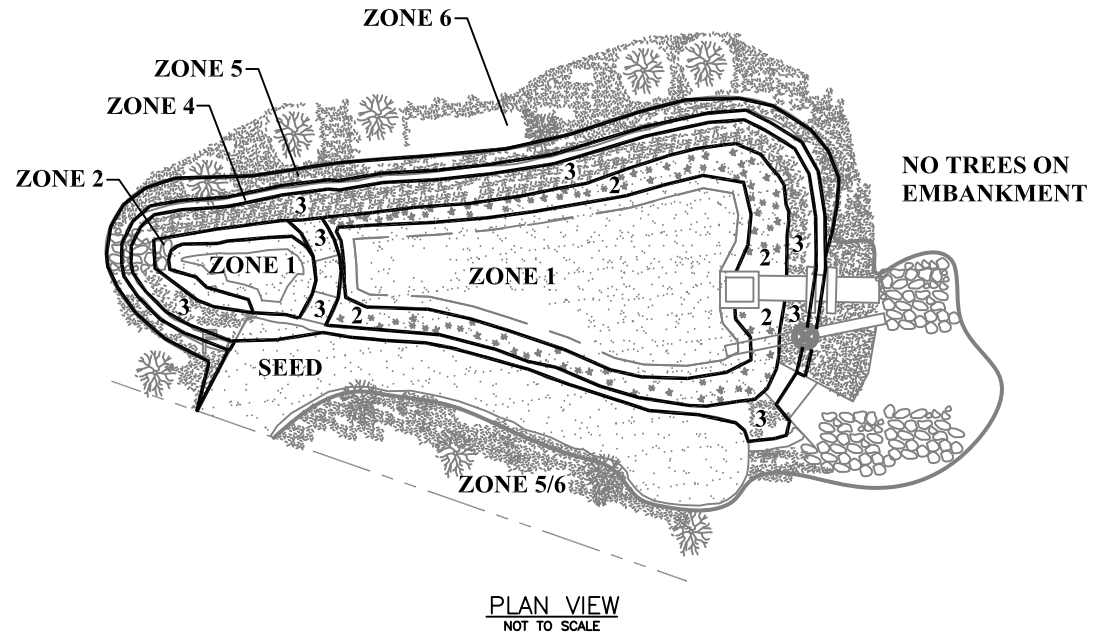


WATER QUALITY ORIFICE DETAIL

NOT TO SCALE

NOTES:

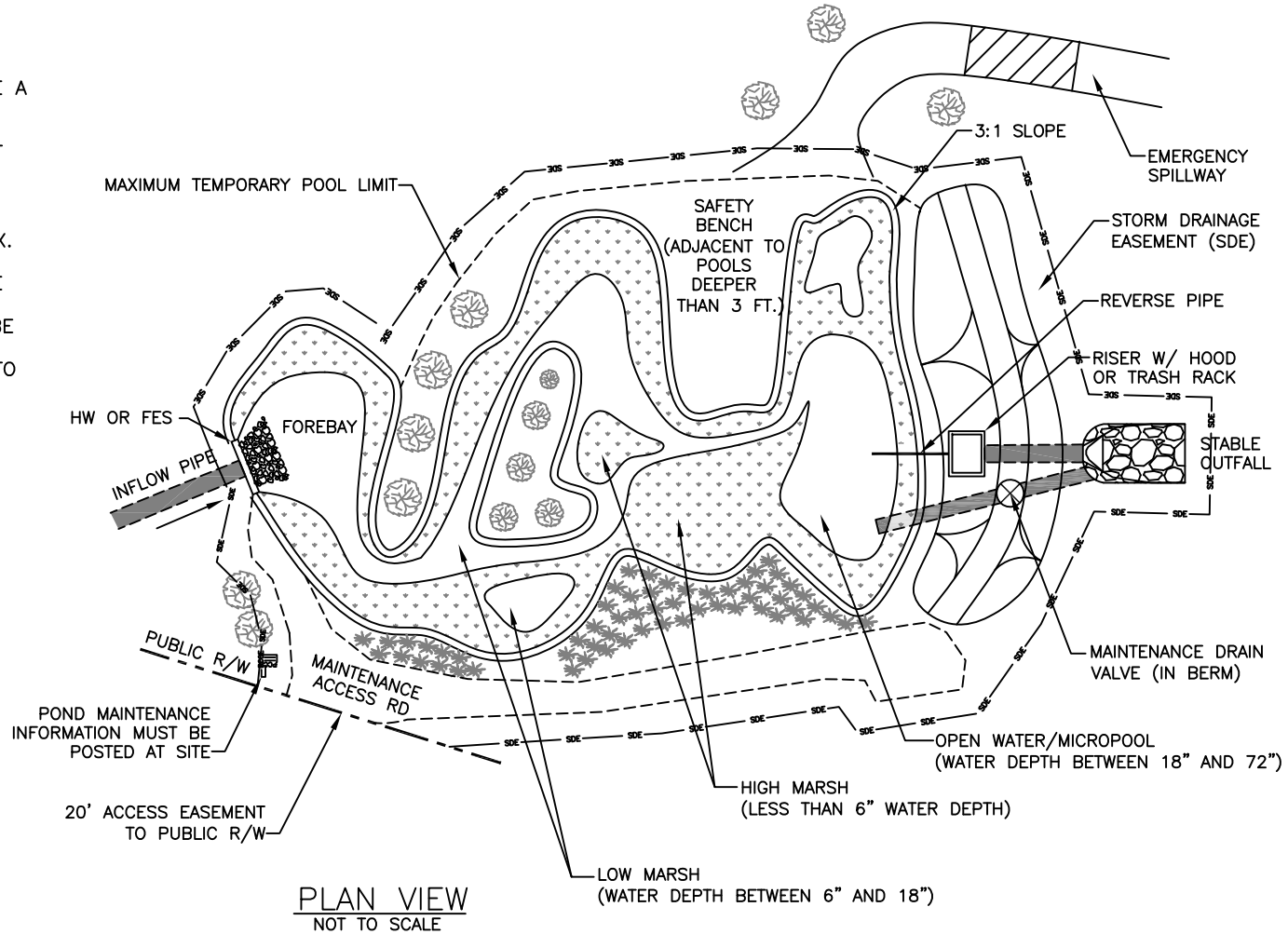
1. PLANTINGS ZONES AND PLANT SELECTION PER THE BMP DESIGN MANUAL, CHAPTER 6 & APPENDICES.
2. ALL PLANTINGS SHALL BE LOCAL NATIVE SPECIES.
3. IRRIGATION MAY BE PROVIDED FOR INITIAL ESTABLISHMENT AND DRY SEASONS.



NOT TO SCALE

NOTES:

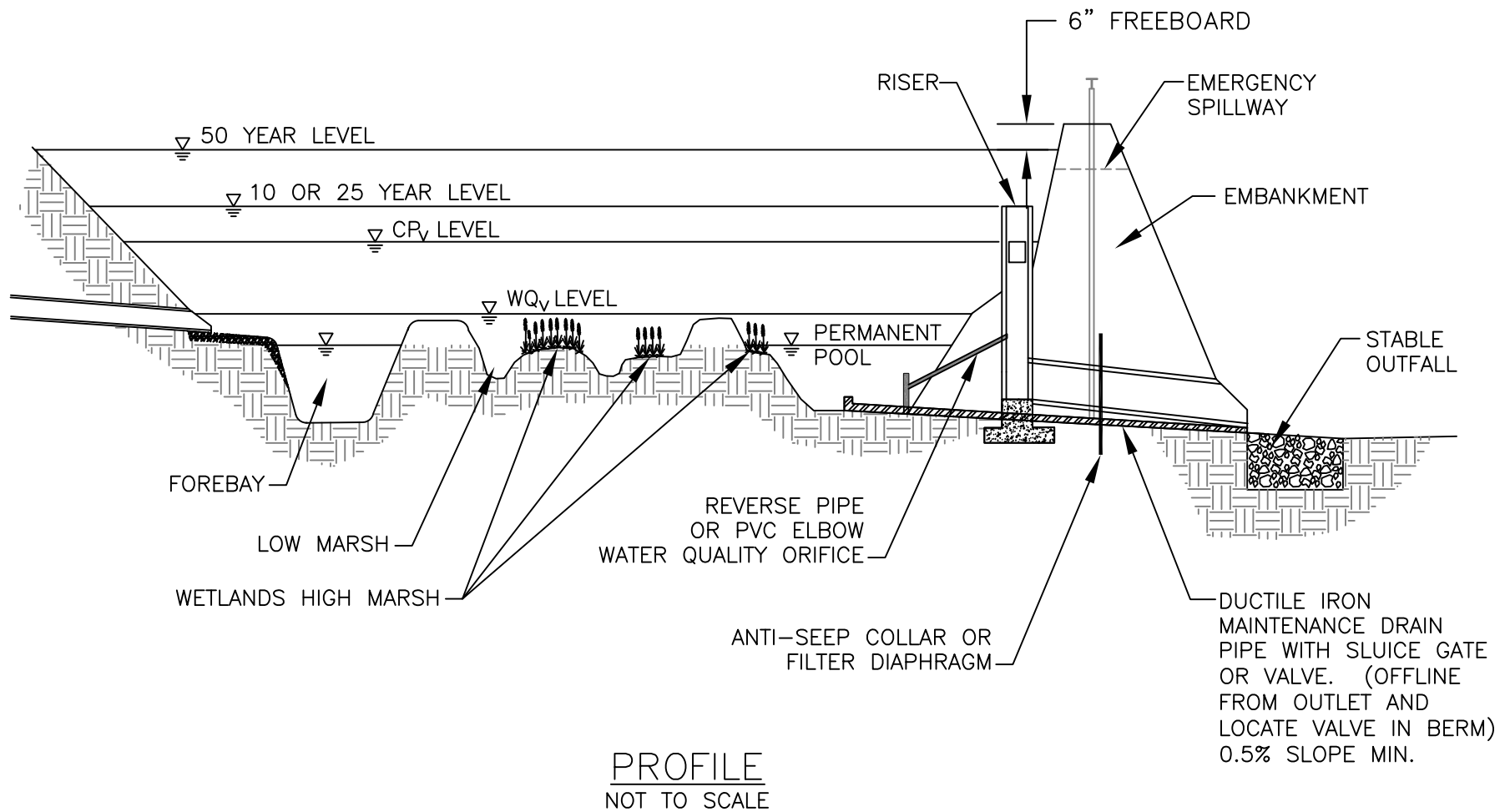
1. 4-6 INCH LAYER OF AMENDED SOIL IS REQUIRED ON ANY MARSH AREA WHERE PLANTINGS ARE REQUIRED (SEE SOIL PARAMETERS IN BMP DESIGN MANUAL)
2. ALL WETLANDS SHALL HAVE A MINIMUM 20-FOOT ACCESS EASEMENT CONNECTING TO A DEDICATED PUBLIC RIGHT OF WAY. ACCESS ROAD SHALL HAVE MIN. 12' STABILIZED WIDTH, MAX. LONG. GRADE OF 15%, MAX. CROSS-SLOPE 5%. IN ADDITION, A 10-FOOT WIDE PERMANENT MAINTENANCE ACCESS EASEMENT MUST BE PROVIDED AROUND THE PERIMETER OF ALL BMPs TO ALLOW FOR ADEQUATE MAINTENANCE AND REPAIR.



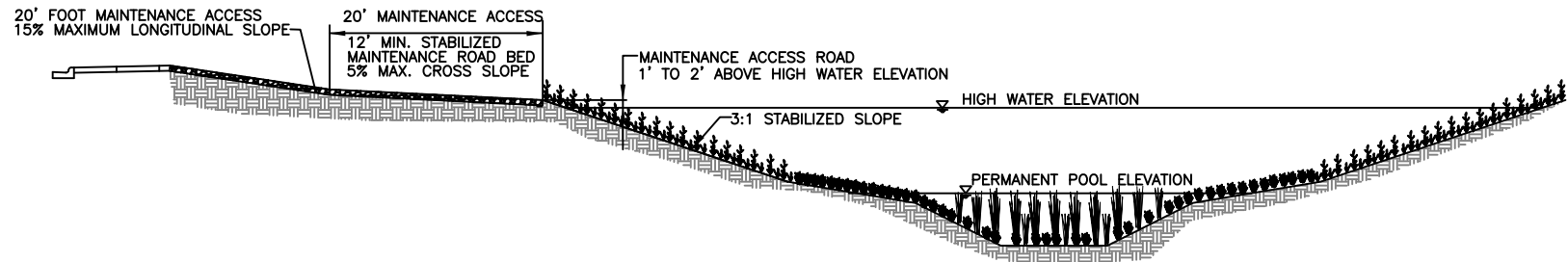
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NOTES

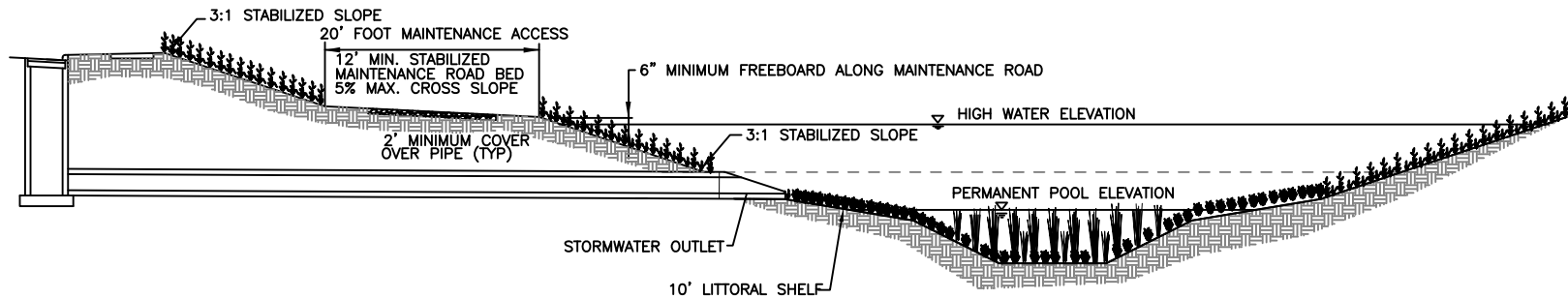
1. 4-6 INCH LAYER OF AMENDED SOIL IS RECOMMENDED IN ANY AREA WHERE PLANTINGS ARE REQUIRED (SEE SOIL PARAMETERS IN BMP DESIGN MANUAL).



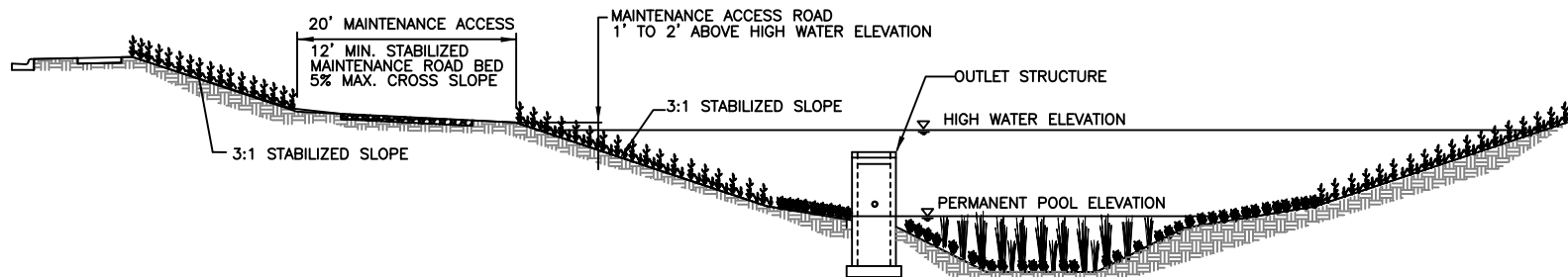
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SECTION AT MAINTENANCE ROAD ACCESS AND FOREBAY

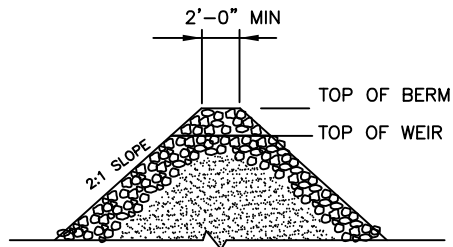


SECTION AT STORMWATER OUTFALL

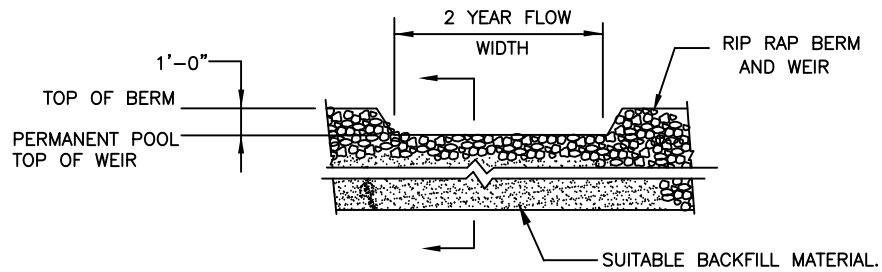


SECTION AT OUTLET STRUCTURE

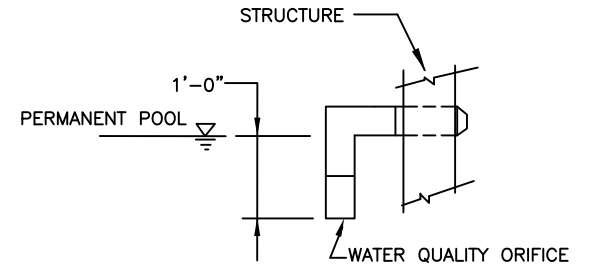
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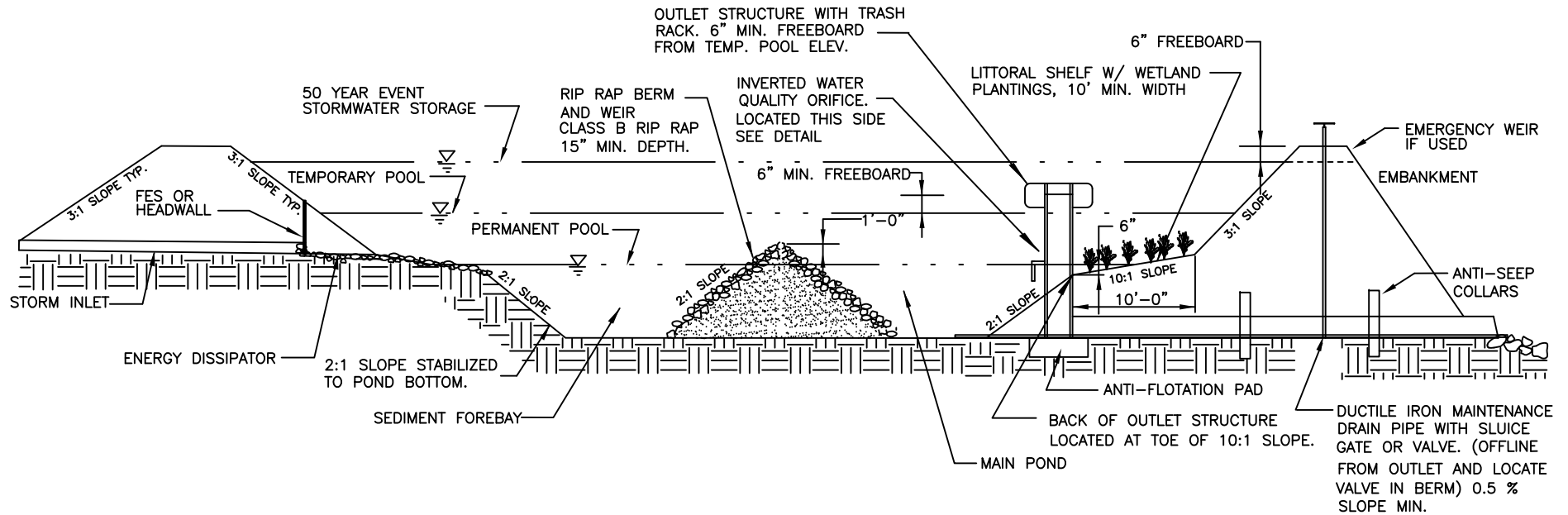
BERM AND WEIR SECTION



BERM AND WEIR DETAIL



WATER QUALITY ORIFICE DETAIL

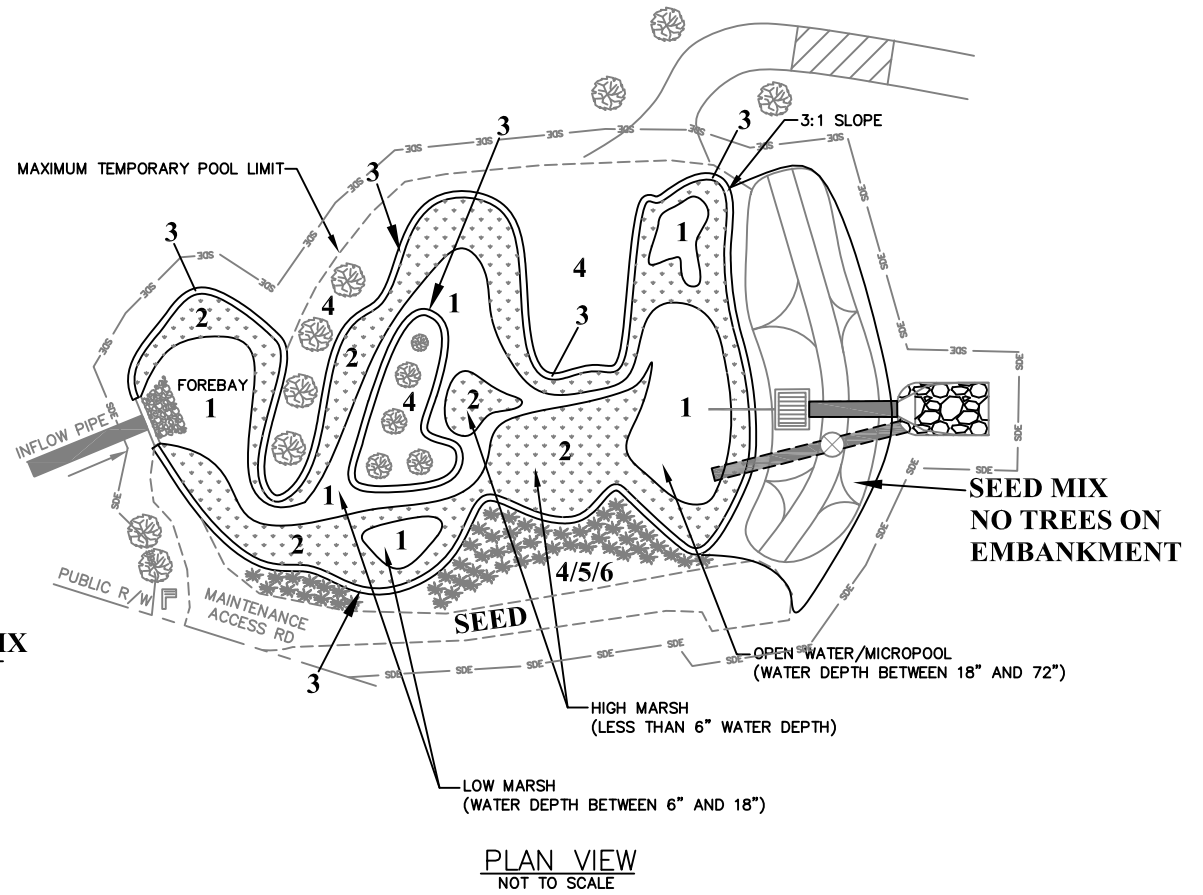
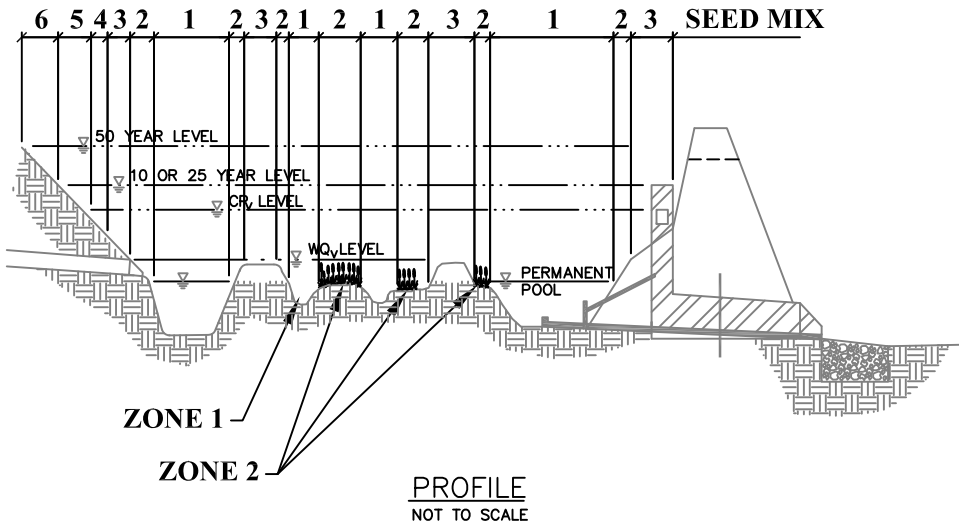


CROSS SECTION
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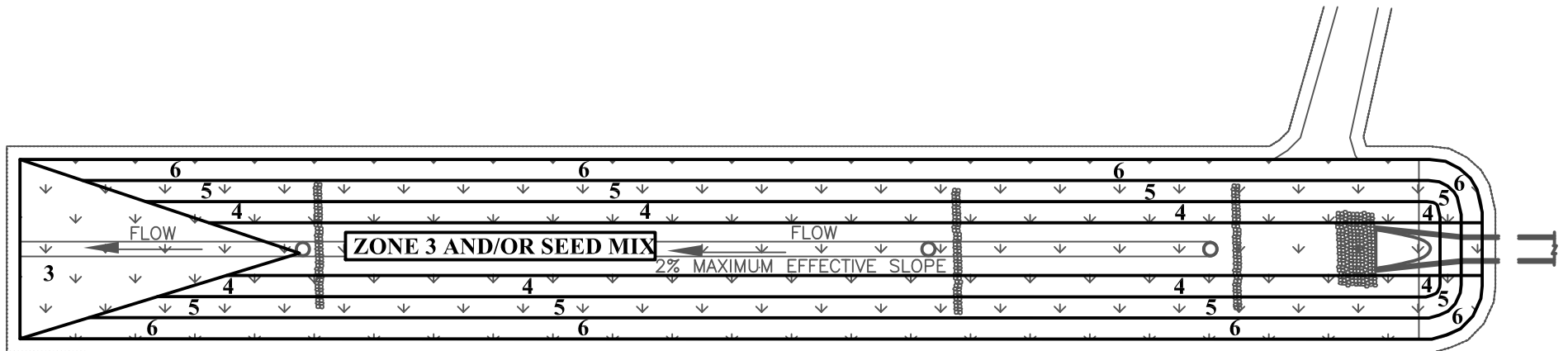
NOT TO SCALE

NOTES

1. PLANTINGS ZONES AND PLANT SELECTION PER THE BMP DESIGN MANUAL, CHAPTER 6 & APPENDICES.
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3. IRRIGATION MAY BE PROVIDED FOR INITIAL ESTABLISHMENT AND DRY SEASONS.



NOT TO SCALE

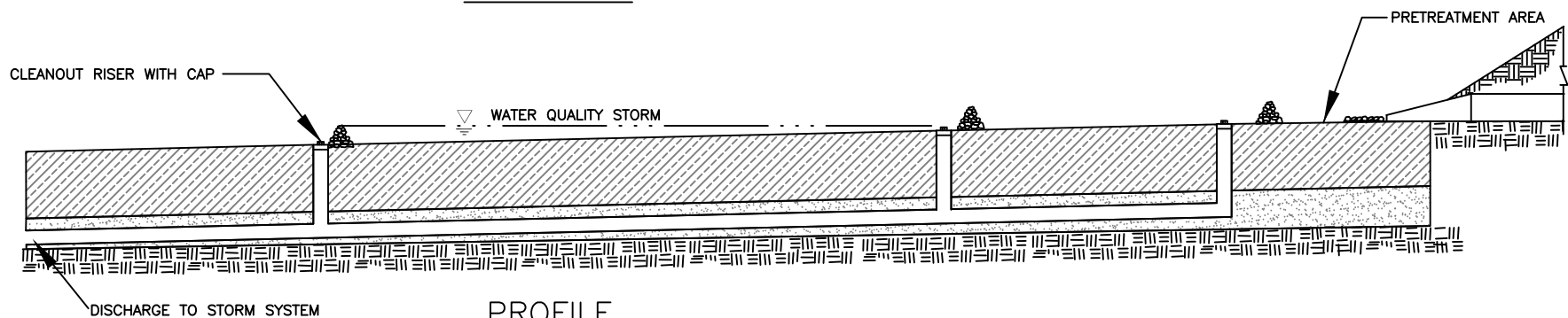
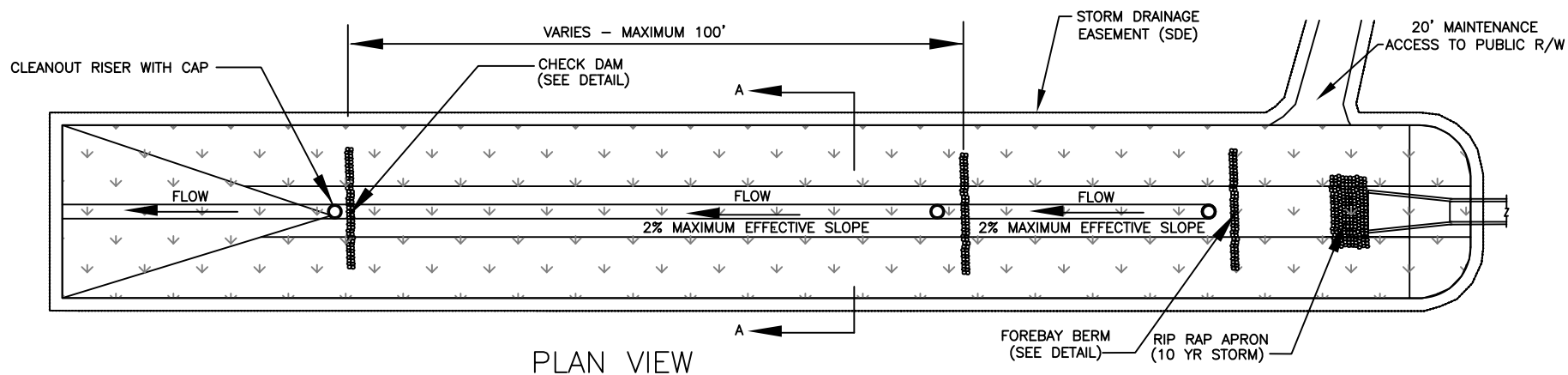


PLAN VIEW

NOTES

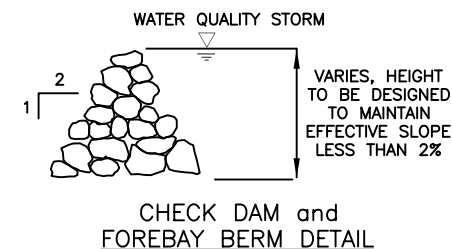
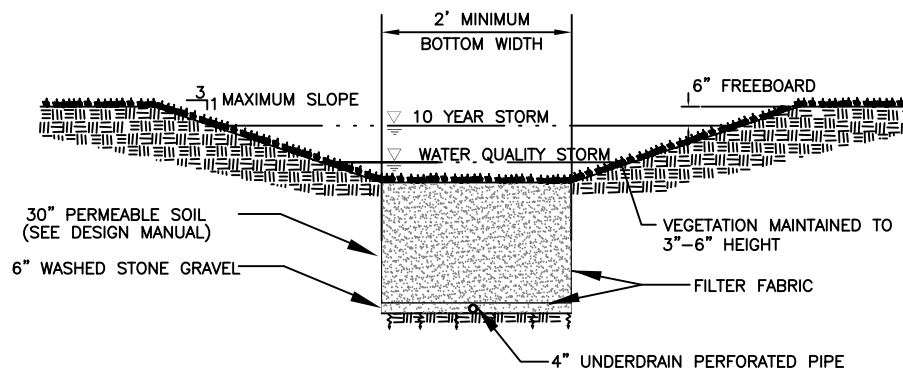
1. PLANTING ZONES AND PLANT SELECTION PER THE BMP DESIGN MANUAL, CHAPTER 6 & APPENDICES.
2. ALL PLANTINGS SHALL BE LOCAL NATIVE SPECIES.
3. IRRIGATION MAY BE PROVIDED FOR INITIAL ESTABLISHMENT AND DRY SEASONS.

NOT TO SCALE

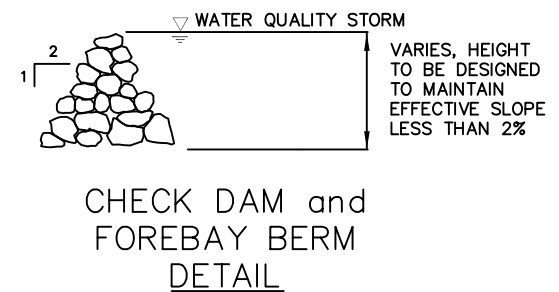
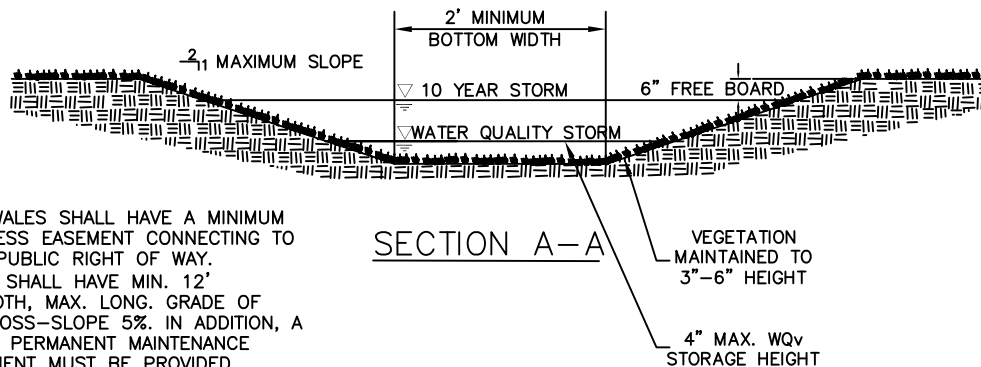
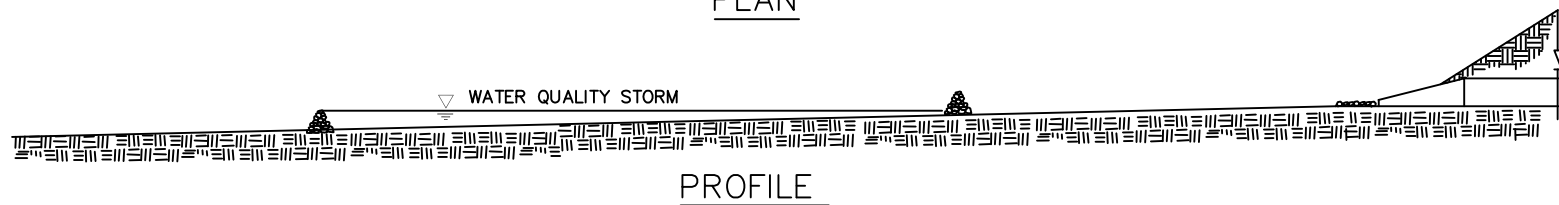
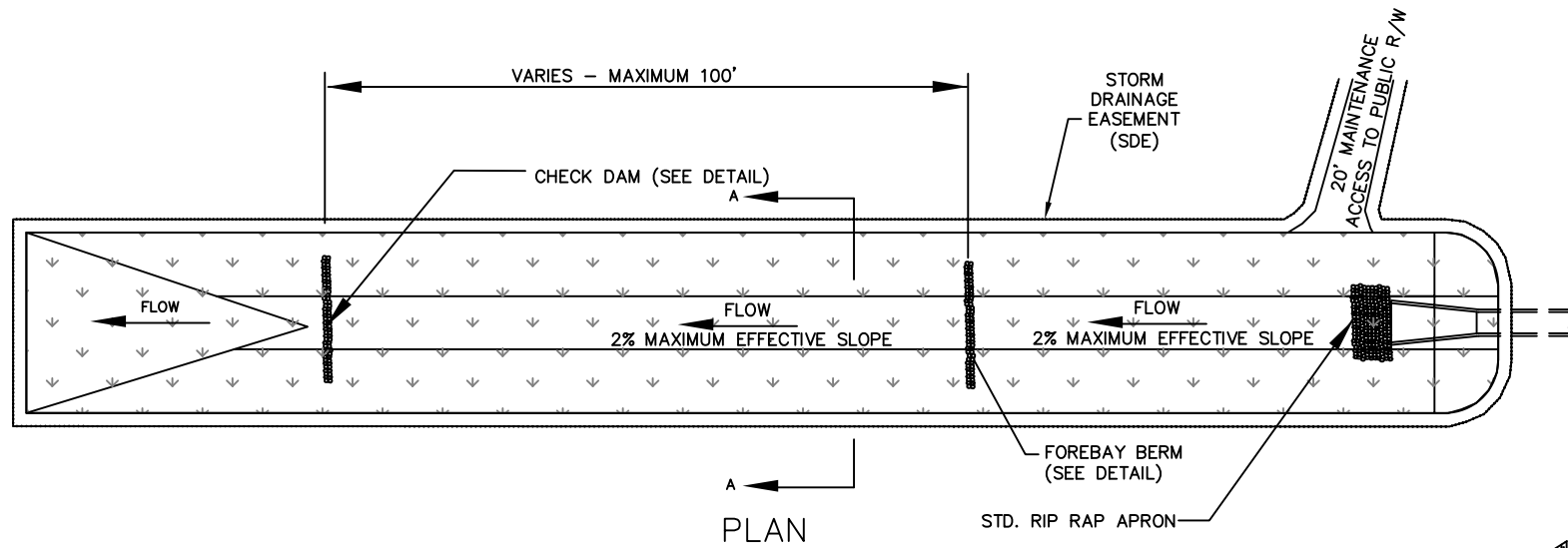


NOTES:

1. ALL ENHANCED GRASS SWALES SHALL HAVE A MINIMUM 20-FOOT ACCESS EASEMENT CONNECTING TO A DEDICATED PUBLIC RIGHT OF WAY. ACCESS ROAD SHALL HAVE MIN. 12' STABILIZED WIDTH, MAX. LONG. GRADE OF 15%, MAX. CROSS-SLOPE 5%. IN ADDITION, A 10-FOOT WIDE PERMANENT MAINTENANCE ACCESS EASEMENT MUST BE PROVIDED AROUND THE PERIMETER OF ALL BMPs TO ALLOW FOR ADEQUATE MAINTENANCE AND REPAIR.



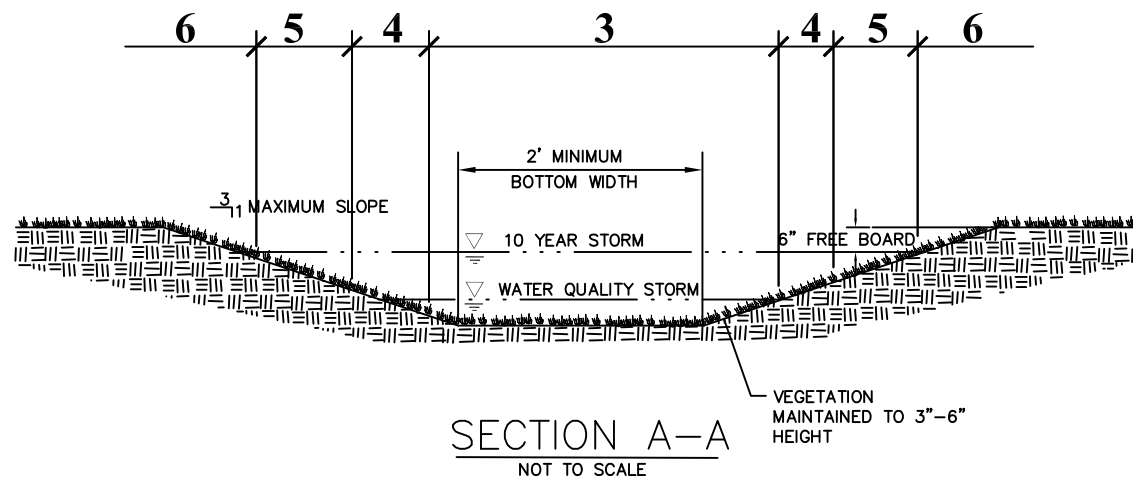
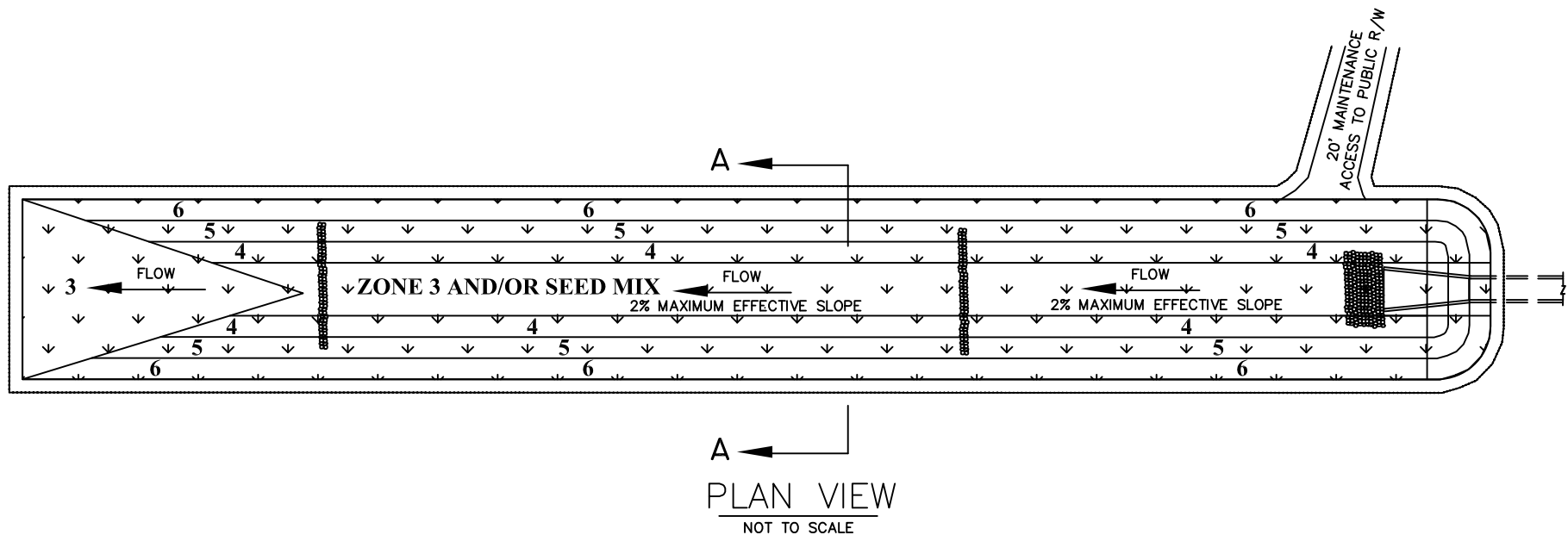
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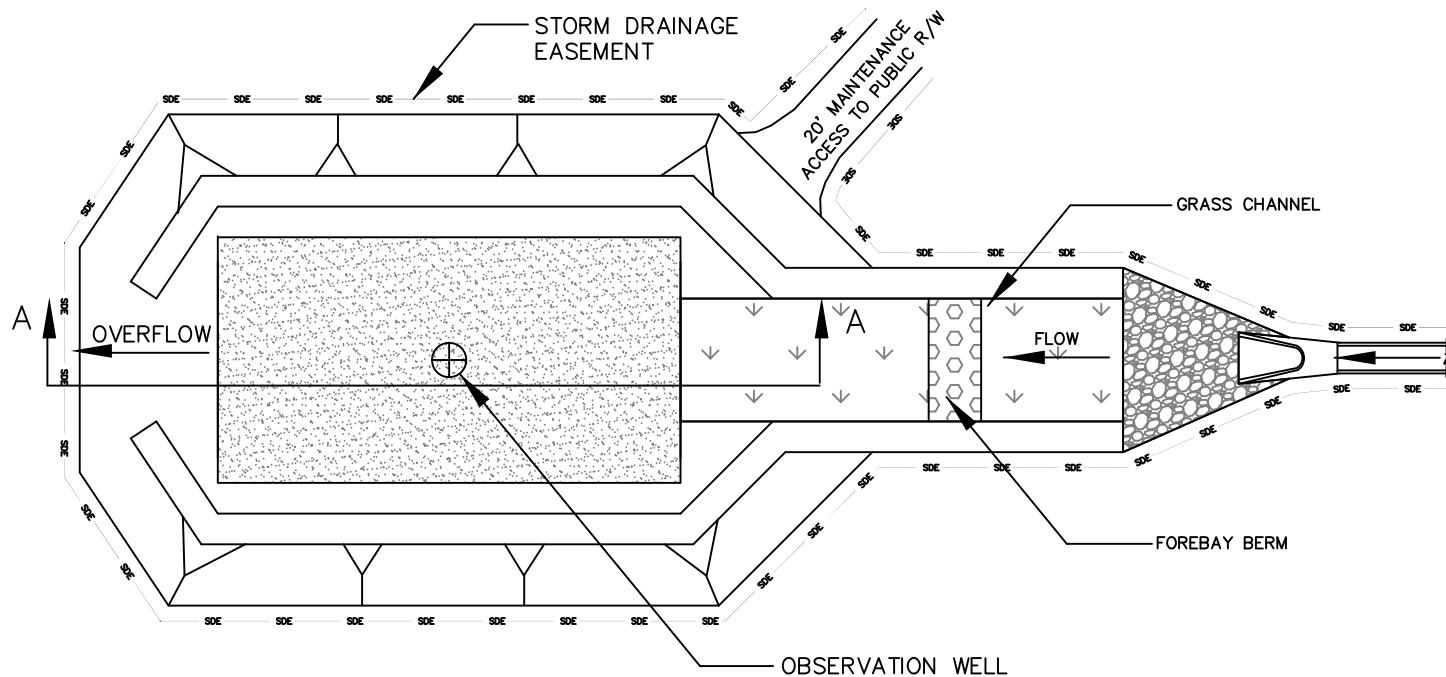
NOTES:

- ALL GRASS SWALES SHALL HAVE A MINIMUM 20-FOOT ACCESS EASEMENT CONNECTING TO A DEDICATED PUBLIC RIGHT OF WAY. ACCESS ROAD SHALL HAVE MIN. 12' STABILIZED WIDTH, MAX. LONG. GRADE OF 15%, MAX. CROSS-SLOPE 5%. IN ADDITION, A 10-FOOT WIDE PERMANENT MAINTENANCE ACCESS EASEMENT MUST BE PROVIDED AROUND THE PERIMETER OF ALL BMPs TO ALLOW FOR ADEQUATE MAINTENANCE AND REPAIR.

NOT TO SCALE



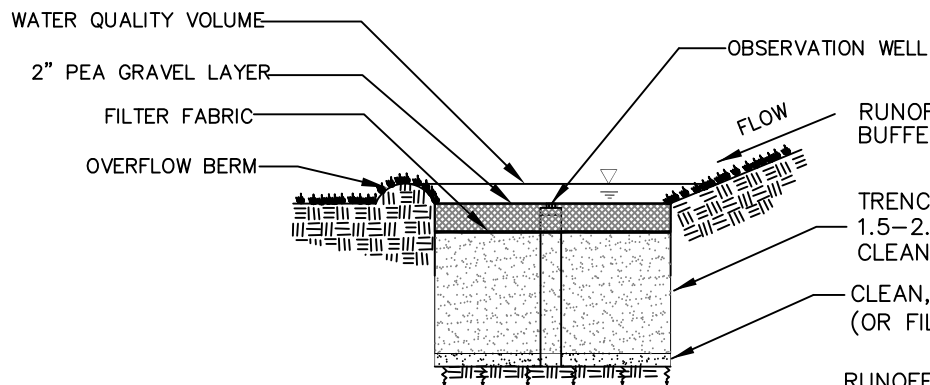
NOT TO SCALE



PLAN
NOT TO SCALE

NOTES:

1. ALL INFILTRATION TRENCHES SHALL HAVE A MINIMUM 20-FOOT ACCESS EASEMENT CONNECTING TO A DEDICATED PUBLIC RIGHT OF WAY. ACCESS ROAD SHALL HAVE MIN. 12' STABILIZED WIDTH, MAX. LONG. GRADE OF 15%, MAX. CROSS-SLOPE 5%. IN ADDITION, A 10-FOOT WIDE PERMANENT MAINTENANCE ACCESS EASEMENT MUST BE PROVIDED AROUND THE PERIMETER OF ALL BMPS TO ALLOW FOR ADEQUATE MAINTENANCE AND REPAIR.
2. 5 ACRE MAXIMUM DRAINAGE AREA.



SECTION A-A
NOT TO SCALE

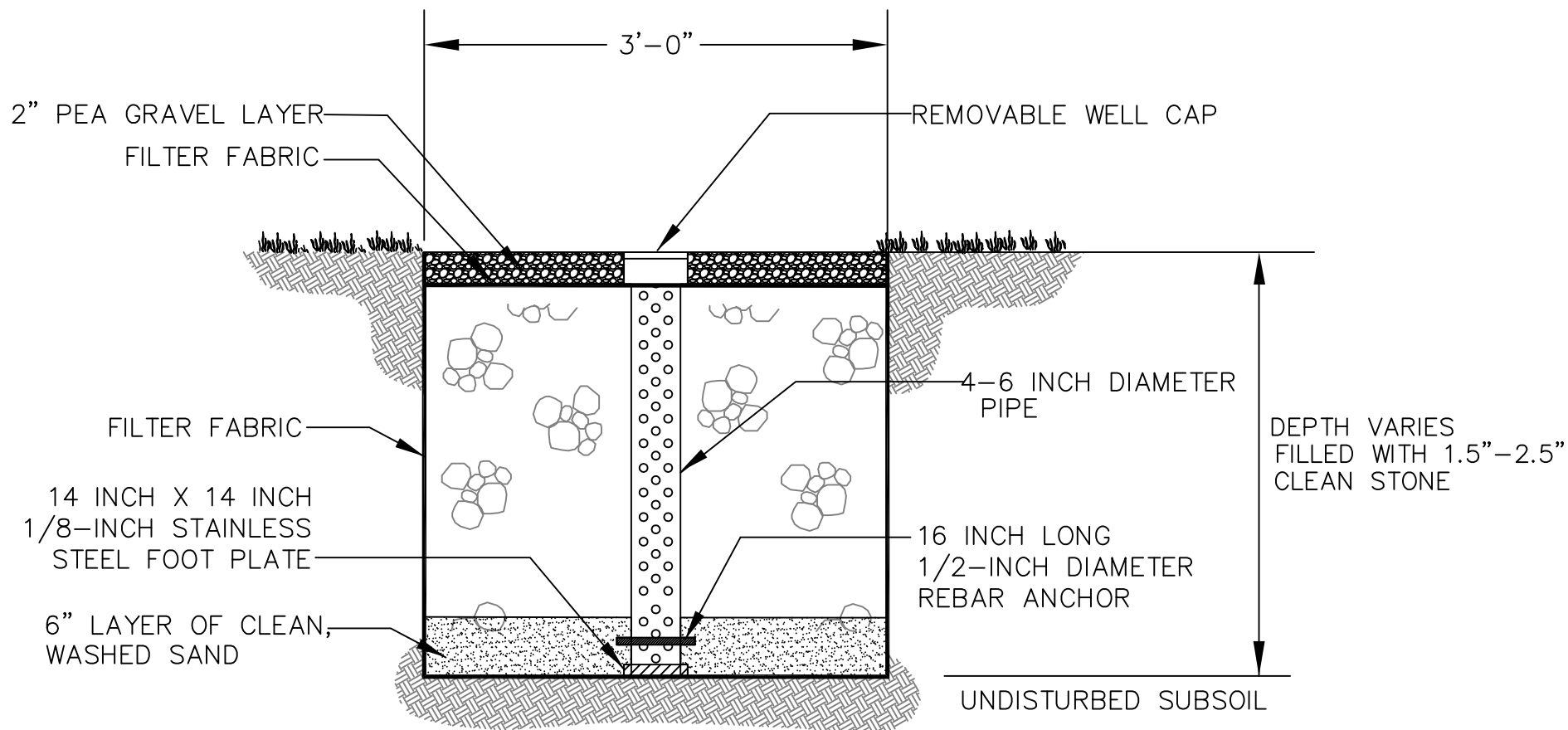
RUNOFF FILTERS THROUGH GRASS BUFFER STRIP (20' MINIMUM)

TRENCH 3-8 FEET DEEP
1.5-2.5 INCH DIAMETER
CLEAN STONE

CLEAN, WASHED SAND, 6" DEEP
(OR FILTER FABRIC EQUIVALENT)

RUNOFF EXFILTRATES THROUGH
UNDISTURBED SUBSOILS WITH 0.5
INCHES PER HOUR MINIMUM RATE.

NOT TO SCALE

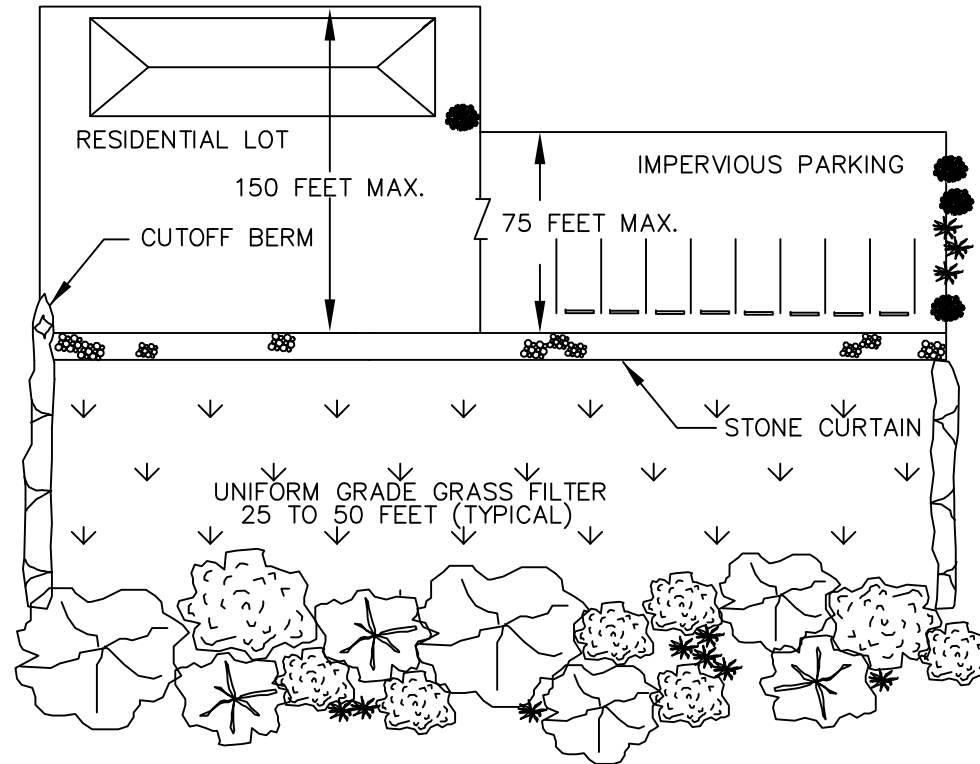


PERFORATION HOLES TO BE 1/2 INCH DIAMETER
AT 3 INCH MINIMUM VERTICAL SPACING

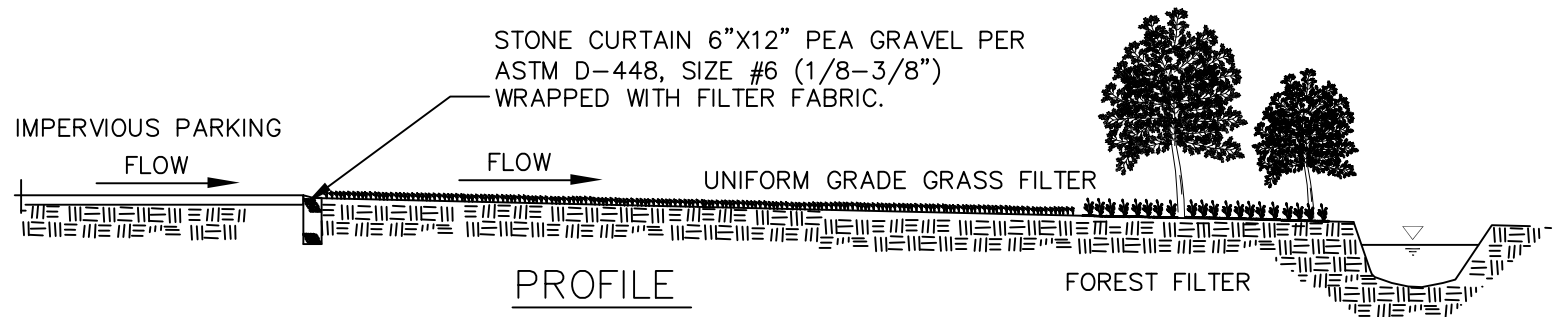
NOT TO SCALE

NOTES:

1. MAXIMUM SLOPE 2% FOR FILTER STRIP AND 5% FOR BUFFER STRIP.
2. 5 ACRE MAXIMUM DRAINAGE AREA.
3. ALL FILTER/BUFFER STRIPS SHALL HAVE A MINIMUM 20 FOOT ACCESS EASEMENT CONNECTING TO A DEDICATED PUBLIC RIGHT OF WAY. ACCESS ROAD SHALL HAVE MIN. 12' STABILIZED WIDTH, MAX. LONG. GRADE OF 15%, MAX. CROSS-SLOPE 5%. IN ADDITION, A 10-FOOT WIDE PERMANENT MAINTENANCE ACCESS EASEMENT MUST BE PROVIDED AROUND THE PERIMETER OF ALL BMPS TO ALLOW FOR ADEQUATE MAINTENANCE AND REPAIR.

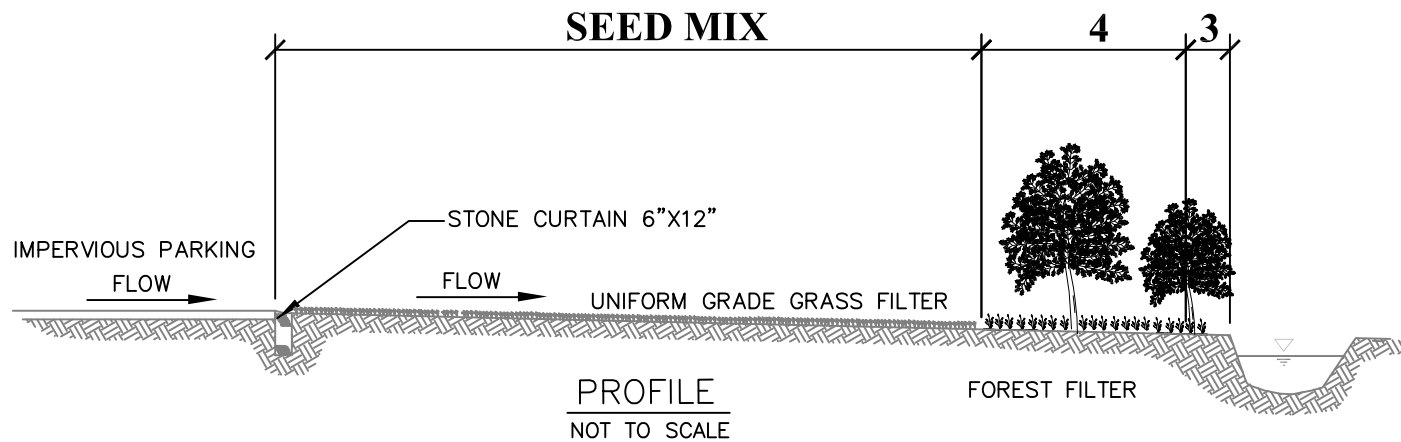
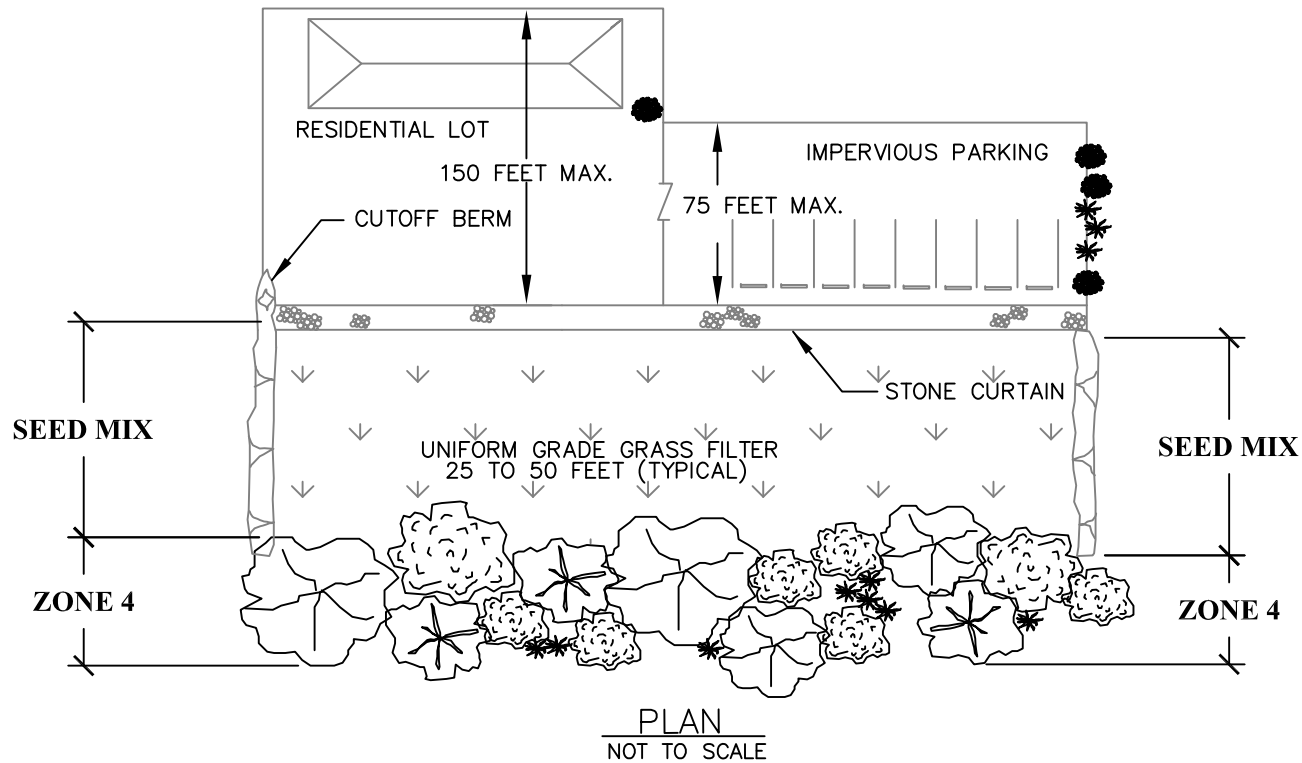


PLAN

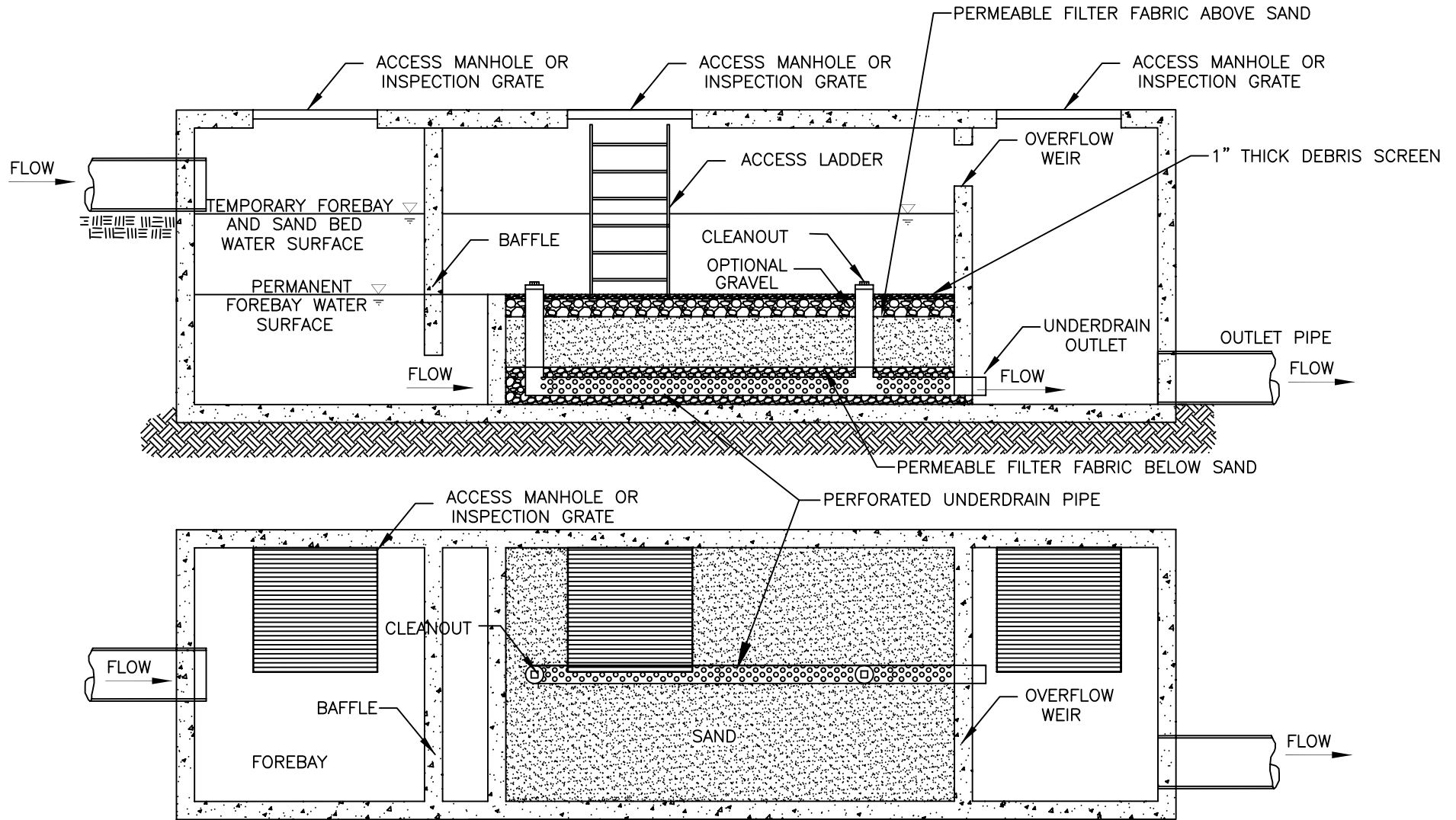


PROFILE

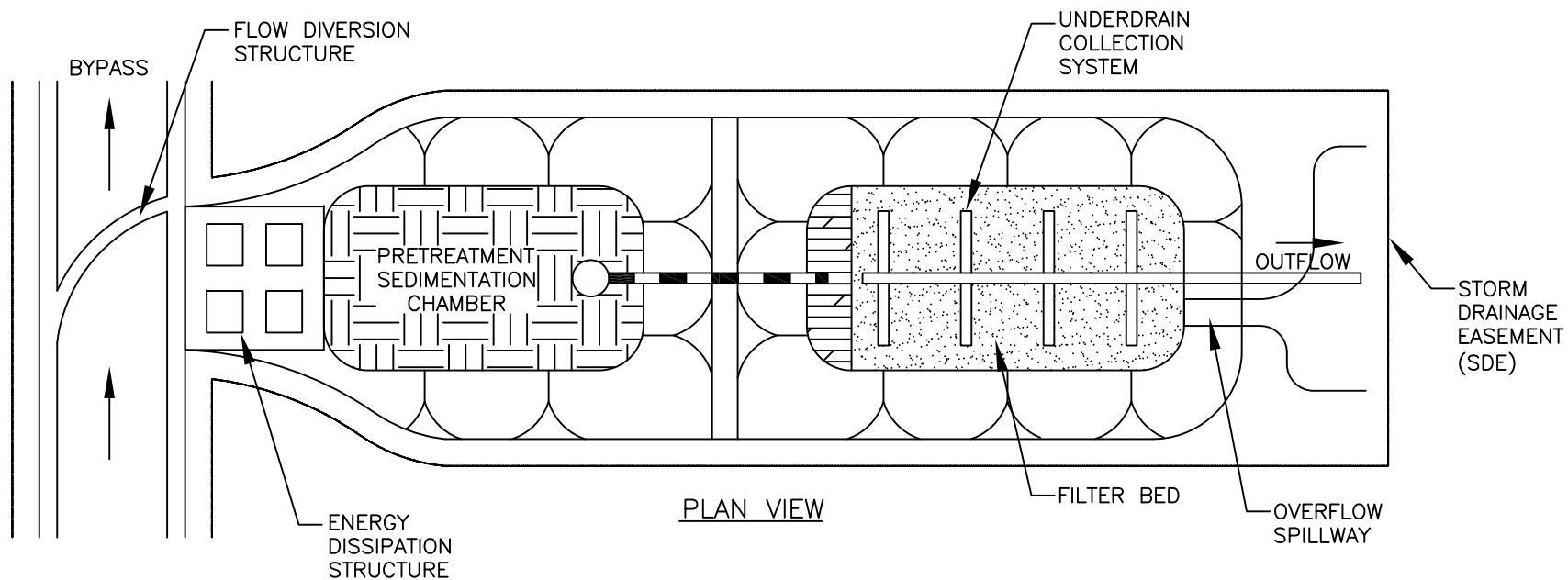
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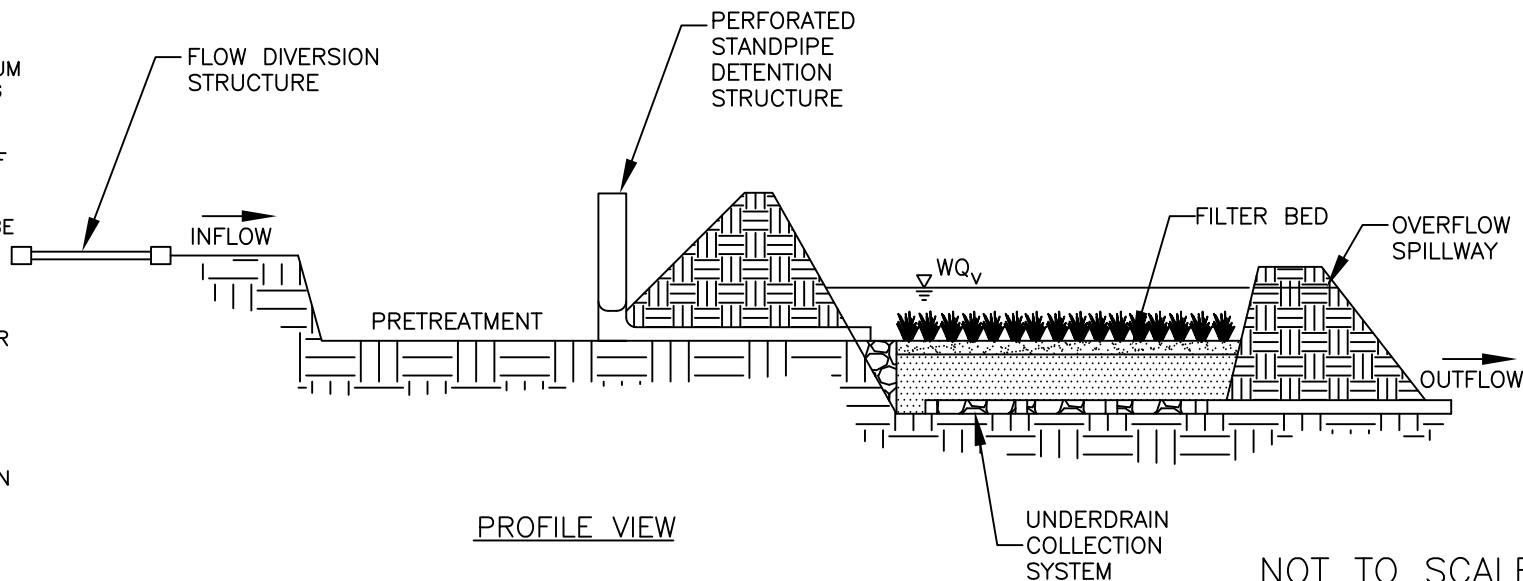
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PLAN VIEW

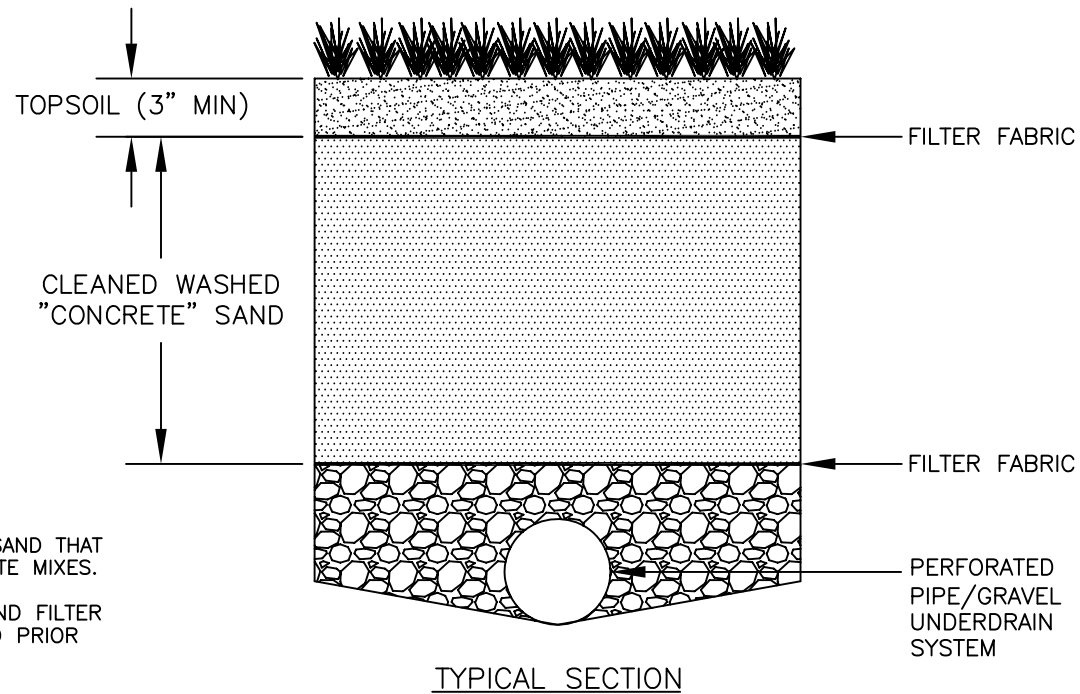
NOTES:

1. ALL SAND FILTERS SHALL HAVE A MINIMUM 20 FOOT ACCESS EASEMENT CONNECTING TO A DEDICATED PUBLIC RIGHT OF WAY. ACCESS ROAD SHALL HAVE MIN. 12' STABILIZED WIDTH, MAX. LONG. GRADE OF 15%, MAX. CROSS-SLOPE 5%. IN ADDITION, A 10-FOOT WIDE PERMANENT MAINTENANCE ACCESS EASEMENT MUST BE PROVIDED AROUND THE PERIMETER OF ALL BMPS TO ALLOW FOR ADEQUATE MAINTENANCE AND REPAIR.
2. ALL DRAINAGE AREAS TO A SAND FILTER FACILITY ARE TO BE STABILIZED PRIOR TO INSTALLATION OF SAND.
3. CLEAN OUTS IN THE UNDERDRAIN SYSTEM ARE TO BE PROVIDED EVERY 100' AND AS DETERMINED BY THE TOWN REVIEWER.



PROFILE VIEW

NOT TO SCALE



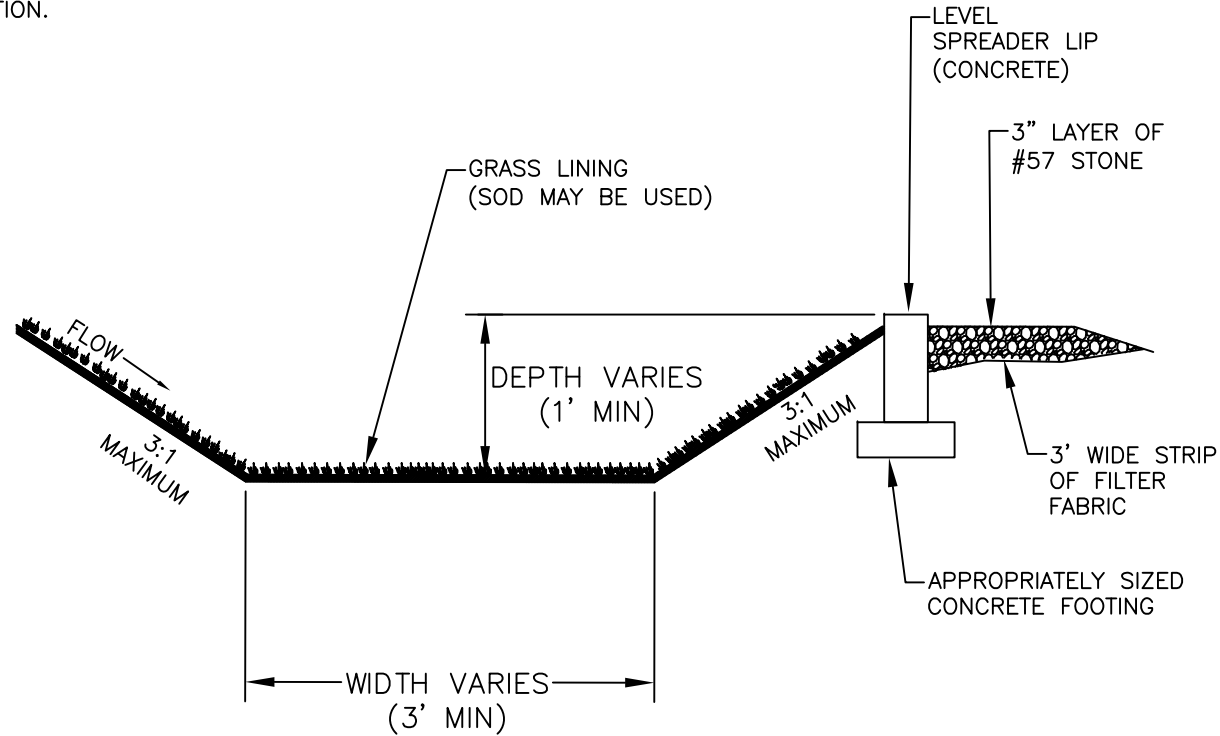
NOTES:

1. "CONCRETE" SAND REFERS TO SAND THAT IS COMMONLY USED IN CONCRETE MIXES.
2. ALL DRAINAGE AREAS TO A SAND FILTER FACILITY ARE TO BE STABILIZED PRIOR TO INSTALLATION OF SAND.

NOT TO SCALE

NOTES:

1. ALL CONCRETE SHALL BE 3600 PSI AT 28 DAYS.
2. ALL DRAINAGE AREAS TO A LEVEL SPREADER FACILITY ARE TO BE STABILIZED PRIOR TO INSTALLATION.



SECTION A-A

NOT TO SCALE

STD. & SPEC. #	TITLE	SPECIAL REQUIREMENTS & NOTES
6.17	ROLLED EROSION CONTROL PRODUCTS	—
6.51	HARDWARE CLOTH & GRAVEL INLET PROTECTION	—
6.60	TEMPORARY SEDIMENT TRAP	WEIR TOP WIDTH 10' MIN., BOTTOM 7' MIN.
6.61	SEDIMENT BASIN	FLASH BOARD RISER NOT PERMITTED
6.64	SKIMMER SEDIMENT BASIN	1ST BAFFLE: RIP RAP & WASHED STONE BERM 2ND BAFFLE: STANDARD BAFFLE 3RD BAFFLE: HARDWARE CLOTH SURROUNDING THE SKIMMER

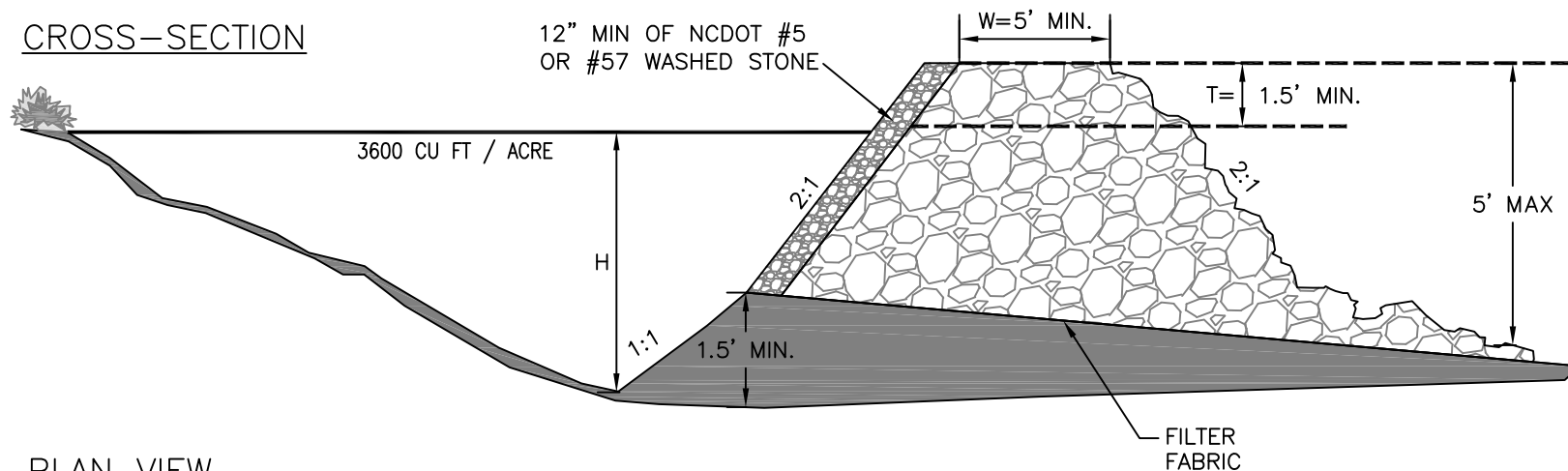
THE STANDARDS & SPECIFICATIONS SHOWN ARE FROM THE "NORTH CAROLINA EROSION AND SEDIMENT CONTROL PLANNING AND DESIGN MANUAL" (NCESCPDM) PREPARED BY NC DEPT. OF ENVIRONMENT AND NATURAL RESOURCES (NCDENR).

NOT TO SCALE

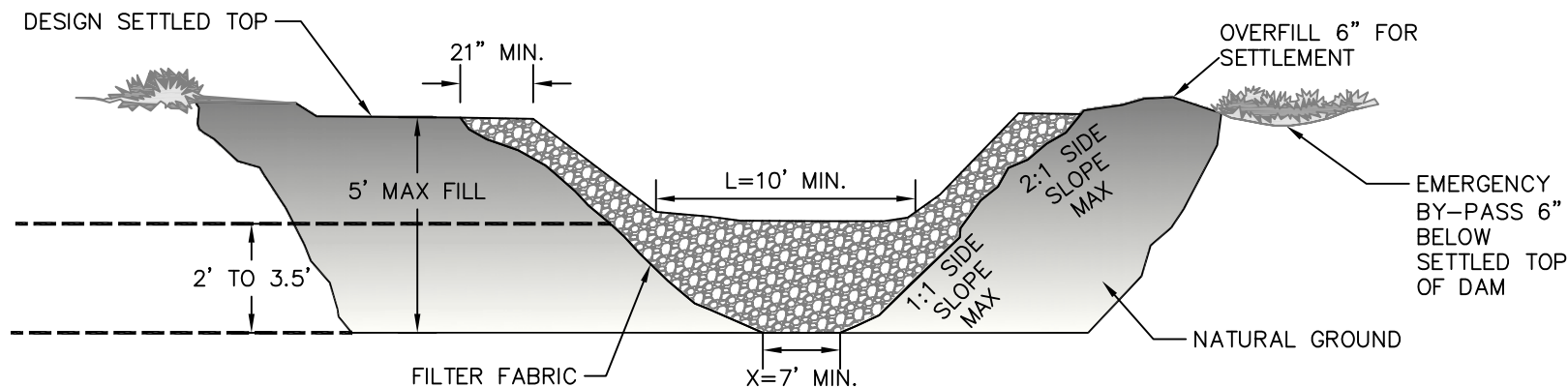
TEMPORARY SEDIMENT TRAP DESIGN CRITERIA	
DRAINAGE AREA (ACRES)	< 5 AC.
MIN. LENGTH TO WIDTH RATIO	2:1
MIN. VOLUME REQUIRED	3600 (CU. FT. PER AC. DISTURBED)
SURFACE AREA REQUIRED	435 (SQ. FT. PER CFS Q10)

NOTE:
PLEASE REFER TO
NCESCPDM SECTION #6.60
FOR ADDITIONAL DESIGN
SPECIFICATIONS REGARDING
TEMPORARY SEDIMENT
TRAPS.

CROSS-SECTION



PLAN VIEW



DATA BLOCK

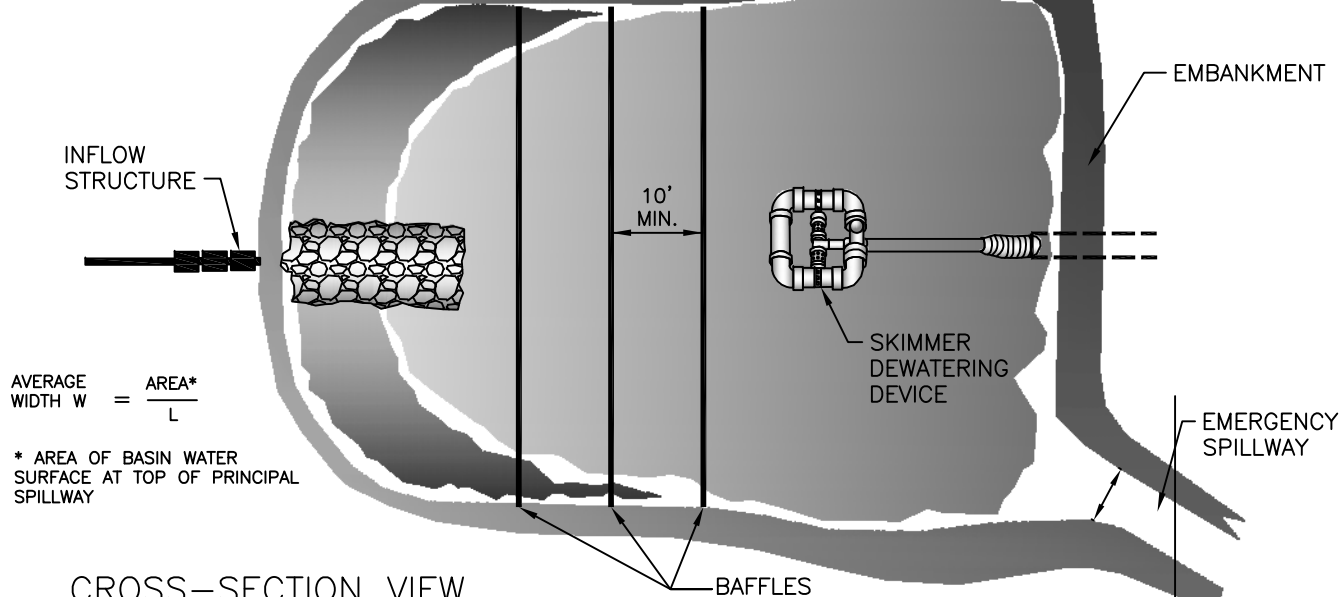
TRAP NO.	DRAINAGE AREA (ACRES)	DENUDED AREA (ACRES)	Q ₁₀	TRAP VOLUME		TRAP SURFACE AREA		CLEANOUT DEPTH H/2 (FT.)	H (FEET)	L (FEET)	T (FEET)	W (FEET)	X (FEET)
				REQUIRED (C.F.)	PROVIDED (C.F.)	REQUIRED (C.F.)	PROVIDED (C.F.)						

NOT TO SCALE

SKIMMER SEDIMENT BASIN DESIGN CRITERIA

DRAINAGE AREA (ACRES)	< 10 AC.
MIN. LENGTH TO WIDTH RATIO	2:1
MAX. LENGTH TO WIDTH RATIO	6:1
MIN. VOLUME REQUIRED	1800 (CU. FT. PER AC. DISTURBED)
SURFACE AREA REQUIRED	325 (SQ. FT. PER CFS Q10)

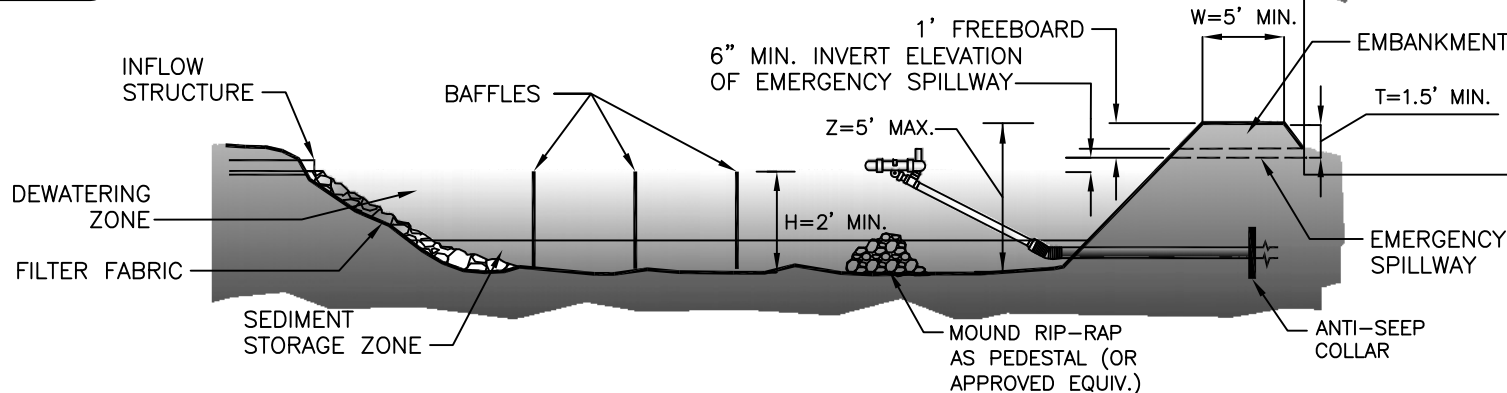
PLAN VIEW



NOTES:

1. REFER TO NCESCPDM SECTION #6.64 FOR ADDITIONAL DESIGN SPECIFICATIONS REGARDING SKIMMER SEDIMENT BASINS.
2. REFER TO STD. 524.1 FOR BAFFLE SPACING AND INSTALLATION.

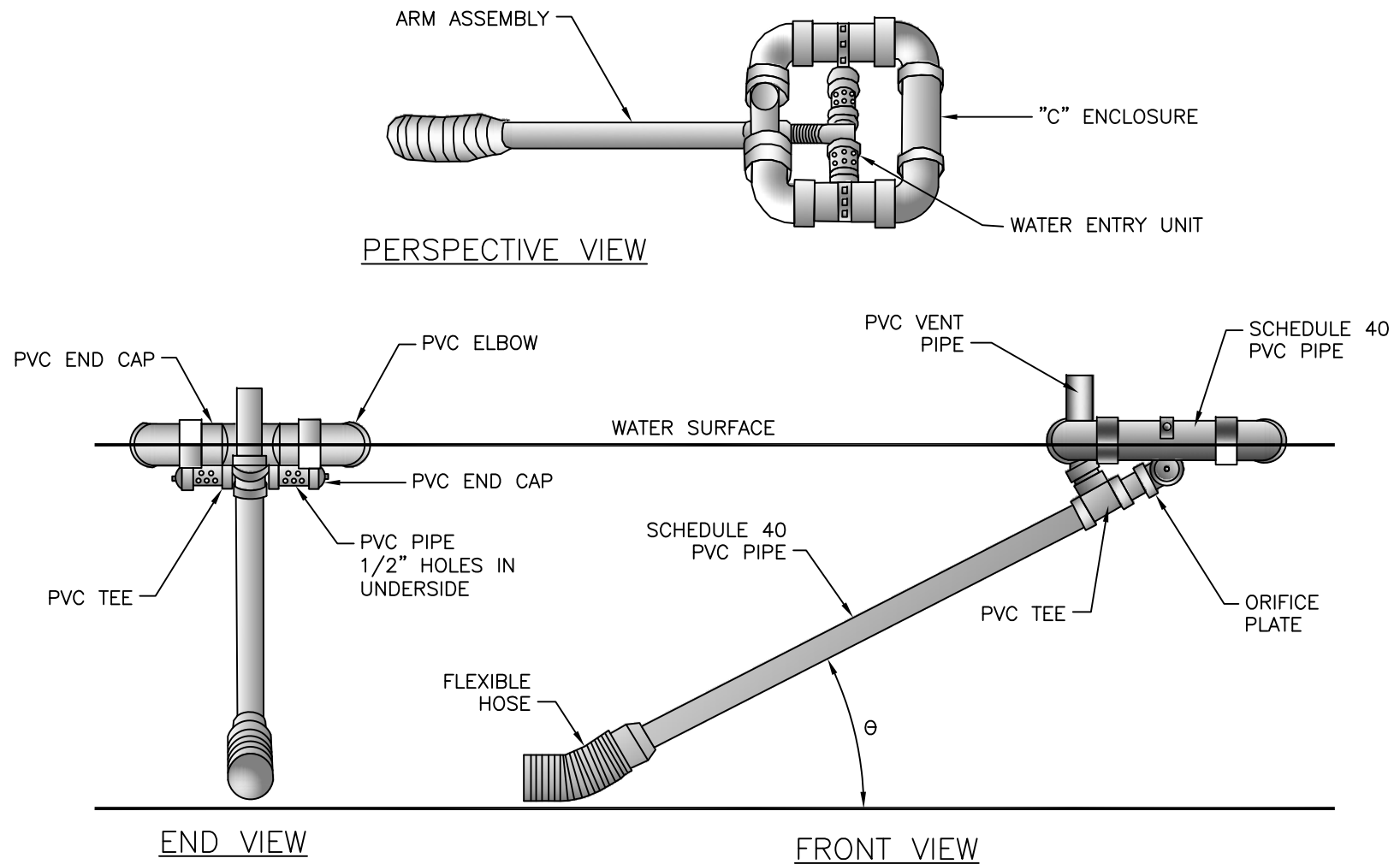
CROSS-SECTION VIEW



DATA BLOCK

BASIN NO.	DRAINAGE AREA (ACRES)	DENUDED AREA (ACRES)	Q ₁₀	TRAP VOLUME		TRAP SURFACE AREA		CLEANOUT DEPTH H/2 (FT.)	H (FEET)	L (FEET)	T (FEET)	W (FEET)	Z (FEET)	SKIMMER PIPE DIAMETER	SKIMMER ORIFICE DIAMETER
				REQUIRED (C.F.)	PROVIDED (C.F.)	REQUIRED (C.F.)	PROVIDED (C.F.)								

NOT TO SCALE

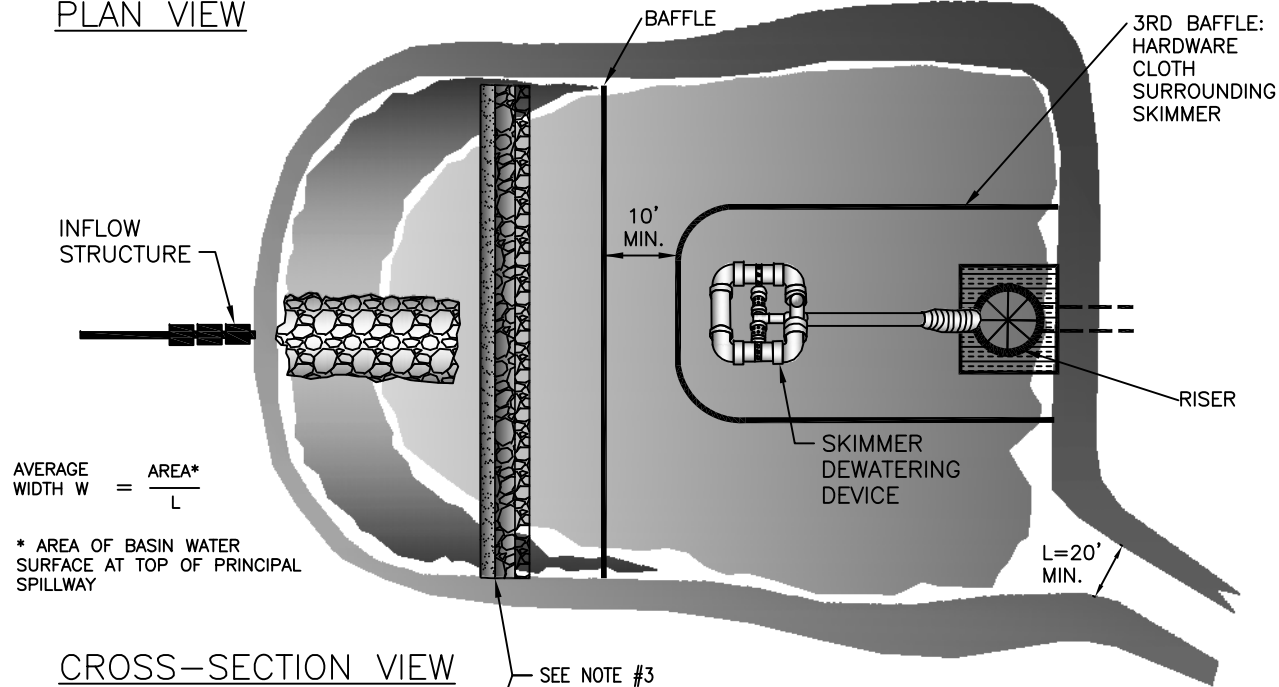


SCHEMATIC OF SKIMMER TAKEN FROM PENNSYLVANIA EROSION AND SEDIMENT POLLUTION CONTROL MANUAL, MARCH 2000.

NOT TO SCALE

SEDIMENT BASIN DESIGN CRITERIA	
DRAINAGE AREA (ACRES)	>10 AC. <100 AC.
MIN. LENGTH TO WIDTH RATIO	2:1
MAX. LENGTH TO WIDTH RATIO	6:1
MIN. VOLUME REQUIRED	1800 (CU. FT. PER AC. DISTURBED)
SURFACE AREA REQUIRED	435 (SQ. FT. PER CFS Q10)

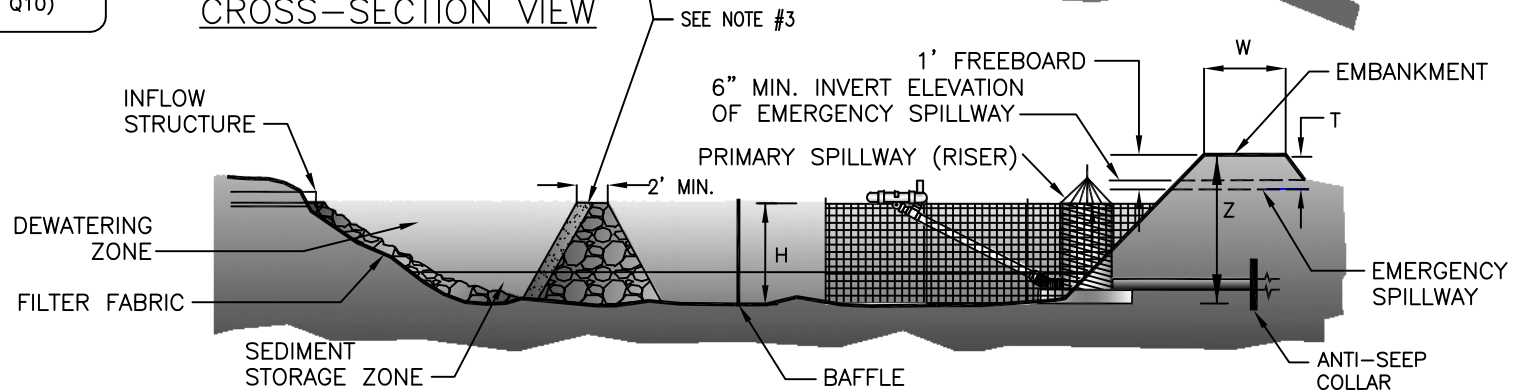
PLAN VIEW



NOTES:

1. REFER TO NCESCPDM SECTION #6.61 FOR ADDITIONAL DESIGN SPECIFICATIONS REGARDING SEDIMENT BASINS.
2. REFER TO STD. 524.1 FOR BAFFLE SPACING AND INSTALLATION.
3. FIRST BAFFLE IS TO BE CONSTRUCTED OF RIP-RAP AND #5 WASHED STONE, WITH A MIN. HEIGHT OF 3' AND MIN. TOPWIDTH OF 2'.

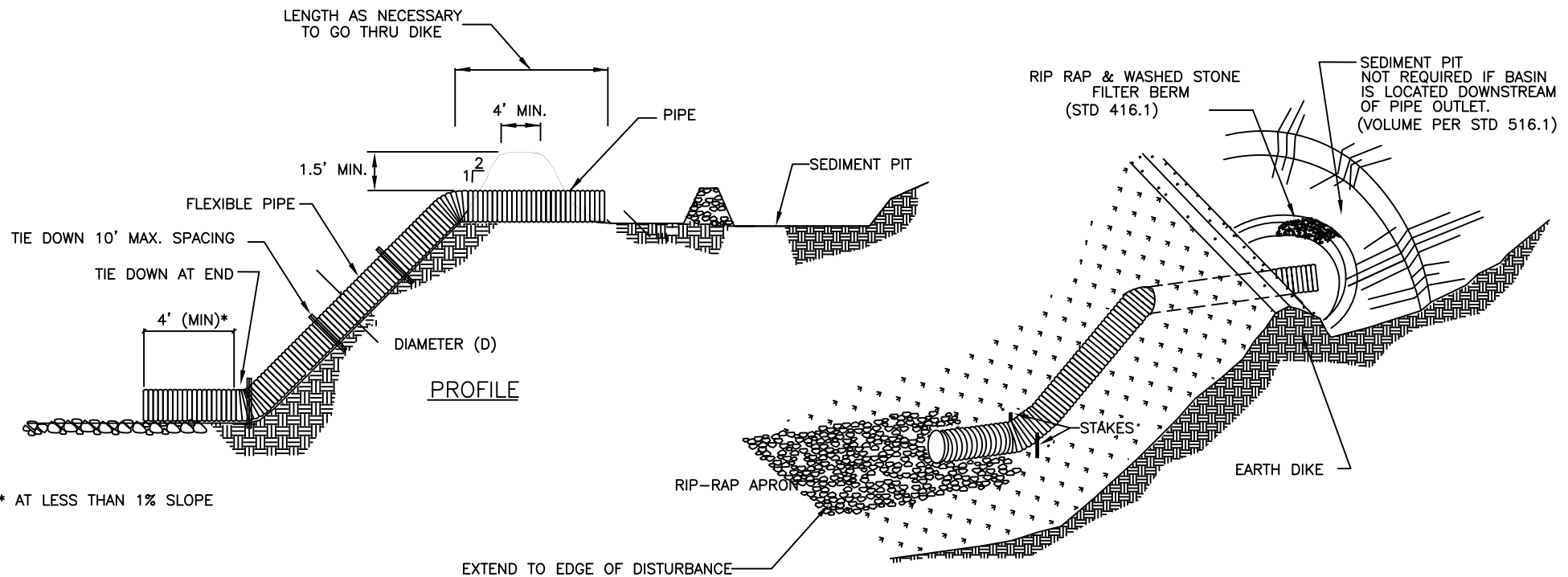
CROSS-SECTION VIEW



DATA BLOCK

BASIN NO.	DRAINAGE AREA (ACRES)	DENUDED AREA (ACRES)	Q ₁₀	TRAP VOLUME		TRAP SURFACE AREA		CLEANOUT DEPTH H/2 (FT.)	H (FEET)	L (FEET)	T (FEET)	W (FEET)	Z (FEET)	SKIMMER PIPE DIAMETER	SKIMMER ORIFICE DIAMETER
				REQUIRED (C.F.)	PROVIDED (C.F.)	REQUIRED (C.F.)	PROVIDED (C.F.)								

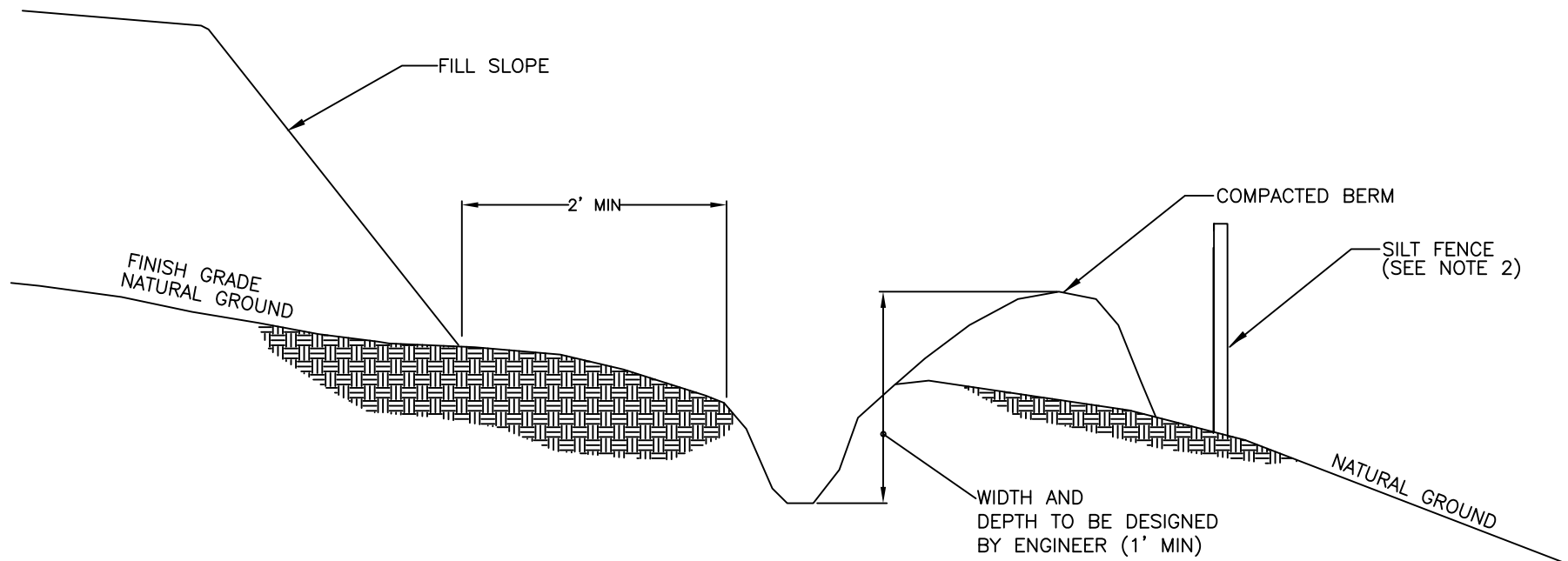
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CONSTRUCTION SPECIFICATIONS:

1. THE TOP OF THE EARTH DIKE OVER THE INLET PIPE AND THOSE DIKES CARRYING WATER TO THE PIPE SHALL BE AT LEAST 1.5 FEET HIGHER AT ALL POINTS THAN THE TOP OF THE INLET PIPE.
2. THE PIPE SHALL BE FLEXIBLE WITH WATER TIGHT CONNECTING BANDS. FLEXIBLE PIPE SHOULD BE STAKED ON EITHER SIDE.
3. A RIP RAP APRON SHALL BE PROVIDED AT THE OUTLET, IF EMPTYING INTO A DISTURBED AREA.
4. THE SOIL AROUND AND UNDER THE INLET PIPE AND ENTRANCE SECTION SHALL BE HAND TAMPED IN 4" LIFTS TO THE TOP OF THE EARTH DIKE.
5. FOLLOW-UP INSPECTION AND ANY NEEDED MAINTENANCE SHALL BE PERFORMED AFTER EACH STORM BY THE FINANCIALLY RESPONSIBLE PARTY OR HIS AGENT.
6. OUTLET PIPE SHOULD BE TAKEN OVER OR THROUGH ANY SILT FENCE, TAKING CARE NOT TO VOID THE EFFECTIVENESS OF THE SILT FENCE.

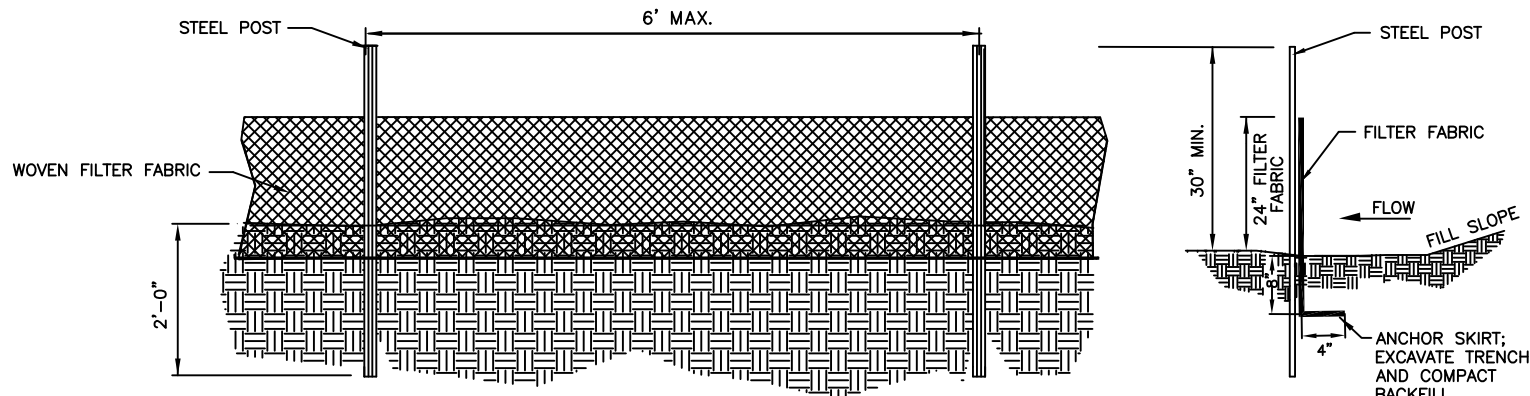
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NOTE:

1. DITCH SHOULD HAVE LONGITUDINAL SLOPE OF 1%.
2. SILT FENCE MAY BE REQUIRED BEHIND BERM

NOT TO SCALE



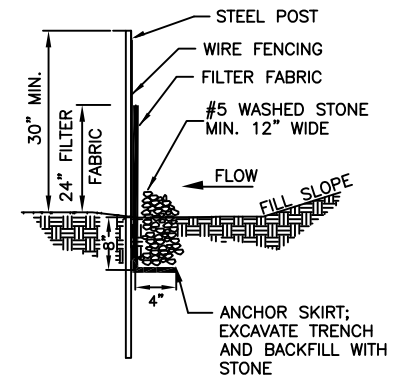
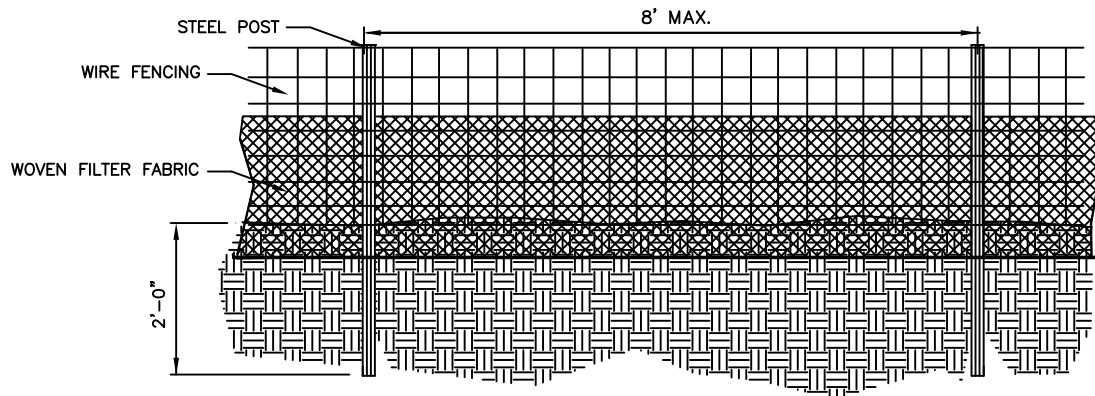
GENERAL NOTES:

1. FILTER FABRIC FENCE SHALL BE A MINIMUM OF 32" IN WIDTH AND SHALL HAVE A MINIMUM OF 6 LINE WIRES WITH 12" STAY SPACING.
2. WOVEN FILTER FABRIC BE USED WHERE SILT FENCE IS TO REMAIN FOR A PERIOD OF MORE THAN 30 DAYS.
3. STEEL POSTS SHALL BE 5'-0" IN HEIGHT AND BE OF THE SELF-FASTENER ANGLE STEEL TYPE.
4. TURN SILT FENCE UP SLOPE AT ENDS.
5. ORANGE SAFETY FENCE IS REQUIRED AT BACK OF SILT FENCE WHEN GRADING IS ADJACENT TO SWIM BUFFERS, STREAMS OR WETLANDS (REFER TO SWIM BUFFER GUIDELINES). THE COLOR ORANGE IS RESERVED FOR VISUAL IDENTIFICATION OF ENVIRONMENTALLY SENSITIVE AREAS.
6. DRAINAGE AREA CAN NOT BE GREATER THAN 1/4 ACRE PER 100 FT OF FENCE.
7. SLOPE LENGTHS CAN NOT EXCEED CRITERIA SHOWN IN TABLE 6.62A NORTH CAROLINA EROSION AND SEDIMENT CONTROL PLANNING AND DESIGN MANUAL.
8. DO NOT INSTALL SEDIMENT FENCE ACROSS STREAMS, DITCHES, WATERWAYS OR OTHER AREAS OF CONCENTRATED FLOW.

MAINTENANCE NOTES:

1. FILTER BARRIERS SHALL BE INSPECTED BY THE FINANCIALLY RESPONSIBLE PARTY OR HIS AGENT IMMEDIATELY AFTER EACH RAINFALL AND AT LEAST DAILY DURING PROLONGED RAINFALL. ANY REPAIRS NEEDED SHALL BE MADE IMMEDIATELY.
2. SHOULD THE FABRIC DECOMPOSE OR BECOME INEFFECTIVE PRIOR TO THE END OF THE EXPECTED USABLE LIFE AND THE BARRIER STILL IS NECESSARY, THE FABRIC SHALL BE REPLACED PROMPTLY.
3. SEDIMENT DEPOSITS SHOULD BE REMOVED WHEN DEPOSITS REACH APPROX. HALF THE HEIGHT OF THE BARRIER. ANY SEDIMENT DEPOSITS REMAINING IN PLACE AFTER THE SILT FENCE IS REMOVED SHALL BE DRESSED TO CONFORM TO THE EXISTING GRADE, PREPARED AND SEEDED.

NOT TO SCALE



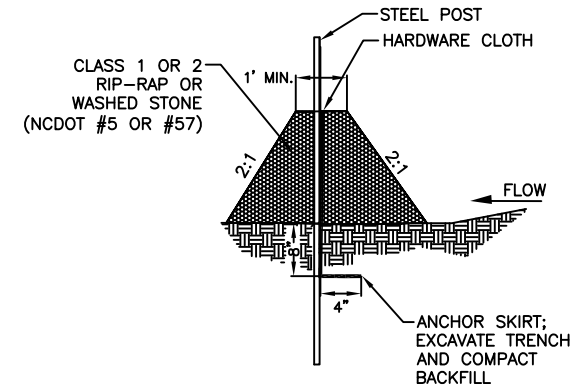
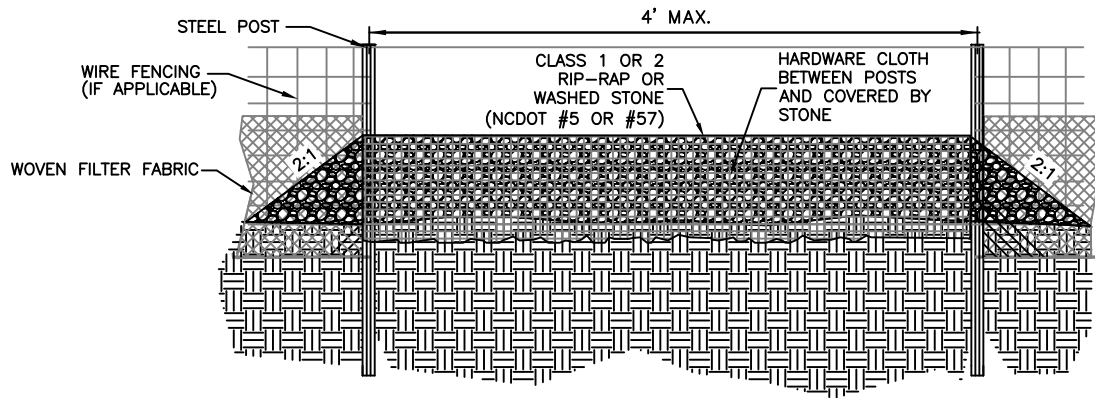
GENERAL NOTES:

1. FILTER FABRIC FENCE SHALL BE A MINIMUM OF 32" IN WIDTH AND SHALL HAVE A MINIMUM OF 6 LINE WIRES WITH 12" STAY SPACING.
2. WOVEN FILTER FABRIC BE USED WHERE SILT FENCE IS TO REMAIN FOR A PERIOD OF MORE THAN 30 DAYS.
3. STEEL POSTS SHALL BE 5'-0" IN HEIGHT AND BE OF THE SELF-FASTENER ANGLE STEEL TYPE.
4. WIRE FENCING SHALL BE AT LEAST #10 GAGE WITH A MINIMUM OF 6 LINE WIRES WITH 6" STAY SPACING.
5. TURN SILT FENCE UP SLOPE AT ENDS.
6. WIRE AND WASHED STONE IS REQUIRED TO BE SHOWN ON PLANS AT THE TOE OF SLOPES GREATER THAN 10 FEET VERTICAL (2:1 SLOPE)
7. ORANGE SAFETY FENCE IS REQUIRED AT BACK OF SILT FENCE WHEN GRADING IS ADJACENT TO SWIM BUFFERS, STREAMS OR WETLANDS (REFER TO SWIM BUFFER GUIDELINES). THE COLOR ORANGE IS RESERVED FOR VISUAL IDENTIFICATION OF ENVIRONMENTALLY SENSITIVE AREAS.
8. DRAINAGE AREA CAN NOT BE GREATER THAN 1/4 ACRE PER 100 FT OF FENCE.
9. SLOPE LENGTHS CAN NOT EXCEED CRITERIA SHOWN IN TABLE 6.62A NORTH CAROLINA EROSION AND SEDIMENT CONTROL PLANNING AND DESIGN MANUAL.
10. DO NOT INSTALL SEDIMENT FENCE ACROSS STREAMS, DITCHES, WATERWAYS OR OTHER AREAS OF CONCENTRATED FLOW.

MAINTENANCE NOTES:

1. FILTER BARRIERS SHALL BE INSPECTED BY THE FINANCIALLY RESPONSIBLE PARTY OR HIS AGENT IMMEDIATELY AFTER EACH RAINFALL AND AT LEAST DAILY DURING PROLONGED RAINFALL. ANY REPAIRS NEEDED SHALL BE MADE IMMEDIATELY.
2. SHOULD THE FABRIC DECOMPOSE OR BECOME INEFFECTIVE PRIOR TO THE END OF THE EXPECTED USABLE LIFE AND THE BARRIER STILL IS NECESSARY, THE FABRIC SHALL BE REPLACED PROMPTLY.
3. SEDIMENT DEPOSITS SHOULD BE REMOVED WHEN DEPOSITS REACH HALF THE HEIGHT OF THE BARRIER. ANY SEDIMENT DEPOSITS REMAINING IN PLACE AFTER THE SILT FENCE IS REMOVED SHALL BE DRESSED TO CONFORM TO THE EXISTING GRADE, PREPARED AND SEEDED.

NOT TO SCALE



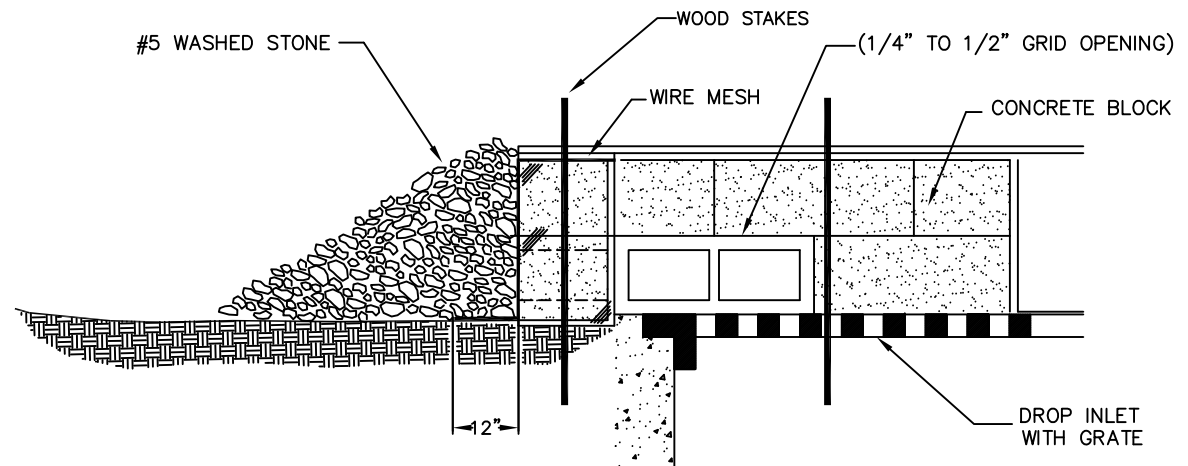
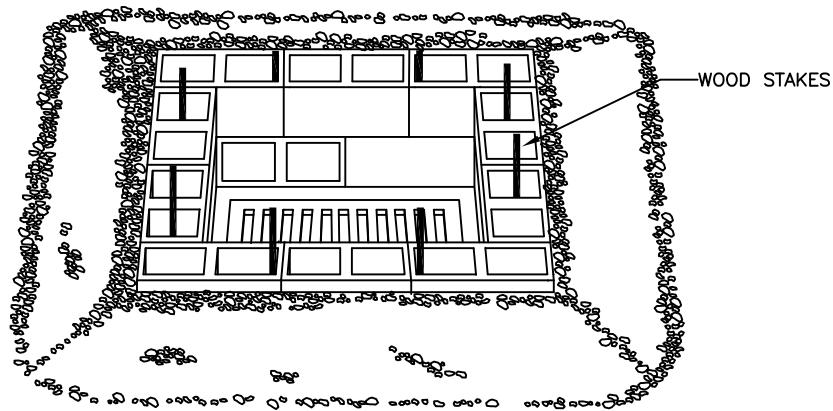
GENERAL NOTES:

1. SEDIMENT FILTER OUTLET AND HARDWARE CLOTH SHALL BE 16 INCHES HIGH BUT NO TALLER THAN 18 INCHES.
2. NCDOT #5 OR #57 WASHED STONE MAY BE USED WITH HARDWARE CLOTH OR CLASS 1 OR 2 RIP-RAP SHALL BE USED AND COVERED WITH 4 INCHES OF NCDOT #5 OR #57 WASHED STONE ON THE UPSLOPE SIDE WITH HARDWARE CLOTH.
3. HARDWARE CLOTH SHALL BE ANCHORED TO THE STEEL POSTS SECURELY USING APPROPRIATE ANCHORS. HARDWARE CLOTH SHALL BE KEYED IN A MINIMUM OF 12 INCHES IN LENGTH AND BACKFILLED PROPERLY AS SHOWN IN ABOVE DETAIL. HARDWARE CLOTH TO BE SAME AS STD. #512.1 (19 GAUGE, 1/4" SPACING).
4. POSTS SHALL BE NO MORE THAN 4 FEET APART.
5. SITE OUTLETS AT ANY POINT SMALL CONCENTRATED FLOWS ARE ANTICIPATED AND AT THE DIRECTION OF THE INSPECTOR.
6. ONE ACRE MAXIMUM DRAINAGE AREA PER OUTLET.

MAINTENANCE NOTES:

1. FILTER OUTLETS SHALL BE INSPECTED BY THE FINANCIALLY RESPONSIBLE PARTY OR HIS AGENT IMMEDIATELY AFTER EACH RAINFALL AND AT LEAST DAILY DURING PROLONGED RAINFALL. ANY REPAIRS NEEDED SHALL BE MADE IMMEDIATELY.
2. THE STONE SHALL BE REPLACED PROMPTLY AFTER ANY EVENT THAT HAS CLOGGED OR REMOVED IT.
3. SEDIMENT DEPOSITS SHOULD BE REMOVED WHEN DEPOSITS REACH HALF THE HEIGHT OF THE BARRIER. ANY SEDIMENT DEPOSITS REMAINING IN PLACE AFTER THE SILT FENCE OUTLET IS REMOVED SHALL BE DRESSED TO CONFORM TO THE EXISTING GRADE, PREPARED AND SEEDED.

NOT TO SCALE



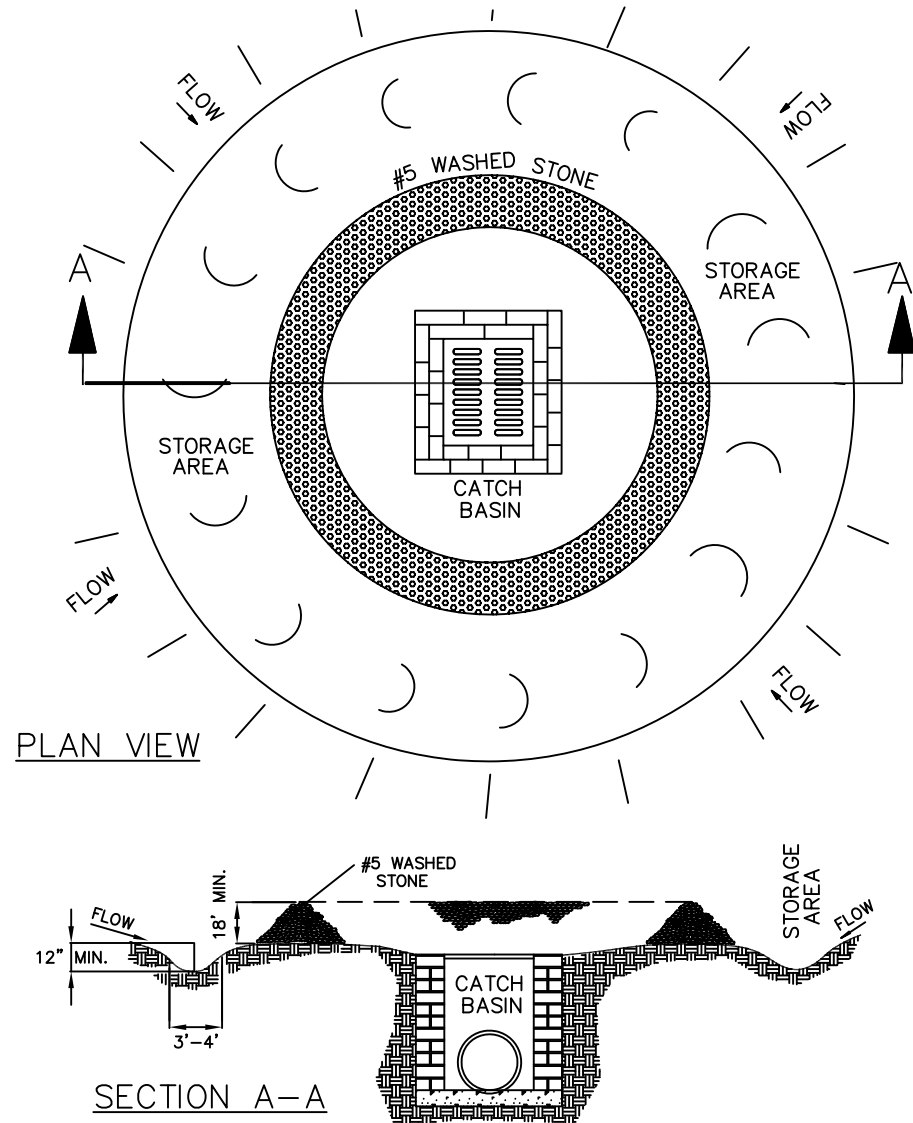
SPECIFIC APPLICATION:

THIS METHOD OF INLET PROTECTION IS APPLICABLE WHERE HEAVY FLOWS ARE EXPECTED AND WHERE OVERFLOW CAPACITY IS NECESSARY TO PREVENT EXCESSIVE PONDING AROUND THE STRUCTURE.

NOT TO SCALE

GENERAL NOTES:

1. SEDIMENT SHALL BE REMOVED AND TRAP RESTORED TO ITS ORIGINAL DIMENSIONS WHEN THE SEDIMENT HAS ACCUMULATED TO 1/2 THE DESIGN DEPTH OF THE TRAP.
2. REMOVED SEDIMENT SHALL BE DEPOSITED IN A SUITABLE AREA AND IN SUCH A MANNER THAT IT WILL NOT ERODE.
3. THE STRUCTURE SHALL BE INSPECTED BY THE FINANCIALLY RESPONSIBLE PARTY OR HIS AGENT AFTER EACH STORM EVENT AND REPAIRS MADE AS NECESSARY.
4. CONSTRUCTION OPERATIONS SHALL BE CARRIED OUT IN SUCH A MANNER THAT EROSION AND WATER POLLUTION ARE MINIMIZED.
5. THE SEDIMENT TRAP SHALL BE REMOVED AND THE AREA STABILIZED WHEN THE DRAINAGE BASIN HAS BEEN PROPERLY STABILIZED.
6. ON LARGER DRAINAGE AREAS RIP RAP MAY BE REQUIRED UNDER THE WASHED STONE.



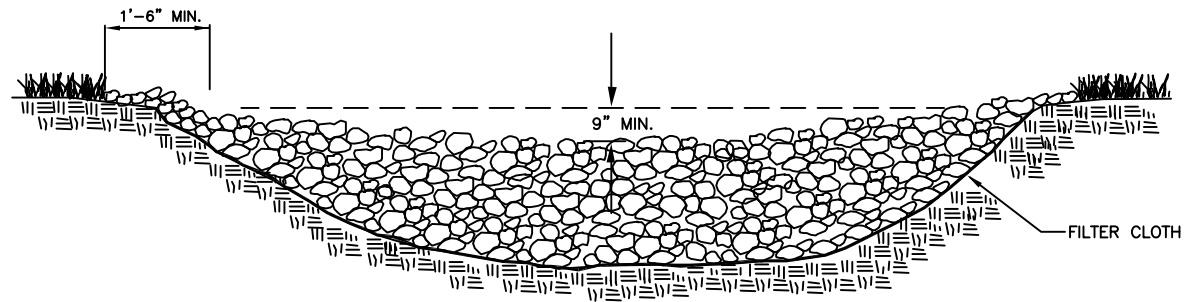
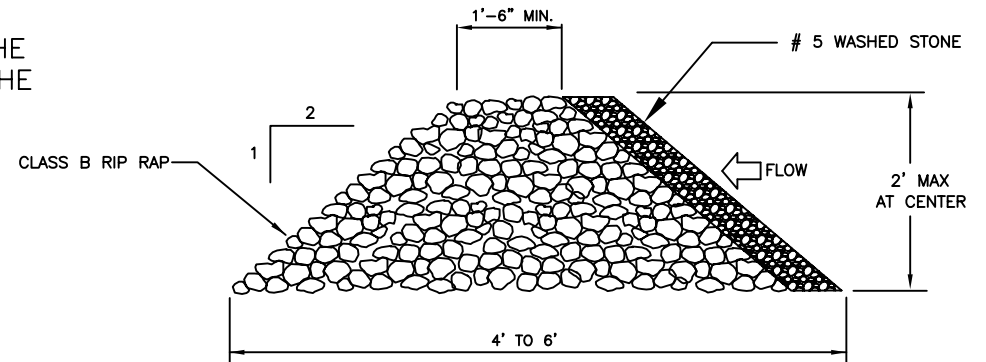
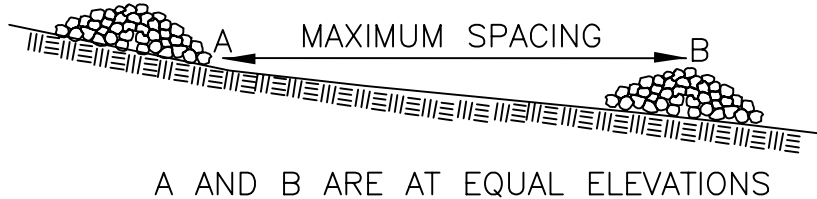
NOT TO SCALE

1. UNIFORMLY GRADE A SHALLOW DEPRESSION APPROACHING THE INLET.
2. DRIVE 5-FOOT STEEL POSTS 2 FEET INTO THE GROUND SURROUNDING THE INLET. SPACE POSTS EVENLY AROUND THE PERIMETER OF THE INLET, A MAXIMUM OF 4 FEET APART.
3. SURROUND THE POSTS WITH WIRE MESH HARDWARE CLOTH. SECURE THE WIRE MESH TO THE STEEL POSTS AT THE TOP, MIDDLE, AND BOTTOM. PLACING A 2-FOOT FLAP OF THE WIRE MESH UNDER THE GRAVEL FOR ANCHORING IS RECOMMENDED.
4. PLACE CLEAN GRAVEL (NC DOT #5 OR #57 STONE) ON A 2:1 SLOPE WITH A HEIGHT OF 16 INCHES AROUND THE WIRE, AND SMOOTH TO AN EVEN GRADE.
5. ONCE THE CONTRIBUTING DRAINAGE AREA HAS BEEN STABILIZED, REMOVE ACCUMULATED SEDIMENT, AND ESTABLISH FINAL GRADING ELEVATIONS.
6. COMPACT THE AREA PROPERLY AND STABILIZED IT WITH GROUND COVER.



GENERAL NOTES:

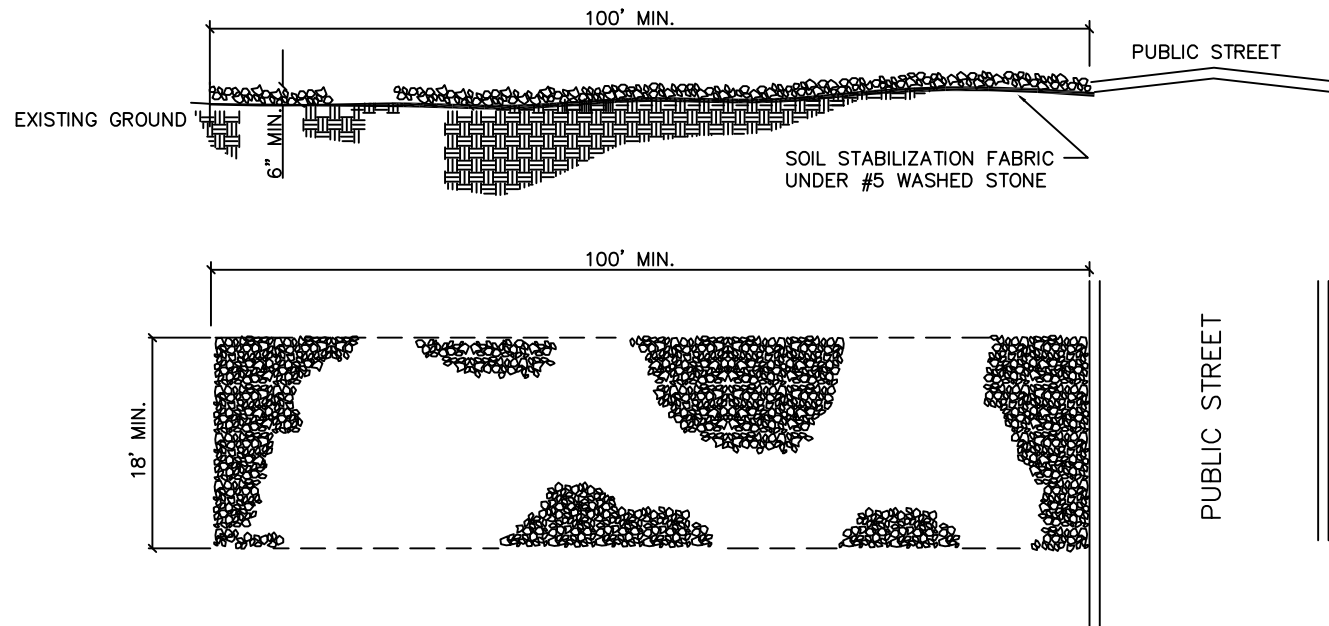
1. RIPRAP SIZE TO BE DESIGNED BY ENGINEER.
2. CHECK DAMS MAY BE USED IN SLOPING DITCHES OR CHANNELS TO SLOW VELOCITY OR TO CREATE SEDIMENT TRAPS.
3. ENSURE THAT MAXIMUM SPACING BETWEEN DAMS PLACES THE TOE OF THE UPSTREAM DAM AT THE SAME ELEVATION AS THE DOWNSTREAM DAM (SEE DIAGRAM BELOW).



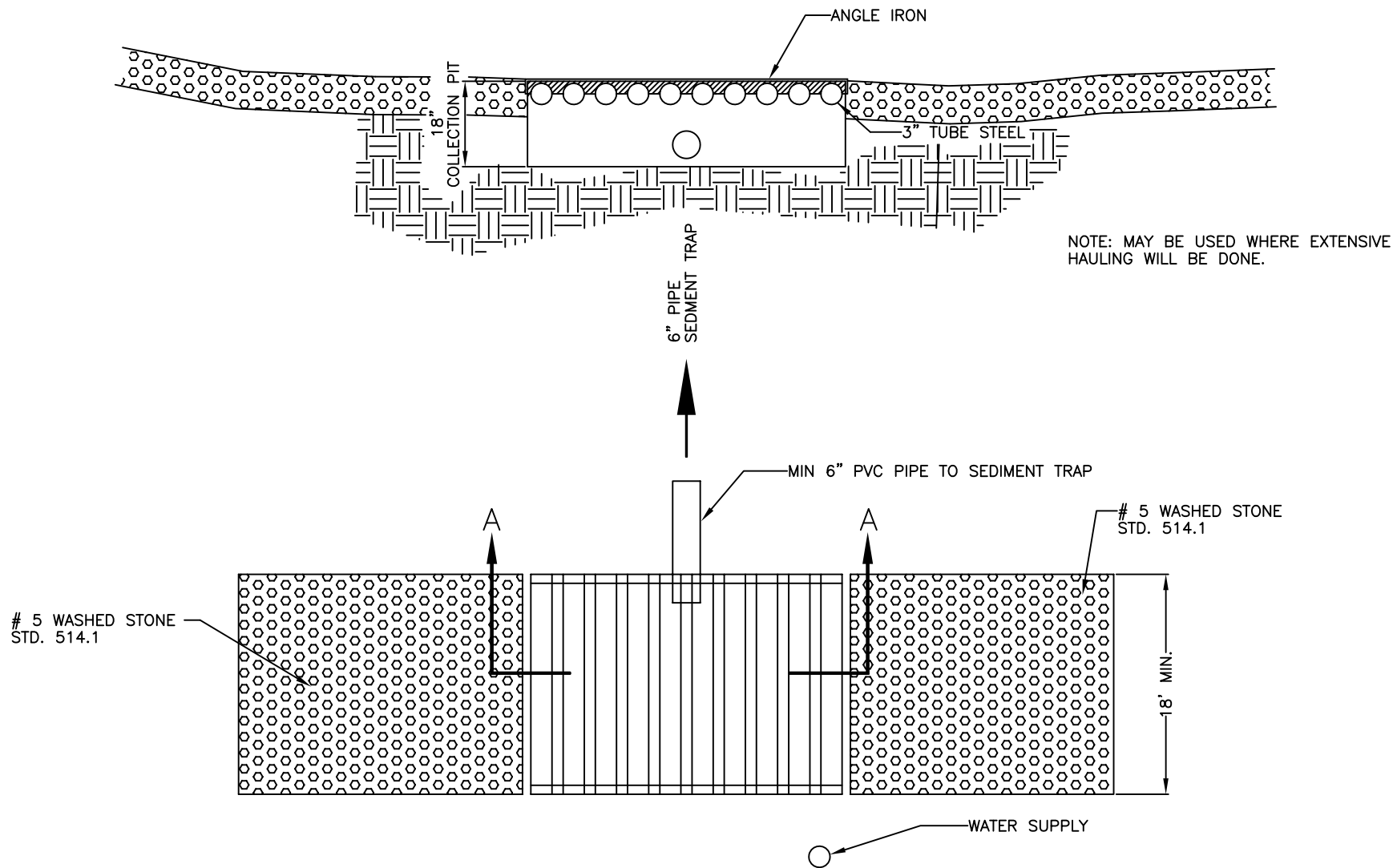
NOT TO SCALE

NOTES:

1. A STABILIZED ENTRANCE PAD OF #5 WASHED STONE OR RAIL ROAD BALLAST SHALL BE LOCATED WHERE TRAFFIC WILL ENTER OR LEAVE THE CONSTRUCTION SITE ONTO A PUBLIC STREET.
2. FILTER FABRIC OR COMPACTED CRUSHER RUN STONE SHALL BE USED AS A BASE FOR THE CONSTRUCTION ENTRANCE.
3. THE ENTRANCE SHALL BE MAINTAINED IN A CONDITION WHICH WILL PREVENT TRACKING OR FLOWING OF SEDIMENT ONTO PUBLIC STREETS OR EXISTING PAVEMENT. THIS MAY REQUIRE PERIODIC TOP DRESSING WITH ADDITIONAL STONE AS CONDITIONS WARRANT AND REPAIR OR CLEANOUT OF ANY MEASURES USED TO TRAP SEDIMENT.
4. ANY SEDIMENT SPILLED, DROPPED, WASHED, OR TRACKED ONTO PUBLIC STREETS MUST BE REMOVED IMMEDIATELY.
5. WHEN APPROPRIATE, WHEELS MUST BE CLEANED TO REMOVE SEDIMENT PRIOR TO ENTERING A PUBLIC STREET. WHEN WASHING IS REQUIRED, IT SHALL BE DONE IN AN AREA STABILIZED WITH CRUSHED STONE WHICH DRAINS INTO AN APPROVED SEDIMENT BASIN SEE STD. NO. 515.1.
6. TOWN MAY REQUIRE A STANDARD COMMERCIAL DRIVEWAY (STD. 108.1 – 116.1) TO ACCESS THE CONSTRUCTION SITE IF THE DRIVEWAY IS ON A THOROUGHFARE.

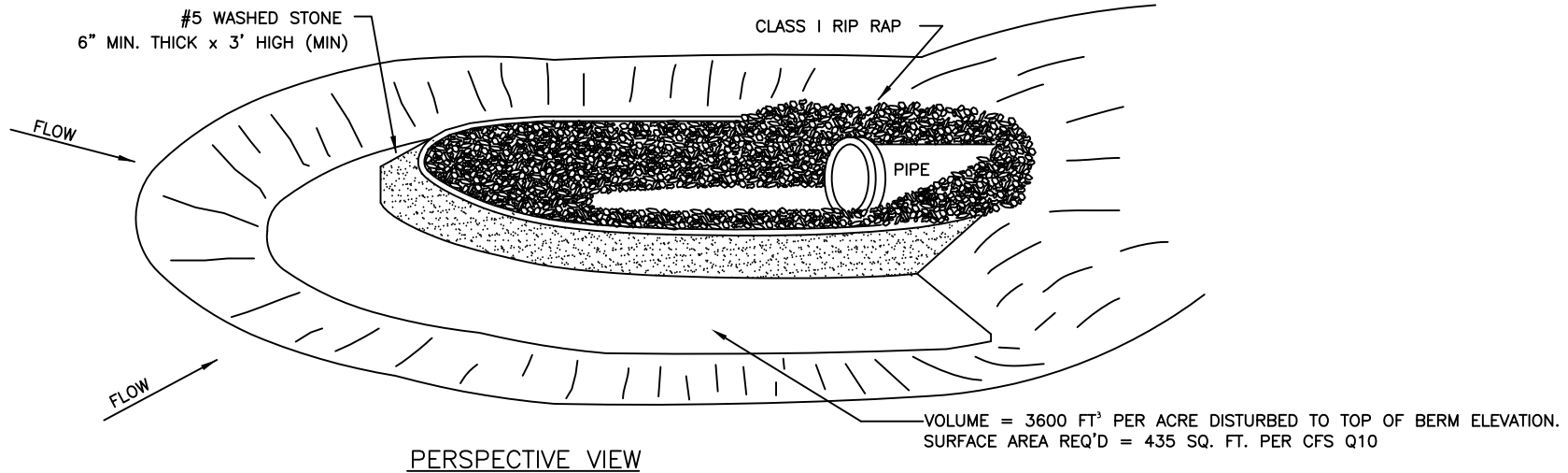


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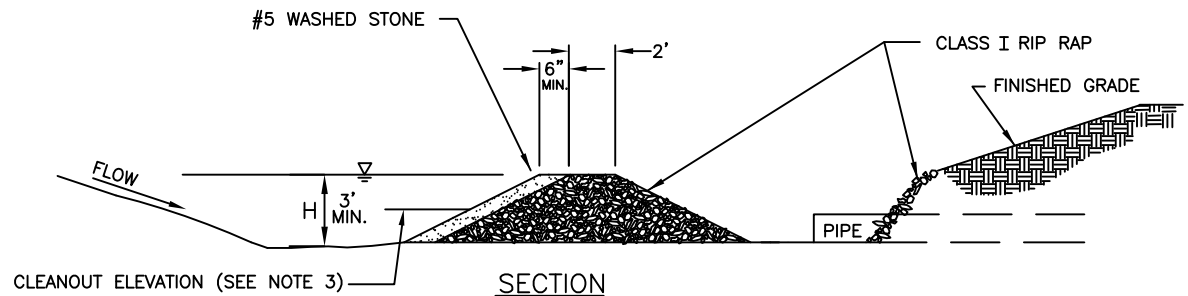
NOT TO SCALE

BASIN NO.	DRAINAGE AREA (ACRES)	DENUDED AREA (ACRES)	BASIN VOLUME		BASIN SURFACE AREA		CLEANOUT DEPTH H/2 (FT.)	H (FEET)
			REQUIRED (S.F.)	PROVIDED (S.F.)	REQUIRED (S.F.)	PROVIDED (S.F.)		

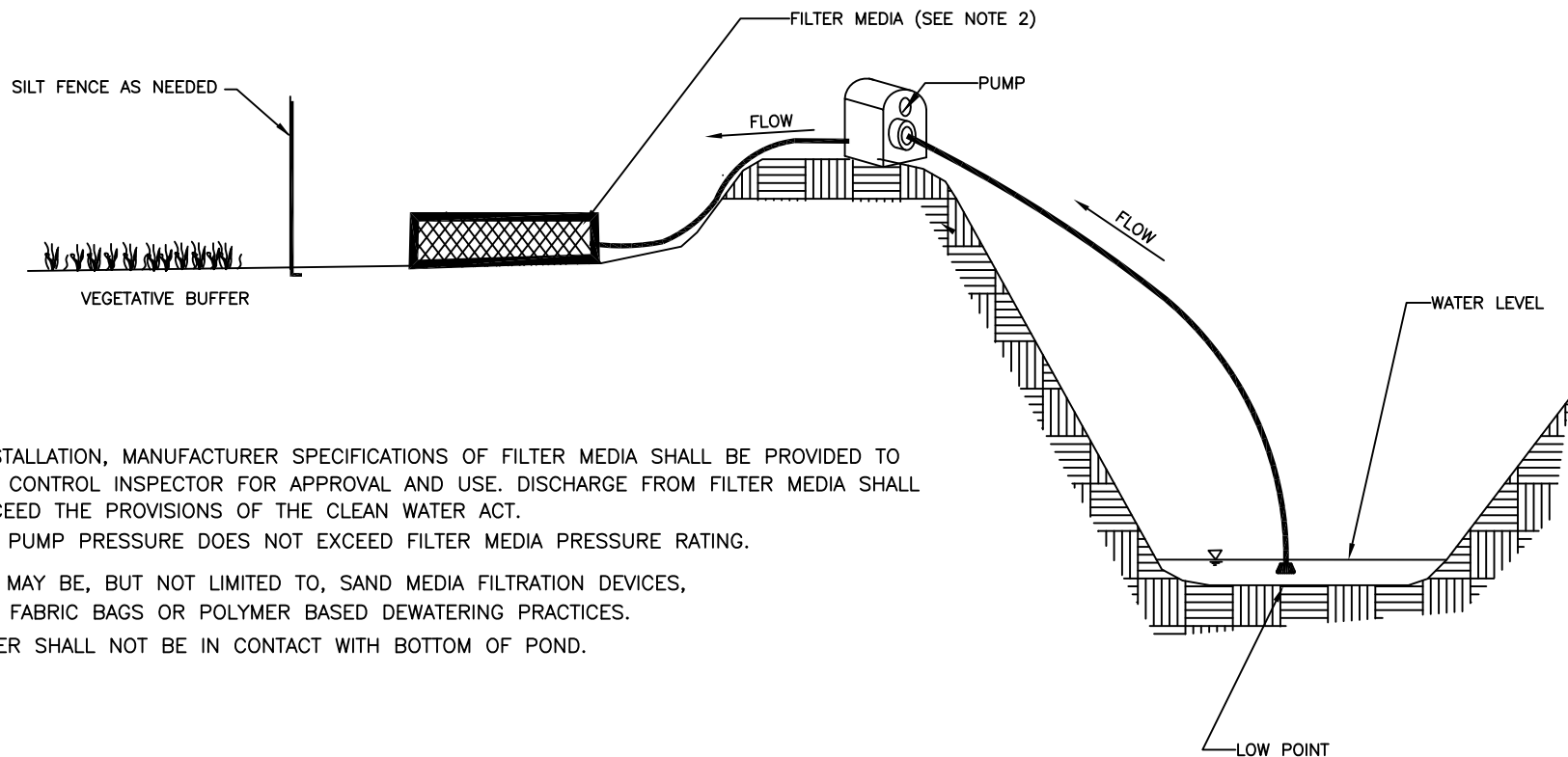


GENERAL NOTES:

1. GRAVEL AND RIP RAP FILTER BERM BASIN SHOULD BE USED TO PROTECT EXISTING PIPE INVERTS THAT DRAIN 5 ACRES OR LESS.
2. DIMENSIONS SHOWN ARE THE MINIMUM ACCEPTED UNLESS OTHERWISE NOTED.
3. CLEANOUT PRIOR TO SEDIMENT REACHING HALF OF BERM HEIGHT.



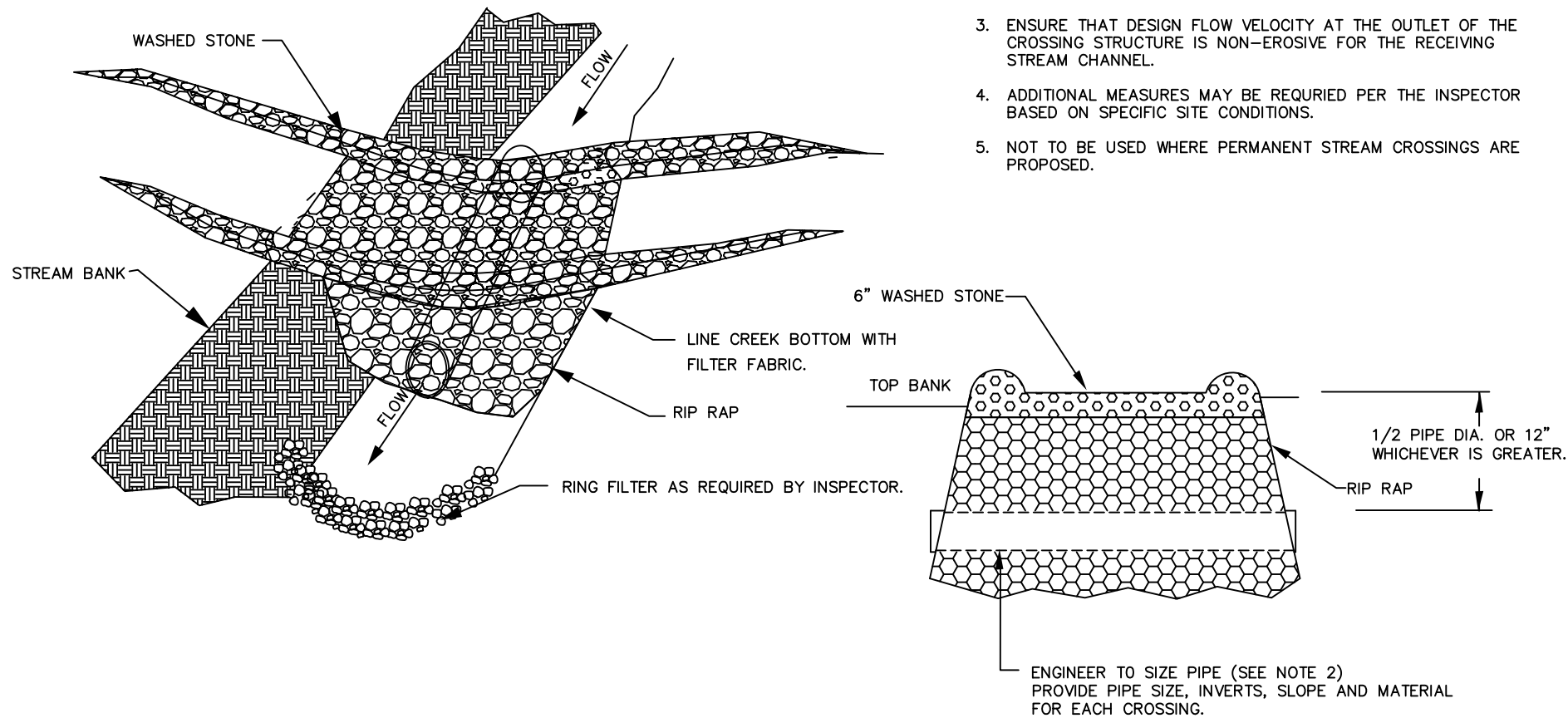
NOT TO SCALE



NOTE:

1. PRIOR TO INSTALLATION, MANUFACTURER SPECIFICATIONS OF FILTER MEDIA SHALL BE PROVIDED TO THE EROSION CONTROL INSPECTOR FOR APPROVAL AND USE. DISCHARGE FROM FILTER MEDIA SHALL MEET OR EXCEED THE PROVISIONS OF THE CLEAN WATER ACT.
2. ENSURE THAT PUMP PRESSURE DOES NOT EXCEED FILTER MEDIA PRESSURE RATING.
3. FILTER MEDIA MAY BE, BUT NOT LIMITED TO, SAND MEDIA FILTRATION DEVICES, RATED FILTER FABRIC BAGS OR POLYMER BASED DEWATERING PRACTICES.
4. PUMP STRAINER SHALL NOT BE IN CONTACT WITH BOTTOM OF POND.

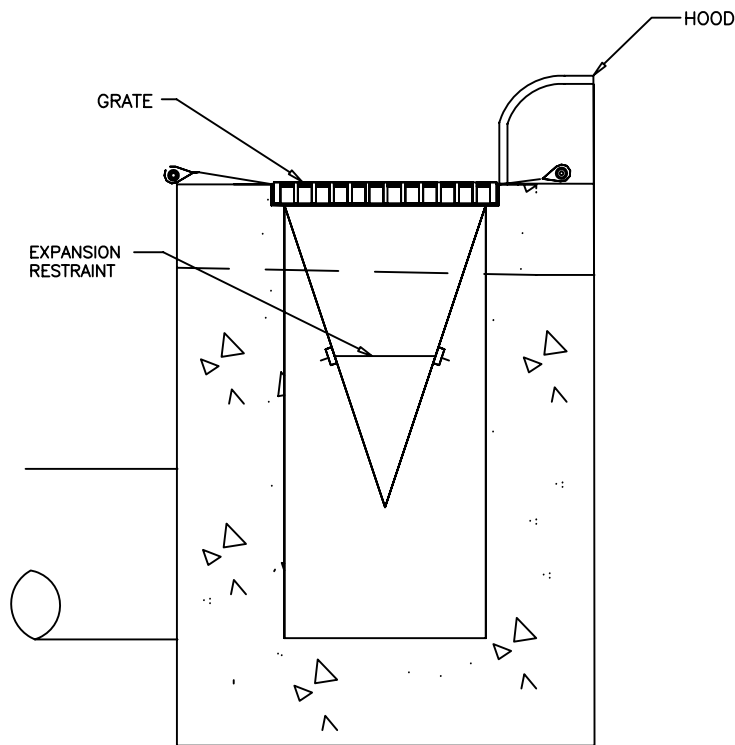
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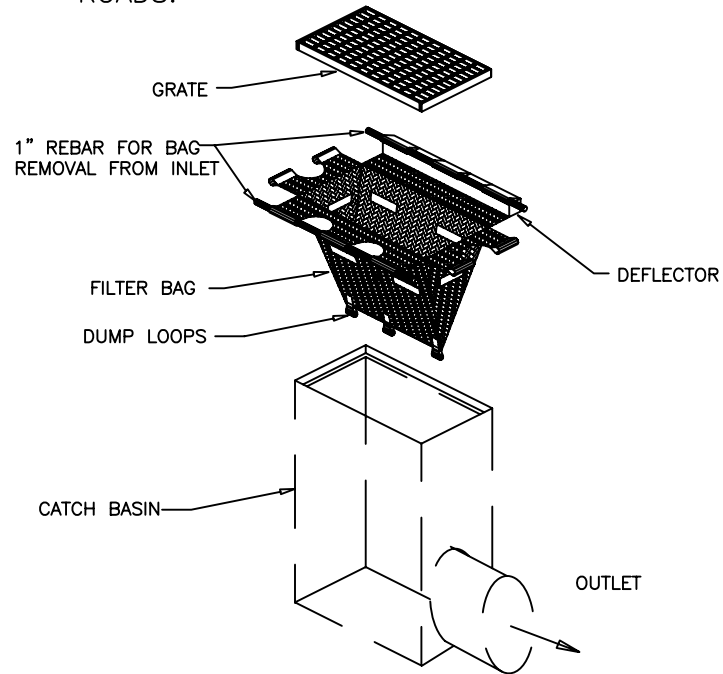
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NOTES

1. INLET MAINTENANCE SHALL BE DOCUMENTED IN PROJECT LOG BOOK.
2. FILTER TYPES SHALL BE APPROVED BY THE INSPECTOR PRIOR TO INSTALLATION.
3. FILTER BAGS MAY BE REMOVED WHEN SITE IS STABILIZED AT THE DIRECTION OF THE ENGINEER.
4. FILTER BAGS SHALL BE REMOVED PRIOR TO STREET ACCEPTANCE.
5. FILTER BAGS SHALL BE CLEANED OR REPLACED ON A REGULAR BASIS (NOT BE MORE THAN HALF FULL AT ANY TIME).
6. FILTER BAGS SHALL NOT BE ALLOWED IN EXISTING TOWN OR NCDOT ROADS.

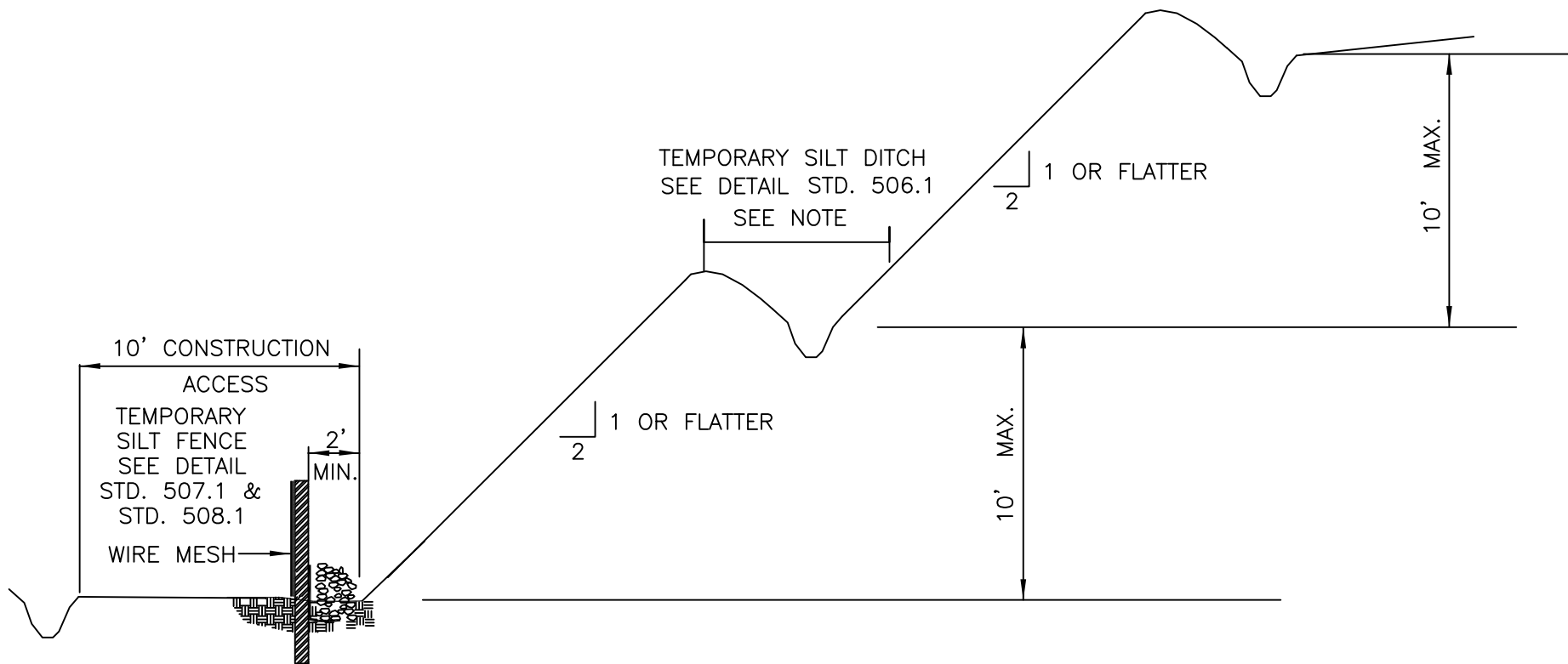


SECTION



INSTALLATION

NOT TO SCALE



NOTE:

DIVERSION DITCH SHOULD FLOW INTO SEDIMENT BASIN ROCK CHECK DAM, OR SLOPE DRAIN

NOT TO SCALE

GENTLE SLOPES

STEEP SLOPES

SEEDING MIXTURE	80 lbs/acre of tall fescue	100 lbs/acre tall fescue 30 lbs/acre Sericea lespedeza (unscarified after August 15) 10 lbs/acre Kobe lespedeza
SEEDING DATES	FALL: August 25 – October Late winter: February 15 – April 15 To extend spring seeding into June, add 15 lbs/acre hulled Bermudagrass Overseeding of Kobe lespedeza over fall-seeded tall fescue is very effective.	FALL: August 25 – October 15 Late winter: February 15 – April 15 To extend spring seeding into June, add 15 lbs/acre hulled Bermudagrass Overseeding of Kobe lespedeza over fall-seeded tall fescue is very effective.
SEEDING AMENDMENTS	Apply lime and fertilizer per soil tests, or 4000 lbs/acre limestone and 1000 lbs/acre 10-10-10 fertilizer.	Apply lime and fertilizer per soil tests, or 4000 lbs/acre limestone and 1000 lbs/acre 10-10-10 fertilizer.

NOTE 1

Ground Cover—Protective cover must be established on all disturbed areas within 21 calendar days after land disturbing activity is completed or has temporarily ceased.

NOTE 2

Graded slopes and fills—Protective cover must be established on all graded slopes and fills within 21 calendar days after a phase of grading is completed or has temporarily ceased.

NOT TO SCALE

TEMPORARY SEEDING FOR WARM AND COOL SEASON

EARLY SUMMER SEASON

STEEP SLOPES

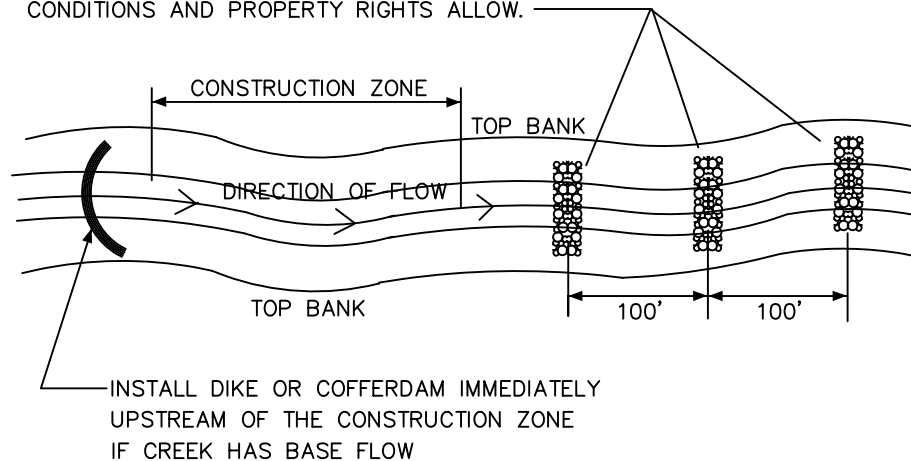
SEEDING MIXTURE	40 lbs/acre of German millet 80 lbs/acre of tall fescue	120 lbs/acre Rye (grain) 80 lbs/acre tall fescue
SEEDING DATES	May 1 – August 15 Refertilize if growth is not fully adequate. Apply 4000 lbs/acre straw or equivalent hydroseeding.	October 25 – December 30 Between December 30 – February 15, add 50 lbs/acre of annual Kobe lespedeza. Apply 4000 lbs/acre straw or equivalent hydroseeding.
SEEDING AMENDMENTS	Apply lime and fertilizer per soil tests, or 2000 lbs/acre limestone and 750 lbs/acre 10-10-10 fertilizer.	Apply lime and fertilizer per soil tests, or 2000 lbs/acre limestone and 750 lbs/acre 10-10-10 fertilizer.

NOT TO SCALE

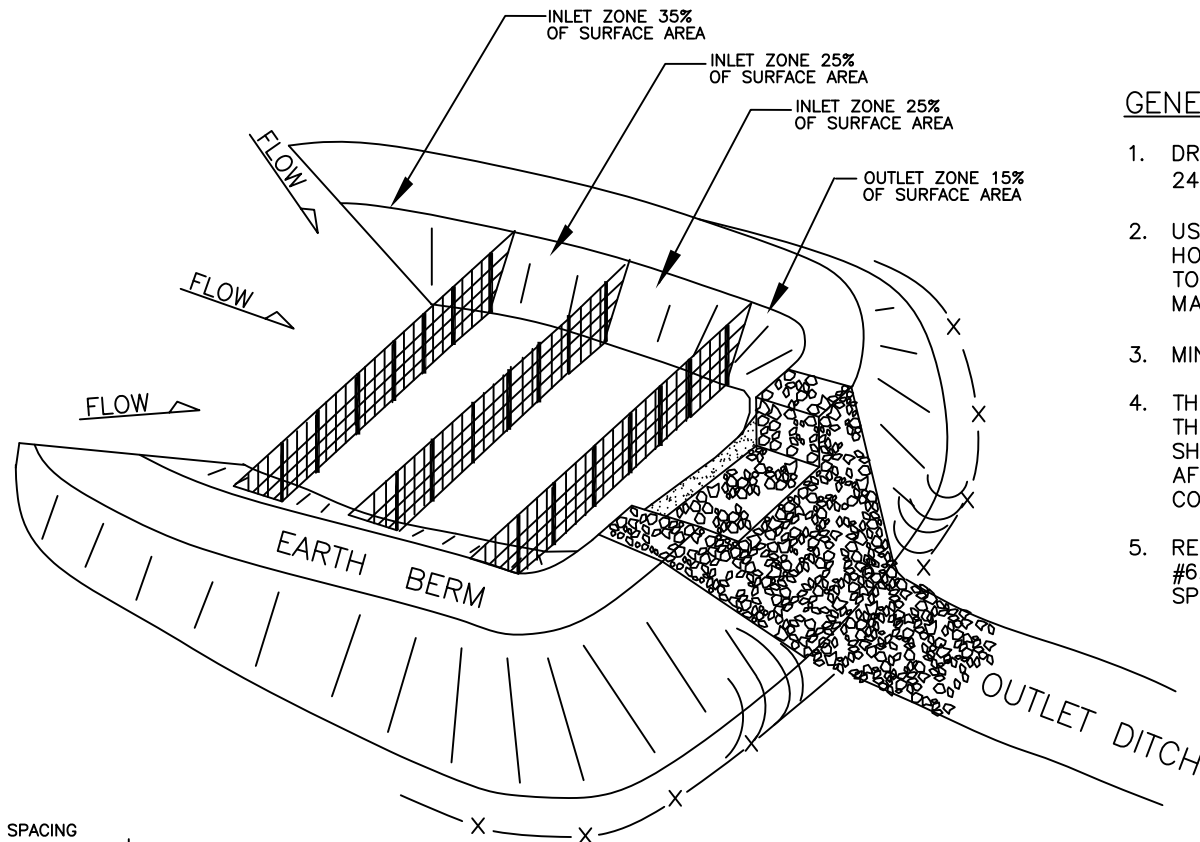
NOTES:

1. WORK IN CREEK SHALL BE PLANNED TO MINIMIZE THE NUMBER OF DAYS OF DISTURBANCE.
2. THE CONTRACTOR IS TO OBSERVE THE LOCAL WEATHER FORECASTS AND NOT BEGIN WORK IN THE CREEK UNLESS AT LEAST THREE DAYS WITHOUT RAIN IS ANTICIPATED.
3. ALL DISTURBED CREEK BED AND BANKS ARE TO BE STABILIZED PRIOR TO THE END OF EACH WORK DAY.
4. FOR LARGER CREEKS, CONSTRUCTION SHOULD OCCUR ON ONE SIDE OF THE CREEK AT A TIME. THE FIRST SIDE SHOULD BE STABILIZED BEFORE BEGINNING CONSTRUCTION ON THE OPPOSITE SIDE.
5. A TEMPORARY PIPE OR PUMP MAY BE INSTALLED TO CONTROL CREEK FLOW DURING CUNSTRUCTION.

CONSTRUCT THREE ROCK CHECK DAMS (STD. 513.1) AT 100-FOOT SPACING DOWN STREAM FROM THE CONSTRUCTION ZONE IF CONDITIONS AND PROPERTY RIGHTS ALLOW.

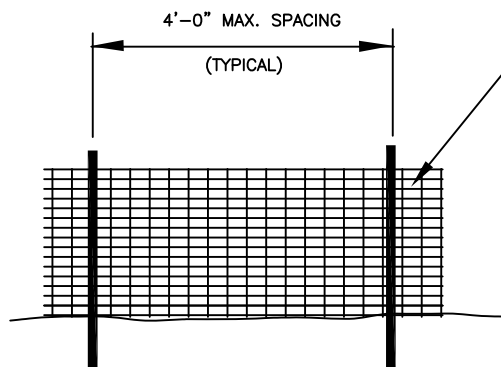


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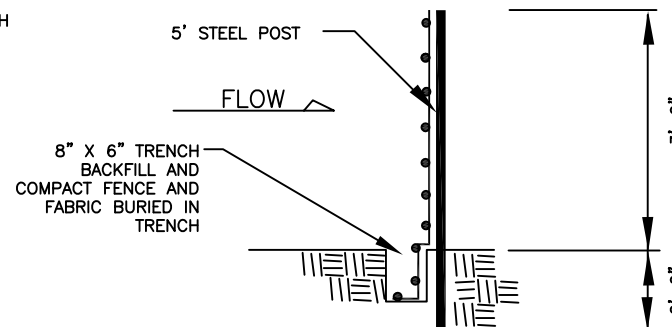


GENERAL NOTES:

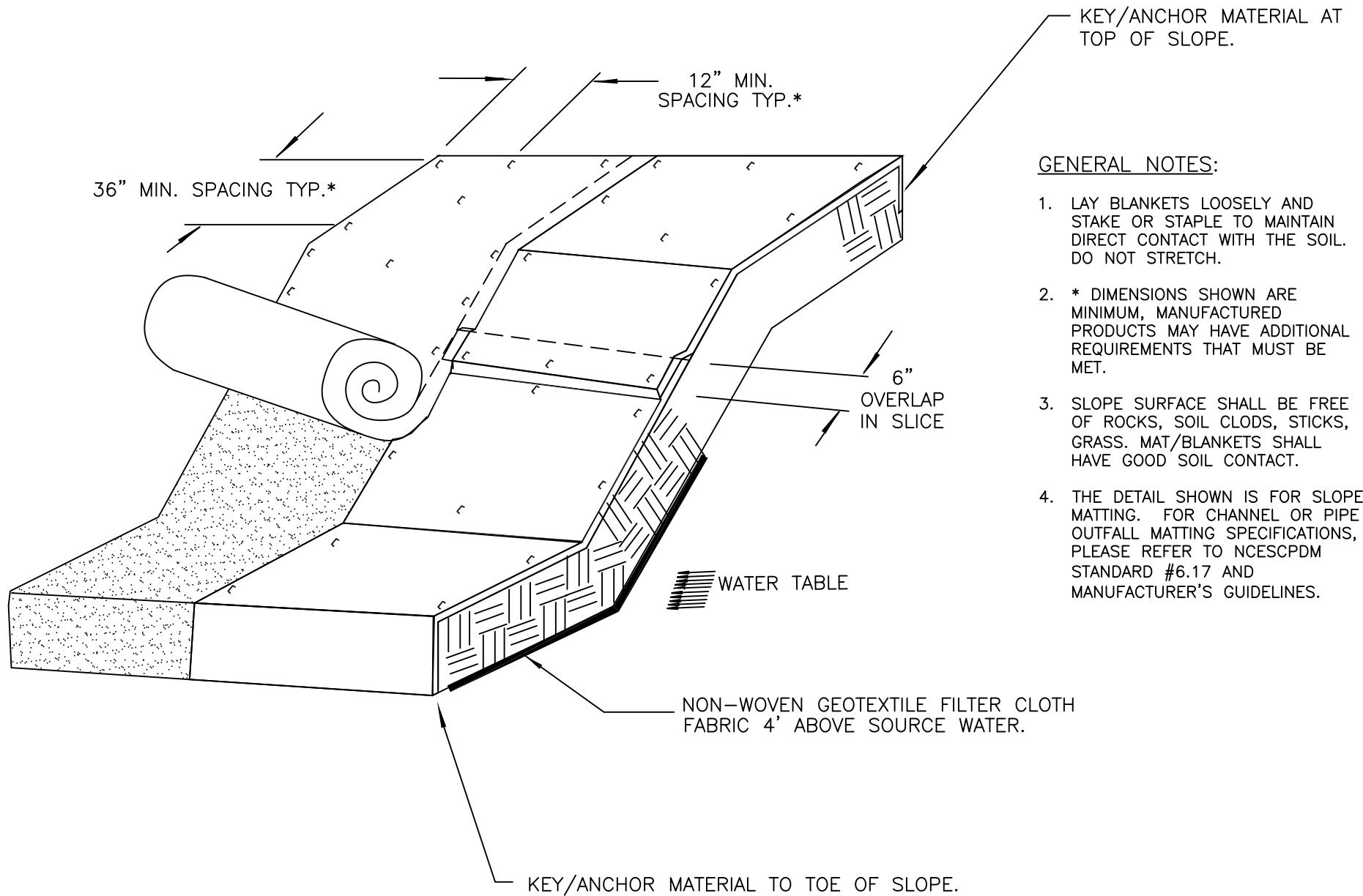
1. DRIVE 5' STEEL POST AT LEAST 24" INTO SOLID GROUND.
2. USE STAPLES 1' APART HORIZONTALLY AND VERTICALLY TO ATTACH THE POROUS MATERIAL TO THE WIRE FENCE.
3. MINIMUM BAFFLE SPACING IS 10'.
4. THE FLOOR OF THE BASIN IN THE OUTLET ZONE AND BERMS SHOULD BE SEEDED IMMEDIATELY AFTER THE BASIN IS CONSTRUCTED.
5. REFER TO NCESCPDM SECTION #6.65 FOR ADDITIONAL SPECIFICATIONS.



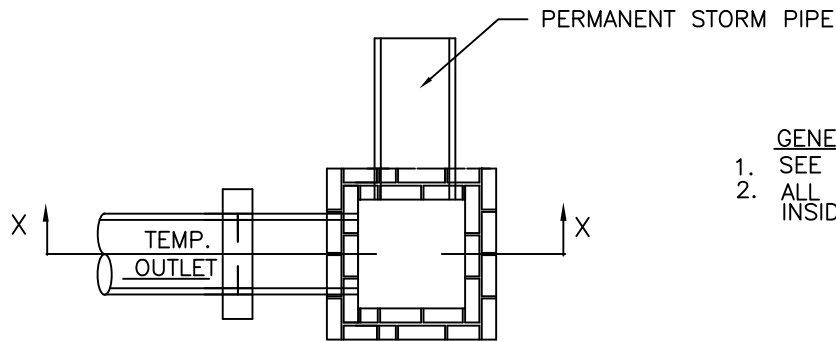
POROUS MATERIAL
(COMBINED JUTE MESH
AND NAG C125
COCONUT LINER OR
EQUIVALENT).



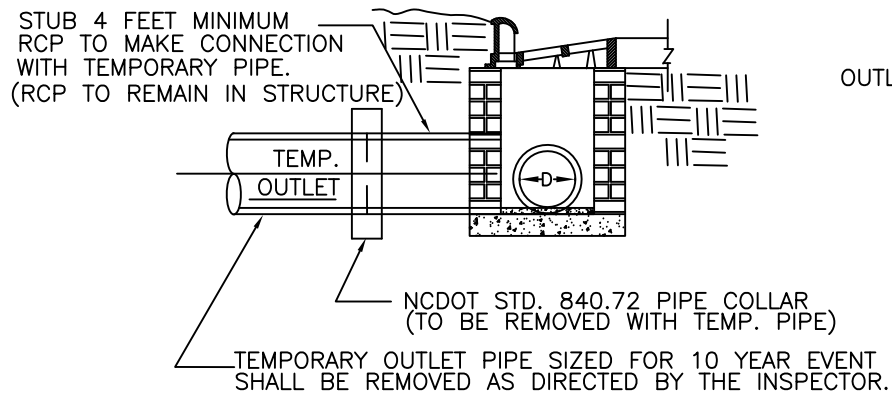
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NOT TO SCALE



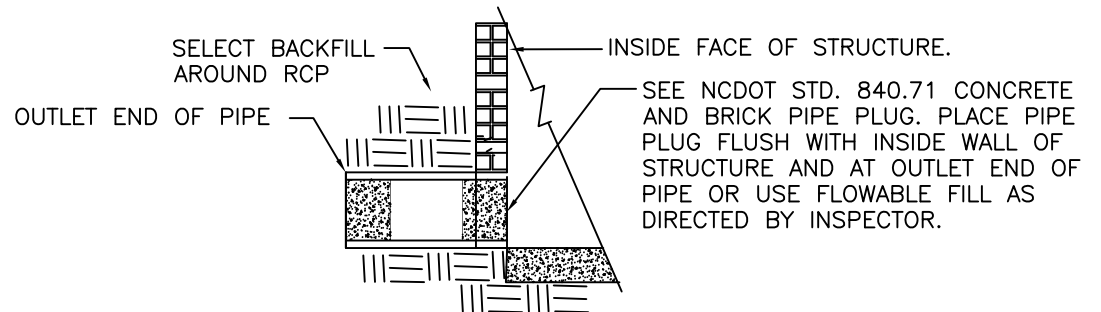
PLAN



SECTION X-X
ACTIVE SYSTEM

GENERAL NOTES:

1. SEE APPROPRIATE STANDARD FOR CATCH BASIN, MANHOLE, JUNCTION BOX USED.
2. ALL PIPE IN STORM DRAIN STRUCTURES SHALL BE STRUCK EVEN WITH THE INSIDE WALL, GROUTED AND BRUSHED SMOOTH.

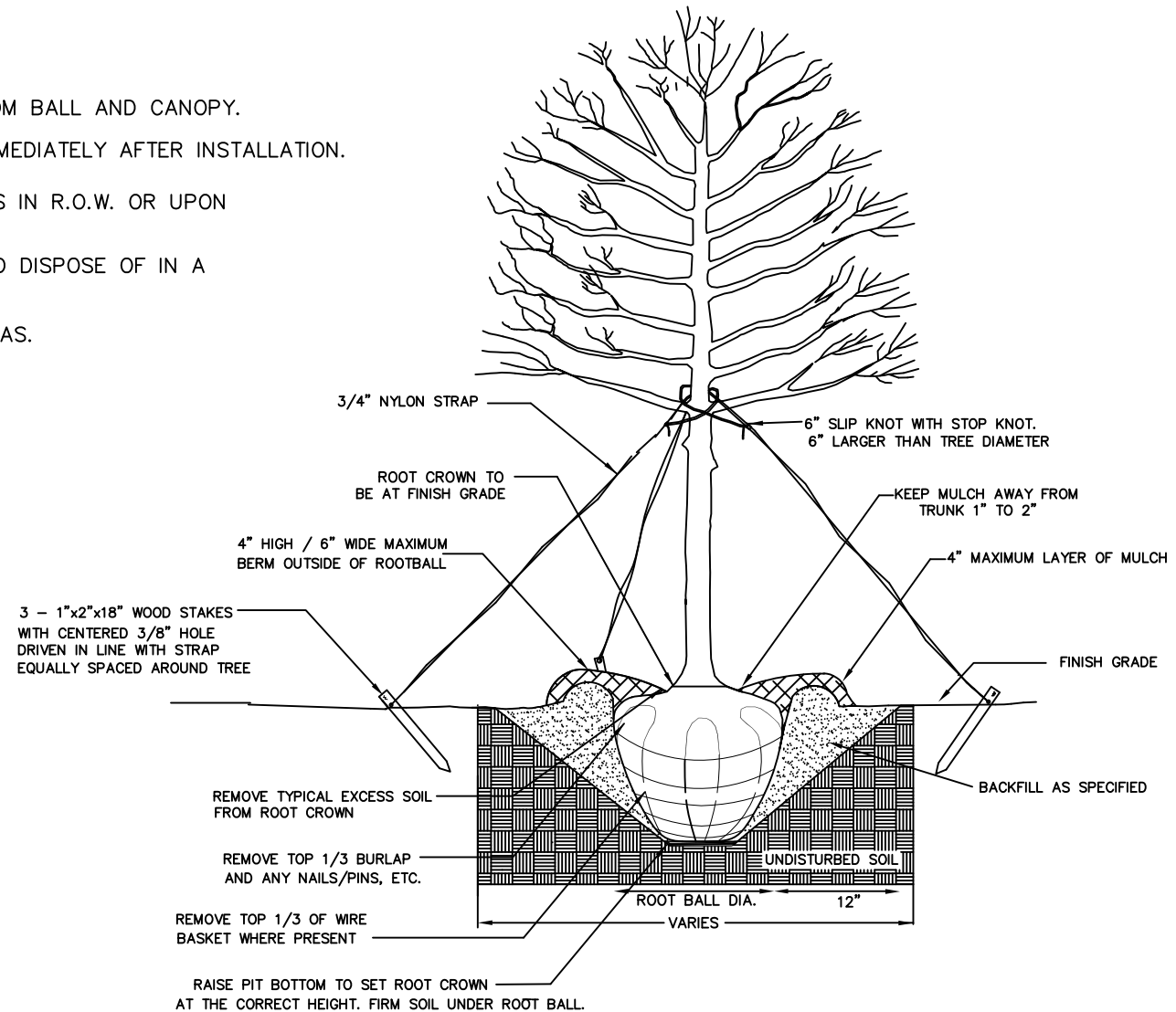


PIPE PLUG DETAIL
AFTER REMOVAL OF TEMPORARY PIPE

NOT TO SCALE

NOTES:

1. REMOVE WIRE AND NYLON TWINE FROM BALL AND CANOPY.
2. SOAK ROOT BALL AND PLANT PIT IMMEDIATELY AFTER INSTALLATION.
3. STAKING IS REQUIRED FOR ALL TREES IN R.O.W. OR UPON REQUEST OF ARBORIST.
4. REMOVE EXCESS SOIL FROM SITE AND DISPOSE OF IN A LEGAL MANNER.
5. RESEED UNMULCHED, DISTURBED AREAS.



ALL TREES SHALL MEET AMERICAN STANDARD FOR NURSERY STOCK (ANSI Z60.1-2004)
FOR EXAMPLE:

CALIPER	HEIGHT (RANGE)	MAX. HEIGHT	MIN. ROOT BALL DIA.	MIN. ROOT BALL DEPTH
2"	12-14'	16'	24"	16"
3"	14-16'	18'	32"	21"

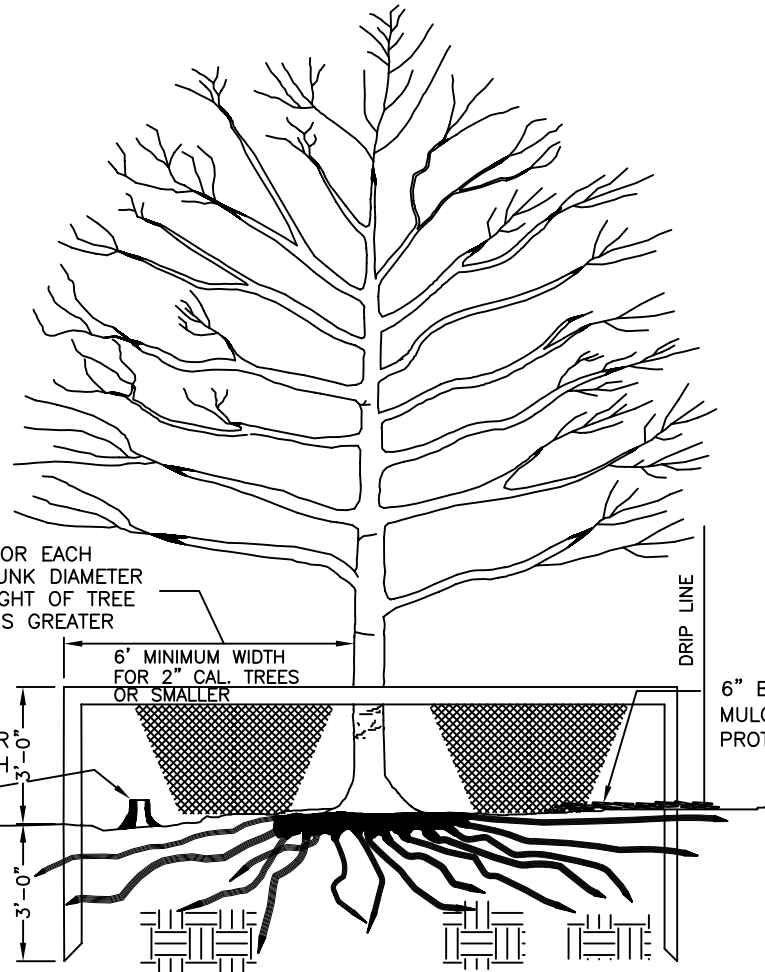
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A circular diagram showing a cross-section of a tree trunk and a limb. An arrow points to a diagonal cut made at the base of the limb, labeled "TREE LIMB REMOVAL".

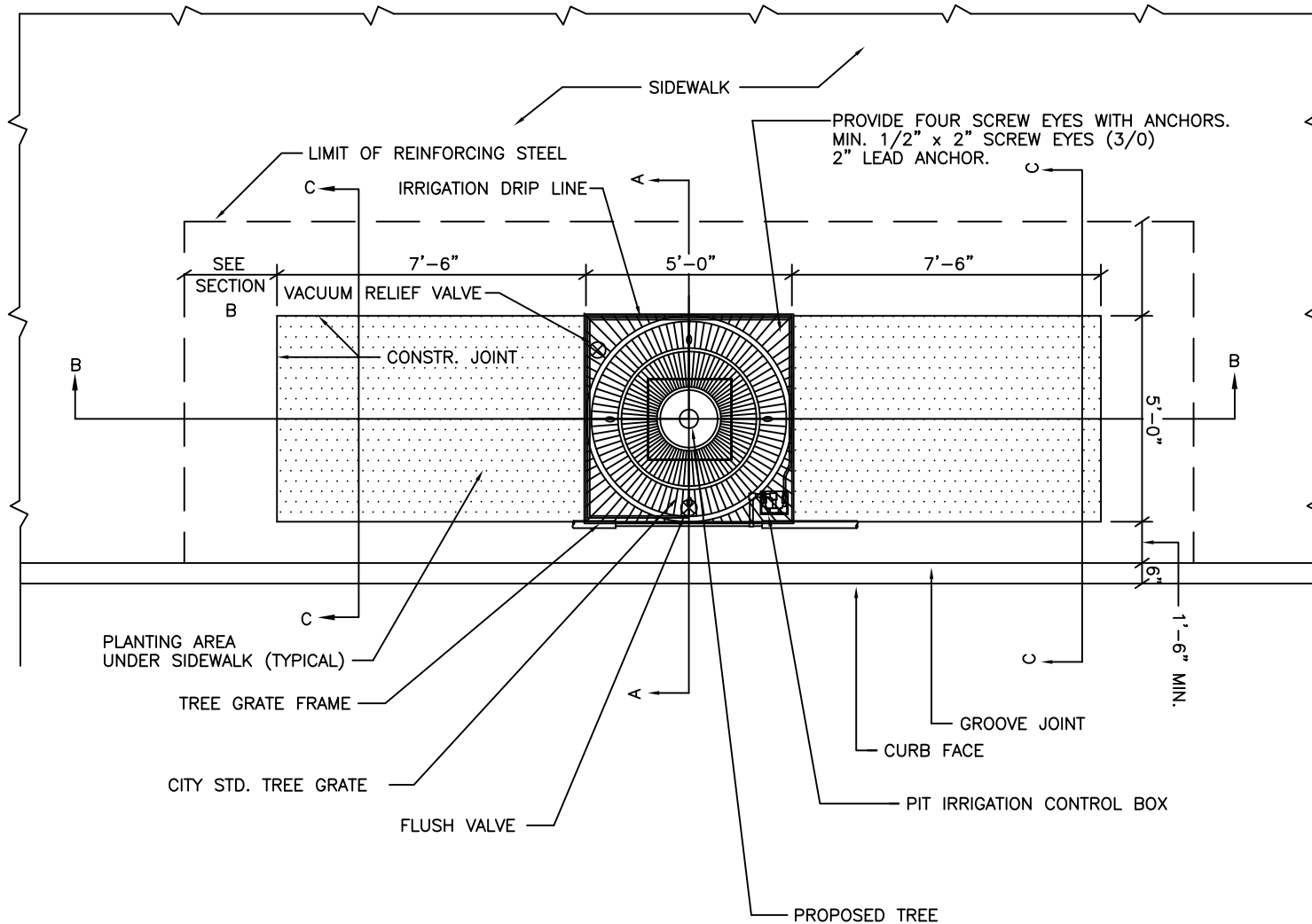
DEAD TREES AND SCRUB OR UNDERGROWTH SHALL BE CUT FLUSH WITH ADJACENT GRADE. NO GRUBBING-ALLOWED UNDER DRIP LINE.

ONE FOOT FOR EACH
INCH OF TRUNK DIAMETER
OR 1/2 HEIGHT OF TREE
WHICHEVER IS GREATER

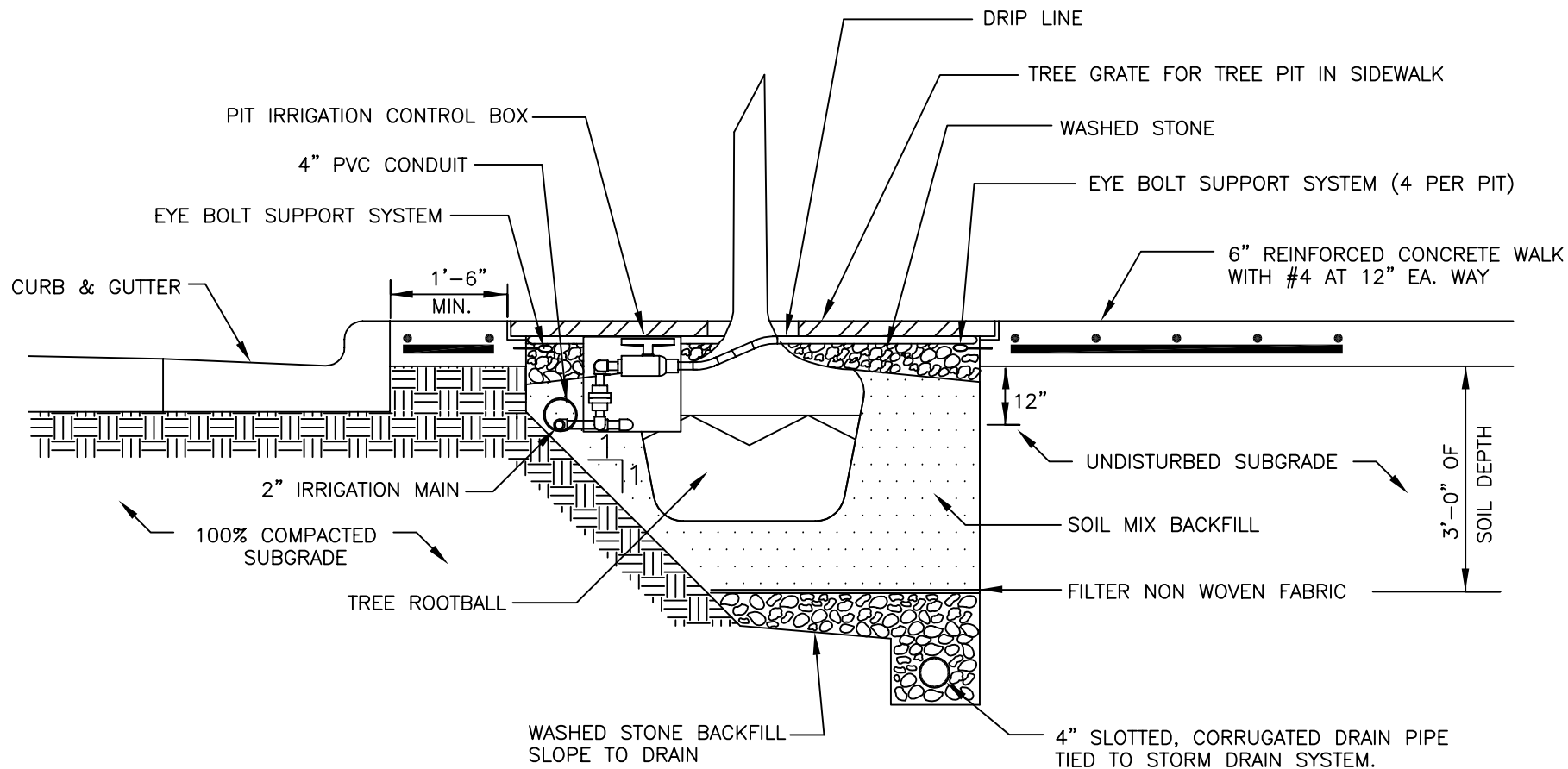
1. REMOVE ALL BARRIERS UPON COMPLETION OF PROJECT.
2. LANDSCAPING PLANS SHALL SHOW THE LOCATIONS OF ALL TREE PROTECTION FENCES.
3. REFER TO CITY OF CHARLOTTE LANDSCAPE CONSTRUCTION STANDARDS SECTION 01000 FOR GENERAL SPECIFICATION REGARDING TREE PROTECTION.



NOT TO SCALE



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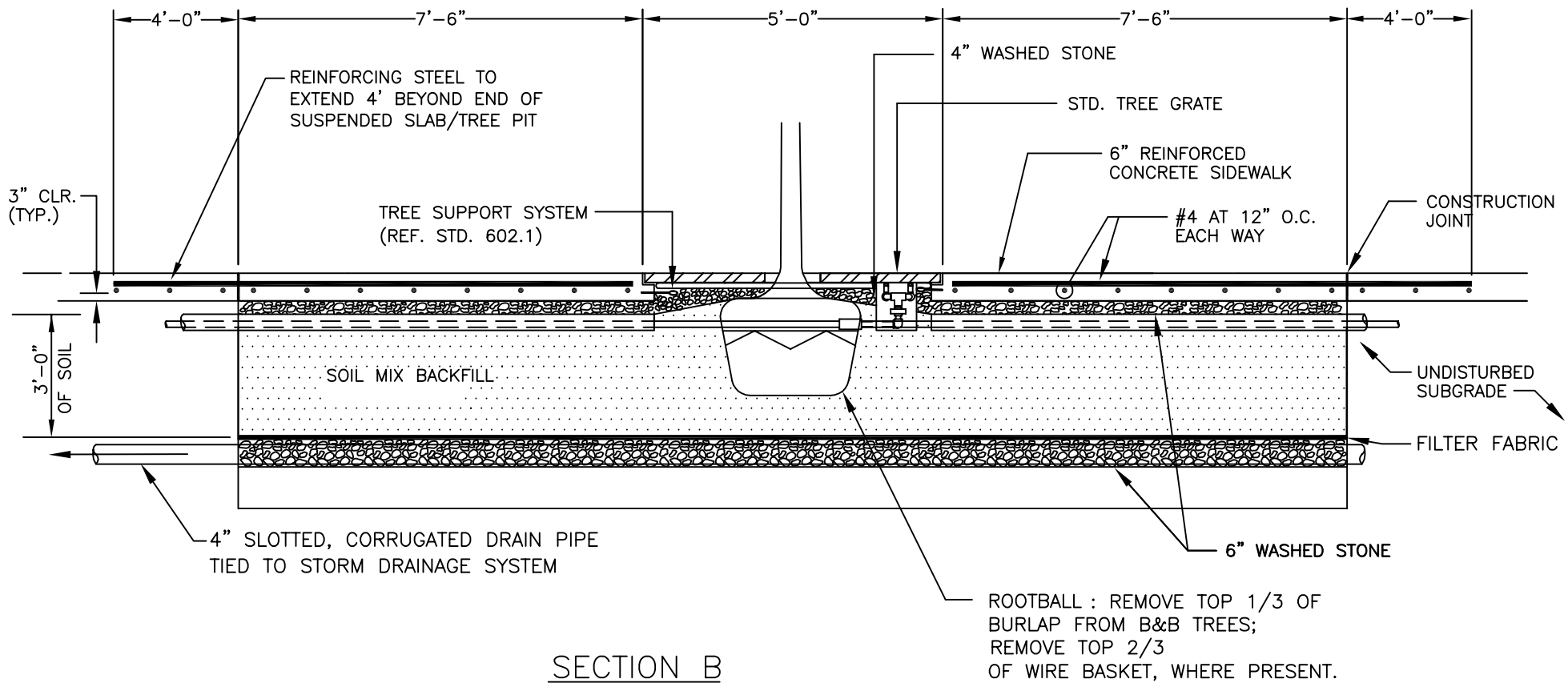


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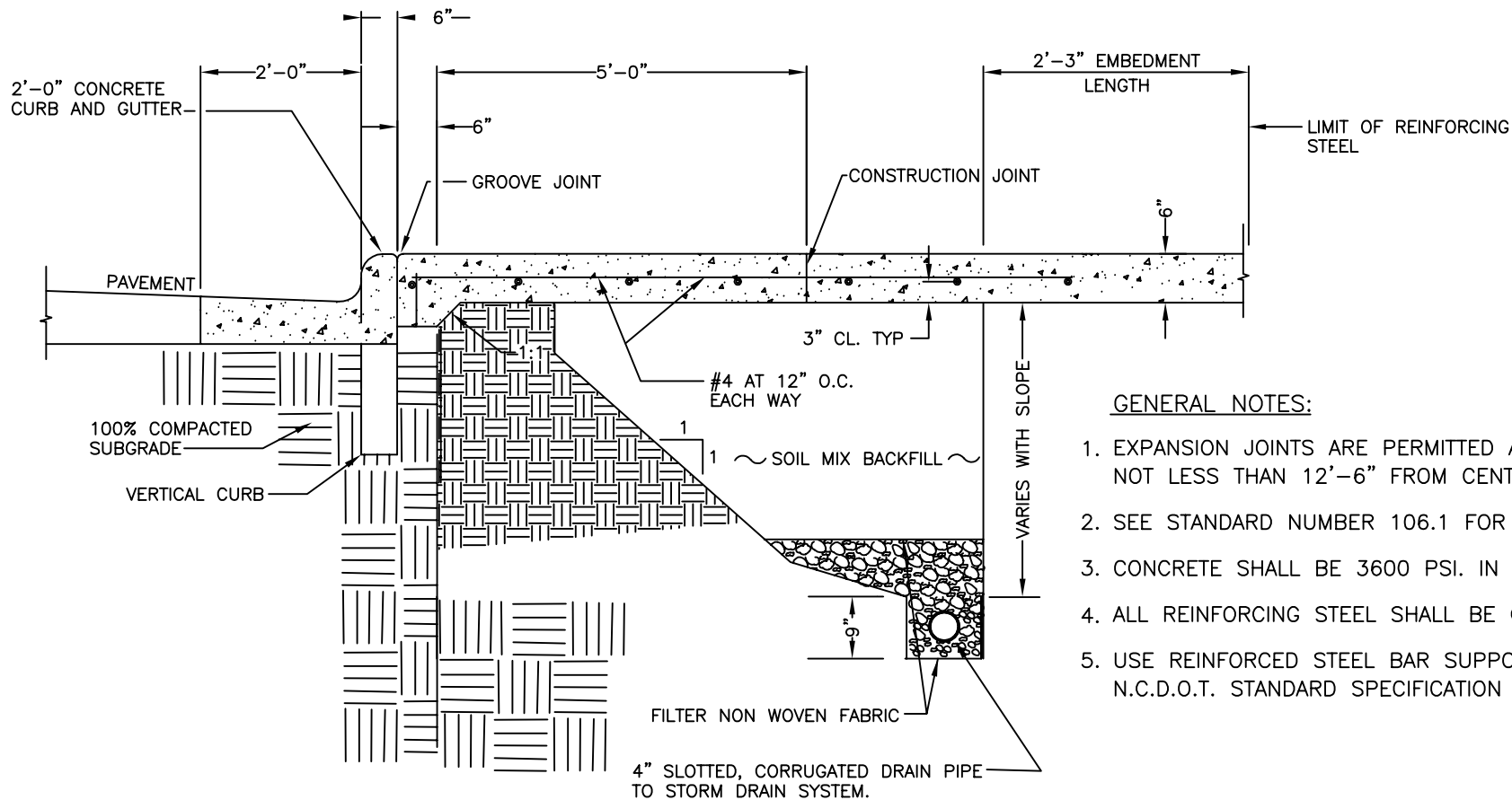
1. IRRIGATION SHOWN IS OPTIONAL.

SECTION A

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NOT TO SCALE



GENERAL NOTES:

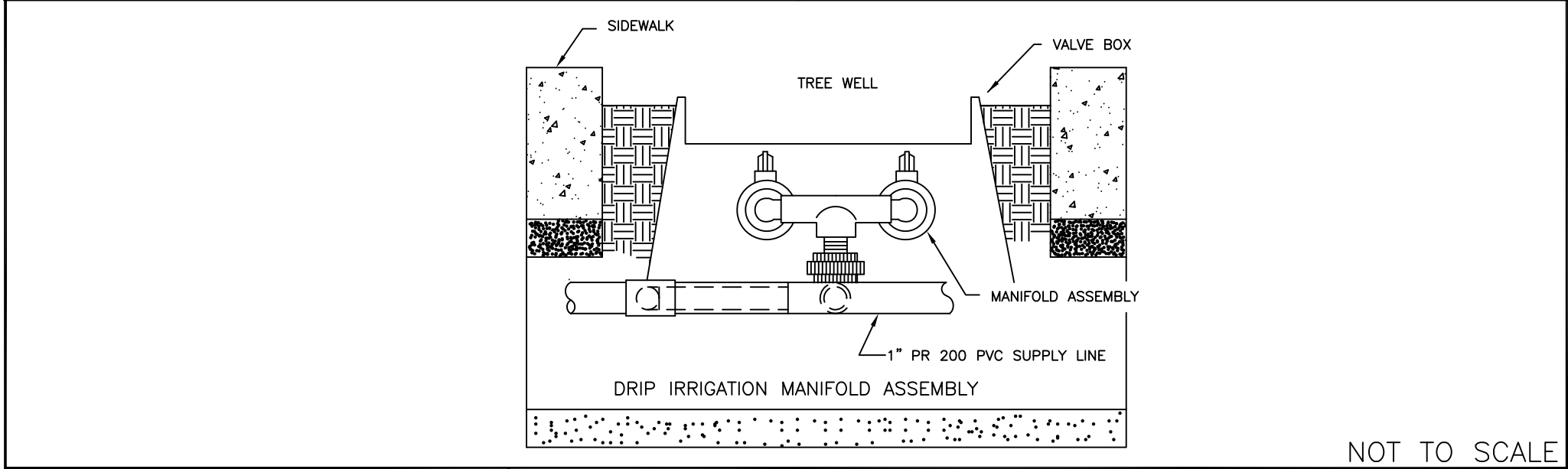
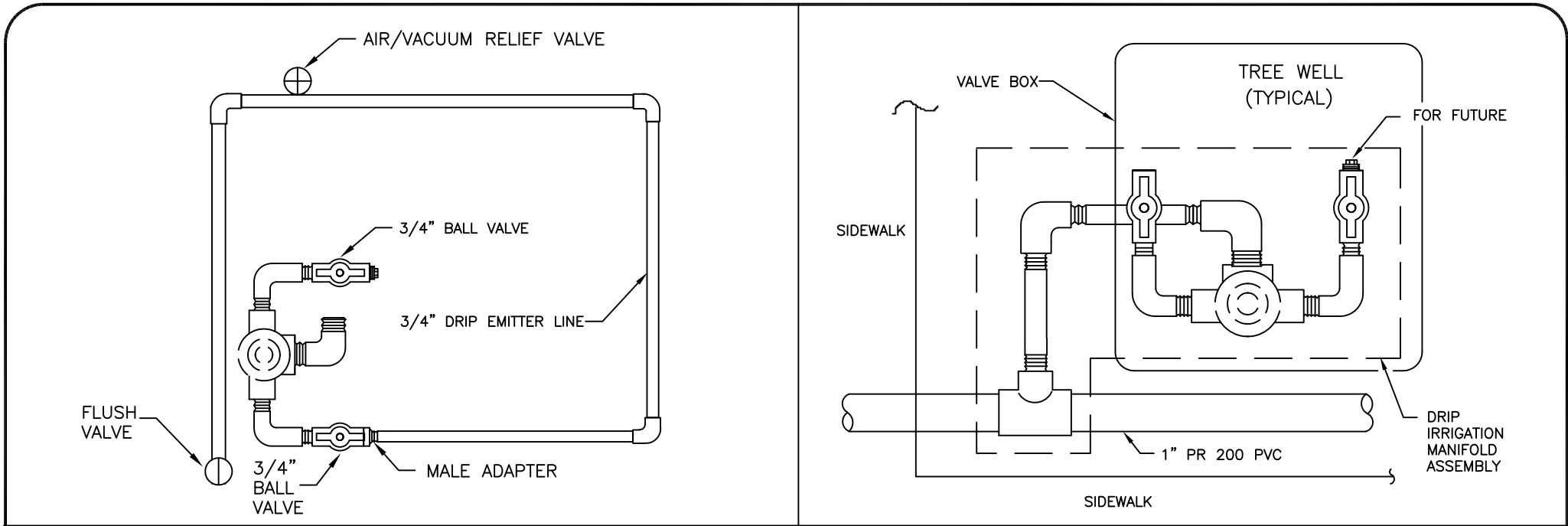
1. EXPANSION JOINTS ARE PERMITTED AT 40' MIN. SPACING AND NOT LESS THAN 12'-6" FROM CENTER OF TREE GRATE.
2. SEE STANDARD NUMBER 106.1 FOR DETAIL OF GROOVE JOINT.
3. CONCRETE SHALL BE 3600 PSI. IN 28 DAYS.
4. ALL REINFORCING STEEL SHALL BE GRADE 60.
5. USE REINFORCED STEEL BAR SUPPORTS IN COMPLIANCE WITH N.C.D.O.T. STANDARD SPECIFICATION 970-4.

SECTION C

NOTE

A DRAINAGE SYSTEM IS REQUIRED AS SHOWN FOR ALL IRRIGATED PLANTING AREAS LOCATED ADJACENT TO STREET.

NOT TO SCALE

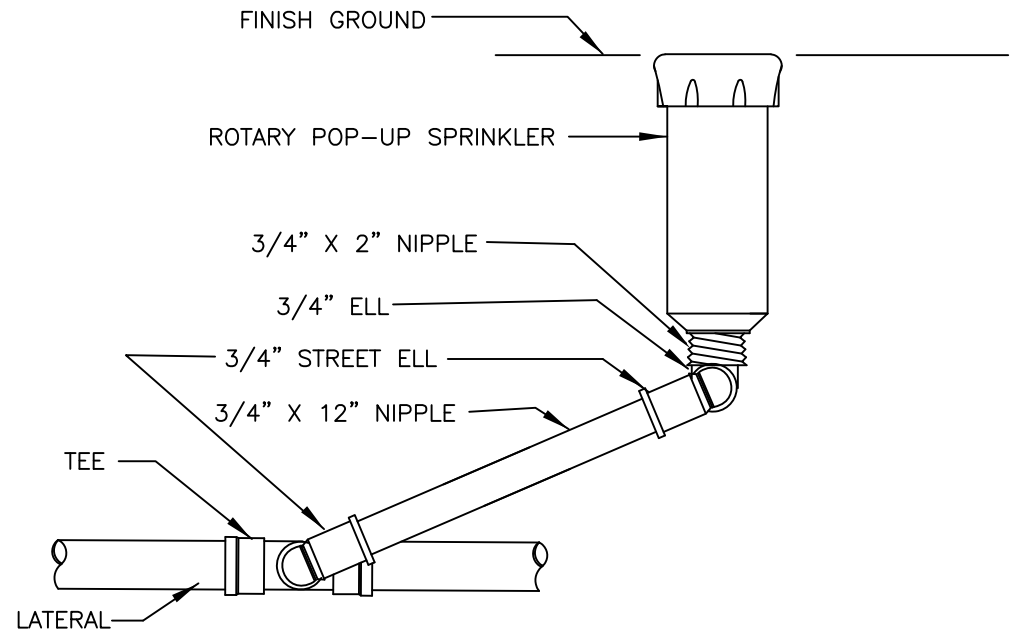
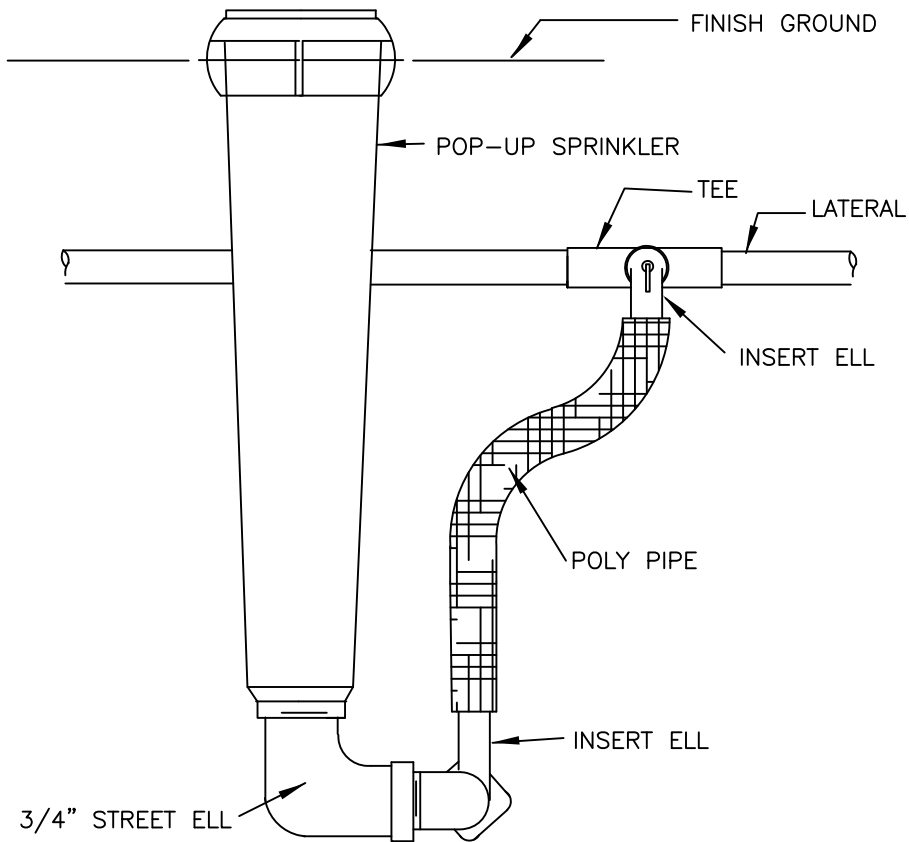


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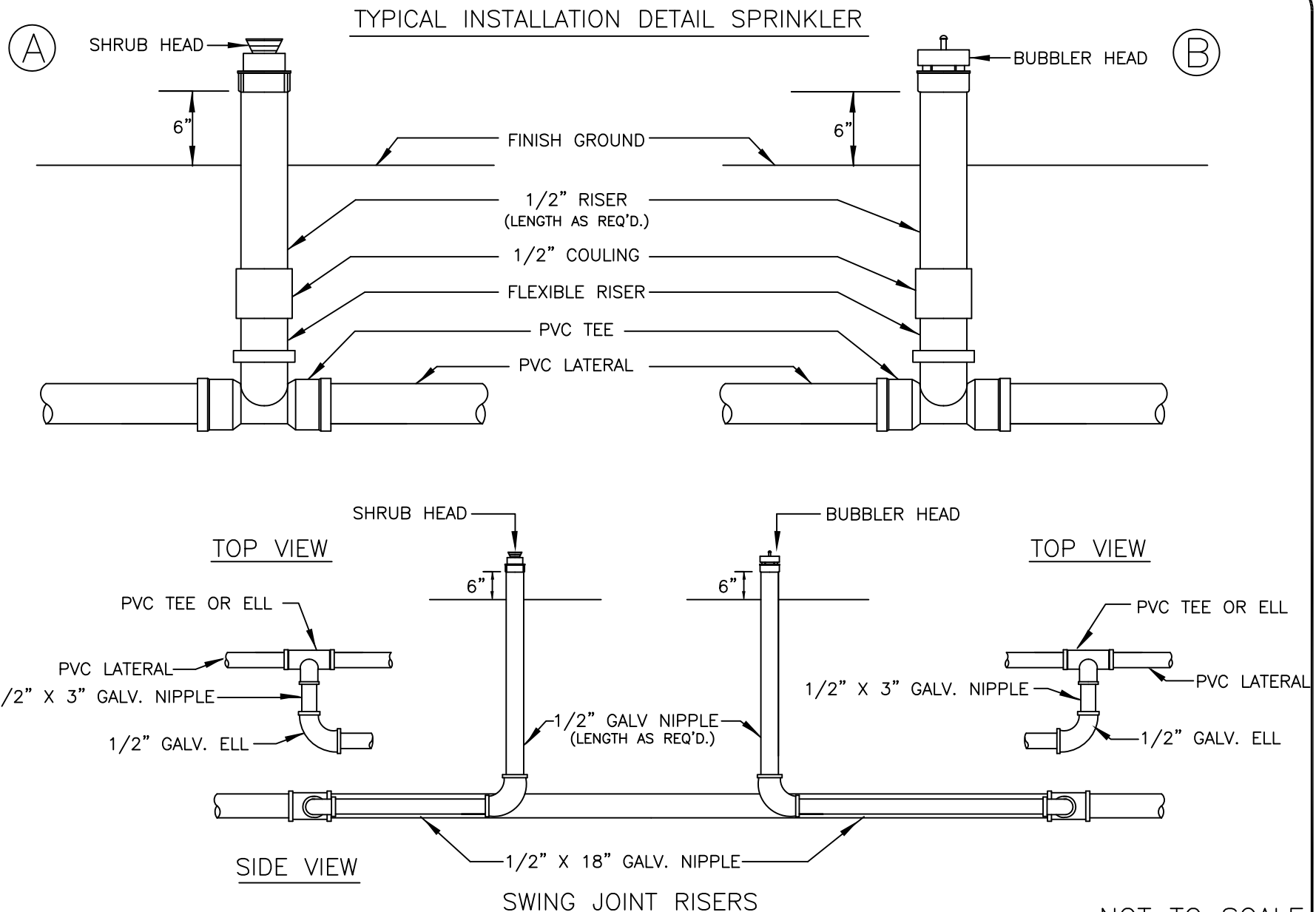
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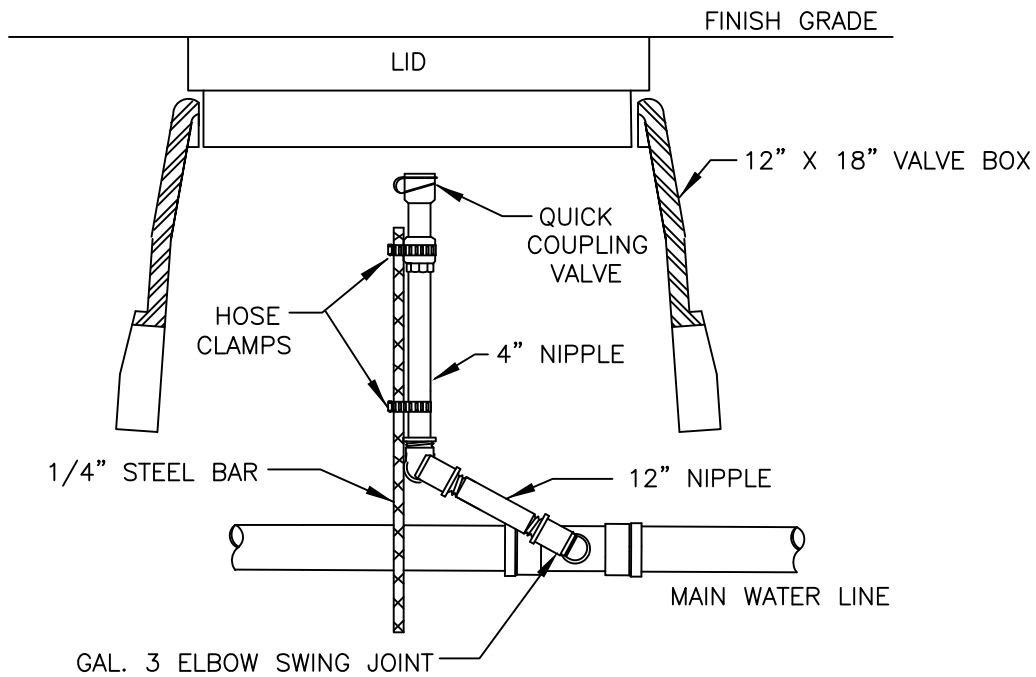
TYPICAL INSTALLATION DETAIL SPRINKLER

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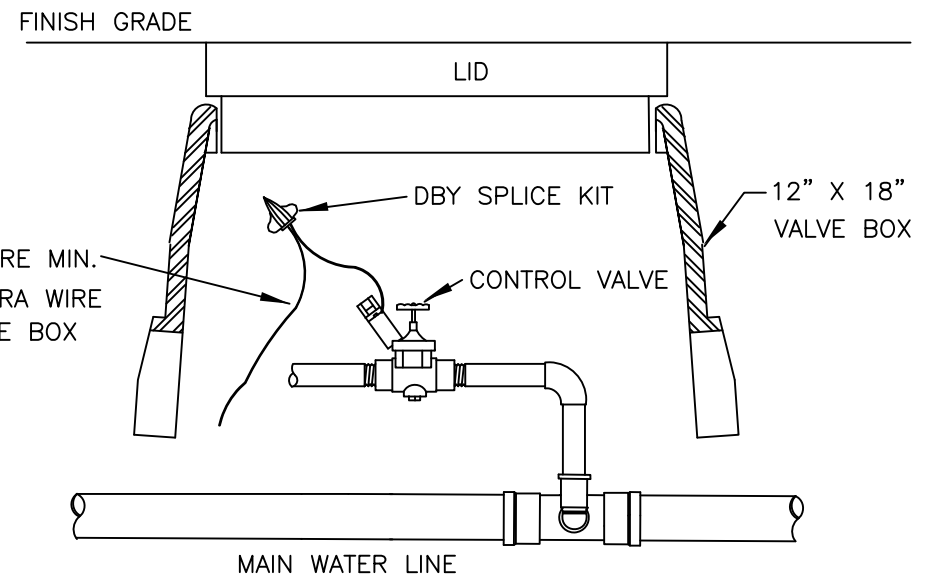


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QUICK COUPLING VALVE

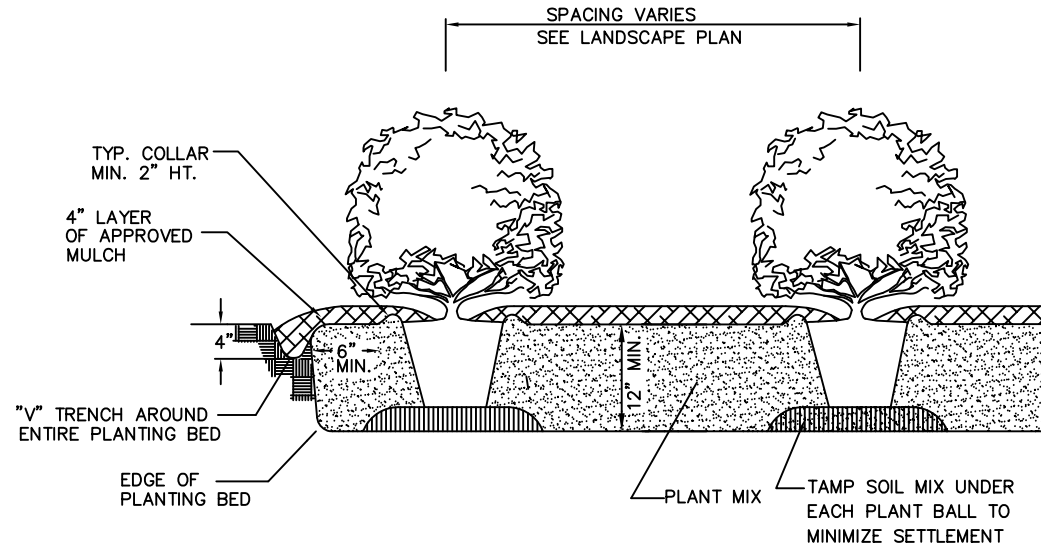


CONTROL VALVE

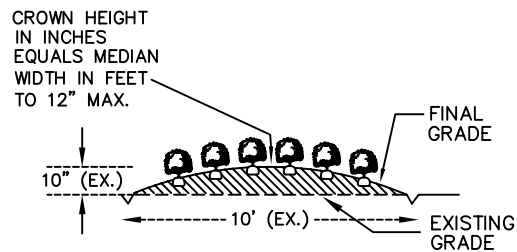
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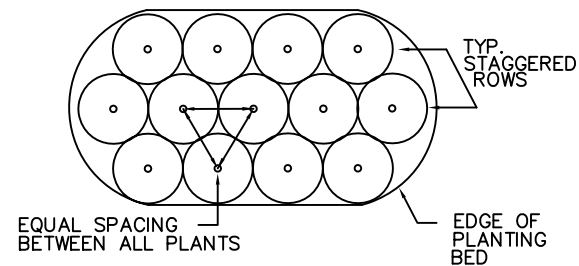
1. SCARIFY ROOT MASS OF CONTAINERIZED PLANT MATERIAL.
2. INSTALL CONTAINERIZED PLANTS AT FINISHED GRADE
3. TAMP PLANTING MIX FIRMLY AS PIT IS FILLED AROUND EACH PLANT BALL.
4. OMIT COLLAR AROUND EACH SHRUB WHEN IRRIGATION SYSTEM IS PRESENT.
5. SOAK EACH PLANT BALL AND PIT IMMEDIATELY AFTER INSTALLATION.



TYPICAL PLANTING BED DETAIL



TYPICAL BED CROWNING

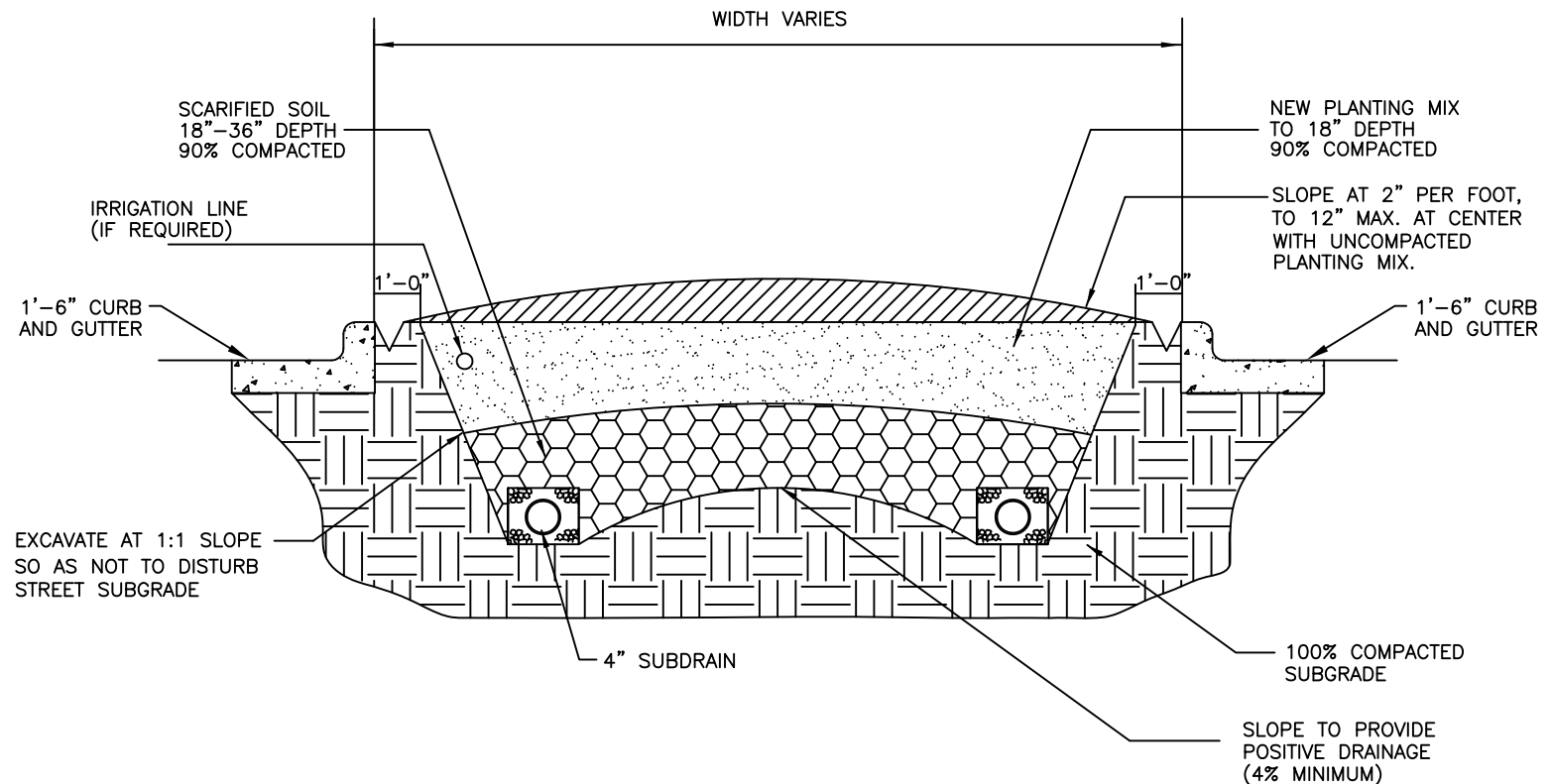


TYPICAL PLANTING BED PLAN

NOT TO SCALE

NOTES:

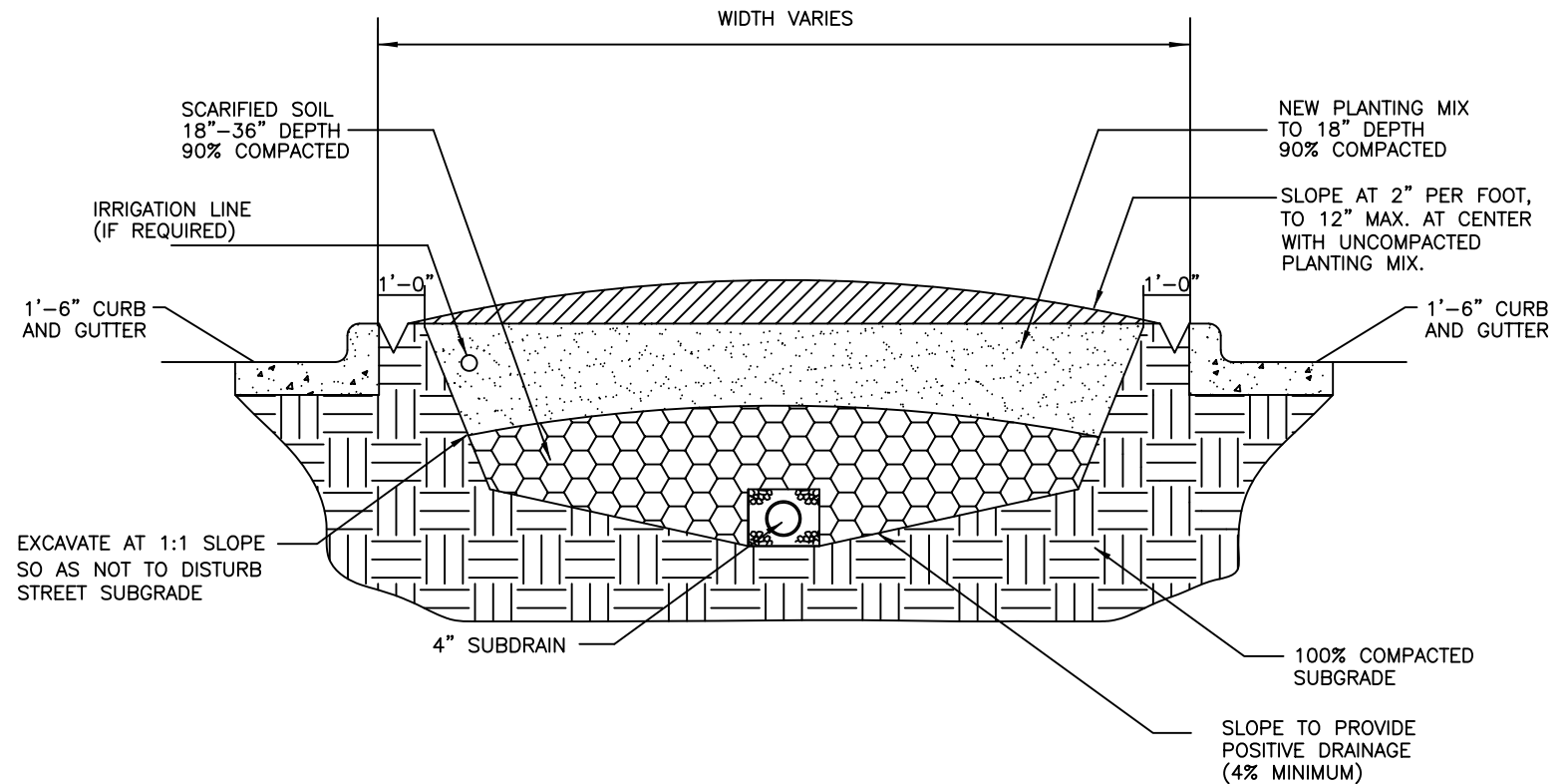
1. FOR NEW PLANTING AREAS, REMOVE ALL PAVEMENT, GRAVEL, SUB-BASE AND CONSTRUCTION DEBRIS BEFORE PREPARING SOIL AND PLANTING TREES.
2. REMOVE SOIL TO A DEPTH OF 18". SCARIFY, TILL OR OTHERWISE LOOSEN THE REMAINING SOIL TO A DEPTH OF 18". ADD NEW PLANTING MIX AS SPECIFIED.
3. SUBSURFACE DRAINAGE SHALL BE INSTALLED IN ALL MEDIANS AND TIED INTO EXISTING STORM DRAIN SYSTEM. A 4 INCH PERFORATED, CORRUGATED PVC DRAIN SHALL BE INSTALLED IN EACH MEDIAN AT THE BOTTOM OF THE EXCAVATED AREA. DRAIN SHALL BE COVERED WITH A MINIMUM 6 INCHES OF #57 WASHED STONE, THEN WRAPPED WITH A SPECIFIED NON-WOVEN GEOTEXTILE FABRIC. SPECIAL CARE SHALL BE EXERCISED WHEN FILLING MEDIANS WITH SOIL SO NOT TO CRUSH OR DAMAGE THE DRAINAGE SYSTEM.



NOT TO SCALE

NOTES:

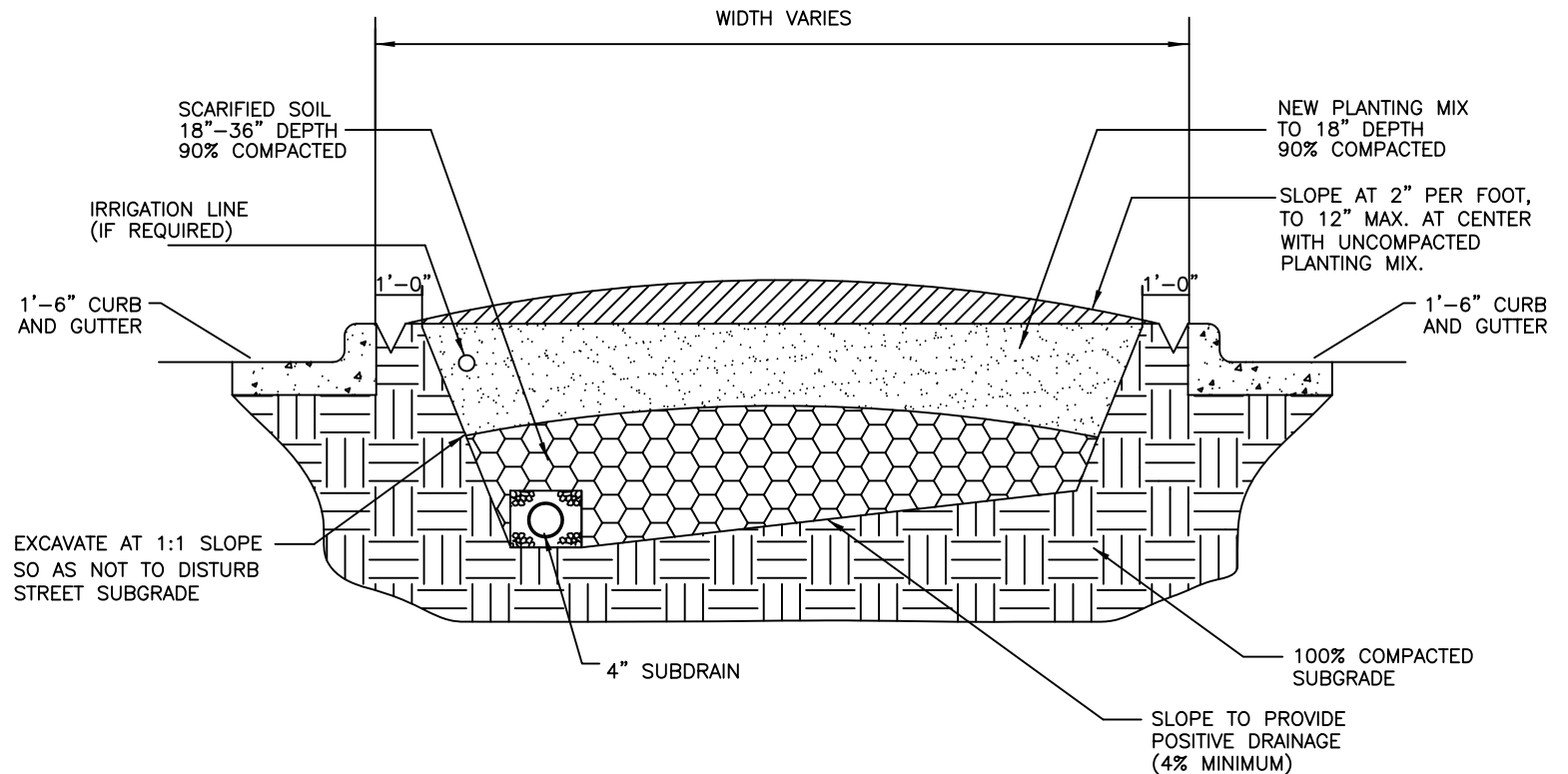
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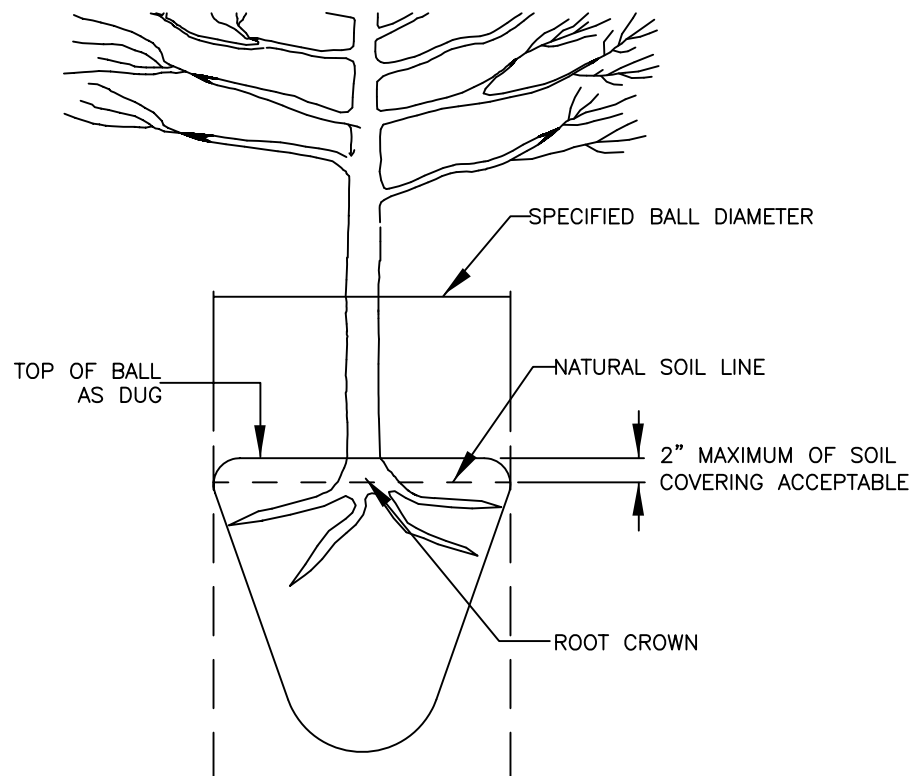
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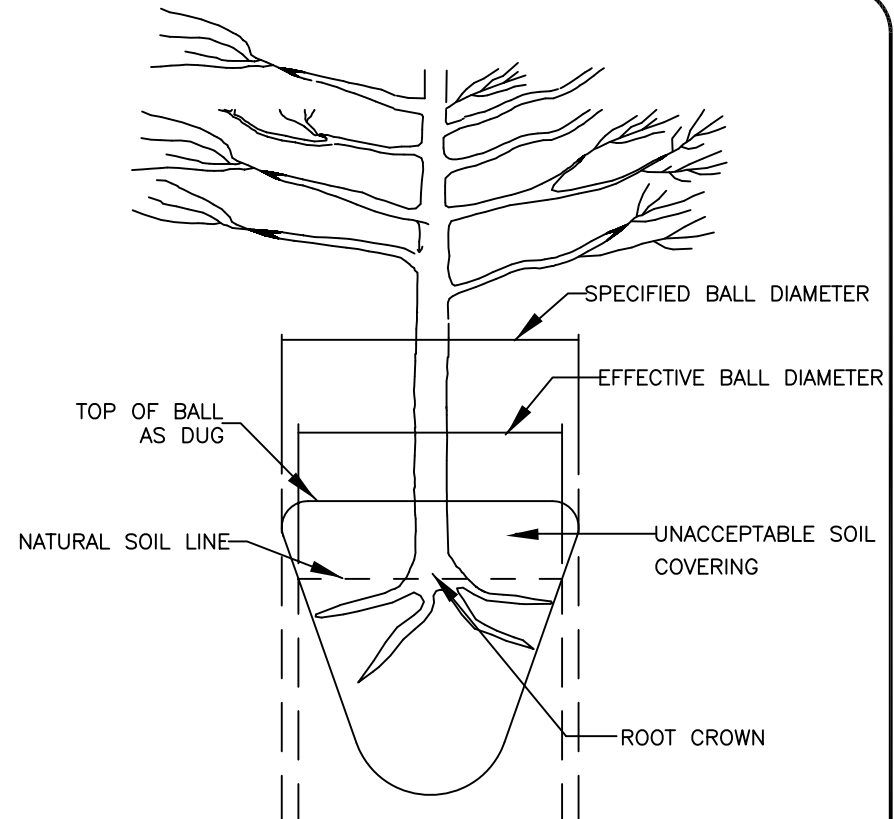
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NOT TO SCALE



ACCEPTABLE CONDITION
(AS DELIVERED)



UNACCEPTABLE CONDITION
(AS DELIVERED)

NOTE:

A ROOT COLLAR EXCAVATION FOR ALL TREES SPECIFIED WILL BE INSPECTED BY THE INSPECTOR TO ENSURE THAT TREES WERE NOT PLANTED/GROWN TOO DEEPLY AT SOURCE (NURSERY). LANDSCAPE CONTRACTOR SHALL HAVE SUPPLIER MARK GROUND LEVEL LINE ABOVE ROOT BALL. IF INSPECTOR DETERMINES THAT THERE IS EXCESSIVE SOIL OVER THE ROOT CROWN, THESE TREES WILL BE REJECTED.

NOT TO SCALE

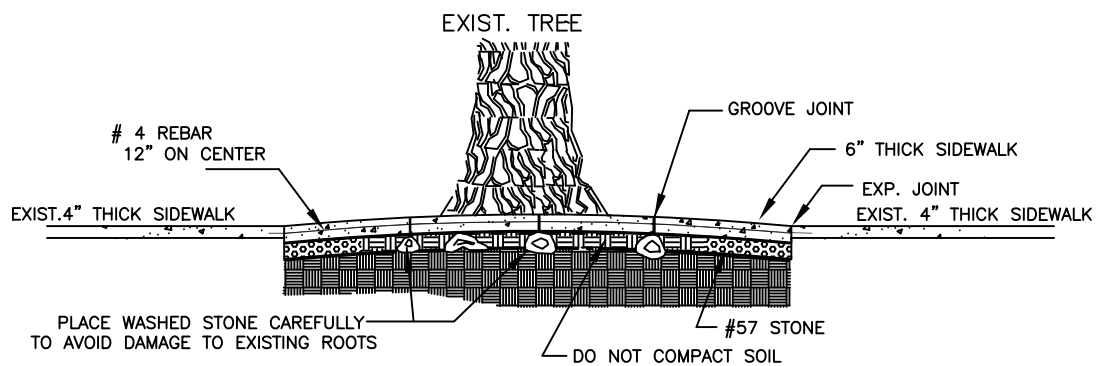
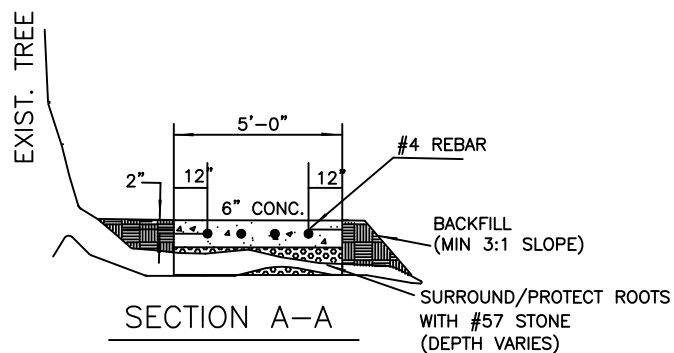
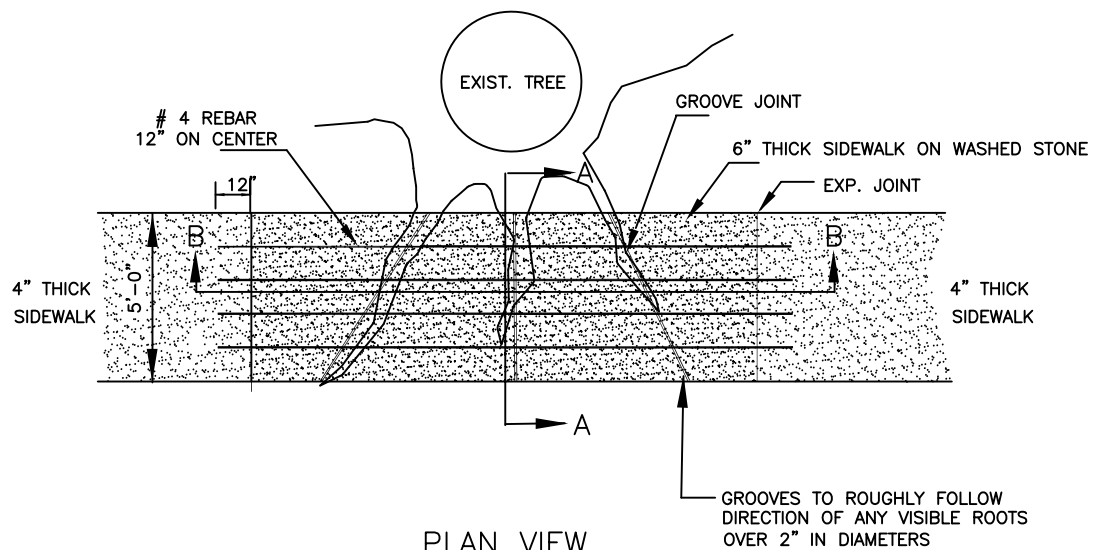
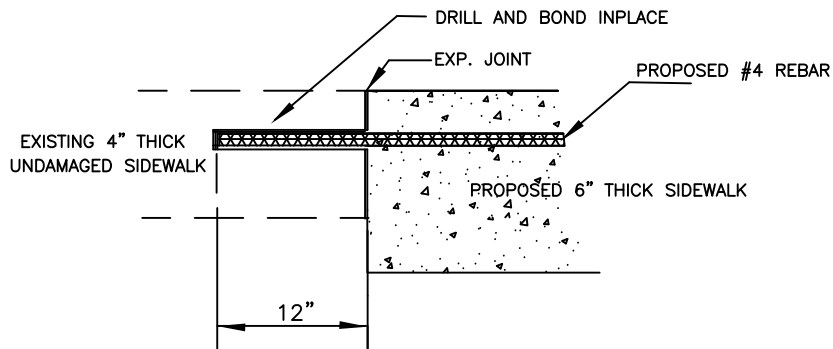
PLANTINGS IN STREET RIGHT-OF-WAY

GENERAL NOTES

1. TREE GRATES AND ASSOCIATED IRRIGATION SYSTEMS REQUIRED BY TOWN ARE TO COMPLY WITH THE TOWN ZONING REQUIREMENTS. ALL INSTALLATIONS OF IRRIGATION SYSTEMS WITHIN THE RIGHT-OF-WAY OF STATE MAINTAINED STREETS REQUIRE AN ENCROACHMENT AGREEMENT EXECUTED THROUGH NCDOT. TREES INSTALLED WITHIN TOWN MAINTAINED RIGHT-OF-WAY LIMITS SHALL BE PER TOWN REQUIREMENTS. TREES INSTALLED WITHIN THE RIGHT OF WAY OF STATE MAINTAINED STREETS REQUIRE A PLANTING PERMIT FROM NCDOT. NCDOT'S REVIEW AND APPROVAL PROCESS MAY INCLUDE ADDITIONAL REQUIREMENTS.
2. A DRAINAGE SYSTEM IS REQUIRED FOR ALL IRRIGATED PLANTING AREAS LOCATED ADJACENT TO STREETS. ALL IRRIGATION/DRAINAGE SYSTEMS REQUIRE AN ENCROACHMENT AGREEMENT EXECUTED BY NCDOT FOR STATE-MAINTAINED ROADS. CONTACT NCDOT FOR ADDITIONAL INFORMATION REGARDING COST, SUBMITTAL AND LIABILITY INSURANCE COVERAGE REQUIREMENTS.
3. LANDSCAPE INSPECTIONS MAY INCLUDE THE FOLLOWING:
 - SUBDRAINAGE INSPECTION
 - TREE PIT/WELL OR PLANTING STRIP INSPECTION
 - SOIL MIX APPROVALS/INSPECTIONS
 - TREE APPROVALS/INSPECTIONS
 - TREE PLANTING INSPECTION
 - IRRIGATION INSPECTION
 - FINAL WALK THROUGH

ALL INSPECTIONS SHALL BE PERFORMED BY TOWN AND/OR TOWN REPRESENTATIVE.

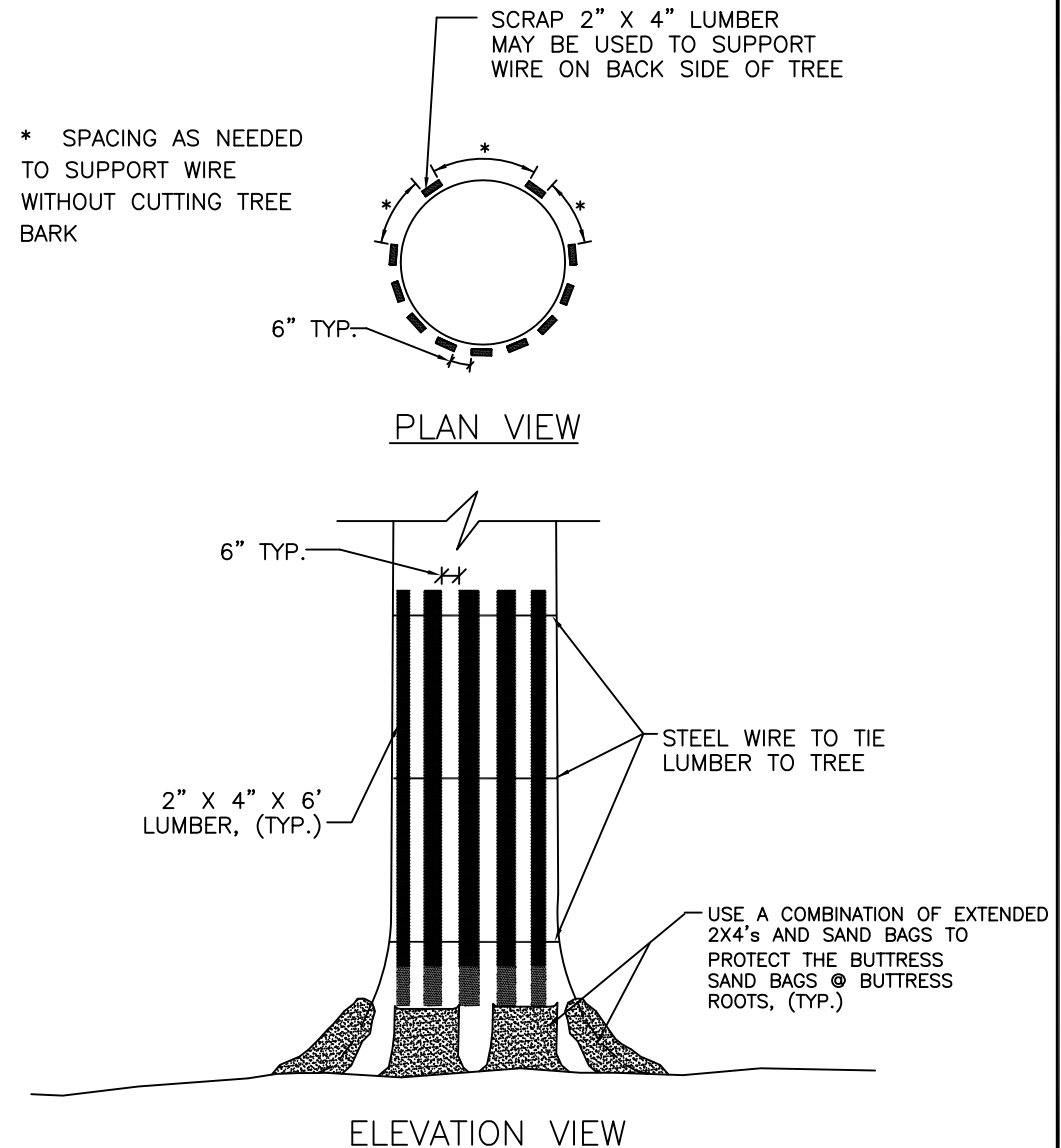
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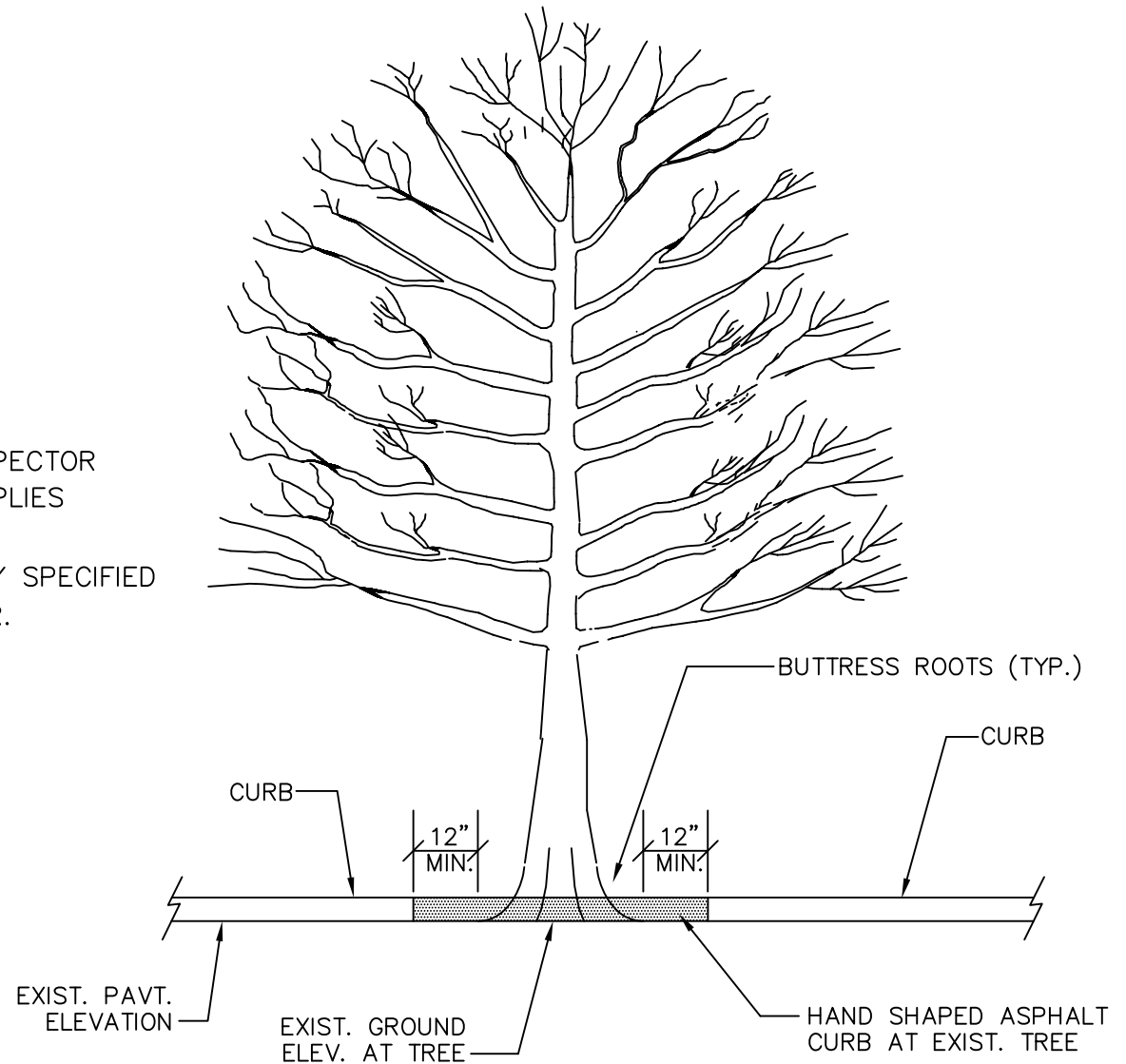
1. THIS TREE BUMPER DETAIL SHALL BE USED WHEN WORKING WITHIN 10' OF AN EXISTING TREE TO BE PROTECTED.
2. ALL TREES SHALL BE SAVED UNLESS NOTED OTHERWISE ON THE PLANS OR DIRECTED BY THE ENGINEER.
3. LUMBER, WIRE, AND SANDBAGS MAY BE REUSED FOR OTHER TREES.
4. THE INTENT OF THIS DETAIL IS TO PROTECT EXISTING TREES FROM DAMAGE DURING CONSTRUCTION ESPECIALLY FROM BACKHOE ARM SWING. AN ALTERNATE APPROACH MAY BE USED IF APPROVED IN WRITING BY THE ENGINEER AFTER CONSULTATION WITH THE TOWN ARBORIST OR HIS DULY AUTHORIZED REPRESENTATIVE.



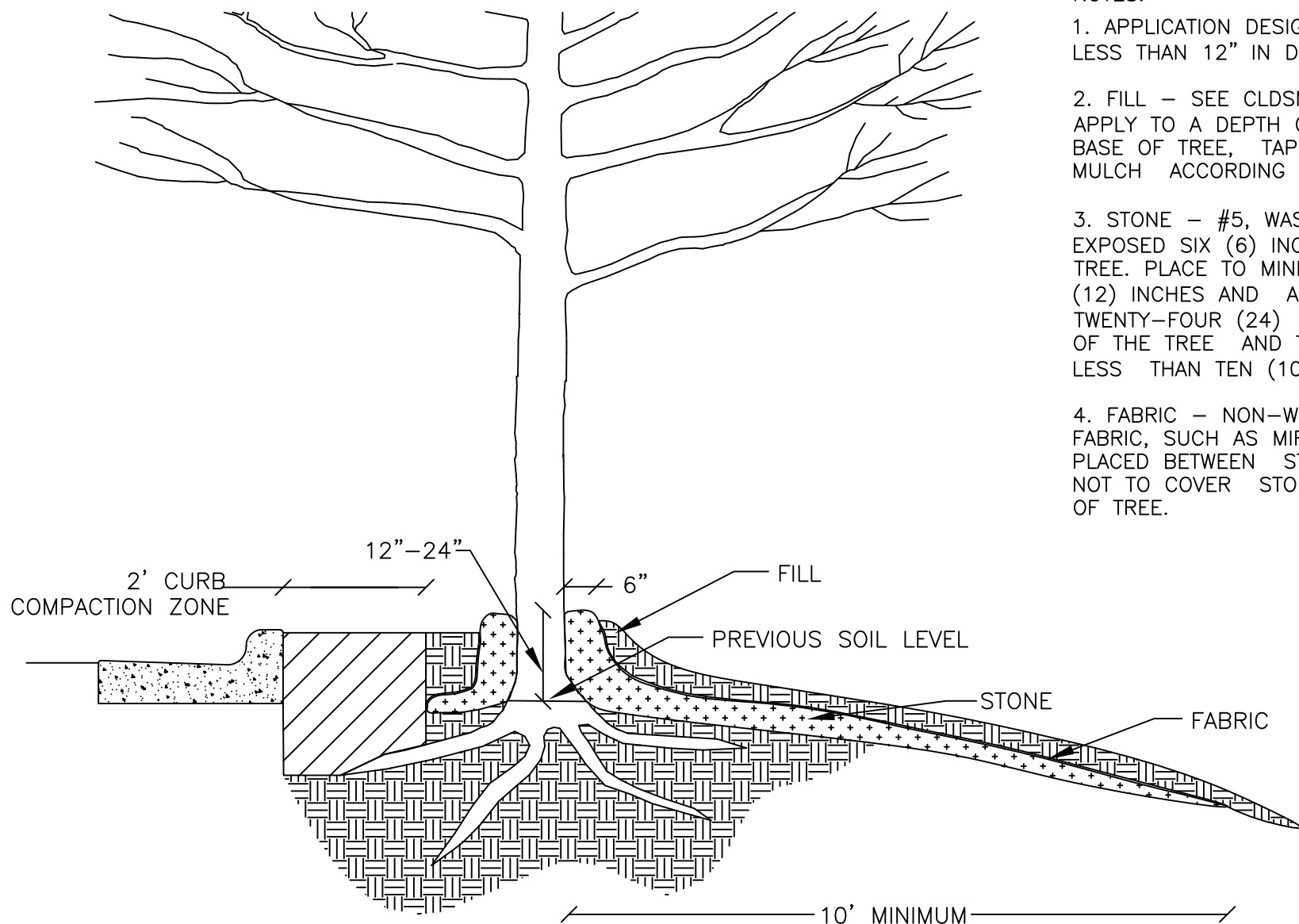
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NOTES:

1. CONTRACTOR SHALL USE EXTREME CAUTION WHEN WORKING NEAR EXISTING TREES.
2. WHERE EXISTING TREES ARE WITHIN 4' OF THE PROPOSED BACK OF CURB, THE PROPOSED CURB SHALL END A MINIMUM OF 12" FROM THE TREE'S BUTTRESS ROOTS.
3. CONTRACTOR SHALL COORDINATE WITH THE INSPECTOR TO IDENTIFY TREES FOR WHICH THIS DETAIL APPLIES PRIOR TO CONSTRUCTION NEAR THE TREE(S).
4. NO TREES SHALL BE REMOVED UNLESS CLEARLY SPECIFIED ON THE PLANS OR IDENTIFIED BY THE ENGINEER.
5. AVOID FILL PLACEMENT NEAR TREE.



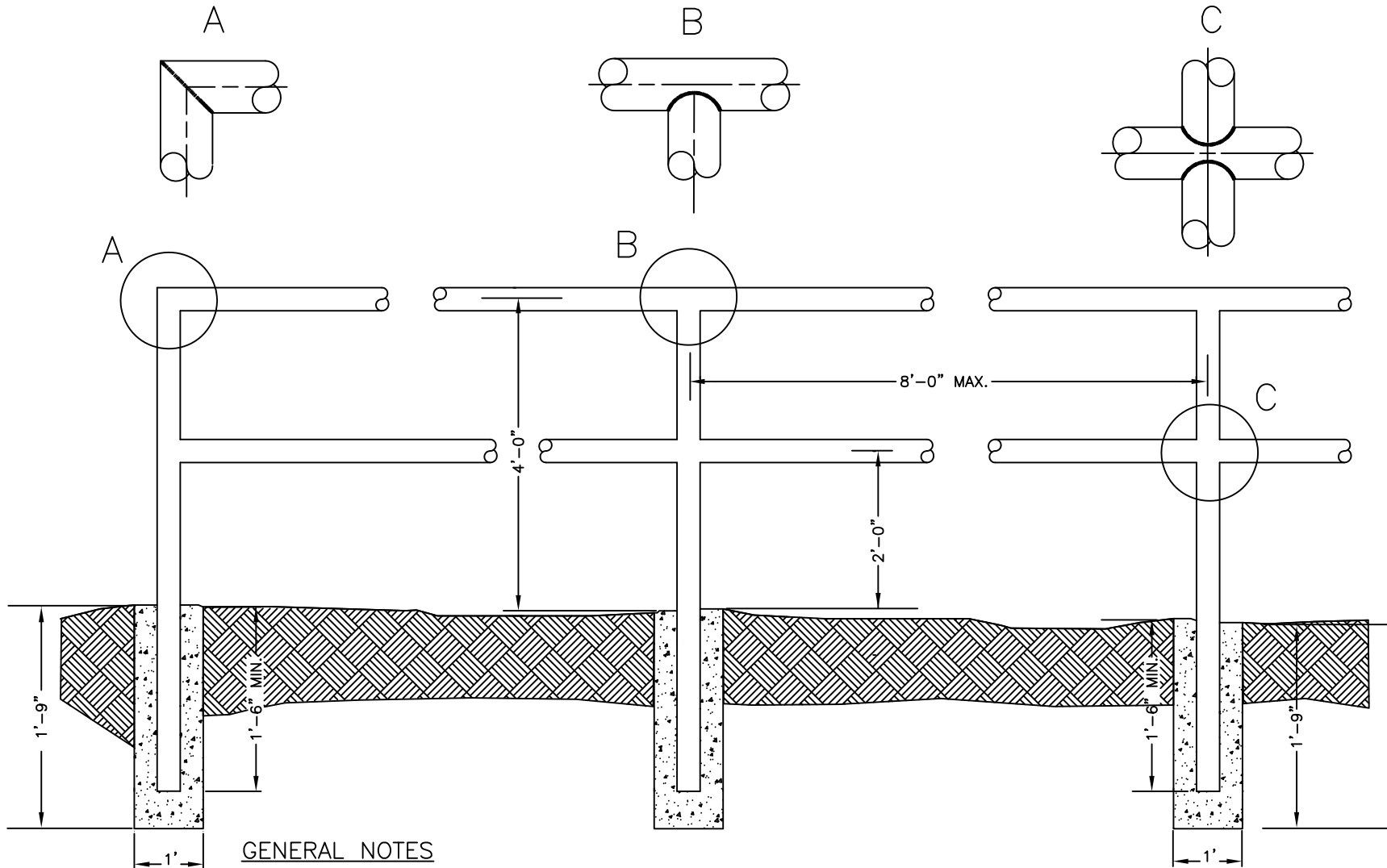
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NOTES:

1. APPLICATION DESIGNED FOR TREES NO LESS THAN 12" IN DIAMETER.
2. FILL — SEE CLDSM — PLANTING MIX. APPLY TO A DEPTH OF FOUR (4) INCHES AT BASE OF TREE, TAPER TO GRADE. SEED AND MULCH ACCORDING TO CLDSM.
3. STONE — #5, WASHED. MAINTAIN EXPOSED SIX (6) INCH WIDTH AT TRUNK OF TREE. PLACE TO MINIMUM DEPTH OF TWELVE (12) INCHES AND A MAXIMUM OF TWENTY-FOUR (24) INCHES AT THE BASE OF THE TREE AND TAPER OUTWARD TO NO LESS THAN TEN (10) FEET.
4. FABRIC — NON-WOVEN GEOTEXTILE FABRIC, SUCH AS MIRAFI OR EQUIVALENT, PLACED BETWEEN STONE AND FILL. IT IS NOT TO COVER STONE EXPOSED AT TRUNK OF TREE.

NOT TO SCALE



GENERAL NOTES

1. ALL CONCRETE TO BE 3600 PSI COMPRESSIVE STRENGTH.
2. TYPE OF PIPE TO BE USED IS 1- $\frac{5}{8}$ INCH MAXIMUM OUTER DIAMETER BLACK IRON, LOW CARBON PIPE OR GALVANIZED.
3. ALL JOINTS TO HAVE A $\frac{1}{2}$ INCH FILLED WELD AT ALL JOINTS.
4. AFTER INSTALLATION, PAINT ASSEMBLY WITH BLACK ALL WEATHER ENAMEL.
5. ALTERNATIVE HANDRAIL DESIGNS TO BE SUBMITTED TO THE TOWN ENGINEER FOR REVIEW.
6. REFER TO DETAIL 701.1 FOR WARRANTIES.

NOT TO SCALE

WARRANTS

HANDRAIL SHALL BE INSTALLED UNDER ANY OF THE FOLLOWING CIRCUMSTANCES IN BOTH NEW CONSTRUCTION AND IN RETROFITTING OR RECONSTRUCTION OF EXISTING ROADWAYS OR SITES:

1. WHEN THE CULVERT-CROSSING DETAIL (STD. 133.1 & 134.1) APPLIES.
2. IN ANY OF THE FOLLOWING COMBINATIONS OF DROPOFF AND OFFSET FROM SIDEWALK:
 - a. 18" OR LARGER DROPOFF WITHIN 2 FEET OF THE EDGE OF THE SIDEWALK
 - b. 36" OR LARGER DROPOFF WITHIN 4 FEET OF THE EDGE OF THE SIDEWALK
 - c. 60" OR LARGER DROPOFF WITHIN 6 FEET OF THE EDGE OF THE SIDEWALK

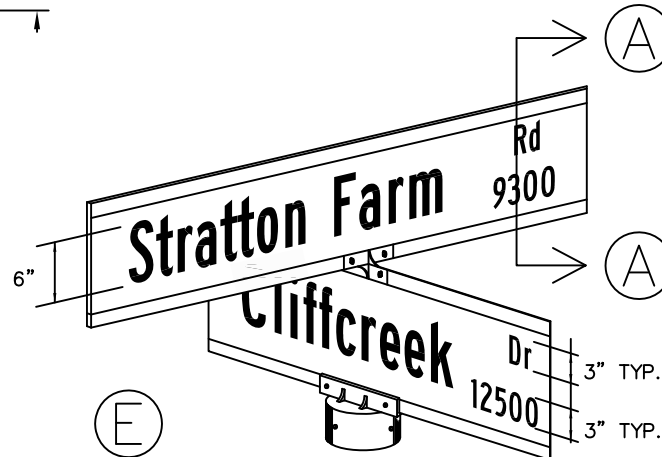
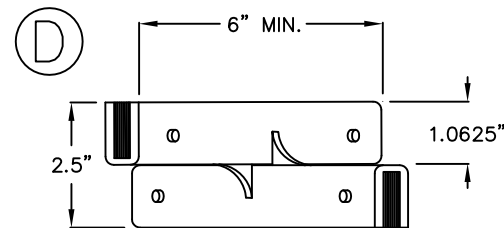
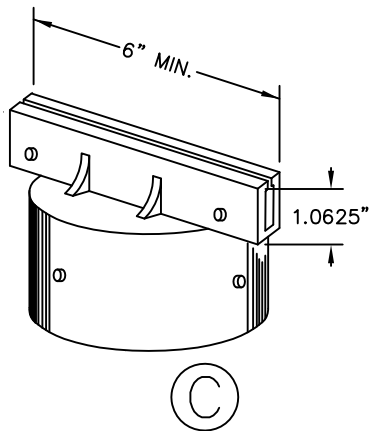
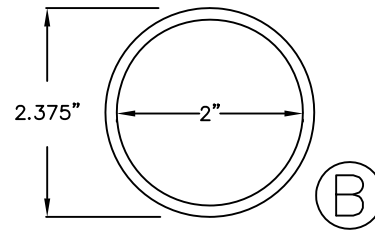
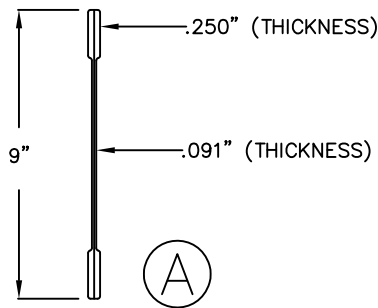
THESE CLEARANCES ASSUME THAT THE CROSS-SLOPE OF THE BERM BETWEEN THE SIDEWALK AND THE DROPOFF IS 6:1 OR FLATTER.

3. AT THE TOP OF ANY DROPOFF WHERE PEDESTRIANS CAN REASONABLY BE EXPECTED IN THE VICINITY.
4. AT THE DIRECTION OF TOWN ENGINEER BASED ON FIELD CONDITIONS.

FOR PURPOSES OF THIS STANDARD, THE TERM "SIDEWALK" IS USED GENERICALLY AND SHALL MEAN ANY SEPARATE PATH OR SURFACE TO BE USED FOR BICYCLE AND/OR PEDESTRIAN TRANSPORTATION. EXAMPLES INCLUDE, BUT ARE NOT LIMITED TO, SIDEWALKS, BIKE PATHS, SHARED-USE PATHS, PEDESTRIAN PATHS, AND GREENWAYS.

DEFINITIONS

— DROPOFF — A SLOPE OF 2:1 OR STEEPER. EXAMPLES INCLUDE HEADWALLS, RETAINING WALLS, AND CULVERTS, ETC.



NOTES:

1. BLADES SHALL BE EXTRUDED ALUMINUM 6063T5 OR 6063T6 ALLOY .080" THICK. (SEE DETAIL A) POST SHALL BE 10'-0" IN LENGTH, TUBULAR 2.375 O.D. GLOSS GALVANIZED STEEL CONTINUOUS MILL DIPPED, WITH NO RAW ENDS; OR 40, 1540 WALL ALUMINUM (SEE DETAIL B).
2. CAP TO BE ALUMINUM #380 ALLOY OR EQUAL SLOTTED FOR .25" EXTRUDED BLADE; 2.375" I.D. BASE, DIE CAST AND POLISHED. CAP SHALL BE TAPPED TO RECEIVE AND INCLUDE 3 STAINLESS STEEL SET SCREWS FOR POST MOUNTING AND 2 STAINLESS STEEL SET SCREWS FOR BLADE MOUNTING. SET SCREWS TO HAVE ALLEN HEADS (SEE DETAIL C).
3. BLADE SPACER BRACKET SHALL MEET SAME SPECIFICATIONS AS THE CAP WITH 2 SCREWS TO EACH BLADE MOUNTING (SEE DETAIL D).
4. THE FACE OF ALL BLADES SHALL HAVE WHITE LETTERS WITH GREEN BACKGROUND WITH RETROREFLECTIVE CHARACTERISTICS MEETING MINIMUM ASTM D-4956 TYPE III STANDARDS. THE PRIMARY LETTERS SHALL BE MIXED CASE WITH 6" FHWA SERIES B FONT AND PREFIX/SUFFIX LETTERS AND BLOCK NUMBERS SHALL BE 3" MIXED CASE FHWA SERIES B FONT. (SEE DETAIL E)
5. PREFERRED MATERIALS:
-WHITE: 3M HIGH INTENSITY GRADE PRISMATIC SERIES 3930 (3930 WHITE)
-GREEN: 3M ELECTRO CUT FILM SERIES 1170 (1177C GREEN)
6. ALL LETTERS SHALL BE SERIES B-2000 FROM THE 2004 STANDARD HIGHWAY SIGNS MANUAL (AND ANY REVISION THERETO) PUBLISHED BY THE FEDERAL HIGHWAY ADMINISTRATION.
7. ALL STREET NAME SIGNS ARE SUBJECT TO APPROVAL BY THE TOWN ENGINEER. BLOCK NUMBERS SHALL BE PROVIDED ON SIGNS AND CORRESPOND TO OFFICIALLY APPROVED ADDRESSES.
8. REFER TO DRAWING 703.1 FOR HUNTERSVILLE LOGO REQUIREMENTS.
9. FOR SIGNS LONGER THAN 48 INCHES IN LENGTH, THE 6 INCH MINIMUM IN C AND D IS TO BE 12 INCHES.

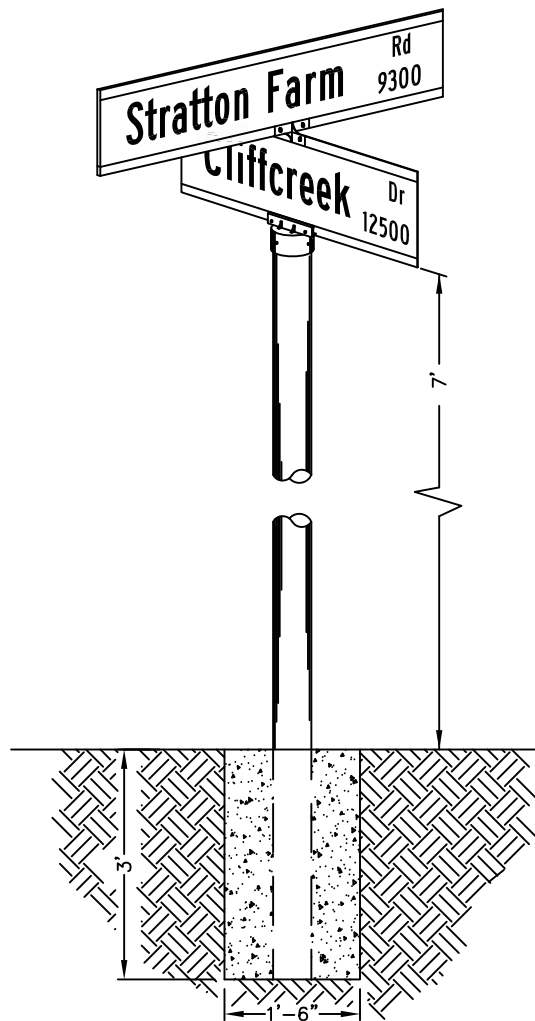
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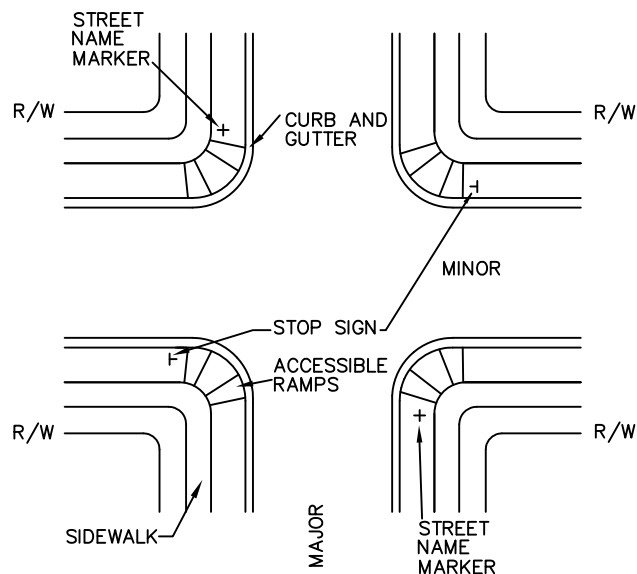
1. HUNTERSVILLE DECORATIVE "H" LOGO SHALL BE USED ON STREETNAME MARKERS LOCATED ON MAJOR AND MINOR THOROUGHFARES WITHIN TOWN LIMITS AS DEFINED BY THE MECKLENBURG-UNION METROPOLITAN PLANNING ORGANIZATION (MUMPO) THOROUGHFARE PLAN. BOTH MAIN STREET AND SIDE STREET MARKERS SHALL BEAR THE DECORATIVE "H" LOGO.
2. THE DECORATIVE "H" SHALL BE GREEN WITH A WHITE BACKGROUND WITH RETROREFLECTIVE CHARACTERISTICS MEETING MINIMUM ASTM D-4956 TYPE III STANDARDS.
3. PREFERRED MATERIALS:
 -WHITE: 3M HIGH INTENSITY GRADE PRISMATIC SERIES 3930 (3930 WHITE)
 -GREEN: 3M ELECTRO CUT FILM SERIES 1170 (1177C GREEN)
4. THE DECORATIVE "H" LOGO DESIGN SHALL BE ACQUIRED FROM HUNTERSVILLE ENGINEERING AND PUBLIC WORKS DEPARTMENT.



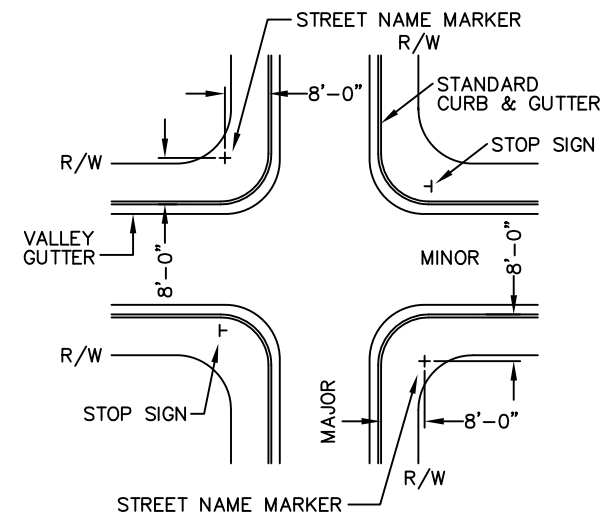
INSTALLATION OF STREET NAME SIGN



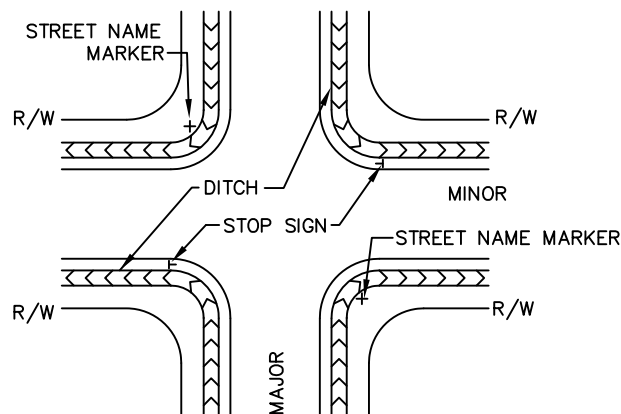
INTERSECTION with SIDEWALK, CURB, and GUTTER



INTERSECTION with CURB and GUTTER



INTERSECTION with DITCHES, and NO CURB and GUTTER



NOTES

1. TWO STREET NAME MARKERS ARE REQUIRED IF THE MAJOR STREET HAS 3 OR MORE LANES.
2. ANY VARIANCE FROM THIS STANDARD MUST BE APPROVED BY THE TOWN ENGINEER.

NOT TO SCALE

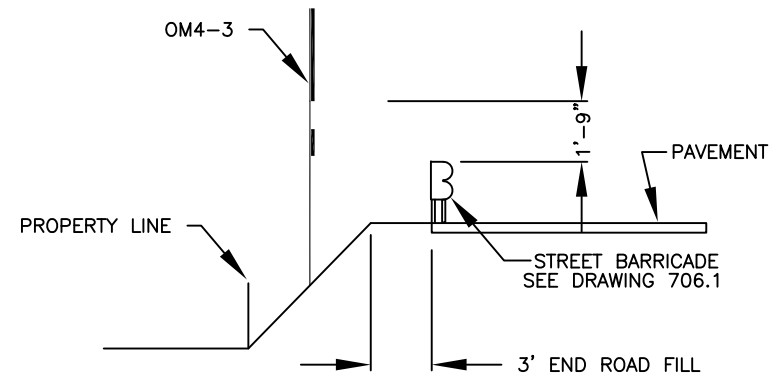
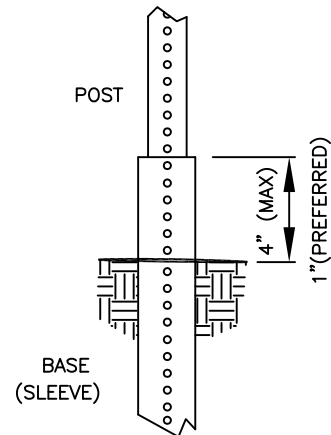
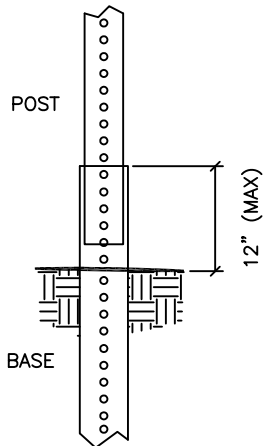
NOTES:

1. WHEN AN END OF ROADWAY OR STUBBED STREET REQUIRES A GUARDRAIL SECTION, END OF ROADWAY MARKER SIGNS (MUTCD OM4-3, 24"x24", SOLID RED) SHALL BE PROVIDED.
2. SIGNS ARE TO BE PLACED BEHIND THE BARRICADE (SEE DRAWINGS 707.1 & 708.1), EVENLY SPACED WITH ONE SIGN PLACED AT THE CENTERLINE LOCATION AND ADDITIONAL SIGNS AT 6' O.C. (MINIMUM OF 3 SIGNS, MAXIMUM OF 5 SIGNS).
3. WHEN BARRICADE IS USED ON A STREET STUB, THE SIGN AT THE CENTERLINE SHALL BE SUPPLEMENTED WITH A STREET CONNECTIVITY SIGN. SEE DRAWING 708.1.
4. ALL SIGNS/MARKERS SHALL MEET OR EXCEED MUTCD STANDARDS FOR RETROREFLECTIVITY.

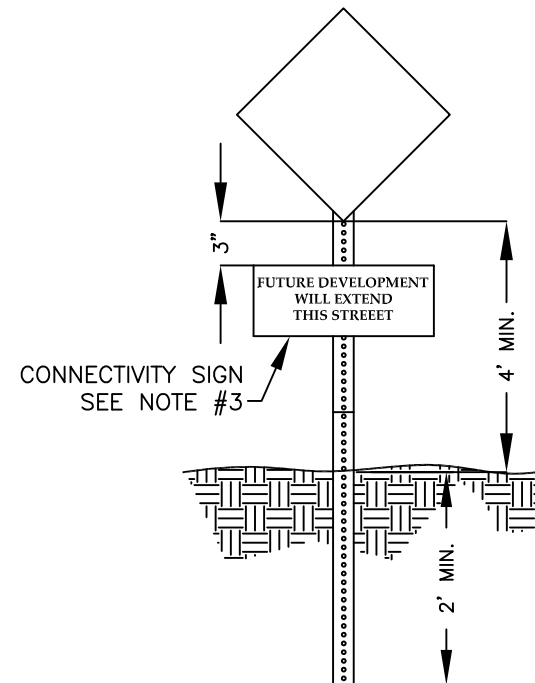
CROSS
SECTION
OF POST
(2 LB./FT.)



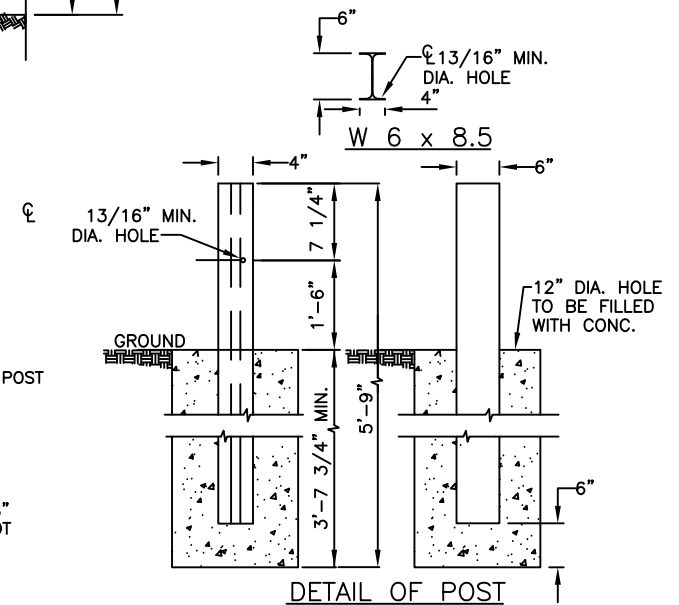
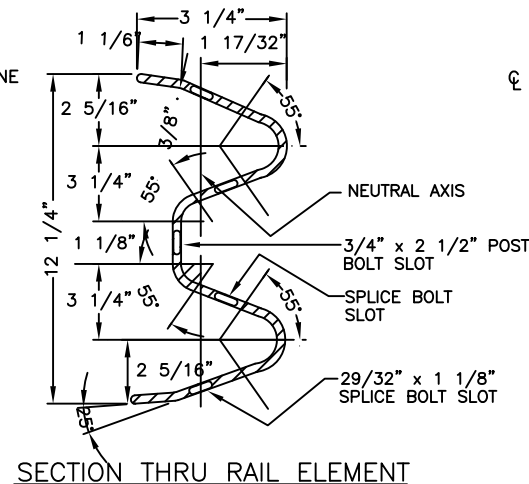
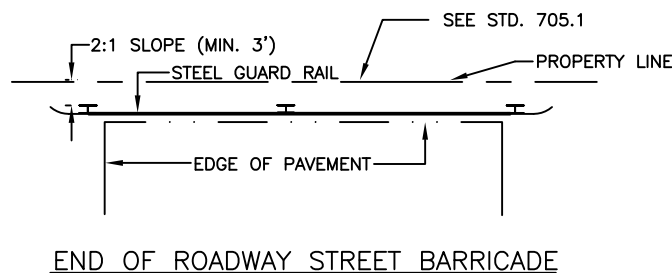
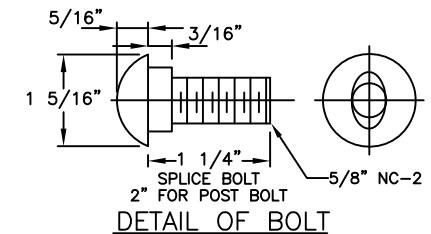
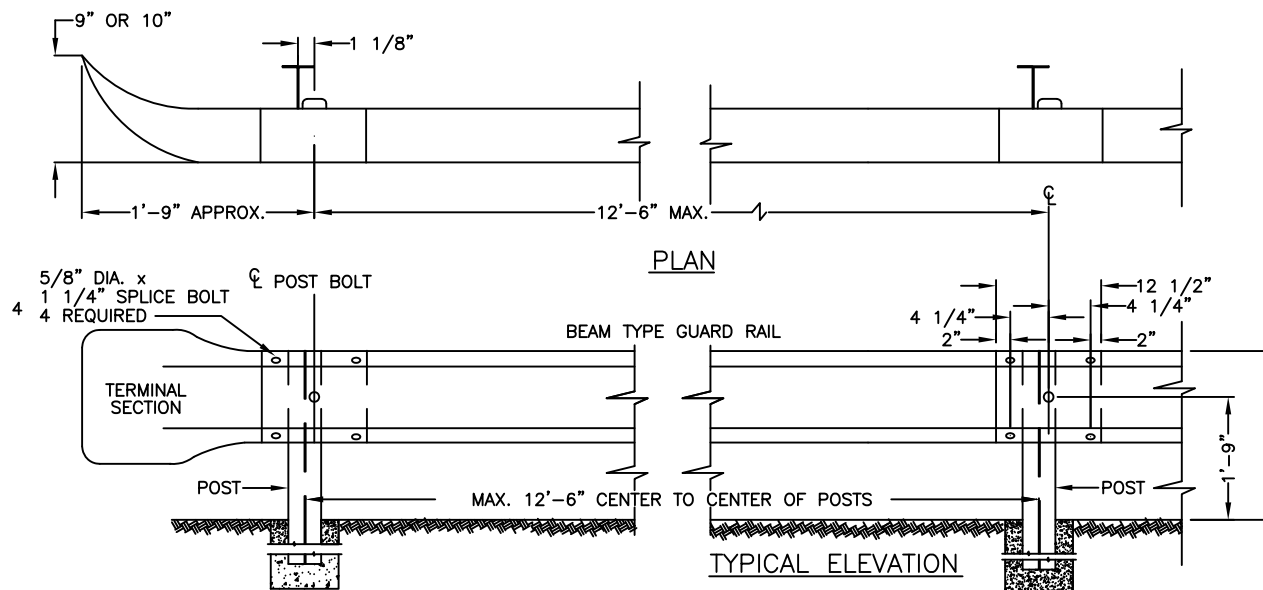
CROSS
SECTION
OF POST
(14 GAUGE)



SIGN LOCATION DETAIL



NOT TO SCALE

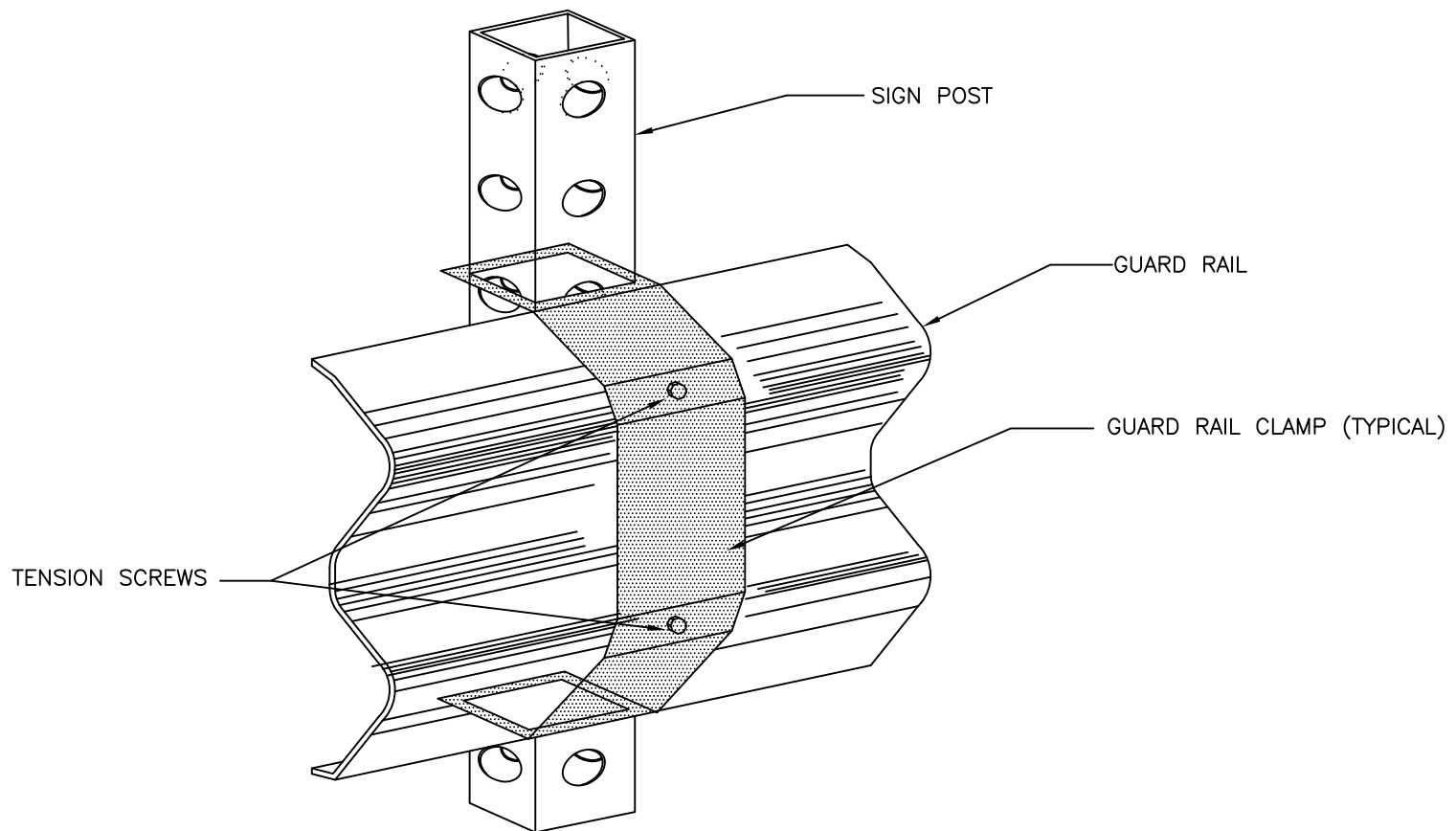


NOTES:

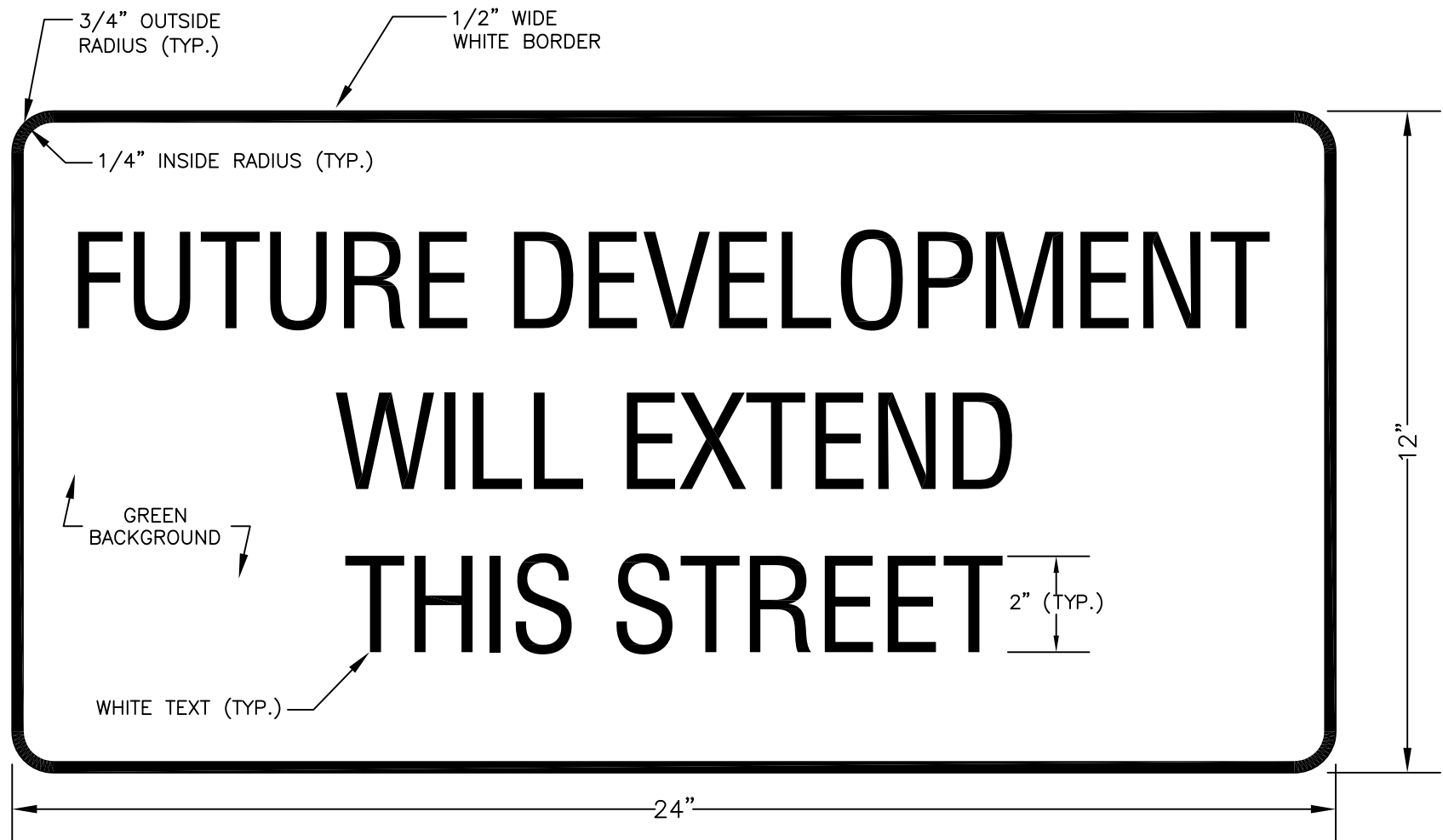
— THIS DETAIL IS NOT A GUARDRAIL DETAIL.
 FOR ROADSIDE GUARDRAIL, REFER TO
 NCDOT STANDARD DRAWINGS 862.01-862.03

— REFER TO STANDARD DRAWINGS 705.1
 AND 707.1 FOR ADDITIONAL INFORMATION.

NOT TO SCALE



NOT TO SCALE



NOTES:

1. SIGN SHALL MEET OR EXCEED MUTCD STANDARDS FOR RETROREFLECTIVITY
2. SIGN MATERIAL SHALL BE 0.080" THICK ALUMINUM
3. ALL LETTERS SHALL BE SERIES B-2000 FROM THE 2004 STANDARD HIGHWAY SIGNS MANUAL (AND ANY REVISION THERETO) PUBLISHED BY THE FEDERAL HIGHWAY ADMINISTRATION.

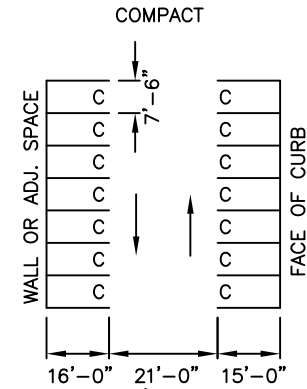
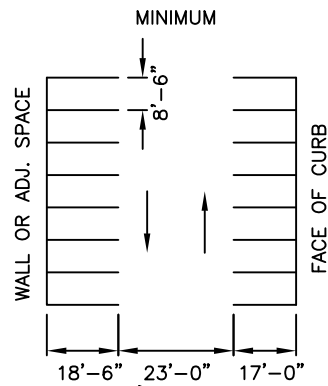
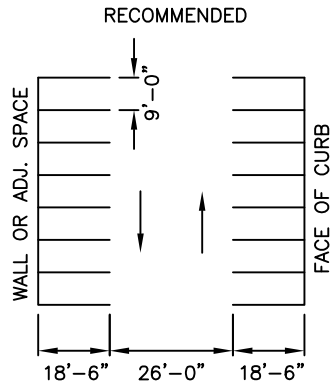
NOT TO SCALE

GENERAL NOTES:

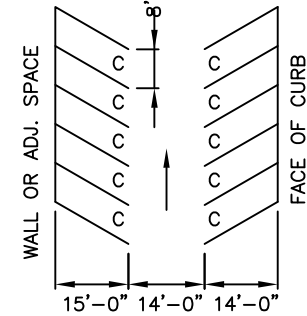
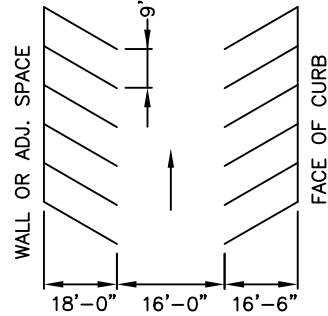
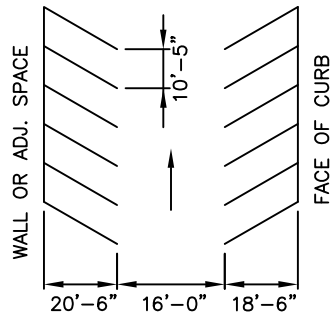
1. STEEL BEAM TYPE GUARD RAILS SHALL BE INSTALLED AT THE END OF ALL DEAD-END STREETS, EXCEPT CUL-DE-SAC STREETS WHICH HAVE BEEN APPROVED WITH A PERMANENT TURNAROUND.
2. FOR STREETS 28' IN WIDTH, THE GUARD RAIL SHALL CONSIST OF TWO (2) 12'-8" SECTIONS OR ONE (1) 25' SECTION, THREE (3) STEEL POSTS, AND TWO (2) TERMINAL SECTIONS. FOR STREETS GREATER THAN 25' IN WIDTH, THE GUARD RAIL SHALL SPAN THE ENTIRE WIDTH OF THE STREET.
3. GUARD RAIL SHALL CONSIST OF RAIL ELEMENTS FABRICATED TO DEVELOP CONTINUOUS BEAM STRENGTH AND INSTALLED AS SHOWN.
4. MINIMUM THICKNESS OF GUARD RAIL SHALL BE 12 GAGE U.S. STANDARD. THE RAIL ELEMENT INCLUDING SPLICES SHALL HAVE A MINIMUM ULTIMATE TENSILE STRENGTH OF 80,000 LBS. GUARD RAIL PARTS FURNISHED SHALL BE INTERCHANGEABLE WITH SIMILAR PARTS REGARDLESS OF THE SOURCE OF MANUFACTURER. THE HOLES FOR CONNECTING BOLTS SHALL BE PUNCHED OR DRILLED. BURNING OF THE HOLES FOR CONNECTING BOLTS SHALL NOT BE PERMITTED.
5. THE GUARD RAIL, BOLTS, NUTS, STEEL POSTS, AND ALL OTHER METAL PARTS SHALL BE GALVANIZED TO CONFORM TO THE REQUIREMENTS FOR THE COATING CLASS, (2.5 OUNCES PER SQUARE FOOT) OF THE CURRENT SPECIFICATIONS FOR ZINC-COATED (GALVANIZED) IRON, AND STEEL SHEETS, COILS, AND CUT LENGTHS, IN ACCORDANCE WITH ASTM 123A.
6. IF THE AVERAGE SPELTER COATING AS DETERMINED FROM THE REQUIRED SAMPLES IS LESS THAN TWO (2) OUNCES OF SPELTER PER SQUARE FOOT, OR IF ANY ON SPECIMEN HAS LESS THAN 1.8 OUNCES OF SPELTER PER SQUARE FOOT OF DOUBLE EXPOSED SURFACE, THE LOT SAMPLED SHALL BE REJECTED. THE FINISHED SHEETS SHALL BE OF FIRST CLASS COMMERCIAL QUALITY, FREE FROM INJURIOUS DEFECTS SUCH AS BLISTERS, FLUX, AND UNCOATED SPOTS.
7. THE GUARD RAIL SHALL BE INSPECTED TO DETERMINE THAT THE MATERIAL, DIMENSIONS, AND WORKMANSHIP ARE IN ACCORDANCE WITH THIS PLAN.
8. WHERE AN END OF ROADWAY REQUIRES GUARD RAIL, END OF ROADWAY MARKER SIGNS SHALL ALSO BE REQUIRED. (REFER TO DRAWINGS 705.1 - 708.1)

NOT TO SCALE

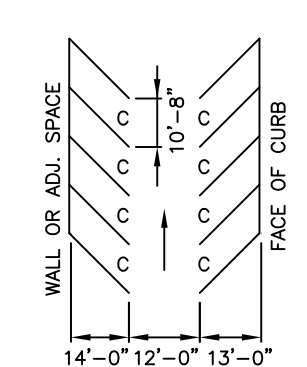
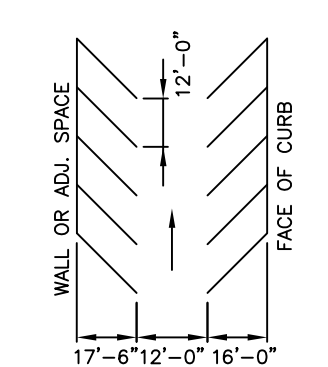
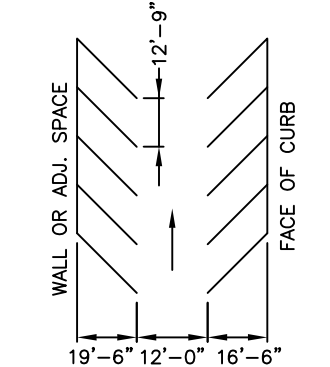
PARKING ANGLE 90°
(TWO WAY OPERATION ONLY)



PARKING ANGLE 60°
(ONE WAY OPERATION ONLY)



PARKING ANGLE 45°
(ONE WAY OPERATION ONLY)



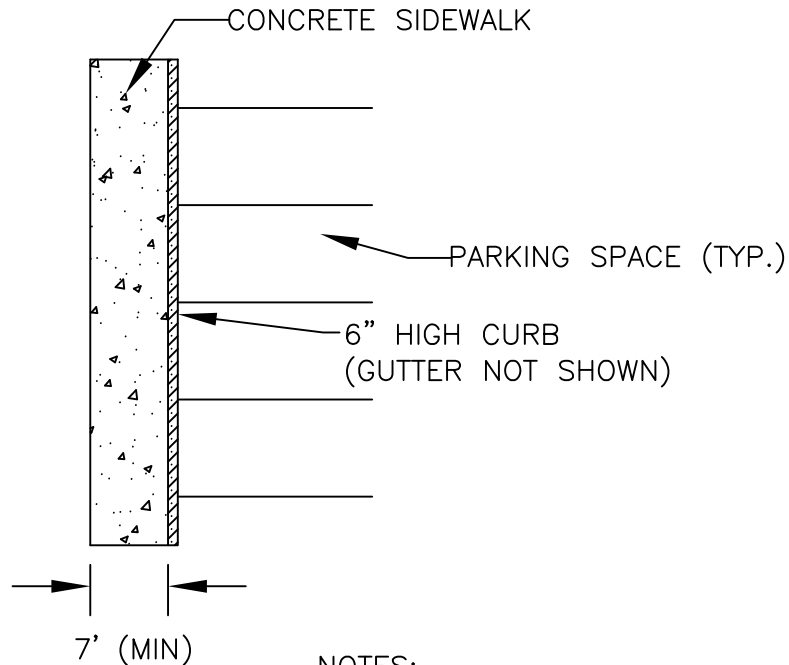
NOTES:

1. FOR ACCESSIBLE PARKING STANDARDS/SIGNAGE SEE STDS. 712.1, 713.1, AND 714.1.
2. PAVEMENT MARKINGS SHALL BE 4" WHITE PAINT.
3. ALTERNATIVE PARKING ANGLES, AISLE WIDTHS, AND OPERATION (TWO-WAY ANGLED PARKING OR REVERSE-ANGLE PARKING) WILL BE CONSIDERED ON A CASE-BY-CASE BASIS.
4. COMPACT SPACES ARE TO BE DESIGNATED WITH A PAINTED C ON EVERY SPACE.

NOT TO SCALE

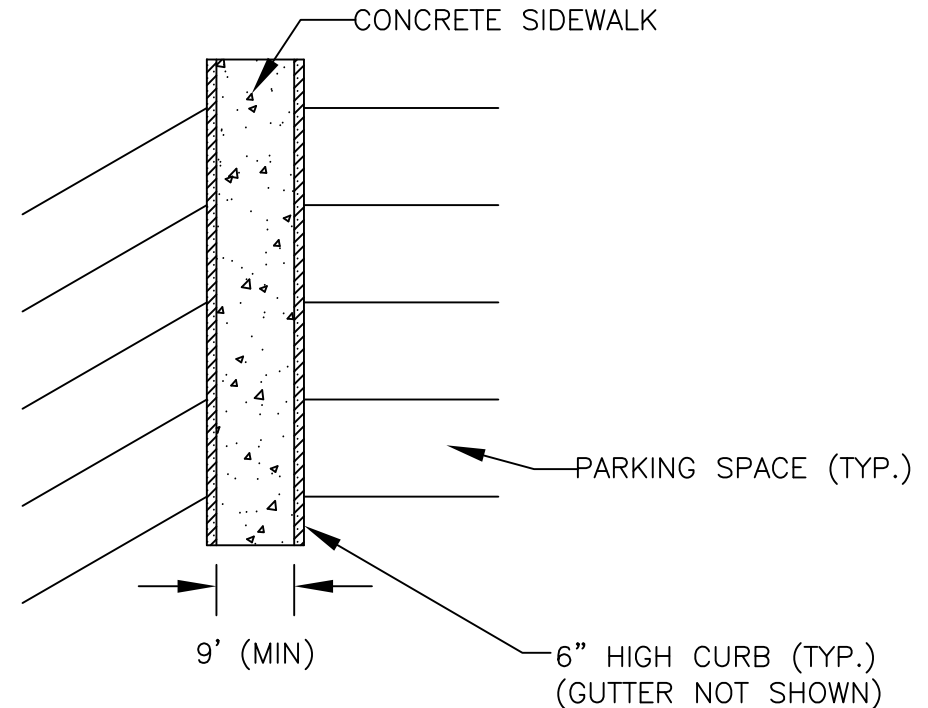
SIDEWALK ADJACENT TO HEAD-IN
OR BACK-IN PARKING SHALL BE
AT LEAST 7 FEET WIDE.

PARKING ON ONE SIDE OF A SIDEWALK



SIDEWALK BETWEEN TWO ROWS OF
HEAD-IN OR BACK-IN PARKING
SHALL BE AT LEAST 9 FEET WIDE.

PARKING ON BOTH SIDES OF A SIDEWALK



NOTES:

1. A 2-FOOT-WIDE GRASS PLANTED AREA LOCATED AT THE BACK OF CURB CAN BE USED IN LIEU OF 2 FEET OF SIDEWALK WIDTH.
2. PARKING AT ANY ANGLE OTHER THAN PARALLEL SHALL BE SUBJECT TO THIS STANDARD.
3. IF MONOLITHIC CURB & SIDEWALK IS USED, ADD 6" TO ALL DIMENSIONS (1' IF PARKING ON BOTH SIDES).
4. WHEELSTOPS IN LIEU OF ADDITIONAL SIDEWALK WIDTH SHALL BE CONSIDERED ON A CASE-BY-CASE BASIS.

NOT TO SCALE

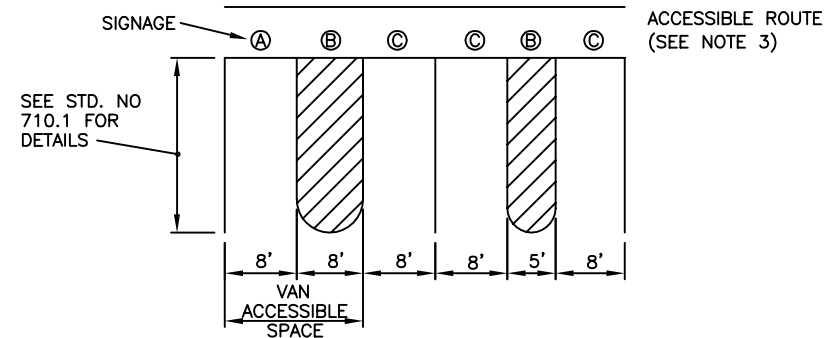
ACCESSIBLE PARKING REQUIREMENTS

TOTAL PARKING SPACES PROVIDED	MINIMUM NUMBER OF ACCESSIBLE SPACES REQUIRED	MINIMUM NUMBER OF ACCESSIBLE SPACES REQUIRED TO BE VAN ACCESSIBLE
1 TO 25	1	1
26 TO 50	2	1
51 TO 75	3	1
76 TO 100	4	1
101 TO 150	5	1
151 TO 200	6	1
201 TO 300	7	2
301 TO 400	8	2
401 TO 500	9	2
501 TO 1000	2% OF TOTAL	1 IN EVERY 6 ACCESSIBLE SPACES
1001 AND OVER	20 PLUS 1 FOR EACH 100 OVER 1000	1 IN EVERY 6 ACCESSIBLE SPACES

SECTION 4.1.2 (5) OF THE AMERICANS WITH DISABILITIES ACT (ADA).
SEE 4.1.2,(5) (d) FOR MEDICAL CARE FACILITIES

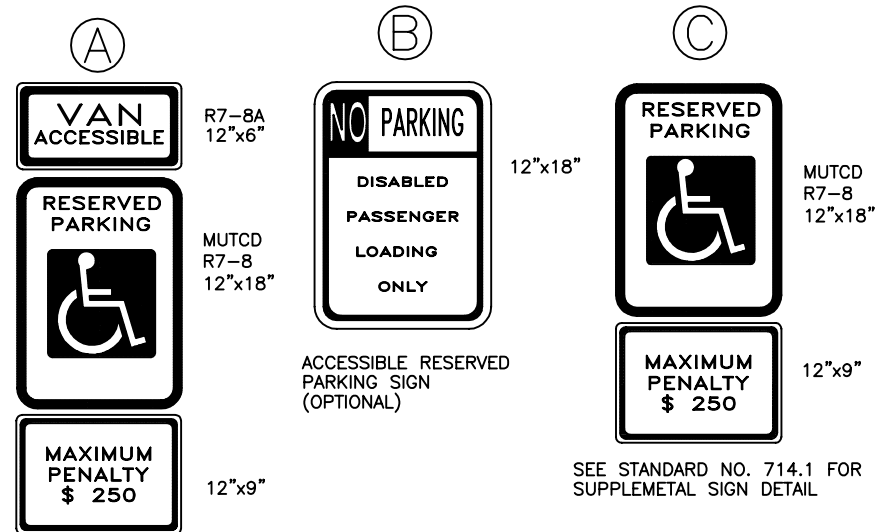
NOTES:

- ALL 12"x18" ACCESSIBLE SIGNS SHALL BE MOUNTED AT SEVEN FEET FROM GRADE TO BOTTOM EDGE OF SIGN FACE (MUTCD). MOUNTING HEIGHT CAN BE REDUCED TO FIVE FEET IF PLACED IN AN AREA BETWEEN SIDEWALK AND BUILDING FACE IN WHICH PEDESTRIANS ARE NOT EXPECTED TO USE.
- REFER TO MANUAL ON UNIFORM TRAFFIC CONTROL DEVICES, (MUTCD) U.S. DEPARTMENT OF TRANSPORTATION AND NORTH CAROLINA DEPARTMENT OF TRANSPORTATION SUPPLEMENT.
- IF ACCESSIBLE ROUTE IS A RAISED SIDEWALK AREA, THEN RAMPS ARE REQUIRED AT LOADING ZONE AREA.



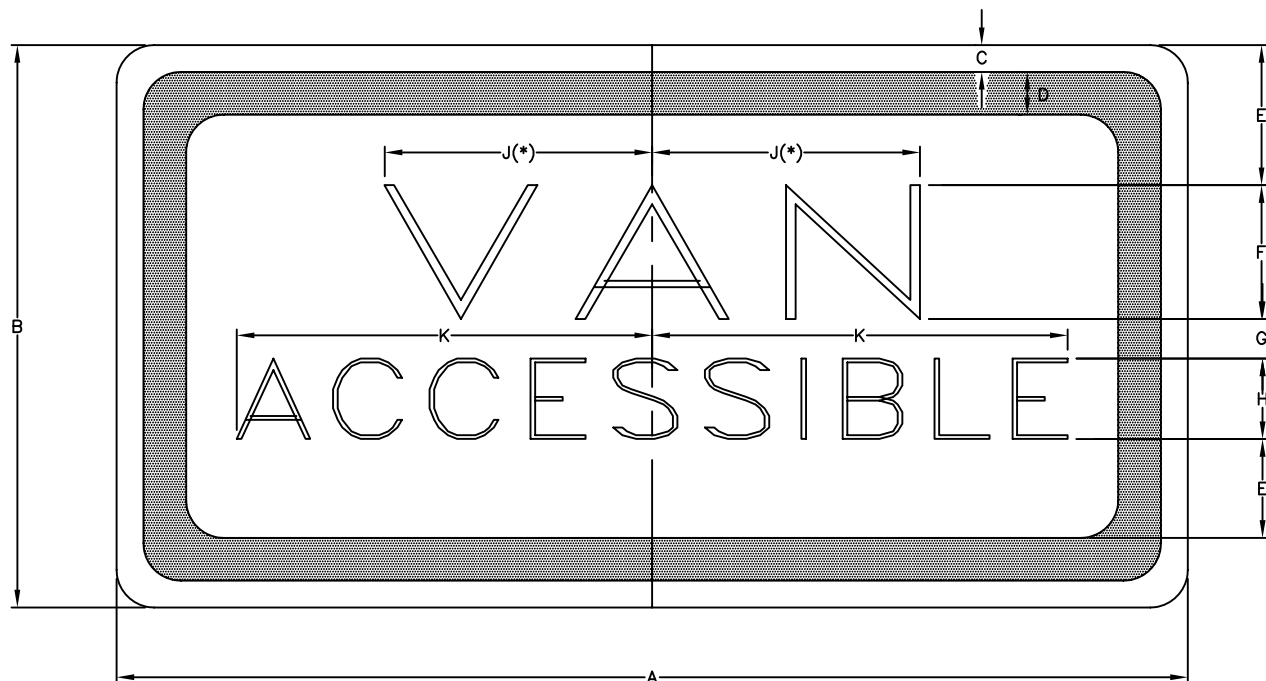
ONE OUT OF EVERY SIX (6) ACCESSIBLE SPACES, BUT NOT LESS THAN ONE, IS REQUIRED TO BE VAN ACCESSIBLE.

PARKING SPACE PAVEMENT MARKINGS



SEE STANDARD NO. 713.1 & 714.1 FOR SUPPLEMENTAL SIGN DETAIL

NOT TO SCALE

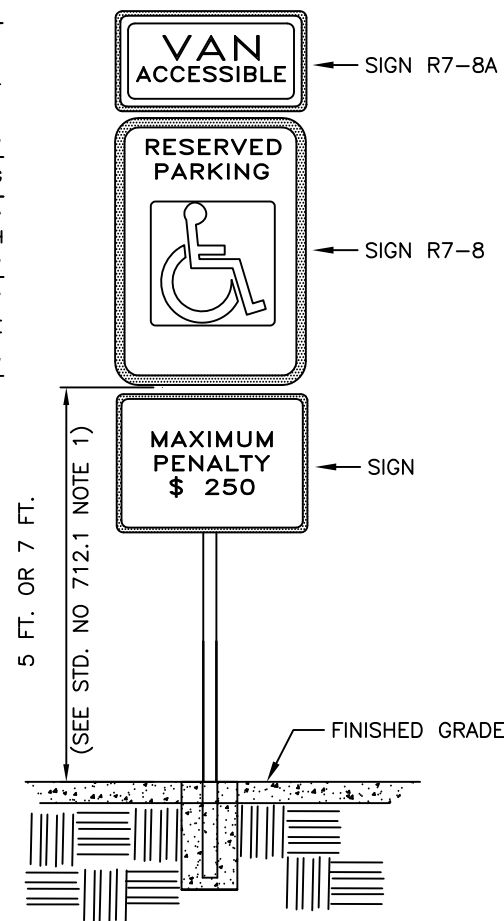


R7-8A

DIMENSIONS (INCHES)										
A	B	C	D	E	F	G	H	J	K	L
12	6	3/8	3/8	1-1/2	1-1/2D	1/2	1D	2-1/2	4	1-1/2

* INCREASE SPACING 50%
D-FHWA (FEDERAL HIGHWAY ADMINISTRATION/USDOT)
SERIES D LETTERS

LEGEND AND BORDER - GREEN
BACKGROUND - WHITE
PRISMATIC SHEETING PREFERRED



NOT TO SCALE



LEGEND AND BORDER - GREEN
BACKGROUND - WHITE

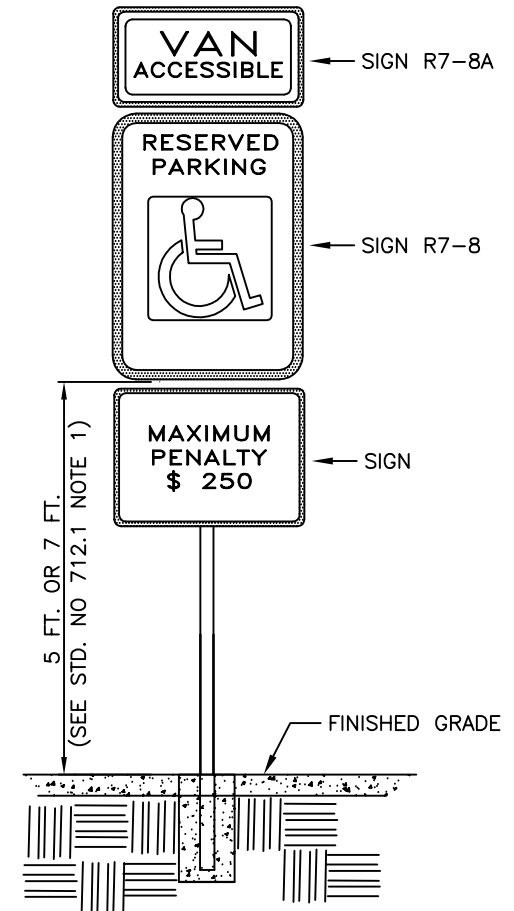
SIGN APPROVED FOR USE
UNDER GENERAL STATUTE 20-37.6

MAXIMUM PENALTY SIGNS ARE REQUIRED TO
ACCOMPANY ALL R7-8 PARKING SIGNS

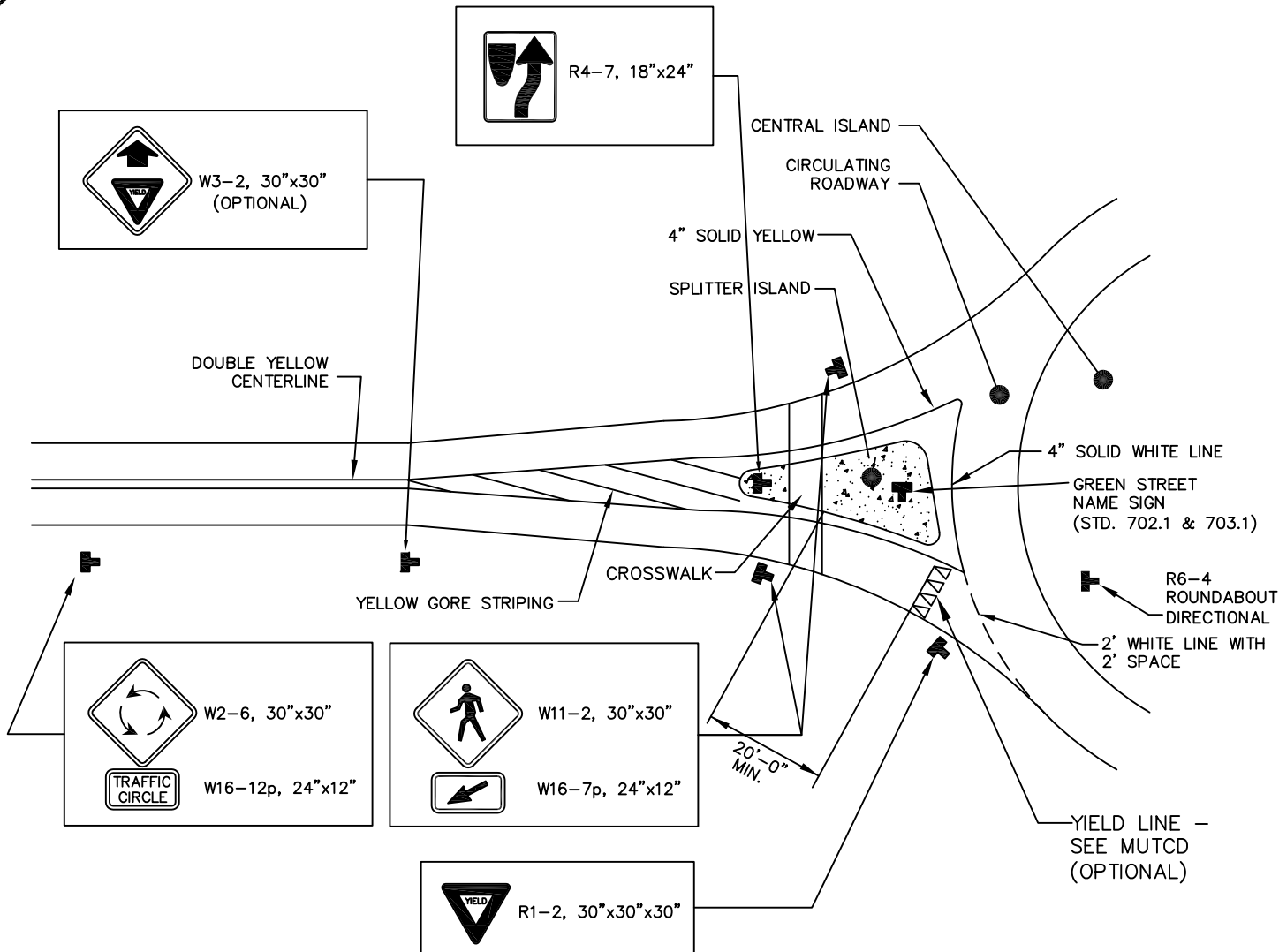
SIGN LETTERING TO BE FHWA D SERIES LETTERS 1.5 INCHES TALL

NOTE:

SUPPLEMENTAL ACCESSIBLE SIGN USED IF
THERE IS ONLY ONE REQUIRED ACCESSIBLE PARKING
SPACE (MUST BE VAN ACCESSIBLE) AND AT EACH
ADDITIONAL REQUIRED VAN ACCESSIBLE SPACE. (SEE
STD. NO. 712.1)



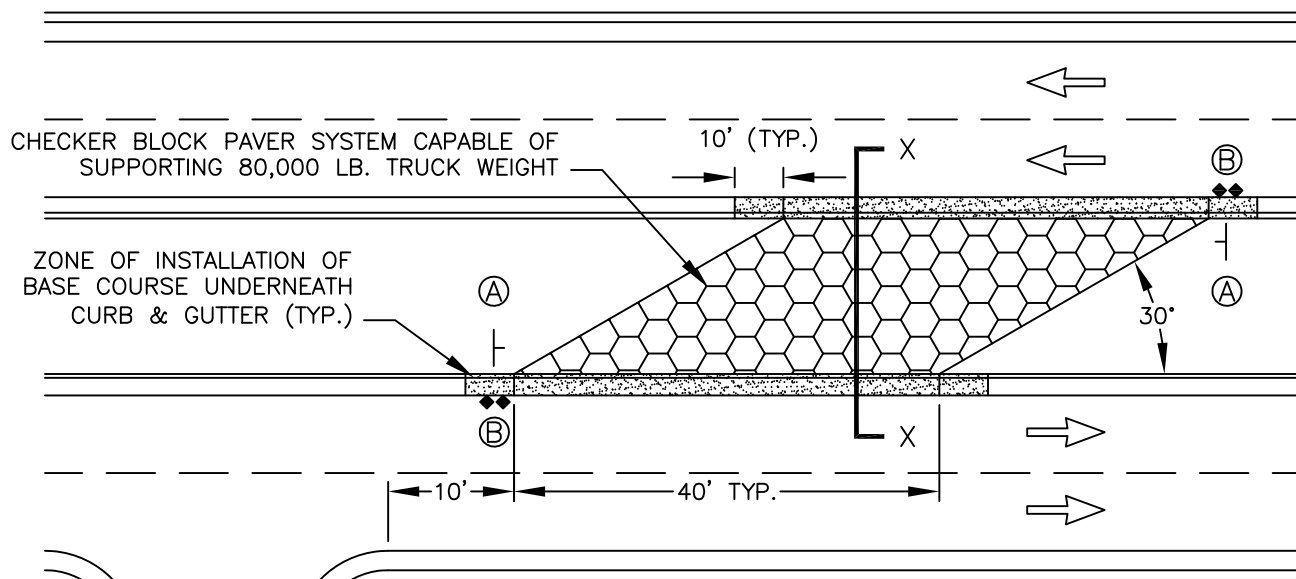
NOT TO SCALE



NOTES:

1. PAVEMENT MARKINGS TO BE PER LATEST EDITION OF THE MANUAL ON UNIFORM TRAFFIC CONTROL DEVICES (MUTCD).
2. SIGNS TO BE LOCATED/SPACED PER MUTCD REQUIREMENTS.
3. "CIRCULAR INTERSECTION" AND "TRAFFIC CIRCLE" SUBPLATE SIGNS, AND KEEP RIGHT SIGN ARE REQUIRED ON THOROUGHFARES. NCDOT AND/OR TOWN WILL DETERMINE IF ONE OR MORE OF THESE ARE NECESSARY ON LOCAL OR COLLECTOR STREETS.
4. "PEDESTRIAN CROSSING" AND ARROW SUBPLATE SIGNS ARE REQUIRED WHEREVER THERE IS A MARKED CROSSWALK OR ON A THOROUGHFARE.
5. "YIELD" SIGNS ARE ALWAYS REQUIRED.
6. PAVEMENT MARKINGS, SPLITTER ISLAND DESIGNS, CROSSWALK, ETC., ARE SHOWN FOR CONTEXT ONLY. REFER TO THE MUTCD AND/OR THE FEDERAL HIGHWAY ADMINISTRATION'S MANUAL ROUNDABOUTS: AN INFORMATIONAL GUIDE FOR MORE DETAIL OR DESIGN INFORMATION.
7. ADDITIONAL SIGNS MAY BE NEEDED ON A CASE-BY-CASE BASIS.
8. ALL PAVEMENT MARKINGS SHALL BE THERMOPLASTIC.

NOT TO SCALE

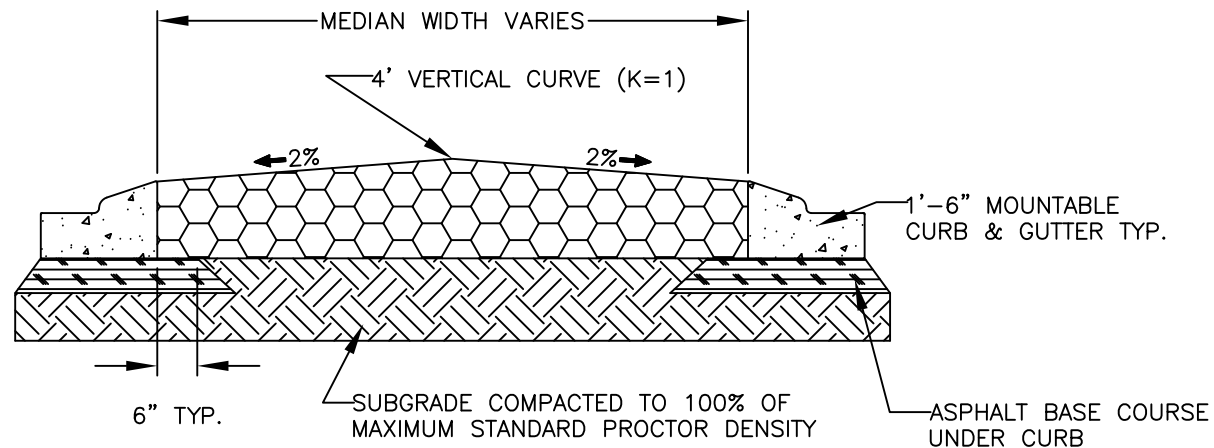


- Ⓐ "NO LEFT TURN" (R3-2, 24"x24")
- Ⓑ YELLOW/YELLOW RAISED PVMT. MARKER 1' O.C. SEE NCDOT STD. #1250.01.

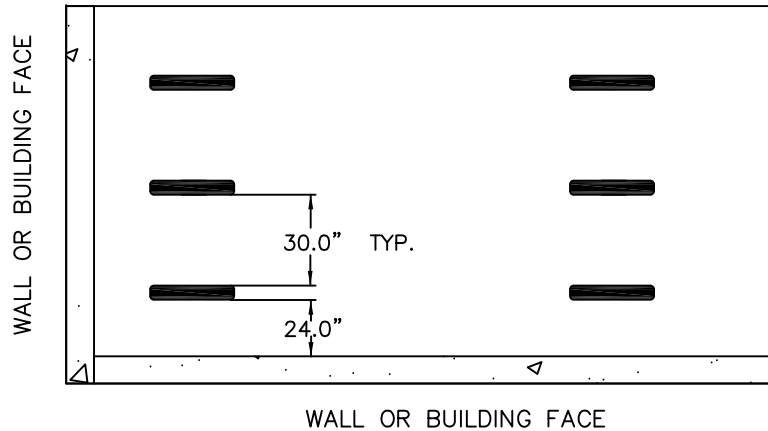
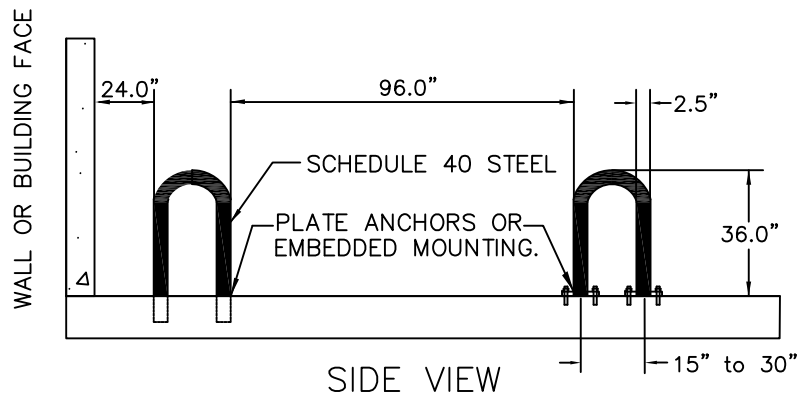
NOTES:

1. CROSSOVER TO BE OFFSET 10' FROM ANY INTERSECTING STREET OR DRIVEWAY OTHER THAN A FIRE DEPARTMENT DRIVEWAY.
2. ASPHALT BASE COURSE UNDERNEATH MOUNTABLE CURB AND GUTTER SHALL EXTEND AT LEAST 10 FEET BEYOND CROSSOVER.

SIDE STREET OR DRIVEWAY



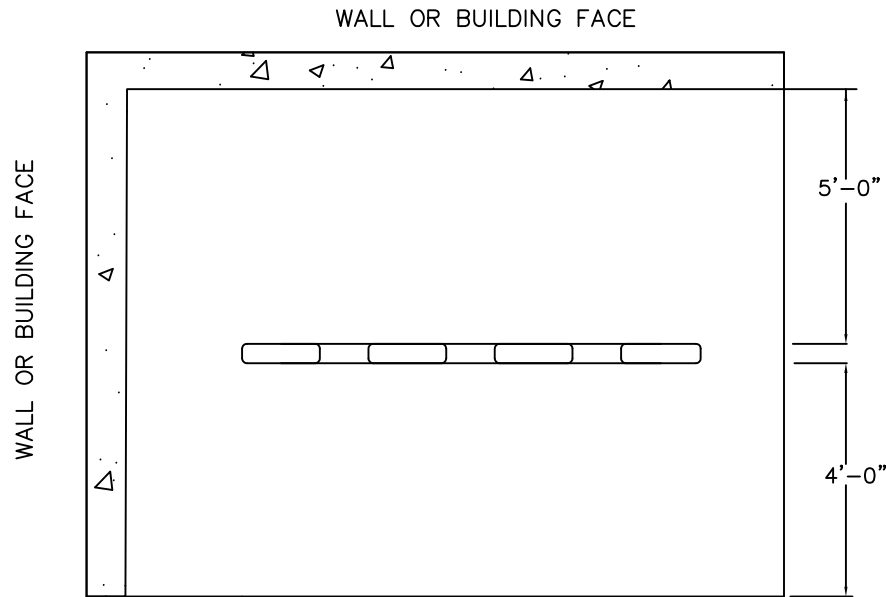
NOT TO SCALE



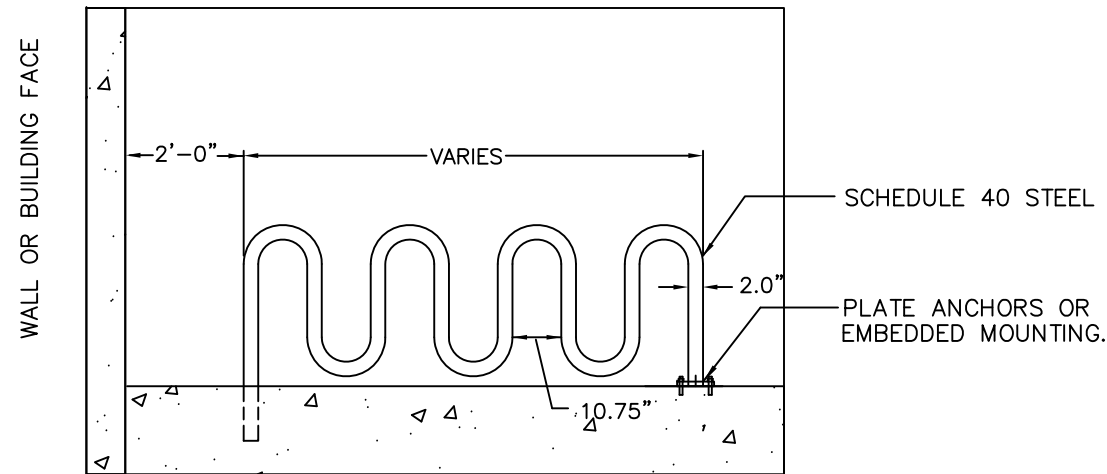
NOTES:

1. BIKE RACKS SHOULD BE INSTALLED AS PER MANUFACTURER'S RECOMMENDED INSTALLATION PROCEDURES.
2. ALTERNATIVE BIKE RACKS OR LOCKERS MAY BE USED BUT ARE SUBJECT TO APPROVAL BY THE TOWN ENGINEER.
3. ALL DIMENSIONS SHOWN ARE MINIMUM.

NOT TO SCALE



PLAN VIEW

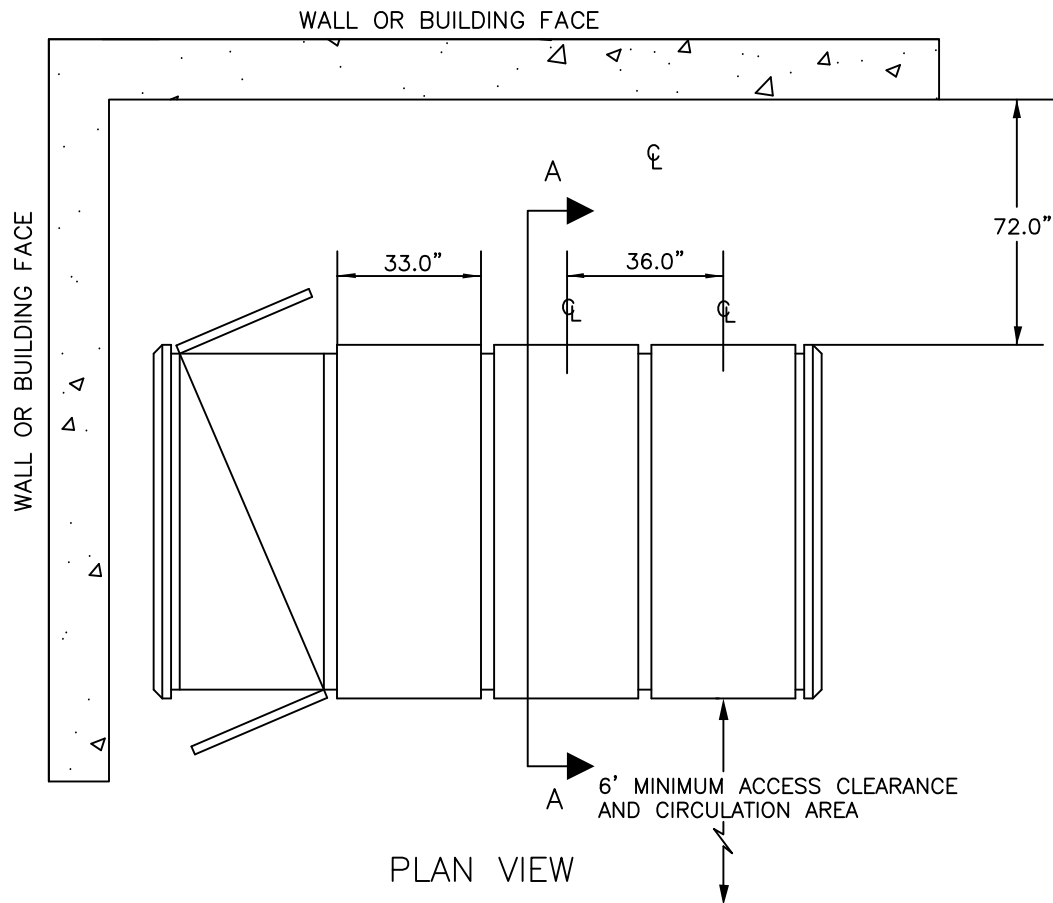


SIDE VIEW

NOTES:

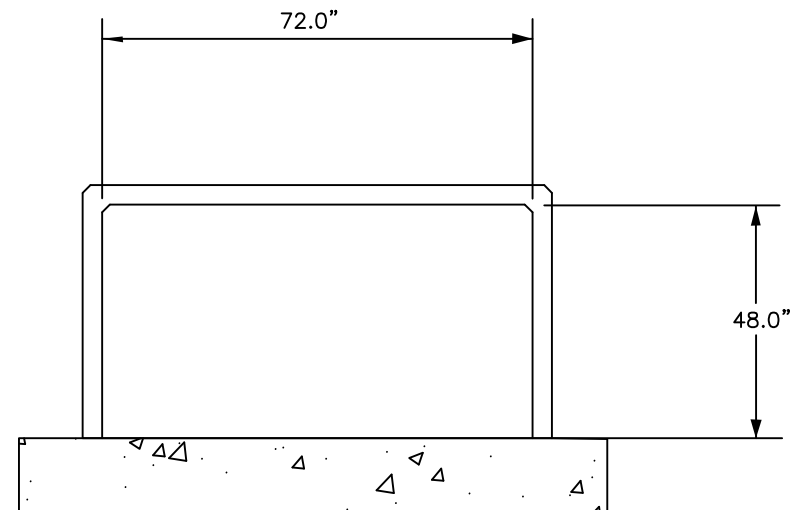
1. BIKE RACKS SHOULD BE INSTALLED AS PER MANUFACTURER'S RECOMMENDED INSTALLATION PROCEDURES.
2. ALTERNATIVE BIKE RACKS OR LOCKERS MAY BE USED BUT ARE SUBJECT TO APPROVAL BY THE TOWN ENGINEER.
3. ALL DIMENSIONS SHOWN ARE MINIMUM.

NOT TO SCALE



NOTES:

- BIKE RACKS SHOULD BE INSTALLED AS PER MANUFACTURER'S RECOMMENDED INSTALLATION PROCEDURES.
- ALTERNATIVE BIKE RACKS OR LOCKERS MAY BE USED BUT ARE SUBJECT TO APPROVAL BY THE TOWN ENGINEER.
- ALL DIMENSIONS SHOWN ARE MINIMUM.
- ALLOW FOR POSITIVE DRAINAGE AWAY FROM LOCKERS.



NOT TO SCALE

Appendix

Town of Huntersville Plan Review Checklist

Revised 10/2015

Project Name _____ Date Reviewed: _____
Reviewer: _____ Telephone: _____
Contact: _____ Telephone: _____

Cite all appropriate standard detail numbers for any structures or specifics used within the plans in reference to the most current copy of the Town of Huntersville Engineering Standards and Procedures Manual.

GENERAL SUBMITTAL REQUIREMENTS

- ___ Existing and proposed paved areas, buildings, structures & utilities.
- ___ Off-site construction or grading requires a letter from the affected property owner granting permission.
- ___ Is site adjacent to imminent NCDOT, Transit or Town project?
If yes, written approval is required from respective authority.
- ___ Zoning: _____ Parcel Tax Number: _____
- ___ Copy of Rezoning Petition in plans _____ CD Notes
- ___ Jurisdiction (Town or ETJ) _____ Vicinity map
- ___ Legible Scale (1:100 max) _____ North arrow
- ___ Parking Summary
- ___ Property Boundary and Adjoining Property Boundaries, Owners, and Zoning
- ___ Pervious and Impervious Area (new and/or existing)
- ___ Adjacent existing streets & drainage _____ Title block w/ site name & location
- ___ Label Contours (2' max. interval, existing & proposed)
- ___ Storm Drainage Calculations (Sealed) _____ SWIM Buffer Present
- ___ Watershed Overlay District (List drainage basin)
- ___ Utility Easements (existing and proposed)
- ___ Surveyed Right of Way _____ Wetlands Permits (Army Corps/DWQ)
- ___ Sealed Retaining Wall Design (Note about shop drawings)
___ **Note:** Retaining walls that are in and/or affect public right-of-way must be reviewed and approved by the Town (Any questions contact Max Buchanan).
- ___ **Note:** If project falls inside CATS corridor buffer, coordinate with John Mrzygod (704-336-2245).
(Any project within 100' of existing or proposed track layout)
- ___ All relevant standard details reference the Town of Huntersville Engineering Standards & Procedures Manual
- ___ Sketch Plan notes (if applicable)
- ___ **PE Certification Statement for Subdivisions and Streets on cover sheet:** *"The Town of Huntersville requires that all streets proposed to be taken over by the Town for maintenance be reviewed, inspected, and certified by a licensed professional engineer registered in the state of North Carolina for adequate construction. Review of street construction by the certifying Engineer is required throughout the construction process. Refer to the Town of Huntersville Engineering Standards and Procedures Manual for additional information including the required certification form."*
- ___ **Retaining wall note on cover sheet:** *"All retaining walls requiring special inspections (>5' in height) shall meet the requirements of the MCSBC Chapter 17 – special inspections as detailed per the Mecklenburg County Code Enforcement Special Inspections Process. Additionally, copies of the retaining wall plans shall be submitted to the Huntersville Engineering Department for review and approval. The Town of Huntersville shall be copied on all statements of special inspections, special inspections final reports, and design principal in responsible charge (DPIRC) letters."*

CHECKLIST COMPLETION ITEMS (Grading Permit not issued until completion of all applicable items)

- ___ NCDOT Driveway Permit: 4 sets of completed applications with original signatures (copies are not allowed)
- ___ Town of Huntersville Driveway Permit (if applicable)
- ___ Town Encroachment Permit for utility cuts in an existing Town-maintained street. Contact the Town Engineering Department at 704-766-22220

Town of Huntersville Plan Review Checklist
Revised 10/2015

- ☐ **All applicable notes are on plans per Mecklenburg County Land Development Plan Notes Checklist:**
<http://charmeck.org/mecklenburg/county/LUESA/WaterandLandResources/LandDevelopment/Documents/PlanReviewNotesChecklist.docx>
- ☐ Health Department approval for proposed septic system locations
- ☐ DWQ, Army Corps, and/or Water Quality (for buffer disturbance, SSMP) approval (if applicable)
- ☐ Approved Floodplain Development Permit for any development within the Community Special Flood Hazard Area
- ☐ All BMP easements and BMP Maintenance Plan (PCO 20), Operation and Maintenance Agreement/Declaration of Covenants (PCO19) are recorded. These must be reviewed, and approved by LUESA staff. Copies of the recorded documents must be received prior to plan approval.
- ☐ BMP Inset Tables:
<http://www.charmeck.org/stormwater/regulations/Pages/PCOSWManagementPermitsFormsCountyandSixTownsOnly.aspx>
- ☐ All applicable items on Post-Construction checklist and/or LID checklist are resolved.
Post-Construction Plan Review Checklist can be found on the web at:
<http://www.charmeck.org/stormwater/regulations/Pages/PCOSWManagementPermitsFormsCountyandSixTownsOnly.aspx>
- ☐ Structural Plans for retaining walls, non-standard structures.
- ☐ Charlotte Water Plans
- ☐ Grading and/or improvements within utility R/W or in other easements shall require a letter of approval from the utility company.
- ☐ Grading and/or improvements within railroad R/W shall require a letter of approval from the railroad company.
- ☐ NCDENR Grading Permit (Required when government funds are used for projects. Examples: Hospital, Police Station, Fire Station, Utility Projects, Mecklenburg County, Towns, Parks, etc.)
- ☐ Aesthetics of utility structures in front of buildings. Locations of all utility structures (i.e. boxes, transformers, backflow preventors, etc.) are shown on the plans. All aesthetic concerns must be resolved prior to plan approval.

DRIVEWAYS AND NEW STREET ACCESSES

- ☐ Driveway Applications (with original signatures)
- ☐ 4 original to NCDOT Newell Office (1 scanned copy submitted in EPM) – if access or improvements to NCDOT-maintained street
- ☐ 1 Town permit application – if access or improvements to Town-maintained street
- ☐ Street Name _____ NC Route/Road Number _____
- ☐ Typical Section(s) with Pavement Schedule
- ☐ Wedging detail shown if required.
- ☐ Left-turn lanes are required for sites serving 50 or more residential lots, on roads with ≥ 4000 ADT, or site-specific for commercial property.
- ☐ Roadway cross-sections on the existing street being accessed, every 50' for length of proposed widening (See NCDOT Cross-Section Guidelines). Include additional cross-sections 50' before and after the widening.
- ☐ Overlay asphalt for entire width for NCDOT-maintained streets (or to the middle of the nearest full lane for Town-maintained streets). Overlay full width of road for symmetrical widening on an existing two-lane roadway.
- ☐ Traffic Control Plan that meets MUTCD and NCDOT and/or WATCH standards.
- ☐ Traffic Control Phasing and Traffic Control Notes.
- ☐ Pavement Marking Plan that meets MUTCD and NCDOT standards.
- ☐ Sight Distance Profile:
- ☐ Intersection sight distance based on ESAPM.
- ☐ Measured 14.5' back from the travel lane.
- ☐ Plan and profile shall be shown for at least the distance of the profile in both directions.
- ☐ Driver eye height and object height is 3.5' above existing/proposed road surface.
- ☐ Horizontal sight distance alignment: Lines of sight in both directions shall be shown and labeled.

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- ☐ Vertical sight distance alignment: Show the centerline roadway profile of the existing street and the grades along the lines of sight. A minimum of 1' shall be provided between the grade and line of sight.
- ☐ Include any potential conflicts. Sight triangle shall be clear of obstructions.
- ☐ Sight distance easements may be required if lines of sight cross adjacent properties (i.e. curves in roadway, limited frontage, etc.).
- ☐ Stopping sight distance along road per ESAPM
- ☐ Sight Triangle/Maintenance Easement, measured along R/W lines, shown at all intersections (35x35 for collector or thoroughfare intersection; 15x15 for local street intersections; 10x70 for connections to NCDOT roads)
- ☐ Clear Zone Requirements per AASHTO Roadside Design Guide.
- ☐ Show all drive entrances to all properties along the existing street. Show all property lines.
- ☐ Right-of-way line and Controlled Access line, if present. Right-of-way must be verified by a Professional Land Surveyor.
- ☐ Right of way adequate for improvements. Right of way acquisition for proposed road improvements shall be acquired prior to plan approval.
- ☐ Driveway Std. # and width
- ☐ Design vehicle turning templates
- ☐ WB 50 – All commercial drives
- ☐ B-40 – Residential Street
- ☐ Tapers for turn lanes per NCDOT Driveway Access manual
- ☐ Existing and proposed storm drainage
- ☐ Parking Spaces per ESAPM 710.1 and Town Ordinance
- ☐ Traffic Circulation within the project (Fire trucks, buses, etc.)
- ☐ Garbage collection method (Town or Private)
- ☐ Corner radii labeled
- ☐ SU 30 – Alleys
- ☐ P – Drive thrus, Parking Decks

IMPROVEMENTS TO EXISTING RIGHT-OF-WAY

- ☐ Offsite R/W requirements shown on plans (for symmetrical widening, off-site frontage improvement tie-ins, etc.).
- ☐ Need R/W agreement (recorded) for road widening and/or improvements where there is no existing R/W
- ☐ Storm drainage associated with street improvements.
- ☐ Show and label cross-section and profile of existing street with proposed improvements (i.e. curb and gutter, sidewalk, pavement section, ditch cross-section, associated drainage, etc.).
- ☐ If the existing street is maintained by NCDOT, a NCDOT Encroachment Agreement is required for the proposed improvements (contact NCDOT, 980-523-0000)
- ☐ Bicycle improvements along roadway may be required (check Bikeway Improvement Plan map or Town Ordinance)
- ☐ Copy of rezoning petition shall be included in the plans. Rezoning petition could list requirements for additional or non-standard right-of-way improvements.
- ☐ As of 4/8/09, the increase in impervious area due to widening of existing roads **MUST** be included in the BUA and **MUST** be included in the sizing of the development's BMPs.
- ☐ On-street parking dimensions meets ESAPM #710.1-712.1 and/or Town Ordinance.
- ☐ **Design exceptions accepted by the Town require notation on the plans.**

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When a proposed street ties into an existing street (classified below), street improvements may be required:

	Arterials	Commercial Arterials	Collectors	Commercial Collectors
Property Frontage Typical	ESAPM #207.1	ESAPM #207.1	ESAPM #203.1	ESAPM #208.1, 209.1
Right-of-way width	Major: 100'* Minor: 70'*	100' Dedicated 50' Reserved	See Town Zoning Ordinance	See Town Zoning Ordinance
Curb & gutter (ESAPM#100.1, 101.1, 102.1)	2'6"	2'6"	2'6"	2'6"
Sidewalk, Planting Strip	5', TBD	5', TBD	5', 7'	5', 7'
Minimum Shoulder	8' (6' for NC-maintained)	8' (6' for NC-maintained)	8'	8'

- R/W widths different for urban arterials (see MUMPO Thoroughfare Map)

PUBLIC STREETS

- _____ Cross-sections must conform to ESAPM standards.
- _____ Accessible ramps required at all intersections. [ESAPM #123.1-134.1]
- _____ Taper from local to local-limited. [ESAPM #135.1]
- _____ Bicycle improvements along roadway may be required.
- _____ AASHTO sight triangles at intersections.

CUL-DE-SACS

- _____ Residential [ESAPM #211.1] with appropriate pavement radius*
- _____ Commercial [ESAPM #212.1]
- _____ Provide spot elevations along lip of cul-de-sac to insure positive drainage for profiles $\leq 2\%$
- ◆ 42' recommended to allow for emergency vehicles turning with minimal reverse movement. R/W must be adjusted to accommodate larger pavement radius.

PRIVATE STREETS and ALLEYS

- _____ Maximum grade 10%
- _____ Minimum horizontal radius at CL determined by design vehicle
- _____ Show limits of private street
- _____ Vertical curve K-values (crest/sag 10/20)
- _____ Cross-section per ESAPM # 210.1, 215.1-218.1
- _____ ESAPM sight triangles at intersections with public streets.
- _____ Parking bays allowed on only one side of the street unless a median is provided.

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HORIZONTAL AND VERTICAL REQUIREMENTS

For use in designing Residential and Retail/Mixed-Use Public Streets

Posted Speed Limit	25	30	35	40	45
Stopping Sight Distance* (feet)	155	225	285	350	415
Intersection Sight Distance - Left-Turn Movement From Stop*and ** (feet)	280	365	425	485	545
Intersection Sight Distance - Right-Turn From Stop*and ** (feet)	240	315	370	420	475
Minimum Horizontal Radius (Normal Crown) (feet)	200	430	675	980	1470
Minimum K value for Crest Vertical Curves	11	24	37	56	81
Minimum K value for Sag Vertical Curves	25	43	58	75	94
Maximum Longitudinal Grade	10 percent				
Maximum Longitudinal Grade within 125 feet of intersection (measured from intersecting street nearest edge of pavement of travel way)	5 percent				
Intersection Angle Range	75 to 105 degrees				

* Values will need to be adjusted for grades of more than +/- 3 percent

** Values to be adjusted for streets with more than two total lanes; measurements to be taken 14.5' from travel lane

- _____ Show profiles with vertical data and storm drainage for all proposed streets (Public and Private).
- _____ Profiles for streets to be barricaded must extend 300 feet beyond the phase/property line or to the proposed future end of the street, whichever is less [ESAPM #705.1 through 709.1]
- _____ Non-symmetrical vertical curves must have L1 or L2 $\geq$ one-third of total length
- _____ Thoroughfares - refer to AASHTO "A Policy on Geometric Design of Highways and Streets"

INTERSECTIONS

- _____ Minimum curb and R/W radius (measured from face of curb)
 - _____ Residential Local Street 20
 - _____ Residential Local Street to Residential Alley 10
 - _____ Residential Collector 25
 - _____ Retail/Mixed-Use Local 25
 - _____ Retail/Mixed-Use Collector 25
 - _____ Industrial Local and Collector 35
- _____ Minimum offset to adjacent Local 125
- _____ intersection (ft) Collector 350

FIRE DEPARTMENT ACCESS REQUIREMENTS

- _____ Fire apparatus shall access within 150 feet of all exterior walls. (NC Fire Code 503.1.1)
- _____ Dead-end fire access roads in excess of 150 feet in length shall be provided with an approved area for turning around fire apparatus. (NC Fire Code 503.2.5)
- _____ Fire apparatus access roads shall have an unobstructed width of not less than 20 feet (measured from FOC to FOC) and an unobstructed vertical clearance of not less than 13 feet 6 inches. (NC Fire Code 503.2.1)
- _____ Turning radii required is 42 feet outside and 30 feet inside. (NC Fire Code 503.2.4)
- _____ Fire apparatus access roads shall be designed and maintained to support the imposed loads of fire apparatus (80,000 lbs) and shall be surfaced so as to provide all-weather driving capabilities. (NC Fire Code 503.2.3)
- _____ Multifamily buildings – Emergency ladder access shall be provided from a finished grade that is no more than 15% maximum slope for a distance of 12 feet.
- _____ Buildings over 3 stories must have frontal clearance for rescues.

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- _____ Building Code – Minimum 20' separation between buildings (measured to/from eaves) or firewalls.
- _____ Utility plans must show location of fire hydrants, water lines, and size of water line.
- _____ Fire hydrant required within 750 feet of most remote point of building as the truck travels.
- _____ Fire hydrant required within 200 feet of fire department connection of all sprinklered commercial structures and 750 feet to residential sprinklered structures.
- _____ Six-inch dead end water mains serving a hydrant may not exceed 350 feet.

LANDSCAPING PLAN

- _____ Show street lights and security lighting.
- _____ Show street name signs, regulatory signs (stop signs, speed limit signs, yield signs, etc.)
- _____ Show storm water lines and structures.
- _____ Show mail cluster box units.
- _____ No trees within 40' of stop sign on approach.
- _____ No trees within 40' of radius return.
- _____ No trees within 10' of stormwater structure.
- _____ No trees at or in stormwater drainage easements.
- _____ No trees on top of storm lines.

STORM DRAINAGE (Refer to Storm Water Design Manual [SWDM] Chapters 3-8)

- _____ All design in accordance with the ESAPM and SWDM.
- _____ Provide details of all drainage structures or refer to ESAPM.
- _____ Location, size, invert and rim or grate elevations and pipe material for existing and proposed storm drains and culverts on and immediately adjacent to the site
- _____ Storm system pipes designed for 10-year storm [SWDM 3.5.1]
- _____ Where roadway overtopping is anticipated, drainage that is picked up in a culvert (or other drainage structure) prior to crossing the road must be conveyed in a system designed for the 25-year storm (50-year storm for thoroughfares). Flow calculations must be based on the appropriate design storm to the end of the system or to a relief point lower than the street elevation.
- _____ Discharge leaves site in same direction and relative location as pre-developed condition.
- _____ Structures or flared end sections at all inlets and outlets of all pipe systems other than small outlets to rain gardens. (CMP and HDPE FES are **NOT** allowed) [SWDM 6.5.9]
- _____ Non-standard drainage structures (CBs, DIs, MHs, JBs, HWs, FESs, bottomless culverts, etc.) will require sealed construction drawings.
- _____ Minimum 2' cover on pipes (on Class III RCP, 1' minimum cover on Class IV RCP).
- _____ The minimum allowable slope is 0.5 percent, or the slope which will produce a velocity of 2.5 fps when the storm water in system is flowing full, whichever is greater. [SWDM 5.9.6]
- _____ Maximum velocity in pipes 20 fps (10 fps for CMP) [SWDM 5.9.2 & 6.5.4]
- _____ Maximum discharge velocity at pipe outlets is 5 fps (or the original velocity prior to development) [SWDM 6.5.4]
- _____ Rip rap required for all pipe outfalls (0% slope, Min. L=10', Min. Depth=10"). Filter fabric must be placed underneath. [ESAPM #306.1]
- _____ Corrugated metal pipes must be aluminum or aluminized steel
- _____ Maximum of one (1) acre runoff draining into the street at one point
- _____ No concentrated runoff flowing over Town/County sidewalks except at driveway entrances
- _____ Maximum spread of 6 feet (or half a travel lane) on public streets based on rainfall intensity of 4 in/hr [SWDM 1.3]. An additional 1 ft is allowed for valley curb and 1'-6" curb and gutter, and additional 2 ft for 2'-6" curb and gutter.
- _____ Slope less than 2% across intersections may need additional CBs and spot elevations to prevent intersection flooding
- _____ Catch Basin placement at intersections [ESAPM #122.1]
- _____ Check drainage at stub streets (i.e. spread, future culvert crossings, and 100+1 culvert analysis)

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- _____ Crossdrains/Culverts must meet the following criteria:
- _____ Culverts – Minimum diameter of 18”
 - _____ 100-yr storm does not exceed 6” over the road
 - _____ Cross-drainage on thoroughfare classified roadways – 50-yr storm (25-yr storm for minor roadways)
 - _____ 12” freeboard for culverts $\leq 3'$ in diameter (25-yr storm)
 - _____ 18” for all culverts $> 3'$ in diameter (25-yr storm)
 - _____ Maximum ponding at yard inlets is 1' with 6” freeboard for design storm.
 - _____ HW/D ≤ 1.2
 - _____ Check applicability of Culvert Crossing [ESAPM #133.1, 134.1]
 - _____ Standard Headwall detail is referenced or design sealed by Engineer
 - _____ For bottomless culverts, upstream and downstream footer elevations shown
 - _____ Existing creek/channel typical section
 - _____ Roadway cross-section at culvert crossing @ legible scale
- _____ Pipes in R/W 48” or less in diameter may be HDPE or RCP, >48” may be RCP, >60” may be aluminized steel; minimum 15” (any HDPE pipe used within one of towns shall have the public works approval and bedding material and 3rd party certified.)
- _____ Provide details of curb cuts and flumes
- _____ Properly sized and labeled Storm Drainage Easements (SDE) see widths below. Specify who will maintain these easements.

<u>Pipe Diameter (inches)</u>	<u>Pipe Easement Width (feet)</u>
15-24	15
30-36	20
42-48	25
54+	30 min. (Varies)

_____ **Storm Drainage Easement Note:** “The purpose of the storm drainage easement (SDE) is to provide storm water conveyance. Buildings are not permitted in the easement area. Any other objects which impede storm water flow or system maintenance are also prohibited.”

_____ No section of storm water piping shall be more than 300’

_____ Storm drainage pipe should be no more than 2’ from curb and gutter line

SWIM STREAM BUFFERS

- _____ Buffer widths are shown and labeled on plans and are correct based off POLARIS.
- _____ Plans indicate that all engineered channels, piped outfalls, and rip rap aprons end outside of the buffer.
- _____ Drainage outfalls must be designed to allow water to sheet flow across the buffer to filter out pollutants.
- _____ Energy dissipators, diversion devices or wetland flow must be used to spread concentrated flows and must be located outside the buffer area. Adequate vegetative cover must be retained or re-established below energy dissipators to prevent erosion.
- _____ No outlet drains more than 10 acres drainage area to any one point in a stream buffer.
- _____ Where disturbance to the buffer is shown, a letter from Water Quality is required stating that the disturbance shown is allowable or whether a mitigation plan has been approved.
- _____ All temporary sediment basins and all other erosion control devices are located outside of the minimum buffer width.
- _____ **Buffer Delineation**
- _____ Buffer area is clearly identified on plans in bold, clear type as “S.W.I.M. STREAM BUFFER”
 - _____ The outside boundary of each buffer zone is clearly marked and labeled on plans (stream side, managed use, upland)
 - _____ Field located top of bank from where the buffer width is measured is labeled on plans
 - _____ The outside boundary of the buffer must be permanently marked with an iron pin or other acceptable property corner marker at all street crossings. The iron must be called out on the plans.
 - _____ Show orange fabric fencing on Grading and Erosion Control Plans.
 - _____ Streams and buffer boundaries including the delineation of each zone and a dimension from a

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specific location are shown on all surveys and record plats, including individual record plats for any lots affected.

_____ Where applicable for 100' buffers/50% additional width: Provide calculations for 50% additional width based on the area between the current 100-year FEMA FRINGE BOUNDARY to the 100' SWIM Buffer or the 0.1 FLUM Encroachment line, whichever is less.

DRAINAGE AREA MAPS

- _____ All drainage area maps indicate project limits and are at a legible scale.
- _____ Drainage area maps clearly and legibly depict onsite and off-site areas (colored coded maps encouraged).
- _____ Maps are complete with properly labeled existing/proposed contours and indicate roof drainage direction.
- _____ Off-Site topography for drainage basins must extend to a point where ridgelines can be clearly identified.
- _____ T_c Travel Paths (pre/post) delineated on drainage area maps and labeled.
- _____ Erosion Control Basin map delineate and label the drainage areas for each phase of construction.
- _____ Storm drainage maps are labeled by inlet number and indicate acreage of drainage areas.
- _____ Subdivision Drainage Areas are depicted in the plan set.
- _____ Detention Basin maps show pre and post-development drainage areas and show SCS soil types.

GRADING & EROSION CONTROL

(See the North Carolina Erosion and Sediment Control Planning and Design Manual (NCESCPDM) and the Town of Huntersville Engineering Standards and Procedures Manual [ESAPM], Mecklenburg County Soil Erosion and Sediment Control Ordinance and Policies and Procedures guidelines)

- _____ Completed and notarized Financial Responsibility Form (if the financially responsible party is out of State, a North Carolina agent must be assigned).
- _____ Construction sequence (suggested sequence – see LD Plan Notes Checklist:
<http://charmeck.org/mecklenburg/county/LUESA/WaterandLandResources/LandDevelopment/Documents/PlanReviewNotesChecklist.docx>)
- _____ Urban Forestry Staff approval is required.
- _____ Landscape Management Staff approval is required for tree removal in the R/W Construction sequence (suggested sequence attached)
 - _____ Denuded acres (outline denuded limits and list acreage)
 - _____ Undisturbed buffer areas - dimensions and locations (grading in a buffer requires zoning approval)
 - _____ Swales, berms & temporary diversion ditches [ESAPM #506.1]
 - _____ Location of on-site and/or off-site waste burial areas; location of borrow sites
 - _____ Stabilized gravel construction entrance [ESAPM #514.1]
 - _____ Temporary Silt fence [Used for sheet flow only], [ESAPM #506.1-507.1]
 - _____ High Hazard Temporary Silt fence [Used for sheet flow only], [ESAPM #508.1]
 - _____ Hardware cloth, Rip Rap w/ washed stone for outlet protection
 - _____ Riser basin: Required within watershed areas or drainage areas >50 acres [ESAPM #501.1, 502.1]
 - _____ Detail shown on Plans with Basin dimensions (top of berm, top of storage, bottom of storage, depth of storage)
 - _____ Storage volume required/provided
 - _____ Design of emergency spillway for 10-year storm (Dimensions and calculations)
 - _____ Label basin contours. Tie into existing contours
 - _____ Design of concrete ballast pad (Calculations)
 - _____ Detail of anti-seep collars and locations
 - _____ Design of riser and outlet pipes for 2-year storm (Calculations)

ENHANCED EROSION CONTROL MEASURES

(See Enhanced Erosion Control Measures Notes in LD checklist:

<http://charmeck.org/mecklenburg/county/LUESA/WaterandLandResources/LandDevelopment/Documents/PlanReviewNotesChecklist.docx>)

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[wNotesChecklist.docx](#))

CHANNEL DESIGN (Refer to Storm Water Design Manual Chapter 4)

- _____ Channel cross-sections and calculations _____ Maximum 2:1 side slopes [SWDM 4.3.1]
- _____ Designed for 10-yr storm _____ Minimum 6" freeboard
- _____ The final design of artificial open channels should be consistent with the velocity and shear strength limitations for the selected lining [SWDM 4.3.4]
- _____ Liner specifications and design information from manufacturer
- _____ Transition to channel sections with a minimum of 5:1 taper [SWDM 4.3.1]
- _____ Channels with bottom widths > 10' must have a minimum bottom cross slope of 12:1 [SWDM 4.3.1]
- _____ Properly sized and labeled storm drainage easement (SDE); see widths below

<u>Area in Acreage</u>	<u>Channel Easement Width (feet)</u>
1-45	20
45-120	30
120-500	40
500+	See Std. # 20.30

- _____ Driveway pipes sized for Q_{10} , check for roadway overtopping as required

100+1 FLOOD ANALYSIS [$Q_{100} > 50$ cfs]

- _____ For pipe systems not designed for Q_{100} that receives >50 cfs in Q_{100} , flood protection analysis is required
- _____ Flood study calculations shall be sealed by PE
- _____ Calculations for flow and t_c
- _____ Runoff coefficients shall be based on assumption of full development of parcels per current zoning
- _____ Include digital copy of channel analysis input file in submittal (e-mail or CD acceptable)
- _____ Proposed Stormwater Protection Elevations (SWPE) shall be labeled on site and grading plan for each lot (the SWPE should be based on Overland Relief Channel if applicable)
- _____ Show and label location of 100+1 Stormwater Elevation Line (SWEL) on site and grading plan.
- _____ Minimum 1,200 sf of buildable area per lot
- _____ Show or describe on plans the location and elevation (ref. vertical datum, i.e. NAVD 88) of permanent benchmark used in channel survey. All topographic information shown on grading plans shall also be referenced to this benchmark. All flood studies shall be done using the NAVD88 datum
- _____ Existing channel cross-section geometry used in analysis shall be based on a "Class A" field survey and shall be sealed by RLS
 - _____ When fill is proposed within the 100+1 Stormwater Elevation Line, plotted channel cross-sections shall be included in the subdivision plan set and shall show both existing and proposed cross-section geometry. Sheets to be signed by RLS for existing and PE for proposed conditions.
 - _____ When no grading is proposed within the 100+1 Stormwater Elevation Line, it is not necessary to include plotted cross-sections in the subdivision plan set. A note shall be added to the grading plan specifying that there shall be no grading within the limits of the 100+1 Stormwater Elevation Line.
 - _____ Cross sections taken perpendicular to the stream centerline, maximum 100' stations
 - _____ Plot cross-sections showing the 100+1 elevation on site and grading plan.
 - _____ Cross-sections based on topographic map are subject to approval of the Review Engineer
- _____ In case of a bridge or culvert:
 - _____ Cross-sections should be prepared at 25 feet from inlet and 5 to 20 feet from outlet
 - _____ Road profile at culvert crossing to an elevation 2 feet above the low point of road
 - _____ A physical description of the culvert (type, shape, invert, etc.)
 - _____ Upstream and downstream top of footer elevations for natural bottom culverts
 - _____ Stormwater Protection Elevations for lots upstream of street crossings shall be based on the higher of the elevation determined by analysis or the low point of the road plus 1 foot
 - _____ Maximum height of flow over a street is 6".

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FLOODWAY [Drainage Area > 640 Acres]

- _____ Floodway Cross-sections (Label, Stream Station, Location, Future Conditions Flood Elevation)
- _____ Label the following lines clearly on grading and site plan:
 - _____ Proposed FEMA Flood Fringe Line
 - _____ Future Conditions Flood Fringe Line
 - _____ Community Encroachment Line
 - _____ FEMA Floodway Encroachment Line
- _____ Flood Protection Elevation (FPE) (Community Flood Elevation + 1'/Community Flood Elevation + 2' along Catawba River)
- _____ Contact Bill Tingle at (704) 336-3734 concerning floodland development permit if grading in a Future Conditions Flood Fringe area. Approval required prior to plan approval.

DETENTION [Refer to Storm Water Design Manual - Chapter 7] {Post Construction requirements may differ from the below requirements, refer to Post Construction Administration manual if applicable}

- _____ Detention worksheet completely filled out for each facility
- _____ Design for release rate of pre-developed 2-yr & 10-yr storm (6hr or 24hr storm) [SWDM 7.3.2]
- _____ Routing calculations are required for storage volume and emergency spillway (sealed) [SWDM 7.3.3]
- _____ The same hydrologic procedure shall be used to determine pre- and post-development hydrology.
- _____ Design Calculations must include:
 - _____ Calculations for pre & post-development weighted runoff coefficients
 - _____ Coefficients shall be specified and justified. Soil survey showing soil type shall be included. Good condition shall be used.
 - _____ Time of concentration calculations (use SCS method) [SWDM 3.9.6.1]
 - _____ Time of concentration travel path (clearly label segments on contoured drainage area map)
 - _____ Inflow hydrograph for all design storms for fully developed and pre-developed conditions [See Chapter 3 of SWDM for approved inflow hydrographs]
 - _____ Stage-storage table for proposed storage facilities.
 - _____ Stage-discharge table for all outlet control structures.
- _____ Undetained runoff must be accounted for in calculations.
- _____ All detention volume shall be drained within 72 hours.
- _____ Emergency spillways for ponds shall be designed to pass the 50-yr storm [SWDM 7.3.2] (Lined in fill areas)
- _____ Maximum 6" depth of storage in code-required parking areas, 10" additional parking areas, 15" storage allowed in truck storage and loading areas (depth based on 10-yr storm) [SWDM 7.3.3]
- _____ Limits of 10 yr detention storage elevation in parking lots labeled
- _____ Maximum slope on basins of 2:1 [SWDM 7.3.4]
- _____ Minimum 6" freeboard above 50-year elevation on earthen basins (on basins less than 15' deep) [SWDM 7.3.4]
- _____ Any orifice smaller than 4" in diameter must be protected with a trash rack to prevent blockage [SWDM 7.5.6]. Detail of trash rack shall be included, that has an area at least 10 times larger than the orifice. The spacing of the bars should be ½ size of the orifice minimum.
- _____ Locate & label on site plan the elevation of dam with dimensions and spot elevations.
- _____ Locate & label on site plan the location of primary and emergency outlets.
- _____ Provide detail of outlet restrictive device.
- _____ 2-yr and 10-yr 6 hour storm, good pre-developed condition must be used when applying for Storm Water fee credit.
- _____ No utilities or habitable structures within impoundment areas.
- _____ Detention facilities are not allowed within required buffers [Meck. County Zoning Ordinance 12.302]
- _____ Detention facilities are not allowed within the existing or proposed R/W [Meck. County Zoning Ordinance 12.103]
- _____ **Underground Detention systems**
 - _____ All pipes must maintain a minimum 0.5% slope.

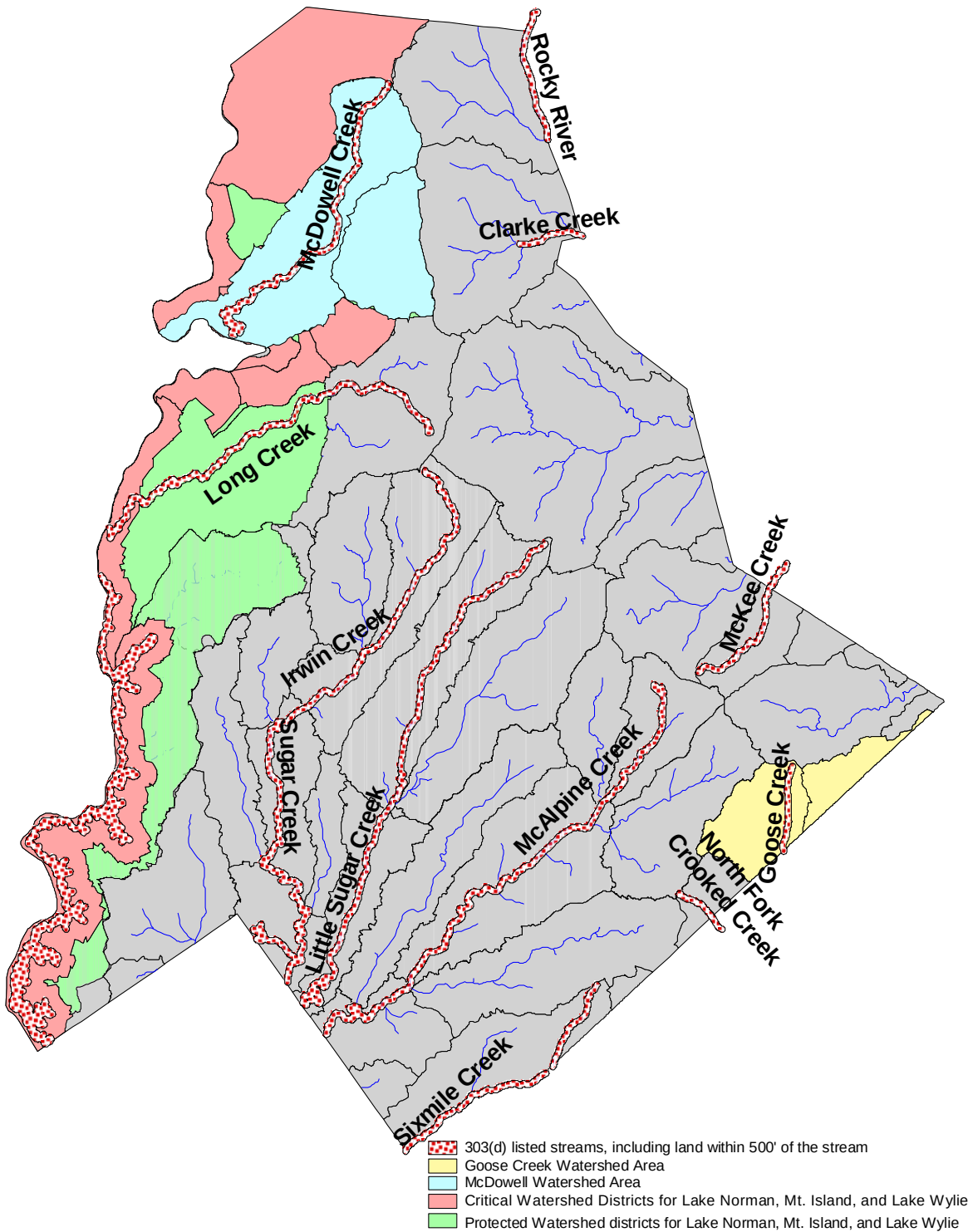
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- _____ All pipe network nodes labeled with spot elevations.
- _____ Adequate access for maintenance at control structures (cleanouts).
- _____ Upper reaches of system provided with access points and proper ventilation.
- _____ Discharge point for 50-year storm is evaluated.

_____ **Regional Detention Basins**

- _____ Drainage Area map with clear delineation of parcels served.
- _____ Pipe systems/channels from site to regional facility/floodway sized for 10-yr storm water runoff.
- _____ Permanent Detention Easements labeled on plan [ESAPM #314.1]
- _____ Off-site detention requires Permanent Detention Easement recorded at Mecklenburg County's Register of Deeds. (Easements must be reviewed by Mecklenburg County LUESA)
- _____ BMP Design (refer to current Charlotte-Mecklenburg BMP Design Manual)

Attachment 1: Enhanced Erosion Control Requirement Areas





DRIVEWAY PERMIT APPLICATION

This application is required for all new connections to Town maintained streets except an individual single family residence.

Location of Property Requesting New or Modification of Access: _____

Nearest Public Street Intersection: _____

Development Name: _____

Proposed Location and Type of Driveway(s):

Driveway #1 (circle all that apply) **New** **Existing** **Residential** **Commercial/Other**

Exact Distance and Direction to Nearest Public Street Intersection: _____

Driveway #2 (circle all that apply) **New** **Existing** **Residential** **Commercial/Other**

Exact Distance and Direction to Nearest Public Street Intersection: _____

Show proposed driveway(s) on attached sketch or site plan. This plan must show details of proposed driveways including proposed width and radius, pipe length and size, adjacent property lines, driveways on adjacent property, existing and proposed buildings, and parking areas and roadway features.

-
- I, the undersigned property owner, request access and permission to construct driveway(s) or street(s) on public right-of-way at the above location.
 - I agree that no signs or objects will be placed on or over the public right-of-way other than those approved by the Town.
 - I agree that the driveway(s) or street(s) will be constructed as shown on the attached plans.
 - I agree that the driveway(s) or street(s) as used in this agreement include any approach tapers, storage lanes or speed change lanes as deemed necessary by the Town Engineer.
 - I agree that if any future improvements to the roadway become necessary, the portion of driveway(s) or street(s) located on public right-of-way will be considered the property of the Town of Huntersville, and I will not be entitled to reimbursement or have any claim for present expenditures for driveway or street construction.
 - I agree that this permit becomes void if construction of driveway(s) or street(s) is not completed within 6 months.
 - I agree to construct and maintain the driveway(s) or street(s) in a safe manner so as not to interfere with or endanger the public travel.
 - I agree to provide during construction proper signs, signal lights, flaggers and other warning devices for the protection of traffic in conformance with the current "Manual on Uniform Traffic Control Devices for Streets and Highways" and Amendments or Supplements thereto. Information as to the above rules and regulations may be obtained from the Town Engineer.
 - I agree to indemnify and save harmless the Town of Huntersville from all damages and claims for damage that may arise by reason of this construction.
 - I agree that the Town of Huntersville will assume no responsibility for any damages that may be caused to such facilities, within the highway right-of-way limits, in carrying out its construction.
 - I agree to provide a Performance and Indemnity Bond in the amount specified by the Town Engineer for any construction proposed on the Town of Huntersville maintained system.
 - **I AGREE TO NOTIFY THE TOWN ENGINEER IN WRITING WHEN THE PROPOSED WORK BEGINS AND WHEN IT IS COMPLETED.**

I agree that the driveway(s) described in this permit application will be constructed according to the approved driveway permit.

Property Owner Name: _____ Phone: () - _____

Property Owner Mailing Address: _____

Applicant’s Signature: _____ Date: _____

Application Approved by Town Engineer: _____ Date: _____
(Signature)

- Instructions: Town of Huntersville Driveway Permit Applications Must Include:
- 1. Two original signed Driveway Application Forms.
 - 2. Two sets of site plans (or detailed sketch on separate page).
 - 3. Mail or hand deliver application forms and site plans to Town Engineer’s attention to:
105 Gilead Road, Ste 300, Huntersville, NC 28078.

If the new driveway is to be located on a state maintained road, contact the North Carolina Department of Transportation at 704-596-6900.

Driveways on all thoroughfare roads must be at least 20 feet from side property line and 40 feet from existing driveways on adjacent property.

(For Official Use Only)

Date Application Received:_____	Application Received by:_____
Permit Fee: \$200	Permit Fee Received by:_____
Is Bond Required:_____	
Bond Amount: _____	Bond Received by:_____
Inspector:_____	Date Inspected:_____



ROAD NAME _____ SUBDIVISION _____ NEAREST INTERSECTION _____

TOWN OF HUNTERSVILLE

RIGHT OF WAY ENCROACHMENT AGREEMENT
FOR ALL ENCROACHMENTS ON TOWN
MAINTAINED ROADWAYS

-AND-

THIS AGREEMENT, made and entered into this the _____ day of _____, 20____, by and between the Town of
Huntersville Engineering & Public Works, party of the first part; and _____ party of the second part,

WITNESSETH

THAT WHEREAS the party of the second part desires to encroach on the right of way of the public road designated as _____, located with the following:

PROJECT DESCRIPTION _____

WHEREAS, it is to the material advantage of the party of the second part to effect this encroachment, and the part of the first part in the exercise of authority conferred upon it by statute, is willing to permit the encroachment within the limits of the right of way as indicated, subject to the conditions of this agreement;

NOW, THEREFORE, IT IS AGREED that the party of the first part hereby grants to the party of the second part the right and privilege to make this encroachment as shown on attached plan sheet(s), specifications and special provisions which are made a part hereof upon the following conditions, to wit:

That the said party of the second part binds and obligates himself to install and maintain the encroaching facility in such safe and proper condition that it will not interfere with or endanger travel upon said highway, nor obstruct nor interfere with the proper maintenance thereof, to reimburse the party of the first part for the cost incurred for any repairs or maintenance to its roadways and structures necessary due to the installation and existence of the facilities of the party of the second part, and if at any time the party of the first part shall require the removal of or changes in the location of the said facilities, that the said party of the second part binds himself, his successors and assigns, to promptly remove or alter the said facilities, in order to conform to the said requirement, without any cost to the party of the first part.

That the party of the second part agrees to provide during construction and any subsequent maintenance proper signs, signal lights, flagmen and other warning devices for the protection of traffic in conformance with the latest Manual on Uniform Traffic Control Devices for Streets and Highways and Amendments or Supplements thereto. Information as to the above rules and regulations may be obtained from the Town Engineer of the party of the first part.

That the party of the second part hereby agrees to indemnify and save harmless the party of the first part from all damages and claims for damage that may arise by reason of the installation and maintenance of this encroachment.

It is clearly understood by the party of the second part that the party of the first part will assume no responsibility for any damage that may be caused to such facilities, within the highway rights of way limits, in carrying out its construction and maintenance operations.

That the party of the second part agrees to restore all areas disturbed during installation and maintenance to the satisfaction of the Town Engineer of the party of the first part. The party of the second part agrees to exercise every reasonable precaution during construction and maintenance to prevent eroding of soil; silting or pollution of rivers, streams, lakes, reservoirs, other water impoundments, ground surfaces or other property; or pollution of the air. There shall be compliance with applicable rules and regulations of the North Carolina Division of Environmental Management, North Carolina Sedimentation Control Commission, and with ordinances and regulations of various counties, municipalities and other official agencies relating to pollution prevention and control. When any installation or maintenance operation disturbs the ground surface and existing ground cover, the party of the second part agrees to remove and replace the sod or otherwise reestablish the grass cover to meet the satisfaction of the Town Engineer of the party of the first part.

That the party of the second part agrees to assume the actual cost of any inspection of the work considered to be necessary by the Town Engineer of the party of the first part.

That the party of the second part agrees to have available at the encroaching site, at all times during construction, a copy of this agreement showing evidence of approval by the party of the first part. The party of the first part reserves the right to stop all work unless evidence of approval can be shown.

Provided the work contained in this agreement is being performed on a completed highway open to traffic; the party of the second part agrees to give written notice to the Town Engineer of the party of the first part when all work contained herein has been completed. Unless specifically requested by the party of the first part, written notice of completion of work on highway projects under construction will not be required.

That in the case of noncompliance with the terms of this agreement by the party of the second part, the party of the first part reserves the right to stop all work until the facility has been brought into compliance or removed from the right of way at no cost to the party of the first part.

That it is agreed by both parties that this agreement shall become void if actual construction of the work contemplated herein is not begun within one (1) year from the date of authorization by the party of the first part unless written waiver is secured by the party of the second part from the party of the first part.

IN WITNESS WHEREOF, each of the parties to this agreement has caused the same to be executed the day and year first above written.

TOWN OF HUNTERSVILLE

BY: _____
Public Works Director

ATTEST OR WITNESS:

Second Party

*Where applicable, please attach plans or drawings describing the desired encroachment.

APPLICATION IDENTIFICATION		N.C. DEPARTMENT OF TRANSPORTATION STREET AND DRIVEWAY ACCESS PERMIT APPLICATION		
Driveway Permit No.	Date of Application			
County:				
Development Name:				
LOCATION OF PROPERTY:				
Route/Road:				
Exact Distance	☐ Miles ☐ Feet	NSEW ☐☐☐☐		
From the Intersection of Route No. _____ and Route No. _____ Toward _____				
Property Will Be Used For: ☐ Residential /Subdivision ☐ Commercial ☐ Educational Facilities ☐ TND ☐ Emergency Services ☐ Other				
Property: _____ ☐ is ☐ is not within _____ City Zoning Area.				
AGREEMENT				
- I, the undersigned property owner, request access and permission to construct driveway(s) or street(s) on public right-of-way at the above location. - I agree to construct and maintain driveway(s) or street entrance(s) in absolute conformance with the current "Policy on Street and Driveway Access to North Carolina Highways" as adopted by the North Carolina Department of Transportation. - I agree that no signs or objects will be placed on or over the public right-of-way other than those approved by NCDOT. - I agree that the driveway(s) or street(s) will be constructed as shown on the attached plans. - I agree that that driveway(s) or street(s) as used in this agreement include any approach tapers, storage lanes or speed change lanes as deemed necessary. - I agree that if any future improvements to the roadway become necessary, the portion of driveway(s) or street(s) located on public right-of-way will be considered the property of the North Carolina Department of Transportation, and I will not be entitled to reimbursement or have any claim for present expenditures for driveway or street construction. - I agree that this permit becomes void if construction of driveway(s) or street(s) is not completed within the time specified by the "Policy on Street and Driveway Access to North Carolina Highways". - I agree to pay a \$50 construction inspection fee. Make checks payable to NCDOT. This fee will be reimbursed if application is denied. - I agree to construct and maintain the driveway(s) or street(s) in a safe manner so as not to interfere with or endanger the public travel. - I agree to provide during construction proper signs, signal lights, flaggers and other warning devices for the protection of traffic in conformance with the current "Manual on Uniform Traffic Control Devices for Streets and Highways" and Amendments or Supplements thereto. Information as to the above rules and regulations may be obtained from the District Engineer. - I agree to indemnify and save harmless the North Carolina Department of Transportation from all damages and claims for damage that may arise by reason of this construction. - I agree that the North Carolina Department of Transportation will assume no responsibility for any damages that may be caused to such facilities, within the highway right-of-way limits, in carrying out its construction. - I agree to provide a Performance and Indemnity Bond in the amount specified by the Division of Highways for any construction proposed on the State Highway system. - The granting of this permit is subject to the regulatory powers of the NC Department of Transportation as provided by law and as set forth in the N.C. Policy on Driveways and shall not be construed as a contract access point. I AGREE TO NOTIFY THE DISTRICT ENGINEER WHEN THE PROPOSED WORK BEGINS AND WHEN IT IS COMPLETED.				
2004-01 NOTE: Submit Four Copies of Application to Local District Engineer, N.C. Department of Transportation 61-03419 TEB 65-04rev.				

SIGNATURES OF APPLICANT

PROPERTY OWNER (APPLICANT)

WITNESS

COMPANY	_____	NAME	_____
SIGNATURE	_____	SIGNATURE	_____
ADDRESS	_____	ADDRESS	_____
	_____ Phone No. _____		_____

AUTHORIZED AGENT

WITNESS

COMPANY	_____	NAME	_____
SIGNATURE	_____	SIGNATURE	_____
ADDRESS	_____	ADDRESS	_____
	_____ Phone No. _____		_____

APPROVALS

APPLICATION RECEIVED BY DISTRICT ENGINEER

_____	_____
SIGNATURE	DATE

APPLICATION APPROVED BY LOCAL GOVERNMENTAL AUTHORITY (when required)

_____	_____	_____
SIGNATURE	TITLE	DATE

APPLICATION APPROVED BY DISTRICT ENGINEER

_____	_____
SIGNATURE	DATE

INSPECTION BY NCDOT

_____	_____	_____
SIGNATURE	TITLE	DATE

COMMENTS:



PE Certification for Subdivisions and Streets

Subdivision: _____ Phase / Map: _____

Street(s):	_____	Length:	_____
	_____	Length:	_____
	_____	Length:	_____
	_____	Length:	_____
	_____	Length:	_____
	_____	Length:	_____
	_____	Length:	_____
	_____	Length:	_____

A final inspection of the streets, and complete storm drainage system for the above referenced subdivision has been performed by _____. I have reviewed the inspection data and have determined that all public infrastructure has been constructed in accordance with the design drawings approved by Mecklenburg County and the Town of Huntersville on _____ and all subsequent revisions. My observations and testing indicate the subgrade, base, and pavement have been constructed in accordance with the guidelines established by the Town of Huntersville, as verified by the attached test results.

Name: _____

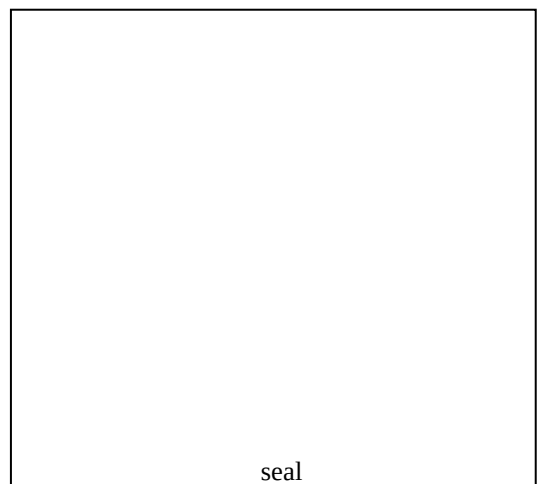
Signature: _____

NC PE#: _____

Date: _____

Received by Town of Huntersville: _____
(initials)

Date: _____





PE Certification for Subdivisions and Streets – Checklist

TYPICAL SECTION

- _____ - Roadway has been constructed in the center of the right of way.
- _____ - Normal crown and/or superelevation have been properly established.
- _____ - Curb & gutter and/or ditches have been constructed properly with no standing water.
- _____ - Backfill of all curb & gutter / sidewalk properly placed and compacted.
- _____ - All slopes have been properly graded.
- _____ - Street Trees have been installed in accordance with approved plan.

ROADWAY CONSTRUCTION

- _____ - Subgrade density tested at minimum of 1 location every 1000' for any section up to 28' wide. 1 test every 300 SY if section is over 28' wide (data included). Nuclear gauge not acceptable for subgrade 100 % of Standard Proctor required for subgrade (top 12"). Embankments only.
- _____ - Subgrade has been proof rolled over the entire section. (pass/fail data & remarks included)
- _____ - Stone depth verified at 1 location per station (data included). Minimum design depth must be achieved over the entire roadway section.
- _____ - Stone density verified by volumetric meter at a minimum of 100% or nuclear gauge at a minimum of 98% compaction with 1 location every 1000' for any section up to 28' wide. 1 test every 300 SY if section is over 28' wide. (data included)
- _____ - Stone base has been proof rolled over the entire section. (Pass/fail data & remarks included)
- _____ - Initial asphalt course(s) was inspected for failures and appropriate repairs made prior to placement of final surface course.

PAVEMENT PLACEMENT

- _____ - Asphalt placement temperatures & mix designs verified.
- _____ - Asphalt depths verified. (minimum depth achieved over entire roadway section)
- _____ - Core samples or nuclear gauge take, minimum of 1 every 2000' or fraction thereof per day. (data / reports included)

DRAINAGE FACILITIES

- _____ - All materials are approved for use. (NCDOT stamp, etc.)
- _____ - All grates/frames/hoods are free of damage and of domestic origin.
- _____ - Masonry drainage structures have been properly constructed – steps installed, inverts poured, pipes cut and grouted, backfill compacted, and free of debris.
- _____ - All pipe inlets, outlets, and channels are properly stabilized and free of erosion.
- _____ - Pipes have been installed with proper cover and slope.
- _____ - Pipe sizes are correct. (per plans)

Name: _____

Signature _____

Date _____

NC PE# _____



seal

TOWN OF HUNTERSVILLE SUBDIVISION PREFINAL CHECKLIST

PROJECT NAME _____

NOTE: STORM DRAIN AS BUILT PLANS ARE REQUIRED TO BE SUBMITTED BEFORE THE ORIGINAL PREFINAL WALK THROUGH.

ITEMS CHECKED	COMPLETED	
	YES	NO
1. CURB AND GUTTER		
2. SIDEWALK		
HANDRAILS WHERE REQUIRED.....		
HANDICAPPED RAMPS.....		
3. CATCH BASINS		
POINTED UP.....		
INVERTED.....		
STEPS.....		
CLEANED.....		
PROPER GRATE.....		
4. STORM DRAIN LINES		
INSTALLED ACCORDING TO APPROVED PLANS.....		
FLASHED.....		
CLEANED.....		
EASEMENT LOCATION.....		
AS BUILTS APPROVED.....		
5. JUNCTION BOXES / MANHOLES		
CORRECT LIDS PLUS RING AND COVER.....		
INVERTED.....		
STEPS.....		
6. DROP INLET / YARD INLET		
INVERTED.....		
STEPS.....		
PROPER GRATE (MEDIAN OR YARD).....		
7. STREET MARKERS		
CORRECT SPELLING.....		
CORRECT BLOCK NUMBERS.....		
CORRECT HEIGHT.....		
CORRECT BLADE SIZE.....		
CORRECT LOCATION.....		
8. STREETS		
PAVEMENT.....		
CORRECT SHOULDER WIDTH.....		
WIDENING STRIP ACCORDING TO DRIVEWAY PERMIT.....		
9. BARRICADES PER STANDARD		

ITEMS CHECKED	COMPLETED	
	YES	NO
10. GUARDRAILS ON STREETS WITH FILL OVER 5 FT. HIGH		
11. CUT AND FILL SLOPES		
STABILIZED ACCORDING TO PLAN.....		
12. RETAINING WALLS		
CONSTRUCTED AND CERTIFIED ACCORDING TO PLAN.....		
13. CHANNELS / GRADED		
PROPER SIDE SLOPES.....		
PROPER TOP AND BOTTOM WIDTHS.....		
RIP RAPPED (PROPER SIZE, THICKNESS, AND APRON SIZE).....		
PROPER LINER MATERIAL / FABRIC.....		
PROPER ARMORING OF INTERSECTING CHANNELS.....		
14. SIDE DITCHES		
PROPER SIDE SLOPES.....		
PROPER TOP AND BOTTOM WIDTH.....		
STABILIZED ACCORDING TO PLAN.....		
15. HEADWALLS AND FLARED END SECTIONS		
INSTALLED ACCORDING TO PLAN.....		
RIP RAPPED APRON INSTALLED ACCORDING TO PLAN.....		
16. SILT FENCE AND POLES REMOVED		
AT INSPECTORS DISCRETION.....		
17. STREET TREES		
CORRECT SPECIES.....		
CALIPER.....		
18. STOP SIGNS AND PARKING SIGNS		
19. STREET LIGHTS where applicable		
20. SITE STABILIZED (Grass seeded, no erosion).....		
21. BMP and LID – Water Quality Structures		
.....		
.....		

**STORM DRAINAGE AS-BUILTS MUST BE SUBMITTED
BEFORE PRE-FINAL INSPECTION.**

LIST APPLICABLE FOR 30 DAYS FROM DATE OF INSPECTION

IN ADDITION TO PUNCHLIST REPAIRS, STORM DRAINAGE AS-BUILTS MUST BE APPROVED AND ROAD MAINTENANCE BOND IN PLACE, WHEN APPLICABLE, BEFORE SUBDIVISION RECEIVES FINAL APPROVAL.

INSPECTOR SIGNATURE _____ DATE _____



Common Punch List Items for Subdivisions and Streets

The following is a list of common items encountered during final inspection of subdivisions and streets prior to Town acceptance. This list is not all inclusive and should serve only as tool to assist the developer in preparation for a final inspection. Other deficiencies may be noted; however, these items should be addressed PRIOR to the final inspection.

Asphalt Pavement

- Good ride quality
- Longitudinal joints in correct location (not under wheel path)
- Transverse and longitudinal joints are smooth.
- Correct cross slope
- No standing water or stains
- Valves and manholes are adjusted to proper height
- No areas of failure (alligator cracking, gouges, etc.)
- Stress cracks must be properly sealed

Pavement Markings

- All pavement marking (stop bars, cross walks, arrows, symbols, lines etc.) have been installed according to the approved plans.
- Retroreflectivity of thermoplastic and paint is adequate.

Signs

- Installed per plans
- Signs compliant with appropriate standard drawings (materials, lettering etc.)
- Correct orientation
- Not damaged

Grading

- No standing water in ditches
- Slopes graded to correct cross slope
- No greater than one inch drop-off at edge of pavement (ditch section roadways)
- Erosion control measures properly removed unless directed otherwise

Drainage

- Drainage structures (catch basins, drop inlets etc.) cleaned out
- Pipes are flush with inside wall of boxes
- Inverts are poured and no standing water
- Pipes sealed properly
- Frames and boxes are sealed where weep was left open
- Grates/frames/hoods are of domestic origin, and are free from damage
- Steps installed
- Pipes are clean
- All pipe inlets and outlets are properly stabilized and free from erosion
- Channels and swales are properly stabilized and free from erosion and there is no standing water

Concrete

- Sidewalk is installed per plans (no missing sections etc.)
- Sidewalk has good finish, joint spacing, no cracks, no footprints, or tripping hazards (raised panels, sewer cleanouts etc.)
- Curb and gutter has good finish, joint spacing and free of standing water
- Curb cracks have been properly sawed and sealed
- Joints with three or more cracks, and joints with excessive damage have been replaced
- Handicapped ramps are in correct locations, have been installed correctly, and have truncated domes
- All forms have been removed and the concrete has been properly backfilled and stabilized

General

- Street trees have been installed per plan and are alive.
- Street lights have been installed per plan
- Guardrail and signage has been installed per standard drawings at the end of stub streets



Final Inspection Request Form

Please note, form must be fully completed. Incomplete forms will not be accepted.

Final Inspection Request

As the authorized agent, I submit this form as my formal request for the Town of Huntersville to begin the Final Inspection process for:

Subdivision Name: _____

Phase: _____ **Map:** _____

Type of Bond currently posted: *(check one)*

____ Subdivision Bond Expires: (date) _____

____ Roadway Maintenance Bond Expires: (date) _____

As a prerequisite to Final Inspection, Storm Drainage and/or BMP as-built drawings for this subdivision **MUST** be approved by Mecklenburg County LUESA.

As-Built Drawings approved on (date) _____

I hereby certify that to the best of my knowledge, with the exception of specific items resulting from the requested final inspection, which will be satisfactorily completed prior to any bond release, all public improvements for the subdivision map indicated above have been constructed in accordance with the approved preliminary plan and corresponding to a record map filed in the office of Mecklenburg County Register of Deeds. I further certify that all work conforms to the construction standards of the Town of Huntersville. I have completed repairs within the subdivision map in accordance with the Town of Huntersville Common Punch List Items For Subdivisions and Streets, and understand that, in the opinion of the inspector, should excessive deficiencies remain, the Final Inspection will be terminated.

Authorized Agent Name: _____

Signature: _____

Mailing Address: _____

City, State, Zip Code: _____

Telephone Number: _____

Email: _____ **Date:** _____

Submit Completed form to: Town Construction Engineer
Town of Huntersville Engineering and Public Works
Post Office Box 664
Huntersville, North Carolina 28070
Telephone: 704-875-7007 Fax: 704-948-0111

TOWN OF HUNTERSVILLE
APPLICATION FOR STREET MAINTENANCE ACCEPTANCE

I hereby certify, as the owner, that to the best of my knowledge the improvements in the _____ subdivision have been constructed in accordance with the preliminary plan and are shown on a record map filed in the Mecklenburg County Register of Deeds Office. All work conforms to the construction standards of the Town of Huntersville Subdivision Ordinance and the Town’s Street Acceptance Policy.

I, therefore, request that the Town of Huntersville consider the following streets in the _____ subdivision for maintenance acceptance. I understand that acceptance of said streets shall be made only by the Huntersville Board of Commissioners in accordance with the Town’s Street Acceptance Policy.

	Street Name	To	From	Approximate Length (in ft.)
1.				
2.				
3.				
4.				
5.				
6.				
7.				
8.				
9.				
10.				

(Attach additional pages if necessary)

Signature of Owner

Date

Address

Phone Number

***TOWN OF HUNTERSVILLE
RESIDENTIAL STREET LIGHTING POLICY***

NEIGHBORHOOD LIGHTING

All Town of Huntersville or DOT maintained neighborhood streets are eligible to be considered for Town maintained street lights. Residents of neighborhoods, or individual streets may submit requests for street lighting. However, the Town of Huntersville may authorize the installation of lights on residential streets with or without a request from residents. Street lights will be installed as funds are available.

RESIDENTIAL NEIGHBORHOOD LIGHTING

Neighborhood lights may be installed on request by the public to the extent funds are approved. For an individual light, the style of fixture to be installed must be consistent with the style of fixture already installed in the neighborhood. Any cost differential from the amount charged to the Town for a standard fixture and a decorative fixture must be paid to the Town, by the requester, as a one-time-only “buy down” charge, prior to authorization.

NEW SUBDIVISION LIGHTING

Subdivision developers are responsible for submitting street light installation plans and funding the installation of street lights for new subdivisions in accordance with the Huntersville Subdivision Ordinance. Street light plans are reviewed and approved by the Town during the subdivision review process. **When streets have been completed and accepted by the Town of Huntersville, the lights for residential developments can then be turned over to the Town, as funds are available.**

REQUEST PROCEDURE

For neighborhood street lights, individual street lights, or to have the Town take over billing, neighborhoods must complete and submit a request form provided by the Town of Huntersville.

The Town will record and log each request at the time it is submitted to the Town Manager’s office. Streetlights are installed on a chronological basis, as funding is available.

After the survey has been logged, a financial spreadsheet estimating the cost of the project shall be completed by the Town Manager, or designated official.

After the request is received and reviewed by Town staff, the contact person on the request form will be notified of any action regarding their request.

For neighborhood lighting requests, the neighborhood will be provided with a street lighting design for signature and approval. At least 75% of the residents on the streets are required to approve the plan before authorization will be given to the Utility Company to install streetlights. It is recommended that every resident review the street lighting design whether or not that signature is required. Only one signature per household is required.

When a completed petition is submitted to the Town Manager's office, streetlights will be installed, as funding is available.

INSTALLATION STANDARDS

Standard streetlight installations will be on wooden poles with overhead wiring

Intersections should get special attention when planning street light locations. If possible, street lights will be installed on the sidewalk side of the street.

The location of an individual light must be consistent with an overall lighting plan for the street. If there is any problem with the location requested, the applicant will be contacted to select an alternate location.

The type of lights installed on a street or in a development should be consistent, unless approved by the Town Manager.

New poles will be located on public right of way in a manner so as not to interfere with pedestrian or vehicular traffic.

The Town of Huntersville will determine final placement, size and type of lighting.

DECORATIVE LIGHTING

As an option, neighborhoods or homeowner associations may request decorative light fixtures. Any cost differential from the amount charged to the Town for a standard fixture and a decorative fixture must be paid to the Town, by the requester, as a one-time-only "buy down" charge, prior to authorization. This "buy down" charge applies to requests for new decorative streets lights or to take over the billing for existing decorative lights.



Street Lighting Request Form

Name: _____ Date: _____

Address: _____

Telephone: _____ E-mail: _____

Neighborhood/Subdivision Name: _____

Type of Request (check one):

New Neighborhood Street Lighting ☐ Individual Street Light ☐ Take over Billing ☐

Type of Lighting Requested (check one):

Standard (cobra head light on wood pole) ☐ Decorative (anything other than Standard) ☐

List below the number of lights requested by street and the specific location of the light(s) requested. For billing takeover requests or for requests of more than one street light, please attach a sketch of the requested locations. See Example below.

# of Lights	Name of Street	Between
1	First Street	Church St and Walker St or house #101 and house #151
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

Return forms to: Department of Engineering and Public Works
105 Gilead Road, Suite 300
PO Box 664
Huntersville, NC 28070
Fax: 704-992-5528 or mbuchanan@huntersville.org